

› Chapter 2

Competitive markets: Demand and supply

BEFORE YOU START

- Over time, you see prices of goods and services change. Why do you think sellers of goods and services change their prices?
- When you buy goods and services, do you think your choices are always in your best self-interest?
- How do consumers and producers make choices in trying to meet their economic objectives?

In this chapter we examine what lies at the heart of every market-based economy: the forces of demand and supply. We will then see how the interactions of demand and supply arrive at equilibrium market prices, and we will study the special qualities of free competitive markets. This chapter will also examine some recent developments in the fields of consumer and producer behaviour that provide novel explanations on how these fundamental decision-makers make choices.

Photo: Fresh vegetables for sale

2.1 Introduction to competitive markets

LEARNING OBJECTIVES

After studying this section you will be able to:

- define all the terms appearing in **orange bold** in the text (AO1)
- outline the meaning of a **competitive market** (AO1)

Markets

The nature of markets

A market originally was a place where people gathered to buy and sell goods. Such markets still exist today, for example cattle markets, fish markets, fruit and vegetable markets and flea markets, involving a physical meeting place where buyers and sellers meet face to face.

> The term **market** has since evolved to include any kind of arrangement where buyers and sellers of goods, services or resources are linked together to carry out an exchange.

The market may be in a specific place (such as a vegetable market), or it may involve many different places (such as the oil market). Buyers and sellers may meet (say, in a shop), or they may never meet, communicating by fax, phone, internet, classified ads, or any other method which allows them to convey information about price, quantity and quality.

A market can be local, where the buyers and sellers originate from a local area; it may be national, in which case the buyers and sellers are from anywhere within a country; or it may be international, with buyers and sellers from anywhere in the world. For example, small neighbourhood bakeries produce and sell bread and other baked goods for the local community – this is a local market. Local takeaway restaurants also produce for the local market. The labour market, on the other hand, tends to be mostly a national market. By contrast, the world oil market includes oil producers in different countries, and buyers of oil virtually everywhere in the world, as well as wholesalers, retailers and other intermediaries involved in buying and selling oil around the world.

Goods and services are sold in product markets, while resources (factors of production) are sold in resource markets (factor markets).

TEST YOUR UNDERSTANDING 2.1

1 What is a market?

2 Suggest more examples of local, national and international markets.

The meaning of a competitive market

Competition is generally understood to be a process in which rivals compete in order to achieve some objective. For example, firms may compete with each other over who will sell the most output, consumers may compete over who will buy a scarce product, workers compete over who will get the best jobs with the highest salaries, countries compete over which will capture the biggest export markets.

Beyond this everyday sense, competition in microeconomics occurs when there are many buyers and sellers acting independently, so that no one has the ability to influence the price at which a product is sold. This should be contrasted with *market power* (also known as *monopoly power*) which refers to the control that a seller may have over the price of the product it sells. The greater the market power, the greater is the control over price. On the other hand, the greater the degree of competition between sellers, the smaller their market power, and the weaker is their control over the price.

In this chapter we will study **competitive markets** composed of large numbers of sellers and buyers acting independently, so that no one individual seller or small group of sellers has the ability to control the price of the product sold. Instead, the price of the product is determined by the interactions of many sellers and buyers, through the forces of demand and supply.

2.2 Demand

LEARNING OBJECTIVES

After studying this section you will be able to:

- define all the terms appearing in **orange bold** in the text (AO1)
- explain the law of demand (AO2)
- draw a demand curve (AO4)
- explain the relationship between individual demand and market demand (AO2)
- analyse the non-price determinants of demand as causes of demand curve shifts (AO2)
- using diagrams distinguish between movements along the demand curve and shifts of the demand curve (AO4)
- explain the assumptions that underlie the law of demand (HL only) (AO2)
  - the law of diminishing marginal utility
  - the income and substitution effects

Understanding the law of demand and the demand curve

Demand is concerned with the behaviour of buyers. Consumers (or households) are buyers of goods and services in product markets, whereas firms (or businesses) are buyers of factors of production in resource markets. In our analysis of demand and supply we will focus mainly on product markets and therefore on the behaviour of consumers as buyers (though the same general principles described here apply also to the behaviour of firms as buyers in resource markets).

Individual demand

As buyers, consumers are demanders of those items they wish to buy.

The **demand** of an individual consumer indicates the various quantities of a good (or service) the consumer is *willing and able to buy* at different possible prices during a particular time period, *ceteris paribus*.

A consumer's demand for a good can be presented as a demand schedule, or as a table listing quantity demanded at various prices. Table 2.1 shows a consumer's demand schedule for chocolate bars. When the price of chocolate bars is \$5, the consumer is willing and able to buy two chocolate bars in a week. When the price is \$4, the consumer is willing and able to buy four chocolate bars in a week, and so on.

Price of chocolate bars (\$)	Quantity of chocolate bars demanded (per week)
5	2
4	4
3	6
2	8
1	10

Table 2.1: Demand schedule for a consumer

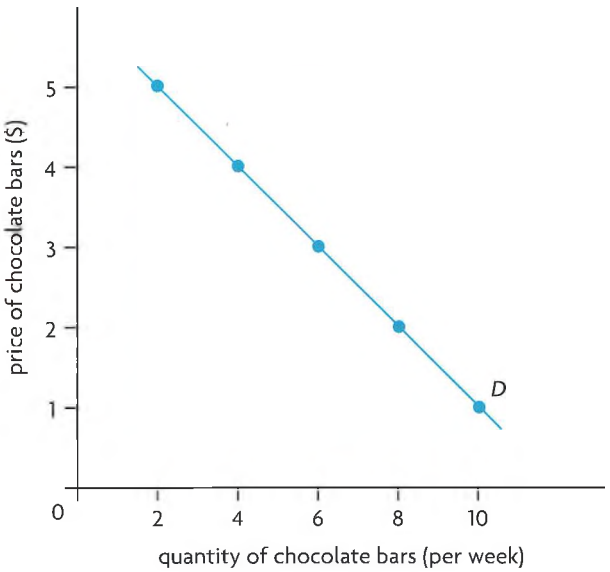


Figure 2.1: Demand curve for an individual consumer

'Willing' means the consumer wants to buy the good; 'able' means that the consumer can afford to buy it. (You may want to buy a Ferrari, but can you afford it? If not, your desire to buy one will not show up as demand for Ferraris. Or, you can afford a Ferrari but you have no desire to own one; again you will not have any demand for Ferraris.)

Ceteris paribus means that all things other than price that can affect how much the consumer is willing and able to buy are assumed to be constant and unchanging (see Chapter 1, and 'Quantitative techniques' chapter in the digital coursebook). In fact, the consumer's demand is affected not only by price, but also by many other things, like income, tastes and prices of related goods. For the moment, we ignore all these and concentrate only on the relationship between the quantity of a good and its price.

The information contained in the demand schedule can be plotted as a graph, shown in Figure 2.1. The price of chocolate bars is plotted on the vertical axis and quantity of chocolate bars on the horizontal axis. The curve in Figure 2.1 is a **demand curve**. Note that even though this is a straight line, it is referred to as a 'curve'.

The demand schedule and demand curve do not tell us anything about how many chocolate bars the consumer will actually buy and what price the consumer will pay. This information will be given to us later through the interaction of demand with supply. The demand information only tells us how many chocolate bars the consumer would be prepared to buy if the price were $5, or $4, and so on.

The law of demand

The demand curve plotted in Figure 2.1 illustrates a very important relationship: as the price of a good falls, the quantity of the good demanded increases. When two variables change in opposite directions, so that as one falls, the other increases, they are said to have a 'negative' (or 'indirect') relationship. This relationship is a 'causal' one, because changes in price *cause* changes in quantity demanded. The negative relationship between price and quantity demanded is known as the law of demand.

According to the **law of demand**, there is a negative relationship between the price of a good and its quantity demanded over a particular time period, *ceteris paribus*: as the price of the good increases, quantity demanded falls; as the price falls, quantity demanded increases, *ceteris paribus*.

The law of demand is most likely to be consistent with your experience. The higher the price of a good, the less of it you are probably willing and able to buy.

From individual demand to market demand

So far we have considered the demand for a good of one individual consumer. *Market demand* shows the total quantities in the market for the good consumers are willing and able to buy at different prices (during a particular period of time, *ceteris paribus*). Market demand is the sum of all individual demands for that good. Figure 2.2 shows how the quantity demanded by consumer A is added to the quantity demanded by consumer B, and so on until all the quantities demanded by all consumers of chocolate bars are added up. (Note that consumer A has a different demand for chocolate bars than consumer B, indicating different preferences). For example, at

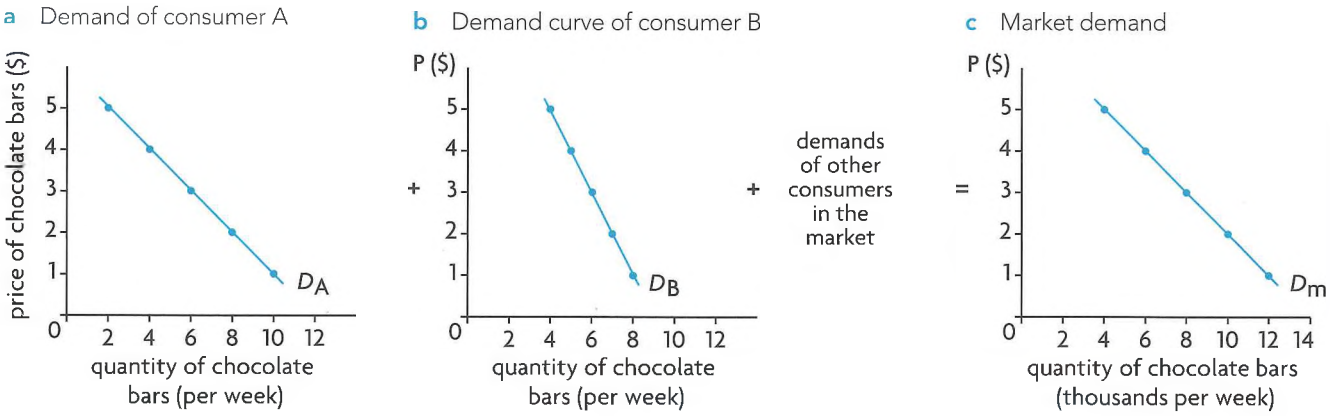


Figure 2.2: Market demand as the sum of individual demands

the price of $4, we add the four bars demanded by consumer A to the five bars demanded by consumer B, and so on to all the quantities demanded by other consumers, to arrive at the sum of 6000 chocolate bars per week. This sum is a point on the market demand curve D_m. When we add individual demands in this way for each of the possible prices, we derive the entire market demand curve D_m, showing the total demand in the chocolate bar market.

Market demand is the sum of all individual demands for a good. The market demand curve illustrates the law of demand, shown by the negative relationship between price and quantity demanded.

Non-price determinants of demand and shifts of the demand curve

The non-price determinants

The **non-price determinants of demand** are the variables other than price that can influence demand. They are the variables assumed to be unchanging by use of the *ceteris paribus* assumption.

Changes in the non-price determinants of demand cause shifts in the demand curve: the entire demand curve moves to the right or to the left. In Figure 2.3, note that the vertical axis is labelled 'P', standing for price, and the horizontal axis is labelled 'Q', standing for quantity. Suppose the original demand curve is given by D_1. If price is P_1, then the demand curve D_1 indicates that quantity Q_1 will be demanded. If the demand curve shifts to the right, to D_2, at the same price P_1 a larger quantity, Q_2, will be demanded. If, on the other hand, the demand curve shifts to the left, from D_1 to D_3, then a smaller quantity, Q_3, will be demanded at the same price P_1.

A rightward shift of the demand curve indicates that more is demanded for a given price; a leftward shift of the demand curve indicates that less is demanded for a given price. A rightward shift of the curve is called an *increase in demand*; a leftward shift is called a *decrease in demand*.

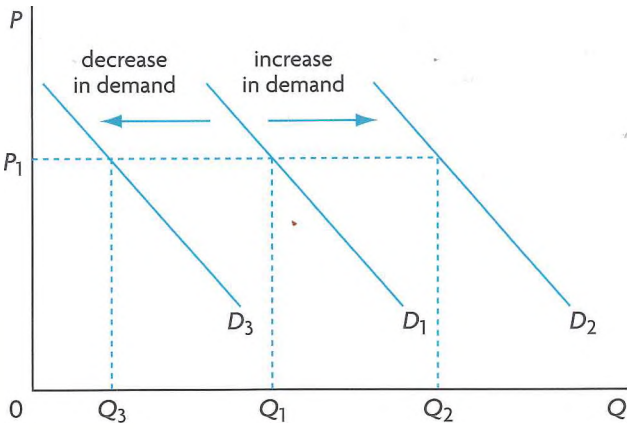


Figure 2.3: Shifts in the demand curve

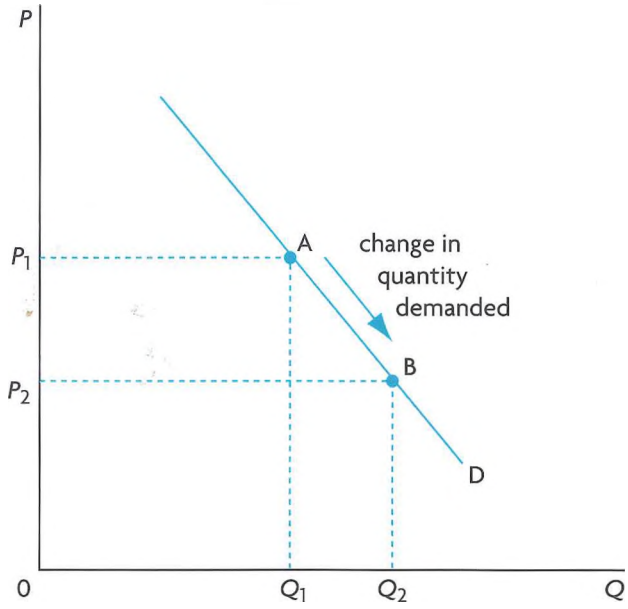
The non-price determinants of market demand include:

- **Income in the case of normal goods.** A good is a **normal good** when demand for it increases in response to an increase in consumer income (demand for the good varies directly with income). Most goods are normal goods. Therefore, an increase in income leads to a rightward shift in the demand curve, and a decrease in income leads to a leftward shift.
- **Income in the case of inferior goods.** While most goods are normal, there are some goods where the demand falls as consumer income increases; the good is then an **inferior good** (the demand for the good varies inversely with income). Examples of inferior goods are second-hand clothes, used cars and bus tickets. As income increases, consumers switch to more expensive alternatives (new clothes, new cars and cars or aeroplanes rather than travelling by bus), and so the demand for the inferior goods falls. Thus an increase in income leads to a leftward shift in the demand curve and a decrease in income produces a rightward shift.
- **Preferences and tastes.** If preferences and tastes change in favour of a product (the good becomes more popular), demand increases and the demand curve shifts to the right; if tastes change against the product (it becomes less popular), demand decreases and the demand curve shifts to the left.
- **Prices of substitute goods.** Two goods are **substitutes (substitute goods)** if they satisfy a similar need. An example of substitute goods is Coca-Cola® and Pepsi®. A fall in the price of one (say, Coca-Cola) results in a fall in the

demand for the other (Pepsi). The reason is that as the price of Coca-Cola falls, some consumers switch from Pepsi to Coca-Cola, and the demand for Pepsi falls. On the other hand, if there is an increase in the price of Coca-Cola, this will result in an increase in the demand for Pepsi as some consumers switch from Coca-Cola to Pepsi. Therefore, for any two substitute goods X and Y, a decrease in the price of X produces a leftward shift in the demand for Y, while an increase in the price of X produces a rightward shift in the demand for Y. In brief, in the case of substitute goods, the price of X and demand for Y change in the same direction (they both increase or they both decrease). Other examples of substitute goods are oranges and apples, Cadbury's and Nestlé chocolate, and milk and yoghurt.

- **Prices of complementary goods.** Two goods are **complements (complementary goods)** if they tend to be used together. An example of complementary goods is computers and computer software. In this case, a fall in the price of one (say, computers) leads to an increase in the demand for the other (computer software). This is because the fall in the price of computers results in a bigger quantity of computers being purchased, and the demand for computer software increases. Therefore, for any two complementary goods X and Y, a fall in the price of X leads to a rightward shift in the demand for Y, and an increase in the price of X leads to a leftward shift in the demand for Y. In the case of complementary goods, the price of X and the demand for Y change in opposite directions (as one increases, the other decreases). More examples of complementary goods are shoes and shoe laces, tennis shoes and tennis rackets, and table-tennis balls and table-tennis rackets. Note that most goods are not related to each other; these are called independent goods. For example, pencils and apples, cars and ice cream, telephones and books are unrelated to one another, and the change in the price of one will have little or no effect on the demand for the other.
- **The number of consumers.** If there is an increase in the number of consumers (demanders), demand increases and therefore the market demand curve shifts to the right; if the number of consumers decreases, demand decreases and the curve shifts to the left. This follows simply from the fact that market demand is the sum of all individual demands.

a A movement along the demand curve, caused by a change in price, is called a 'change in quantity demanded'



b A shift of a demand curve, caused by a change in a determinant of demand, is called a 'change in demand'

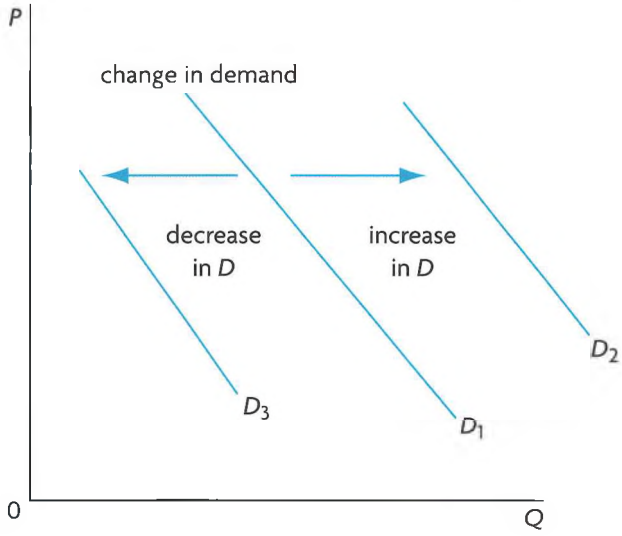


Figure 2.4: Movements along and shifts of the demand curve

Movements along a demand curve and shifts of the demand curve

It is important to distinguish between movements along a demand curve, and shifts of a demand curve. Whenever the price of a good changes, *ceteris paribus*, it leads to a movement along the demand curve. In Figure 2.4(a), if the price falls from P_1 to P_2, the quantity of the good demanded increases from Q_1 to Q_2. A movement along the demand curve from A to B has occurred; this is referred to as an *increase in quantity demanded*. An increase in price gives rise to a *decrease in quantity demanded*, resulting in a movement from B to A.

By contrast, any change in a non-price determinant of demand results in a shift in the entire demand curve, as shown in Figure 2.4(b); this is called a *change in demand*. For example, if there is an increase in the number of buyers, the demand curve shifts rightward from D_1 to D_2; this is called an *increase in demand*, shown in Figure 2.4(b). A decrease in the number of buyers causes a leftward shift of the demand curve from D_1 to D_3; this is called a *decrease in demand*. To summarise:

Any change in price produces a *change in quantity demanded*, shown as a movement on the demand curve. Any change in a non-price determinant of demand leads to a *change in demand*, represented by a shift of the entire demand curve.

TEST YOUR UNDERSTANDING 2.2

1 a Define 'demand'.

b State the law of demand.

c Explain whether the law of demand shows a negative or positive relationship.

d Show the law of demand in a diagram.

e Describe the relationship between individual demand and market demand.

f Distinguish between a 'change in demand' and a 'change in quantity demanded' and explain the cause or causes of each.

CONTINUED

g How would you show the difference between a movement along the demand curve and a shift of the demand curve in a diagram?

h Identify the non-price determinants of demand.

2 Using diagrams, show the impact of each of the following on the demand curve for product A.

a The number of consumers in the market for product A increases.

b Consumer income increases and product A is an inferior good.

c Consumer income decreases and product A is a normal good.

d A news report claims that use of product A has harmful effects on health.

e The price of substitute good B falls.

f The price of complementary good B increases.

Assumptions underlying the law of demand (HL only)

The law of diminishing marginal utility

The law of *diminishing marginal utility* is based on a simple theory of consumer behaviour that explains the negative relationship between price and quantity demanded, or the law of demand.

Utility is the satisfaction that consumers gain from consuming something. It is a subjective concept, because satisfaction is something that depends entirely on personal tastes and preferences, which vary from person to person.

Utility cannot be measured, but for the purposes of developing the theory, we assume that utility is quantifiable (it can be measured in terms of numbers of units), and the unit that we use is the *util*. Utils do not

exist in actual fact, however we assume they exist for the purposes of building the theory.

Total utility is the total satisfaction that consumers get from consuming something, while *marginal utility* is the extra satisfaction that consumers receive from consuming one more unit of a good. Table 2.2 shows how total utility and marginal utility are related to each other. You can see that both total utility and marginal utility are measured in utils.

Table 2.2 shows Anna's preferences when she buys T shirts. The first T shirt provides her with 15 utils of total utility. This is also her marginal utility. When she buys a second T shirt, her total utility increases from 15 utils to 27 utils, but her marginal utility, or extra satisfaction from buying the second T shirt has fallen to 12 utils. The marginal utility of the second unit is simply the total utility of the second unit minus the total utility of the first unit, which is 27 utils – 15 utils = 12 utils. And so on with each extra T shirt she buys. Her total utility increases, but her marginal utility falls. Marginal utility keeps decreasing and at some point it may even become negative, meaning that total utility starts to fall. This pattern illustrates the law of diminishing marginal utility, which is based on the idea that the satisfaction consumers get from consuming more and more units of a good decreases.

The law of diminishing marginal utility explains the law of demand. As we know, according to the law of demand, as the price of a good decreases, quantity demanded increases *ceteris paribus*. The law of diminishing marginal utility shows that if the consumer derives less and less utility from each extra unit of a good consumed, then she or he will buy additional units only if the price of the good falls.

According to the **law of diminishing marginal utility**, as consumption of a good increases, marginal utility, or the extra utility the consumer receives, decreases with each additional unit consumed. This underlies the law of demand, as it shows that a consumer will be willing to buy an additional unit of a good only if its price falls.

Number of T shirts bought per year	Total utility (Utils)	Marginal utility (Utils)
0	0	–
1	15	15
2	27	12
3	36	9
4	42	6
5	45	3
6	45	0
7	42	–3

Table 2.2: Total and marginal utility

The concept of marginal utility is the basic building block of the theory of consumer behaviour, according to which consumers arrange their purchases of many different goods in a way that will allow them to *maximise the total utility* they derive.

The income and substitution effects

The income and substitution effects are an alternative explanation of the law of demand.

As we have seen, the law of demand shows the relationship between the price of a good and quantity of the good demanded, *ceteris paribus*. Any price change causes a change in quantity demanded, shown as a movement along the demand curve. We will now see that the total effect of a price change on quantity demanded can be broken down into two separate effects: the substitution effect and the income effect, with the total effect of a price change being the sum of these two effects.

- The **substitution effect**: If the price of a good falls, the consumer substitutes (buys more) of the now less expensive good. Therefore quantity demanded increases. There is always a negative relationship between price and quantity demanded as a result of the substitution effect: as price falls, the quantity of the good demanded increases; as price increases, quantity demanded falls.

- The **income effect**: Consider again a fall in price. This means that the consumer's *real income* (or purchasing power) has increased. (To understand what this means consider the following example. Say you have \$12 and you want to buy some pencils. When the price is \$4 per pencil, you can buy three pencils. Suppose then that the price of pencils falls to \$3 per pencil. You can now buy four pencils. The sum of money at your disposal (\$12) has not changed, and yet the 'purchasing power' of \$12, or what your money can buy, has increased as a result of the fall in the price of pencils. 'Real income' is the same as 'purchasing power'; it increases as prices fall, and it decreases as prices rise.) Therefore as price falls and real income increases, quantity demanded of the good increases. Once again there is a negative relationship between price and quantity demanded.

The substitution effect and the income effect reinforce each other: a fall in price leads to an increase in quantity demanded, both because of the substitution effect and because of the income effect.

In the case of most goods, whose price is a small fraction of income, the substitution effect is the most important part of the explanation of the law of demand, because being a small fraction of income, the effect that a price change has on real income will be tiny. The income effect becomes important only in the case of purchases of goods whose price take up a large fraction of income.

Note that the above analysis refers to *normal goods*, where income and demand change in the same direction; as real income increases, consumers want to buy more of the good. If you are interested in seeing what happens in the case of *inferior goods* you can read about it in the digital coursebook as Supplementary material.

TEST YOUR UNDERSTANDING 2.3

1 a Using an example, explain the law of diminishing marginal utility.

b How is this law related to the downward sloping demand curve?

2 Explain how the income and substitution effects can explain the downward sloping demand curve.

2.3 Supply

LEARNING OBJECTIVES

After studying this section you will be able to:

- define all the terms appearing in **orange bold** in the text (AO1)
- explain the law of supply (AO2)
- draw a supply curve (AO4)
- explain the relationship between individual supply and market supply (AO2)
- analyse the non-price determinants of supply as causes of supply curve shifts (AO2)
- using diagrams, distinguish between movements along the supply curve and shifts of the supply curve (AO4)
- explain the assumptions that underlie the law of supply (HL only) (AO2)
  - law of diminishing marginal returns
  - increasing marginal cost

Understanding the law of supply and the supply curve

Supply is concerned with the behaviour of sellers, which include firms in the product markets and households in resource markets. As we are focusing on product markets, we will consider the behaviour of firms as sellers (though the same general principles also apply to sellers of factors of production in resource markets).

Individual supply

Firms produce goods and services, and they supply them to product markets for sale. As sellers, therefore, they are suppliers of goods and services.

The **supply** of an individual firm indicates the various quantities of a good (or service) a firm is *willing and able* to produce and supply to the market for sale at different possible prices, during a particular time period, *ceteris paribus*.

A firm's supply of a good can be presented as a supply schedule, or a table showing the various quantities of a good the firm is willing and able to produce and supply at various prices. Table 2.3 shows a firm's supply schedule for chocolate bars. The same information appears as a graph in Figure 2.5, where price is plotted on the vertical axis and quantity on the horizontal axis. The line appearing in the diagram is the **supply curve** of the firm. If the price is $4, the firm supplies 500 chocolate bars in the course of a week; if price were $3, then the firm would supply 400 chocolate bars, and so on.

As in the case of demand, where price is only one thing that determines how much is demanded, so in the case of supply, price is only one thing that influences how much the firm supplies to the market; hence the *ceteris paribus* assumption. For the moment, we will ignore other possible influences on supply and focus only on the relationship between price and quantity.

Price of chocolate bars ($)	Quantity of chocolate bars supplied (per week)
5	600
4	500
3	400
2	300
1	200

Table 2.3: Supply schedule for a firm

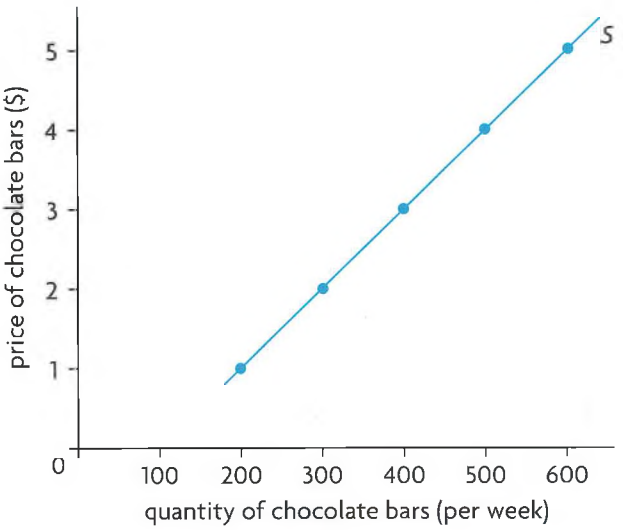


Figure 2.5: Supply curve for a firm

The supply schedule and the supply curve do not tell us anything about how many chocolate bars the firm will actually supply to the market nor what price the firm will receive. The supply information tells us only how many chocolate bars the firm would be prepared to produce and sell if the price were $5, or $4, and so on.

The law of supply

The supply curve in Figure 2.5 illustrates an important relationship: as price increases, quantity supplied also increases. When two variables change in the same direction (as one increases, the other also increases), they are said to have a 'positive' (or 'direct') relationship. This relationship is a 'causal' one, because changes in price *cause* changes in quantity supplied. The positive causal relationship between the two variables, price and quantity supplied, is summarised in the law of supply.

According to the **law of supply**, there is a positive relationship between the quantity of a good supplied over a particular time period and its price, *ceteris paribus*: as the price of the good increases, the quantity of the good supplied also increases; as the price falls, the quantity supplied also falls, *ceteris paribus*.

From individual supply to market supply

Market supply indicates the total quantities of a good that firms are willing and able to supply in the market at different possible prices and is given by the sum of all individual supplies of that good. Figure 2.6 provides an example where at each price, the quantity supplied by firm A is added to the quantity supplied by firm B, and so on, until all the quantities supplied by all firms producing chocolate bars are added up. For example, at the price of $3, firm A supplies 400 bars per week and firm B supplies 300 bars. If we add these quantities together with all the quantities supplied by other firms, we obtain 8000 bars per week, which is a point on the market supply curve, S_m, corresponding to the price of $3. When the firms' supplies are added up this way for each possible price, we derive the market supply curve, S_m.

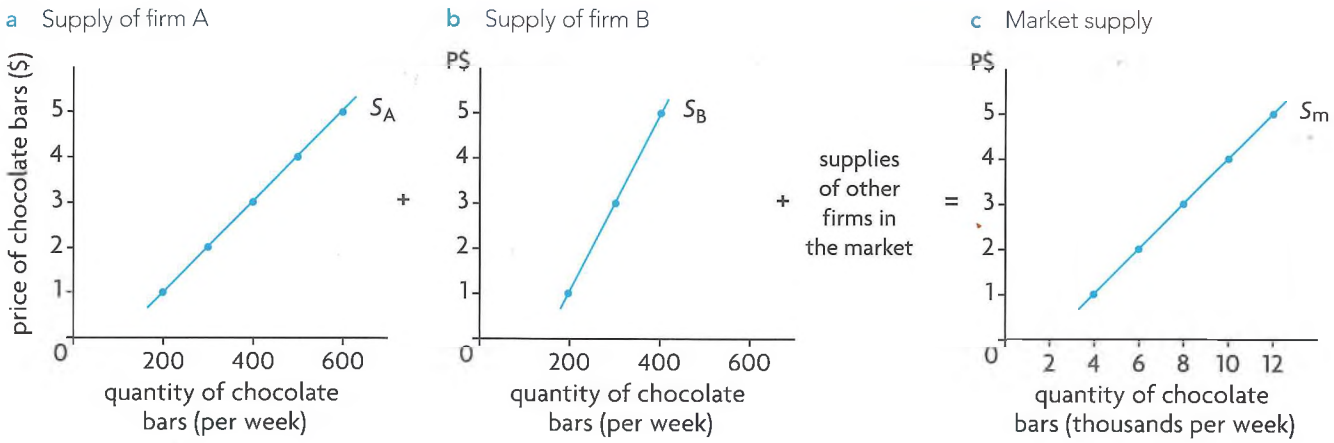


Figure 2.6: Market supply as the sum of individual supplies

Market supply is the sum of all individual firms' supplies for a good. The market supply curve illustrates the law of supply, shown by a positive relationship between price and quantity supplied.

The vertical supply curve

Under certain special circumstances, the supply curve is vertical at some particular fixed quantity, as in Figure 2.7. A vertical supply curve tells us that even as price increases, the quantity supplied cannot increase; it remains constant. The quantity supplied is independent of price. There are two reasons why this may occur:

- There is a fixed quantity of the good supplied because there is no time to produce more of it. For example, there is a fixed quantity of tickets in a theatre, because there is a fixed number of seats. No matter how high the price, it is not possible to increase the number of seats in a short period of time.
- There is a fixed quantity of the good because there is no possibility of ever producing more of it. This is the case with original antiques (for example, Stradivarius violins) and original paintings and sculptures of famous artists. It may be possible to make reproductions, but it is not possible to make more originals.

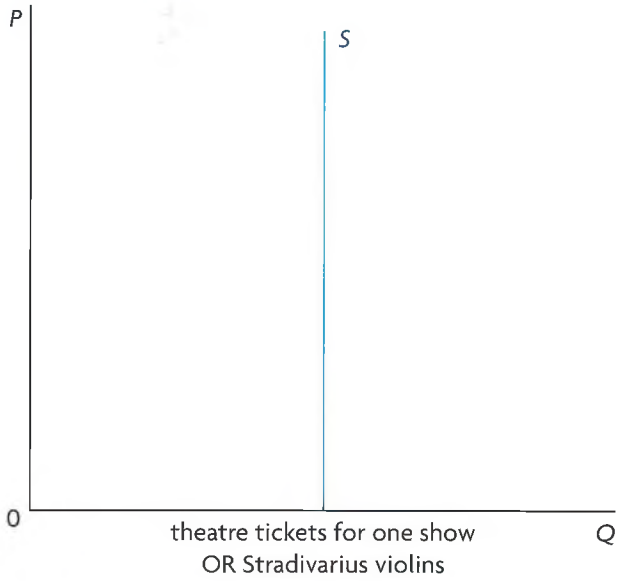


Figure 2.7: The vertical supply curve

Non-price determinants of supply and shifts of the supply curve

The non-price determinants

We now turn to the **non-price determinants of supply**, or the factors other than price that can influence supply. Changes in the determinants of supply cause shifts in the supply curve. A rightward shift means that for a given

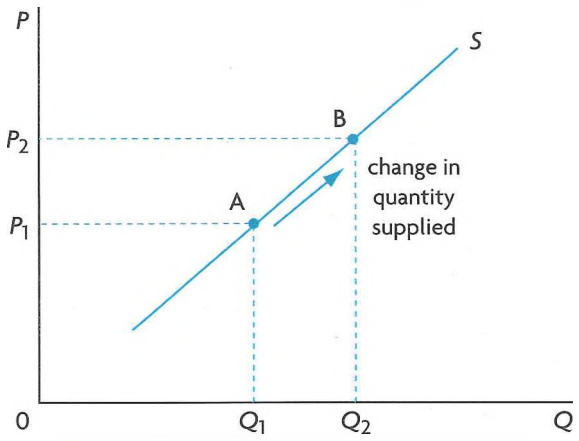
price, supply increases and more is supplied; a leftward shift means that for any price, supply decreases and less is supplied. As Figure 2.8(b) shows, when supply is S_1, quantity Q_1 will be supplied at price P_1. If there is an increase in supply to S_2, at the same price P_1, then Q_2 quantity is supplied. If supply falls to S_3, then Q_3 quantity is supplied at the same price P_1.

A rightward shift of the supply curve indicates that more is supplied for a given price; a leftward shift of the supply curve indicates that less is supplied for a given price. A rightward shift of the curve is called an *increase in supply*; a leftward shift is called a *decrease in supply*.

The non-price determinants of market supply include the following:

- **Costs of factors of production (factor or resource prices).** The firm buys various factors of production (land, labour, capital, entrepreneurship) that it uses to produce its product. Prices of factors of production (such as wages, which are the price of labour) determine the firm's costs of production. If a factor price rises, production costs increase, production becomes less profitable and the firm produces less; the supply curve shifts to the left. If a factor price falls, costs of production fall, production becomes more profitable and the firm produces more; the supply curve shifts to the right.
- **Technology.** A new improved technology lowers costs of production, thus making production more profitable. Supply increases and the supply curve shifts to the right. In the (less likely) event that a firm uses a less productive technology, costs of production increase and the supply curve shifts leftward.
- **Prices of related goods: competitive supply.** Competitive supply of two or more products refers to production of one or the other by a firm; the goods compete for the use of the same resources, and producing more of one means producing less of the other. For example, a farmer, who can grow wheat or corn, chooses to grow wheat. If the price of corn increases, the farmer may switch to corn production as this is now more profitable, resulting in a fall in wheat supply and a leftward shift of the supply curve. A fall in the price of corn results in an increase in wheat supply and a rightward shift of the supply curve.
- **Prices of related goods: joint supply.** Joint supply of two or more products refers to production of goods that are derived from a single product, so that it is not possible to produce more of one without producing more of the other. For example, butter and skimmed milk are both produced from whole milk; petrol (gasoline), diesel oil and heating oil are all produced from crude oil. This means that an increase in the price of one leads to an increase in its quantity supplied and also to an increase in supply of the other joint product(s).

a A movement along the supply curve, caused by a change in price, is called a 'change in quantity supplied'



b A shift of the supply curve, caused by a change in a determinant of supply, is called a 'change in supply'

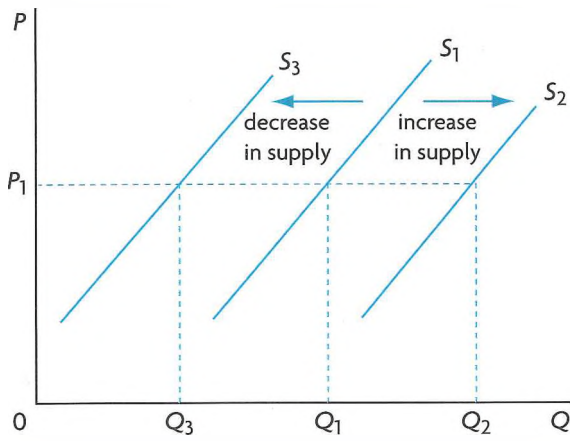


Figure 2.8: Movements along and shifts of the supply curve

- **Producer (firm) price expectations.** If firms expect the price of their product to rise, they may withhold some of their current supply from the market (not offer it for sale), expecting that they will be able to sell it at the higher price in the future; in this case, there is a fall in supply in the present and a leftward shift in the supply curve. If the expectation is that the price of their product will fall, supply increases in the present to take advantage of the current higher price, hence a rightward shift in the supply curve.
- **Taxes (indirect taxes or taxes on profits).** Firms treat taxes as if they were costs of production. Therefore, the imposition of a new tax or the increase of an existing tax represents an increase in production costs, so supply will decrease and the supply curve shifts to the left. The elimination of a tax or a decrease in an existing tax represents a fall in production costs; supply increases and the supply curve shifts to the right. (See Chapter 4.)
- **Subsidies.** A **subsidy** is a payment made to the firm by the government, and so has the opposite effect of a tax. (Subsidies may be given in order to increase the incomes of producers or to encourage an increase in the production of the good produced.) The introduction of a subsidy or an increase in an existing subsidy is equivalent to a fall in production costs, resulting in a rightward shift in the supply curve, while the elimination of a subsidy or a decrease in a subsidy leads to a leftward shift. (See Chapter 4).
- **The number of firms.** An increase in the number of firms producing the good increases supply, resulting in a rightward shift in the supply curve; a decrease in the number of firms decreases supply and produces a leftward shift. This follows from the fact that market supply is the sum of all individual supplies.
- **'Shocks', or sudden unpredictable events.** Sudden, unpredictable events called 'shocks', can affect supply, such as weather conditions in the case of agricultural products, war, or natural/man-made catastrophes. For example, the Louisiana oil spill in 2010 resulted in a decrease in the supply of locally produced seafood.

Movement along a supply curve and shift of the supply curve

Just as in the case of the demand curve, so in the case of the supply curve we distinguish between movements along and shifts of the entire curve. Movements along a supply curve can be caused only by changes in price. In Figure 2.8(a), as price increases from P_1 to P_2, quantity supplied increases from Q_1 to Q_2. There has been a movement along the supply curve from A to B. This is called a *change in quantity supplied*. If there is a change in a non-price determinant of supply, supply will increase or decrease, and the entire curve will shift to the right or to the left, as in Figure 2.8(b). This is called a *change in supply*.

Any change in price produces a *change in quantity supplied*, shown as a movement on the supply curve. Any change in a determinant of supply (other than price) produces a *change in supply*, represented by a shift of the whole supply curve.

TEST YOUR UNDERSTANDING 2.4

1 a Define 'supply'.

b State the law of supply.

c Explain the relationship between price and the quantity supplied.

d Show the law of supply in a diagram.

e Describe the relationship between individual supply and market supply.

f Distinguish between a 'change in supply' and a 'change in quantity supplied' and explain the cause or causes of each.

g Draw two diagrams to show the difference between a movement along the supply curve and a shift of the supply curve.

h Identify the non-price determinants of supply.

CONTINUED

2 Give some examples of goods with a vertical supply curve.

3 Using diagrams, show the impact of each of the following on the supply curve of product A.

- a The number of firms in the industry producing product A decreases.
- b The price of oil, a key input in the production of product A, increases.
- c Firms expect that the price of product A will fall in the future.
- d The government grants a subsidy on each unit of A produced.
- e The price of product B falls, and B is in competitive supply with A.
- f The price of product B increases, and B is in joint supply with A.
- g A new technology is adopted by firms in the industry producing A.

Assumptions underlying the law of supply (HL only)

The law of supply is based on the relationship between production and the costs of production. We begin this section by introducing some new concepts that are essential to understanding the relationship between production and costs.

Law of diminishing marginal returns

The short run and the long run in microeconomics

All firms use *inputs* (or resources, or factors of production) to produce output. The quantities of inputs needed to produce output is determined by a technical relationship between inputs and output. This technical relationship depends on a distinction between the *short run* and the *long run*:

- The **short run** is a time period during which at least one input is fixed and cannot be changed by the firm. When we say fixed, we mean it is unchanging in quantity and quality. For example, if a firm wants to increase output, it can hire more labour and increase materials, tools and equipment, but it cannot easily change the size of its buildings, factories and heavy machinery. The buildings, factories and heavy machinery that are unchanging are *fixed*, whereas the labour and materials are *variable*. As long as the firm has at least one fixed input, it is operating in the short run.
- The **long run** is a time period when all inputs can be changed. Using the example above, in this time period the firm can build new buildings and factories and buy more heavy machinery; it can change all of its inputs. In the long run the firm has no fixed inputs; we say all inputs are *variable*.

Note that the short run and the long run do not correspond to any particular length of time. Some industries may change their fixed inputs over weeks or months while others may do so over many years.

We will make extensive use of the distinction between the short run and the long run in Chapter 7. For now you should note that the ideas developed in this section are concerned *only with the short run.*

The meaning of marginal product

We will now examine the relationship between inputs and output in the short run. Since we are studying the short run, we know the firm has both fixed and variable inputs. For simplicity, let's consider a farm that produces potatoes, where the size of the farm and the agricultural machinery and tools are fixed. The variable inputs are labour and other agricultural inputs (seeds, fertiliser and so on). The only way the farmer can increase the quantity of output in the short run is by increasing the quantity of the variable inputs it uses. We can now distinguish between:

- **total product**, which is the total quantity of output (potatoes) produced by the firm
- **marginal product**, which is the extra or additional output (potatoes) produced by one additional unit of a variable input, which we will assume to be labour; it tells us by how much output increases as labour increases by one worker.

Table 2.4 shows how the total product and marginal product of each worker change as the number of workers increases. We can see that the total product increases continuously. The marginal or extra product added by each worker is simply the addition to total product by each worker. So the first worker adds 20 kilos of potatoes to total product, the second worker adds 30 kilos (= 50 – 20), the third worker 40 kilos (= 90 – 50) and so on.

Number of workers (the variable input)	Total product (kilos of potatoes)	Marginal product (kilos of potatoes)
0	0	-
1	20	20
2	50	30
3	90	40
4	120	30
5	140	20
6	150	0

Table 2.4: Total product and marginal product

Diminishing marginal returns

What is interesting to note in Table 2.4 is that the marginal product of our variable input of labour at first increases, it reaches a maximum with the third worker, and then begins to fall. This pattern of increasing and then falling marginal product is so universally valid it has the status of a law: it is known as the *law of diminishing marginal returns* or for short, the *law of diminishing returns*.

Let's examine the reasoning behind this pattern. When there are zero workers on the land, there is no output at all; it is equal to zero. When one worker is hired, there will be some output and so total product is 20 kilos of potatoes. Marginal product is also equal to 20 kilos. But one worker alone on the farm must do all the ploughing, planting, harvesting, and so on, and so output is quite low. When a second worker is hired, the two workers share the work, and total product increases to fifty kilos, indicating that the output produced by the two together is more than double the output of the first working alone. The additional (or marginal) product due to the second worker (30 kilos) is greater than that of the first (20 kilos). This process is repeated with the addition of the third, and marginal product increases.

With three workers, marginal product is the greatest it can be; when the fourth worker is added, marginal product begins to fall, and falls continuously thereafter. This is the point at which diminishing returns begin. Why does this happen? It happens because of overcrowding: each additional worker has less and less land to work with, and so produces less and less output. Eventually, the conditions on the farm become so crowded that the sixth worker adds zero extra output.

More generally, marginal product will begin to fall at some point not just on a farm with a fixed piece of land, but whenever more and more units of a variable input are added to a fixed input (provided the technology of production is unchanging). For example, in the case of a factory where more and more workers are hired, each extra worker will have fewer and fewer machines and equipment to work with, and so will add less and less output.

Imagine what would happen if diminishing returns did not exist. Using our farm example, it would be possible for food production to increase indefinitely just by continuously adding variable inputs to a fixed piece of land – a clear impossibility!

According to the **law of diminishing marginal returns** as more and more units of a variable input (such as labour) are added to one or more fixed inputs (such as land), the marginal product of the variable input at first increases, but there comes a point when it begins to decrease. This relationship presupposes that the fixed input(s) remain fixed, and that the technology of production is also fixed.

Increasing marginal costs and the firm's supply curve

We will now discover the relationship between diminishing marginal returns, costs of production, and the firm's supply curve.

The meaning of marginal costs

In economics we make frequent use of the concept of **total cost**, which refers very simply to all costs of production incurred by a firm. We can now consider the meaning of a new concept, *marginal cost*.

REAL WORLD FOCUS 2.1

David Ricardo and the end of agricultural output growth

David Ricardo, a famous English economist of the 19th century, believed that agricultural output would eventually stop growing, because as more and more labour and capital inputs were added to land that was fixed in quantity, the additional output of labour and capital would become smaller and smaller until it would no longer be possible for total output to increase further.



Figure 2.9: Chitwan, Nepal. People working in rice field at sunrise

Applying your skills

1 Explain the concept that describes the process Ricardo was referring to.

2 To what extent do you think Ricardo's fears were justified?

3 Suggest reasons for the growth of agricultural output in the real world in spite of a fixed quantity of land.

Marginal cost *is the extra or additional cost of producing one more unit of output.* It tells us by how much total costs increase if there is an increase in output by one unit.

To understand what this means, we can examine Table 2.5. We can see that as the number of units of total product increases, total cost increases, which is what we would expect since more output always involves more cost. Marginal cost shows the addition to total cost arising from the production of one more unit of output. So production of the first unit adds $12 to total cost, the second unit adds $8 to total cost, the third unit $6, and so on.

Total product (number of units)	Total cost ($)	Marginal cost ($)
1	12	12
2	20	8
3	26	6
4	34	8
5	46	12
6	62	16

Table 2.5: Total cost and marginal cost

How marginal costs are related to diminishing marginal returns

You may have noted that marginal cost at first decreases and then increases. In fact this is the opposite of what happens with diminishing marginal returns. This is not a coincidence, in fact the pattern followed by marginal cost is determined entirely by diminishing marginal returns. This can be seen in Figure 2.10.

When marginal product increases, marginal cost decreases; when marginal product is maximum, marginal cost is minimum; and when marginal product falls, marginal cost increases.

Let's examine this relationship more carefully. Remember that at low levels of output, the marginal product of labour, or the output of each extra worker increases. So each worker produces more and more output. Since workers add to costs, and each worker produces more and more output, the cost of producing each additional unit of output falls.

On the other hand, when marginal product or the additional output of each worker becomes less and less, the cost of each extra unit produced (marginal cost) must be increasing.

Similarly, when the additional output produced by an extra worker is the most it can be, then the extra labour

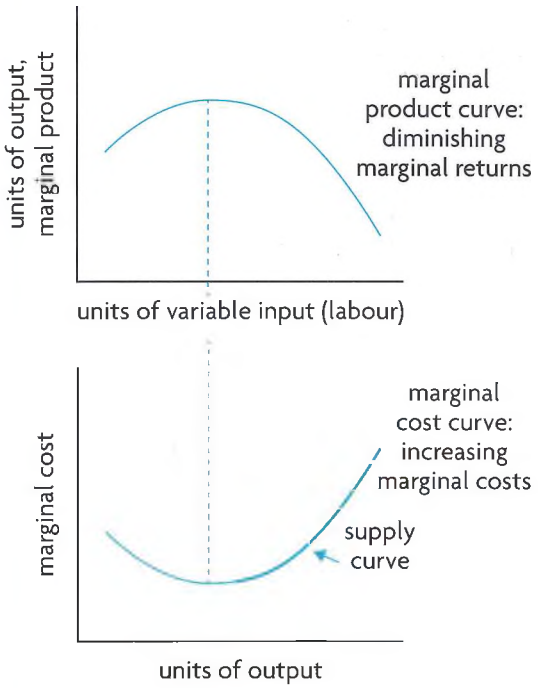


Figure 2.10: Marginal product, marginal cost and the firm's supply curve

cost of producing an additional unit of output is the least it can be.

Marginal cost and the firm's supply curve

As we discussed earlier, the firm's supply curve shows the quantities of a good the firm is willing and able to produce and sell at various prices, *ceteris paribus*. We now know, however, that the firm has costs of production. These costs must be covered through the firm's earnings from selling its output. The firm's earnings per unit of output sold are determined by the price of each unit of the good.

It follows then that the firm will be willing and able to supply some quantity as long as the price is enough to cover its costs. Therefore we can think of the supply curve as showing the price that the firm is willing to accept to produce one more unit of the good. In fact, when marginal cost is increasing, the firm can only produce more output if the price of the good increases to cover the extra cost of each extra unit produced. As a result, a portion of the upward sloping part of the marginal cost curve in Figure 2.10 is the firm's supply curve. This is represented in the bold face portion of the marginal cost curve. The supply curve begins at the point on the marginal cost curve where the firm is making enough revenue so that it is better off producing than shutting down.[1]

> The firm's supply curve is a portion of its marginal cost curve that shows the price-quantity combinations where the extra cost of producing one more unit of output (the marginal cost) is equal to the price of that unit.

TEST YOUR UNDERSTANDING 2.5

1. Explain the law of diminishing marginal returns and outline why this law holds only in the short run.
2. Explain why marginal cost falls as marginal product increases, and why it increases as marginal product falls.
3. Analyse the relationship between a firm's marginal costs and its supply curve.

2.4 Competitive market equilibrium: demand and supply

LEARNING OBJECTIVES

After studying this section you will be able to:

- define all the terms appearing in **orange bold** in the text (AO1)
- draw a diagram to illustrate how demand and supply interact to determine market equilibrium (AO4)
- analyse demand curve and supply curve shifts that give rise to a new market equilibrium, using the concepts of excess demand (shortage) and excess supply (surplus) (AO2)
- draw diagrams to illustrate how shifts in demand and supply give rise to a new equilibrium (AO4)

[1] This occurs when the firm is covering all of its variable costs. An explanation of this topic is beyond the scope of the IB syllabus.

The market demand and market supply for chocolate bars that we considered separately earlier in this chapter show the quantities consumers and firms are *willing and able* to buy and sell at each price. We will now put market demand and market supply together to find out how these interact to determine what happens in the market for chocolate bars.

Market equilibrium

Excess demand (shortages) and excess supply (surpluses)

Figure 2.11 presents the same market demand and supply curves that appeared in Figures 2.2(c) and 2.6(c). The same information appears as a demand schedule and a supply schedule in Table 2.6.

In both Table 2.6 and Figure 2.11 we see that when the price of chocolate bars is $3, quantity demanded is exactly equal to quantity supplied, at 8000 chocolate bars. Note that there is only one price where this can occur. At a higher price, say $4, quantity supplied (10000 bars) is greater than quantity demanded (6000 bars). There is **excess supply**, or a **surplus** of 4000 bars (10000 – 6000). At the even higher price of $5, there is a larger excess supply (surplus) of 8000 bars.

Suppose the price in this market is initially $5. At this price, chocolate producers would be willing and able to produce 12000 bars, but consumers would only be willing and able to buy 4000 bars. What will happen? With unsold output of 8000 bars, producers will lower their price to encourage consumers to buy more chocolate. As the price falls, quantity supplied becomes smaller and quantity demanded becomes bigger. As long as there is a surplus, there will be a downward pressure on the price. The price will keep falling until it reaches the point where quantity demanded is equal to quantity supplied, and the surplus is eliminated.

At a lower price than $3, say $2, quantity demanded (10000 bars) is larger than quantity supplied (6000 bars). There is now **excess demand** or a **shortage** of 4000 bars (10000 – 6000). If price were even lower, at $1, the shortage would be larger, at 8000 bars. At a price of $1, producers would be willing and able to supply only 4000 bars, whereas consumers would be willing and able to buy 12000 bars. Producers will notice that the chocolate bars are quickly sold out, and so begin to raise the price. As they do so, quantity demanded begins to fall and quantity supplied begins to rise. The shortage in the chocolate market exerts an upward pressure on price. The price will keep increasing until the shortage is eliminated; this will happen when quantity supplied is exactly equal to quantity demanded.

Price of chocolate bars ($)	Quantity of chocolate bars demanded (per week)	Quantity of chocolate bars supplied (per week)
5	4000	12000
4	6000	10000
3	8000	8000
2	10000	6000
1	12000	4000

Table 2.6: Market demand and supply schedules for chocolate bars

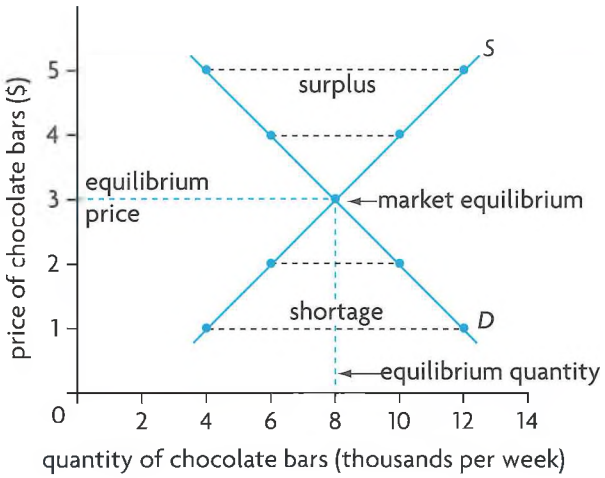


Figure 2.11: Market equilibrium

> The existence of *excess demand* (a *shortage*) or *excess supply* (a *surplus*) in a free market will cause the price to change so that the quantity demanded will be made equal to quantity supplied. In the event of excess demand, price will rise; in the event of excess supply, price will fall.

Market equilibrium

Equilibrium is defined as a state of balance between different forces, such that there is no tendency to change. This is an important concept in economics that we will

encounter repeatedly. When quantity demanded is equal to quantity supplied, there is **market equilibrium**; the forces of supply and demand are in balance, and there is no tendency for the price to change. Market equilibrium is determined at the point where the demand curve intersects the supply curve. The price in market equilibrium is the **equilibrium price**, and the quantity is the **equilibrium quantity**. At the equilibrium price, the quantity consumers are willing and able to buy is exactly equal to the quantity firms are willing and able to sell. This price is known as the *market price*. In the market for chocolate bars in Figure 2.11, the equilibrium price is $3 per chocolate bar, and the equilibrium quantity is 8000 bars. At any price other than the equilibrium price, there is *market disequilibrium*. In a free competitive market, a market disequilibrium cannot last, as demand and supply force the price to change until it reaches its equilibrium level.

competitive market equilibrium, quantity demanded equals quantity supplied, and there is no tendency for the price to change. In a market disequilibrium, there is excess demand (shortage) or excess supply (surplus), and the forces of demand and supply cause the price to change until the market reaches equilibrium.

Changes in market equilibrium

Once a price reaches its equilibrium level, consumers and firms are satisfied and will not engage in any action to make it change. However, if there is a change in any of the non-price determinants of demand or supply, a shift in the curves results, and the market will adjust to a new equilibrium.

Changes in demand (demand curve shifts)

In Figure 2.12(a), D_1 intersects S at point a, resulting in equilibrium price and quantity P_1 and Q_1. Consider a change in a determinant of demand that causes the demand curve to shift to the right from D_1 to D_2 (for example, an increase in consumer income in the case of a normal good). Given D_2, at the initial price, P_1, there is a movement to point b, which results in excess demand equal to the horizontal distance between points a and b. Point b represents a disequilibrium, where quantity demanded is larger than quantity supplied, thus exerting an upward pressure on price. The price therefore begins to increase, causing a movement up D_2 to point c, where excess demand is eliminated and a new equilibrium is reached. At c, there is a higher equilibrium price, P_2, and greater equilibrium quantity, Q_2, given by the intersection of D_2 with S.

A decrease in demand, shown in Figure 2.12(b), leads to a leftward shift in the demand curve from D_1 to D_3 (for example, due to a decrease in the number of consumers). Given D_3, at price P_1, there is a move from the initial equilibrium (point a) to point b, where quantity demanded is less than quantity supplied, and therefore a disequilibrium where there is excess supply equal to the horizontal difference between a and b. This exerts a downward pressure on price, which

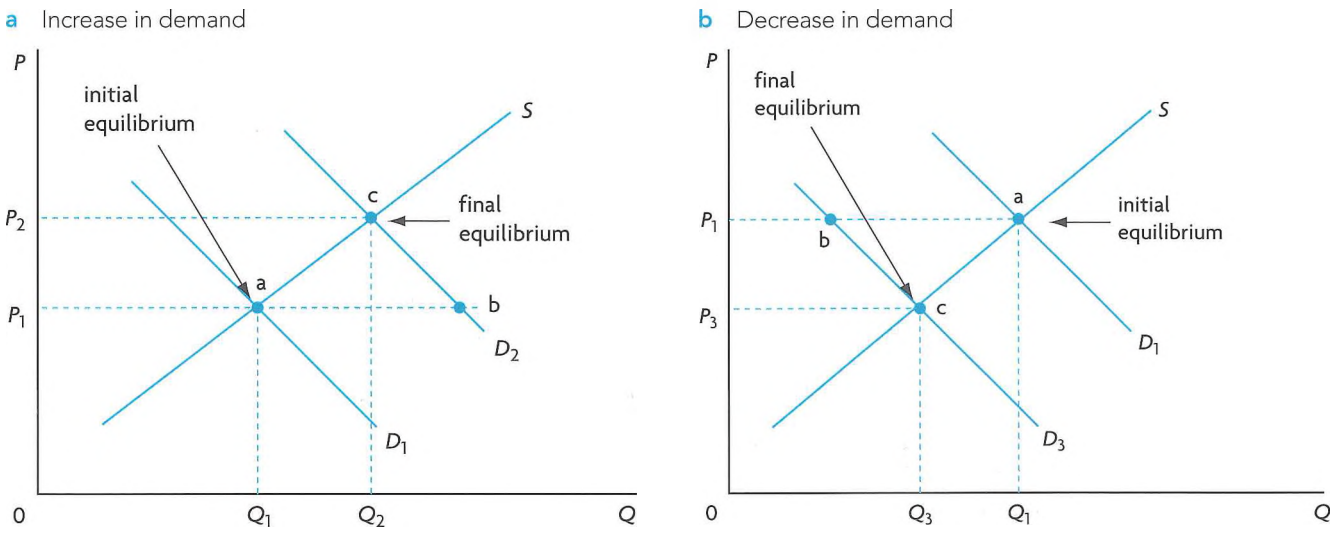


Figure 2.12: Changes in demand and the new equilibrium price and quantity

falls, causing a movement down D_3 to point c, where excess supply is eliminated, and a new equilibrium is reached. At c, there is a lower equilibrium price, P_3, and a lower equilibrium quantity, Q_3, given by the intersection of D_3 with S.

Changes in supply (supply curve shifts)

We now consider supply curve shifts that can arise from changes in the determinants of supply. In Figure 2.13(a), the initial equilibrium is at point a where D intersects S_1, and where equilibrium price and quantity are P_1 and Q_1. An increase in supply (say, due to an improvement in technology) shifts the supply curve to S_2. With S_2 and initial price P_1, there is a move from point a to b, where there is disequilibrium due to excess supply (by the amount equal to the horizontal distance between a and b). Therefore, price begins to fall, and there results a movement down S_2 to point c where a new equilibrium is reached. At c, excess supply has been eliminated, and there is a lower equilibrium price, P_2, but a higher equilibrium quantity, Q_2.

A decrease in supply is shown in Figure 2.13(b) (say, due to a fall in the number of firms). With the new supply curve S_3, at the initial price P_1, there has been a move from initial equilibrium a to disequilibrium point b, where there is excess demand (equal to the distance between a and b). This causes an upward pressure on price, which begins to increase, causing a move up S_3 until a final equilibrium is reached at point c, where the excess demand has been eliminated, and there is a higher equilibrium price P_3 and lower quantity Q_3.

Free and economic goods revisited

You may remember from Chapter 1, Section 1.1, that free goods are goods that are not subject to the condition of scarcity and have a zero opportunity cost, while economic goods are subject to scarcity and have an opportunity cost greater than zero. We are now in a better position to understand the difference between the two by use of demand and supply analysis. Figures 2.14(a) and (b) show a free and economic good respectively. A *free good* is a good for which the quantity supplied is greater than the quantity demanded when the price is zero. Supply is so large relative to demand that there is excess quantity supplied even at a zero price. An *economic good* is a good for which quantity supplied is smaller than quantity demanded when the price is zero. A free good can change into an economic good as a result of a leftward shift in the supply curve (reduced supply) or a rightward shift in the demand curve (increased demand). When demand and supply intersect at a price greater than zero, the good has become an economic good.

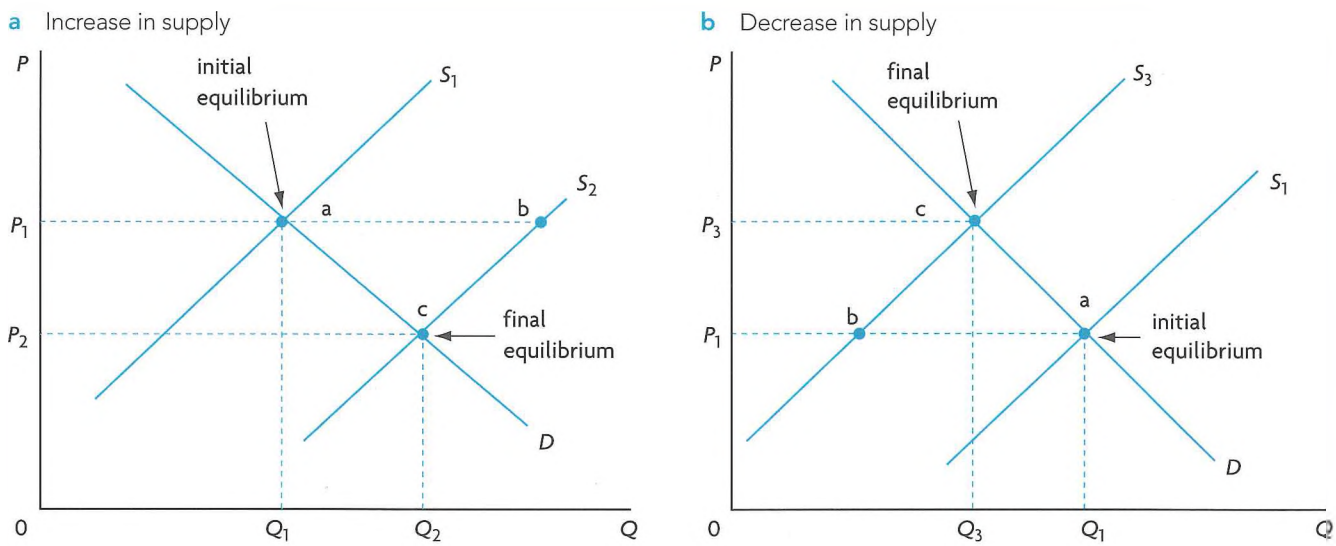


Figure 2.13: Changes in supply and the new equilibrium price and quantity

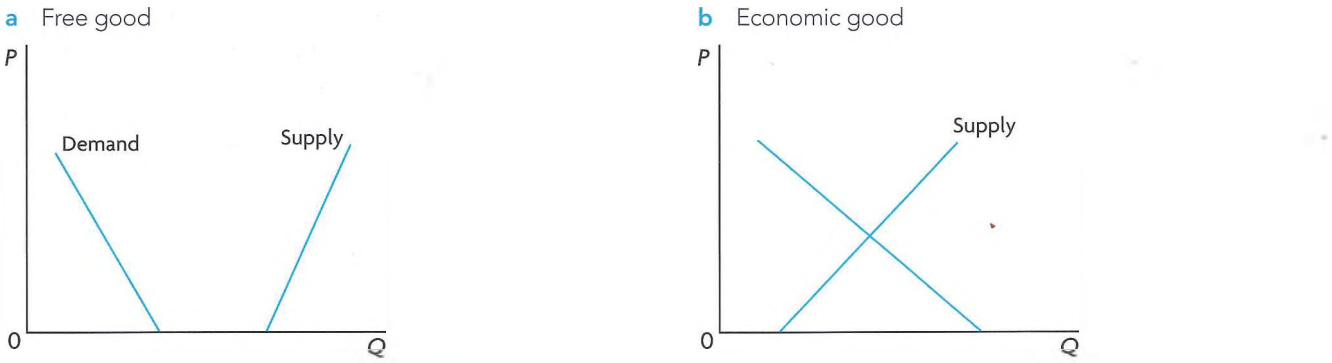


Figure 2.14: Free and economic goods

REAL WORLD FOCUS 2.2

Part I: Avocados and demand for healthy fats

For thousands of years the avocado was held to be a sacred fruit in Central America.Yet this was not so in much of the rest of the world. In 1982, it was reported that an oversupply of avocados in the United States led to a drop in price so large that producers considered selling it as food for dogs (though avocados cause an upset stomach in dogs). An important reason why consumers avoided avocados was their high fat content.

In more recent years, consumers' regard for the avocado has changed dramatically, thanks to the distinction that science has made between the 'good' and 'bad' fat content of foods. Avocados are plentiful in 'good' or healthy (monounsaturated) fats, and in addition contain a host of healthful minerals and vitamins. Consumers around the world are now eating far more avocados than ever before, and their price is rising.

Source: *https://riverheadlocal.com*



Figure 2.15: Green Hass avocado fruit

Part II: Potato chips (crisps) and the supply of potatoes

The summer of 2018 was the hottest in the UK since 1976. As a result there was a significant drop in quantities produced of many crops. By the fall UK farmers noted that potato crops were reduced by one-third. This gave rise to an increase in the price of potato chips (crisps).

Source: *www.independent.co.uk*

Applying your skills

Using diagrams in each case, analyse and explain

1. the effect on avocado prices of consumers' perceptions that avocados are unhealthy due to their high fat content
2. the effect on avocado prices of the information that avocados are very healthy
3. the effect on the price of potato chips (crisps) of extreme weather conditions in the UK.

TEST YOUR UNDERSTANDING 2.6

1 In Figure 2.11, state whether there is excess supply (a surplus) or excess demand (a shortage), and how large this is if price per chocolate bar is:

- a $5,
- b $4,
- c $3,
- d $2, and
- e $1.

2 Use a demand and supply diagram to:

- a show equilibrium price and quantity,
- b show possible disequilibrium prices and quantities,
- c relate disequilibrium prices to excess demand (shortages) and excess supply (surpluses),
- d explain the meaning of 'market equilibrium', and
- e explain the roles of demand and supply in achieving market equilibrium.

3 Use supply and demand diagrams to illustrate the following events.

- a Freezing weather destroys the orange crop and the price of oranges rises.
- b The mass media report on the fat content of cheese and the price of cheese falls.
- c A new technology of production for computers is developed and the price of computers falls.
- d Milk, an input for ice cream production, becomes more expensive and the price of ice cream increases.
- e The mass media report on the health benefits of olive oil and the price of olive oil increases.

4 Assuming a competitive market, use demand and supply diagrams to show in each of the following cases how the change in demand or supply for product A creates a disequilibrium consisting of excess demand or excess supply, and how the change in price eliminates the disequilibrium.

- a Consumer income increases (A is a normal good).
- b Consumer income falls (A is an inferior good).
- c There is an increase in labour costs.
- d The price of substitute good B falls.
- e The number of firms in the industry producing product A increases.
- f A successful advertising campaign emphasises the health benefits of product A.

2.5 The role of the price mechanism and market efficiency

LEARNING OBJECTIVES

After studying this section you will be able to:

- define all the terms appearing in **orange bold** in the text (AO1)
- analyse the price mechanism with respect to its functions including (AO2)
  - resource allocation (signalling and incentive functions)
  - rationing

CONTINUED

- explain the meaning of consumer and producer surplus, as well as social or community surplus (AO2)
- draw a diagram showing maximum social surplus at competitive market equilibrium (AO4)
- explain the conditions for achievement of allocative efficiency at competitive market equilibrium (AO2)
  - social surplus is maximum
  - marginal benefits are equal to marginal costs
- calculate consumer and producer surplus from a diagram (AO4)

Prices determined by the forces of supply and demand in competitive markets are known as the **price mechanism**. The price mechanism serves some important functions that we will now explore.

Functions of the price mechanism

Prices as signals and incentives and the allocation of resources

In Chapter 1 we saw that the condition of scarcity forces societies to make choices. As the production possibilities model illustrated (Figure 1.1), assuming the economy is producing on its production possibilities curve (*PPC*), it must decide on what particular point on the *PPC* it wishes to produce. This involves a choice about *what/how much to produce*, which is a decision on one aspect of resource allocation. It also involves a choice about which of its available resources and in what quantities, it will allocate to produce the combination of goods and services it has chosen. This is a choice on the *how to produce* question of resource allocation.

This brings us to an important question. How does a society make a choice about where to be on its *PPC*? Who decides, and how is this decision carried out? In a market economy, it is simply prices in free markets, resulting from the interactions of demanders and suppliers, which make the decisions and carry them out. We have learned that when markets operate under competitive conditions, market demand and market supply, composed of numerous individual demanders and suppliers, determine equilibrium prices and quantities for goods (and services and resources). At these equilibrium positions, the buying and selling choices of all buyers and sellers are satisfied and are in balance. This market mechanism, working through prices, is known as the *invisible hand of the market*, a phrase first used by Adam Smith, whose contributions to economics were explained in Chapter 1. The invisible hand succeeds in co-ordinating the buying and selling decisions of thousands or millions of decision-makers in an economy without any central authority. The *what/how much to produce* question of resource allocation is answered because firms produce only those goods consumers are willing and able to buy, while consumers buy only those goods producers are willing and able to supply; and the *how to produce* question of resource allocation is answered because firms use those resources and technologies in their production process that they are willing and able to pay for.

How do prices and markets achieve the task of resource allocation?

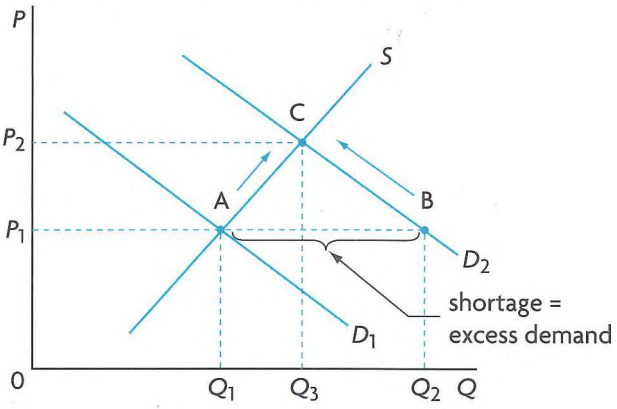
The key to the market's ability to allocate resources can be found in the **signalling** and **incentive** functions of prices in resource allocation. As signals, prices communicate information to decision-makers. As incentives, prices motivate decision-makers to respond to the information.

We will examine the signalling and incentive functions of prices by use of the following two examples.

An example from a product market

Suppose consumers decide they would like to eat more strawberries because of their health benefits (a change in tastes); demand increases and the demand curve shifts to the right from D_1 to D_2 in Figure 2.16(a). At the initial price, P_1, this results in a shortage equal to the difference between Q_2 and Q_1: the quantity demanded Q_2, due to the increase in demand to D_2, is larger than quantity supplied, Q_1. The price of strawberries therefore begins to rise and will continue to rise until the shortage has disappeared. This happens at price P_2 and quantity Q_3, given by the point of intersection of the supply curve with the new demand curve, D_2.

a Adjustment of price to increased demand



b Adjustment of the price of labour to increased labour supply

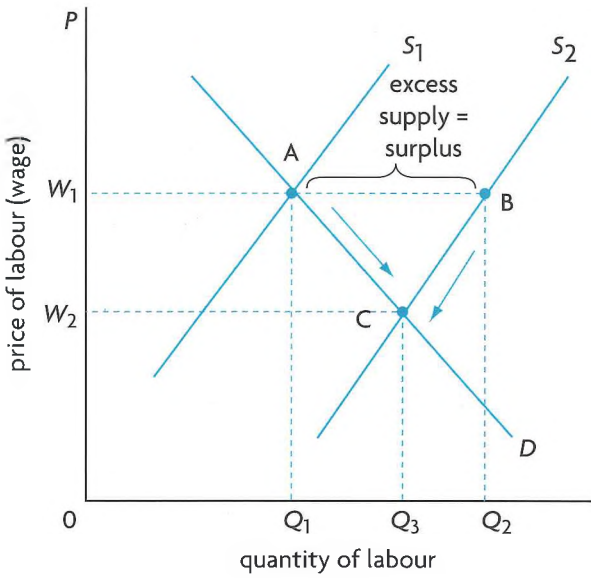


Figure 2.16: Price as a signal and incentive

What has happened? The new, higher price signalled or conveyed information to producers that a shortage in the strawberry market had emerged. The increase in price is also an incentive for producers to increase the quantity of strawberries supplied; at the higher price, strawberry production is more profitable, so producers move along the supply curve from point A to point C, increasing quantity supplied from Q_1 to Q_3. But the new, higher price is a signal and incentive for consumers: it signals that strawberries are now more expensive, and is an incentive for them to buy fewer strawberries. They therefore move along the new demand curve from B to C, buying fewer strawberries than at the original price P_1 (Q_3 is smaller than Q_2). The increase in the price of strawberries resulted in a *reallocation of resources*. More resources are now allocated to strawberry production. (This affects the answer to the *what to produce* question of resource allocation.)

An example from a resource market

The second example is from the labour market (a resource market). The vertical axis in Figure 2.16(b) measures the price of labour (the wage) and the horizontal axis the quantity of labour. Firms are interested in buying labour services, and their demand for labour is given by D. Owners of labour services (workers) supply their labour in the labour market, and the initial supply of labour is shown by S_1.[2]

Assume that because of immigration (foreign workers enter the country), the supply of labour increases, so the labour supply curve shifts to S_2. At the old wage, W_1, there is a surplus of labour shown by the difference between Q_2 and Q_1 of labour. The surplus causes the wage to start falling, and this falls until the surplus has disappeared. The new equilibrium wage is W_2, and the equilibrium quantity of labour Q_3, given by the intersection of D with S_2.

The falling wage has acted as a signal and an incentive. It signalled to firms that there was a surplus in the labour market, and it provided them with an incentive to hire more labour; therefore, they move along the labour demand curve from point A to point C (Q_3 is larger than Q_1). The lower wage is also a signal to workers, providing them with the incentive to move along the new supply curve, S_2, from B to C, where they offer less of their services at the lower wage (Q_3 is less than Q_2). With firms and workers responding to price signals and incentives, there occurred a reallocation of labour resources with firms now producing output with a larger quantity of labour. (This affects the answer to the *how to produce* question of resource allocation.)

[2] The demand curve has the usual downward-sloping shape, because as the wage falls, firms are prompted to hire more labour and so the quantity of labour demanded increases. The supply curve has the usual upward-sloping shape because the higher the wage, the more willing workers will be to supply their labour in the market.

Prices and rationing

Rationing is a method of apportioning or parceling out goods and services among consumers or households. The market mechanism uses *price rationing* for this purpose, which involves the use of prices freely determined in markets. This simply means that whether or not a consumer will get a good is determined by the price of that good. All those who are willing and able to pay the price of a good or service will get it, and all those who are not willing or able to pay the price will not get it. Price, and price alone is the factor that decides. This happens because at the market-determined equilibrium price and quantity, the buying and selling choices of all buyers and sellers in the market for a good are satisfied and are in balance.

Price rationing is simply the market mechanism at work, known as Adam Smith's *invisible hand of the market* (see Chapter 1), which coordinates the countless buying and selling decisions of decision-making units in an economy without any central authority.

When the price mechanism is not working properly, such as in planned economies (see Chapter 1) or if there are price controls (see Chapter 4), then *non-price rationing* systems come into play in order to apportion goods and services among their various users.

The most common non-price rationing method is the waiting line or queue, where consumers get a good on a first come, first served basis. Non-price rationing will be discussed in Chapter 4.

TEST YOUR UNDERSTANDING 2.7

1 a Explain the role of prices in resource allocation.

b Describe how this relates to Adam Smith's *invisible hand*.

2 Consider the market for coffee, and suppose that the demand for coffee falls (because of a fall in the price of tea, a substitute good), leading to a new equilibrium price and quantity of coffee. Using diagrams, explain the role of price as a signal and as an incentive for consumers and for firms in reallocating resources.

3 Consider the labour market, and suppose the supply of labour falls (due to large-scale departure of workers to another country), resulting in a new equilibrium price and quantity of labour. Using diagrams, explain the role of price as a signal and as an incentive for workers (the suppliers of labour) and firms (the demanders of labour) in reallocating resources.

4 Distinguish between price rationing and non-price rationing.

Allocative efficiency in competitive markets

Efficiency, as we saw in Chapter 1, is one of the key concepts running through this course, and very broadly means making the best possible use of resources. We will now study how competitive markets achieve a special type of efficiency called allocative efficiency.

Allocative efficiency refers to producing the quantity of goods mostly wanted by society. Allocative efficiency is achieved when the economy allocates its resources so that the society gets the most benefits from consumption. Since allocative efficiency refers to producing what consumers mostly want, it answers the *what/how much to produce* question in the best possible way.[3]

Reinterpreting demand and supply: marginal benefits and marginal costs

Before we consider allocative efficiency at competitive market equilibrium, we will pause a moment to explore a different interpretation of demand and supply.

The demand curve as a marginal benefit curve

Consumers buy goods and services because these provide them with some benefit, or satisfaction. The greater the quantity of a good consumed, the greater the satisfaction. However, the extra benefit provided by each additional unit increases by smaller and smaller amounts. Imagine you buy a soft drink, which provides you a certain amount of benefit. You are still thirsty, so you buy a second soft drink. Whereas you will enjoy this you will most likely enjoy it less than

[3] At the same time it also answers the *how to produce* question in the best possible way. An explanation of this is beyond the scope of the IB syllabus.

the first; the second soft drink provides you with less benefit than the first. If you buy a third, you will get even less benefit than from the second, and so on with each additional soft drink. The extra benefit that you get from each additional unit of something you buy is called the **marginal benefit** (marginal means extra or additional).

Now remember that the demand curve, as you learned earlier in this chapter, shows the various quantities of a good that a consumer is willing and able to buy at different possible prices, *ceteris paribus*. But it can also be thought of as showing the amount of money that the consumer is willing to pay to get one more unit of the good. Since the extra or marginal benefit that the consumer gets from buying one more unit of the good becomes smaller and smaller as the quantity increases, this means that the price the consumer is willing to pay also gets smaller and smaller as the number of units bought increases.

Since marginal benefit decreases as the quantity of a good consumed increases, consumers will be willing to buy an extra unit of the good only if its price falls. The demand curve can therefore also be called a *marginal benefit (MB) curve*

(Students studying this course at HL will notice that the concept of marginal benefits is very similar to the concept of marginal utility examined earlier. The difference between the two is actually very subtle. Marginal utility as we have seen is something that cannot be measured. Marginal benefit, on the other hand, is interpreted to indicate the consumer's *willingness to pay* for the last or marginal unit bought, which is simply price, and is therefore measurable. The differences between the two concepts are beyond the scope of the IB syllabus.)

The supply curve as a marginal cost curve

In order to be able to produce goods and services, firms have *costs*, which involve payments that must be made in order to acquire factors of production. For example, a farmer must buy seeds and fertilisers, pay workers to work on the land, and so on. The extra cost of producing one more unit of output is called the *marginal cost*. Marginal cost typically increases as the units of output produced increase. (The reasons for this were studied at HL earlier in this chapter in Section 2.3.) This means that the farmer can only produce more output if the price of the good increases to cover the extra cost of each extra unit produced.

The firm's supply curve as you may remember, shows the various quantities of a good that a firm is willing and able to produce and supply at various possible prices, *ceteris paribus*. But in view of the firm's costs of production, the supply curve also shows the price that the firm is willing to accept in order to produce one more unit of the good.

Since marginal cost increases as the quantity of a good produced increases, producers will be willing to produce and sell an extra unit of the good only of its price increases. The supply curve can therefore also be called a *marginal cost (MC) curve*.

Allocative efficiency: *MB* = *MC* at competitive market equilibrium

As we know, market equilibrium occurs at the point of intersection of the demand and supply curves, but depending on how we interpret the demand and supply curves, market equilibrium can be thought of differently. If we interpret the demand curve as a marginal benefit (*MB*) curve, and the supply curve as a marginal cost (*MC*) curve, then market equilibrium occurs where $MB = MC$. The equality of *MB* with *MC* tells us that the extra benefit to society of getting one more unit of the good is equal to the extra cost to society of producing one more unit of the good. When this happens, society's resources are being used to produce the 'right' quantity of the good; in other words, society has allocated the 'right' amount of resources to the production of the good, and is producing the quantity of the good that is mostly wanted by society. *This is none other than allocative efficiency* and is shown in Figure 2.17.

To understand this, consider that if $MB > MC$, then society would be placing a greater value on the last unit of the good produced than it costs to produce it, and so more of it should be produced. If $MC > MB$, then it would be costing society more to produce the last unit of the good produced than the value society puts on it, and so less should be produced. If $MC = MB$, then just the 'right' quantity of the good is being produced.

Putting the above points together, we can conclude that at the point of competitive market equilibrium, where $MB = MC$, the economy achieves allocative efficiency. For allocative efficiency to be achieved for an entire economy, the condition $MB = MC$, must hold in all markets.

Introducing consumer and producer surplus

There is another way we can understand how allocative efficiency is achieved by the competitive market economy; this involves the concepts of consumer surplus and producer surplus.

Consumer surplus

Consumer surplus is defined as the highest price consumers are willing to pay for a good minus the price actually paid. The highest price they are willing to pay is given by the demand curve. The price actually paid is determined at the market equilibrium by supply and demand. Consumer surplus is shown in Figure 2.17 as the shaded area between the demand curve, and the equilibrium price P_e, up to quantity Q_e.

> Consumer surplus is the area under the demand curve and below the price paid by the consumer, up to the quantity purchased.

Consumer surplus indicates that whereas many consumers were willing to pay a higher price to get the good, they actually received it for less. For example, many consumers were willing to pay price P_2 to get quantity Q_a, yet they got Q_a by paying only the lower price P_e. The difference between P_2 and P_e is consumer surplus for quantity Q_a. Similarly, many consumers were willing to pay price P_3 in order to get quantity Q_b, yet they got it by paying only P_e. Again, the difference $P_3 - P_e$ is consumer surplus for quantity Q_b. The same principle applies to all possible prices between the highest price P_1 and the equilibrium price P_e. Therefore, all the consumers who were willing to pay a higher price than P_e to get the good received some benefit over and above what they actually paid for the good. This extra benefit is called consumer surplus.

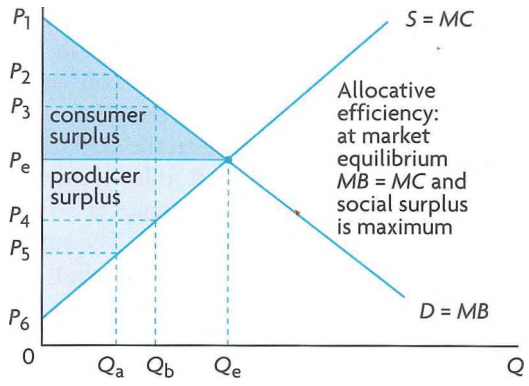


Figure 2.17: Consumer and producer surplus in a competitive market

Producer surplus

Producer surplus is defined as the price received by firms for selling their good minus the lowest price that they are willing to accept to produce the good. The price received is given by the market equilibrium price. The lowest price they are willing to accept is shown by the supply curve. Producer surplus appears in Figure 2.17 as the shaded area between the supply curve, and the equilibrium price P_e, up to quantity Q_e.

> Producer surplus is shown as the area above the firms' supply curve and below the price received by the firm, up to the quantity produced.

As we can see in Figure 2.17, firms that were willing to produce quantity Q_a for price P_5 actually received price P_e. The difference $P_e - P_5$ is producer surplus for quantity Q_a. Similarly, the producer surplus for quantity Q_b is given by the price P_e actually received minus P_4 that the firms were willing to accept for producing Q_b. The same principle applies to all possible prices between the lowest price P_6 and the equilibrium price P_e. Therefore, total producer surplus is shown by the shaded area between the equilibrium price P_e and the supply curve, up to the quantity produced and sold.

Competitive market equilibrium: maximum social surplus and allocative efficiency

At the point of competitive market equilibrium, the sum of consumer and producer surplus is maximum, or

the greatest it can be. To see why, consider what would happen if any quantity less than Q_e were produced in Figure 2.17. If, say, Q_b is produced, the sum of consumer plus producer surplus would be smaller, as this sum would be equal to the shaded area between the demand and supply curves *only up to output* Q_b. It follows, then, that the sum of consumer plus producer surplus is maximised at the point of market equilibrium. The sum of consumer and producer surplus is known as **social surplus** (or **community surplus**).[4]

> At the point of competitive market equilibrium, social surplus, defined as the sum of consumer plus producer surplus, is maximum.

Let's now examine the importance of maximum social surplus at the point of competitive market equilibrium. You will note that competitive market equilibrium is none other than the point where marginal benefit equals marginal cost, or $MB = MC$. Based on our discussion above, we know that when $MB = MC$ there is allocative efficiency. Therefore maximum social surplus is just another way of saying that there is allocative efficiency.

> At the point of competitive market equilibrium shown in Figure 2.17, production of a good occurs where $MB = MC$, which is also *where social surplus, or the sum of consumer plus producer surplus is maximum*. This means that markets are achieving allocative efficiency, producing the quantity of goods mostly wanted by society. Society is making the best possible use of its scarce resources.

When the competitive market realises allocative efficiency, we say that 'social welfare' is maximised. What does this mean? The term **welfare** in economics refers to the amount of consumer and producer surplus. Welfare is clearly maximum when social surplus is maximum, or when $MB = MC$. We can therefore say that:

> In competitive markets, when $MB = MC$, or when social surplus is maximum, social welfare is maximum.

Calculating consumer and producer surplus and social surplus

Figure 2.18(a) is similar to Figure 2.17 only in that it includes figures that allow us to calculate consumer and producer surplus. To see how this is done, note the following two terms.

P intercept = the point on the P axis that is met by a curve. In Figure 2.18(a), the demand curve has a P intercept = 6.5 and the supply curve has a P intercept = 1.

Q intercept = the point on the Q axis that is met by a curve. In Figure 2.18(a), we are not shown any Q intercept. However, in Figure 2.18(b) we see that the supply curve has a Q intercept = 2.

Let's begin with a calculation of consumer and producer surplus in Figure 2.18(a). Consumer surplus, as we know, is the area under the demand curve and below the price paid by consumers, up to the quantity purchased. It is simple to calculate consumer surplus if we think of it as half the area of the rectangle whose one side equals the *P* intercept of the demand curve minus the price paid by consumers, and whose other side equals the number of units purchased.

> **Consumer surplus** =
>
> $$\frac{(P \text{ intercept of D curve minus } P \text{ of consumers}) \times Q \text{ purchased}}{2}$$

$$\text{Therefore consumer surplus} = \frac{(6.5-3)\times 4.5}{2}$$

$$= \frac{3.5\times 4.5}{2} = \$7.88$$

Producer surplus is the area above the supply curve and below the price received by producers, up to the quantity produced. In Figure 2.18(a), it is half the area of the rectangle whose one side equals the price received by producers minus the *P* intercept of the supply curve, and whose other side equals the number of units sold.

> **Producer surplus** =
>
> $$\frac{(P \text{ of producers minus } P \text{ intercept of } S \text{ curve}) \times Q \text{ sold}}{2}$$

[4] Note that the term 'surplus' is used in two senses in this chapter. In one sense it refers to 'excess supply' (Section 2.4) and in the other it refers to benefits received by consumers and producers. You can avoid confusing the two concepts by noting that whenever the term is used on its own, it refers to excess supply.

Therefore:

$$\text{producer surplus} = \frac{(3-1)\times 4.5}{2} = \frac{(2\times 4.5)}{2} = \$4.50$$

Notice that both consumer surplus and producer surplus are given in terms of $, this is because they express a *monetary value* that has been gained by consumers and producers.

It follows then that social surplus, being the sum of consumer and producer surplus = $7.88 + $4.50 = $12.38.

However, it should be noted that the supply curve does not always have a P intercept. This can be seen in Figure 2.18(b), where the supply curve begins on the Q axis, having a Q intercept = 2. In cases like this, the formula given above for producer surplus is not appropriate. Instead, to find the area above the supply curve and under the price up to the quantity produced we need to use the formula for a *trapezium*. A trapezium is a geometrical shape where one pair of opposite sides is parallel but the other pair is not parallel. In Figure 2.18(b), let's call the lengths of the two parallel sides a and b, while the distance between these two sides is called c. The area of a trapezium then is:

$$\text{area of trapezium} = \frac{(a+b)\times c}{2}$$

Therefore considering Figure 2.18(b) we can calculate that:

$$\text{producer surplus} = \frac{(4+2)\times 3}{2} = \frac{6\times 3}{2} = \$9.00$$

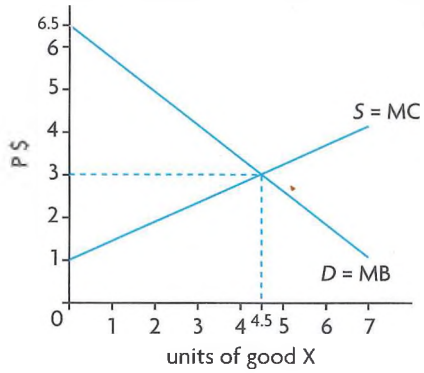
Consumer surplus in Figure 2.18(b), using the formula given earlier is:

$$\text{consumer surplus} = \frac{(6.5-3)\times 4}{2} = \frac{3.5\times 4}{2} = \$7.00$$

Social surplus, being the sum of consumer and producer surplus = $7.00 + $9.00 = $16.00.

As we will discover in later chapters, consumer surplus is always given by the area of a triangle. Producer surplus and social surplus are usually given by the area of a triangle but sometimes they may be the area of a trapezium. Therefore, before you use a formula for a calculation, you must first identify whether you are calculating the area of a triangle or the area of a trapezium.

a Consumer and producer surplus as areas of triangles



b Producer surplus as the area of a trapezium

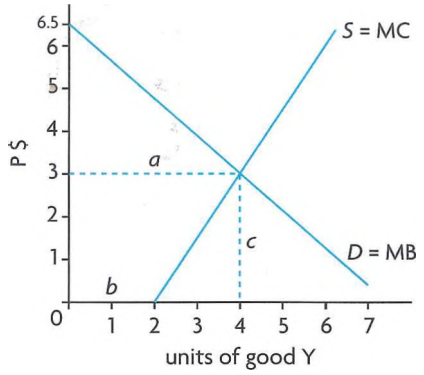


Figure 2.18: Calculating consumer and producer surplus and social surplus

Adam Smith's invisible hand

Adam Smith did not think of consumer and producer surplus, or marginal benefits and marginal costs, but he did come up with the idea that individuals acting in the best self-interest would through the workings of markets, produce an outcome for society that would be in society's best interests. We are now in a far better position to understand the *invisible hand* that was introduced in Chapter 1. It means that the market is able to coordinate the decisions of countless actions of individual economic decision-makers without any central authority, simply through the workings of demand and supply, while at the same time promoting efficiency which encourages the best allocation of scarce resources. This is perhaps the most important contribution of Adam Smith to economic thought.

Allocative efficiency of markets versus government intervention

We have seen that the competitive market succeeds in achieving allocative efficiency, thus ensuring the best

possible use of scarce resources. This idea suggests that there should not be government intervention in markets, as these work very well on their own. However, there are two important issues that arise, calling into question the idea that governments should not intervene.

The first is that efficiency can only arise under a number of very strict and highly unrealistic conditions that are practically never met in the real world. In the real world the market fails to achieve allocative efficiency. Market failures are an important justification for government intervention. When markets fail to achieve allocative efficiency, this means that social surplus is reduced, resulting in what is known as **welfare loss** (also called *deadweight loss*). In other words, social welfare is no longer maximum on account of a portion of it being lost. Welfare loss will be discussed in later chapters.

The second is that the competitive market is unable to provide a satisfactory answer to the *for whom to produce* question, or output and income distribution, thus also inviting some government intervention. The topic of distribution and what can be done to improve outcomes will also be examined in Chapters 12 and 19, as well as in Chapter 6 at HL. It is also discussed in the Theory of knowledge feature 2.1.

These observations do not lessen the significance of the market's potential advantages; they only point out that in the real world, there is a need for government policies to counteract the failings of markets, thus allowing them to realise their potential advantages. There are important reasons why economists study the competitive market extensively. One is that government policies can be assessed with respect to their efficiency consequences (Chapter 4). Another is that it can form the basis for government policies that try to create conditions in the real world that allow actual economies to come closer to achieving economic efficiency (Chapters 5–7). A third is that it provides standards for economic efficiency against which actual outcomes, which are less than perfectly efficient, can be assessed (Chapters 4 and 7).

TEST YOUR UNDERSTANDING 2.8

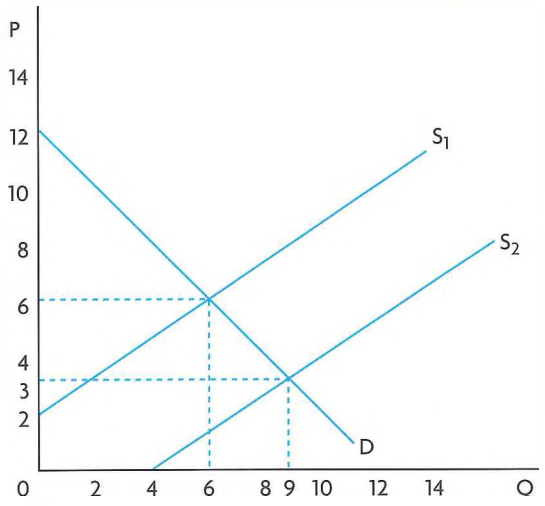
1 Explain the meaning of allocative efficiency.

2 Explain the meaning of
   - a consumer surplus,
   - b producer surplus,
   - c social (or community) surplus, and
   - d maximum 'social welfare'.

3 a Use a demand and supply diagram to illustrate consumer and producer surplus.
   - b Identify the level of output where their sum is maximum.
   - c Explain the condition $MB = MC$, and use your diagram of part (a) to show at which level of output this condition is satisfied.
   - d Outline what the conditions of maximum consumer plus producer surplus, and $MB = MC$ tell us about allocative efficiency, and what they tell us about social welfare.

4 Describe some limitations of the concept of maximum social welfare.

5 a Using the diagram calculate consumer and producer surplus at the initial equilibrium given by S_1 and D.



   - b Suppose the supply curve shifts to S_2. Calculate the new consumer and producer surplus that arise at the new equilibrium.
   - c Calculate the increase or decrease in consumer and producer surplus due to the shift in the supply curve.
   - d Explain what accounts for the changes in consumer and producer surplus in terms of the changes in P and Q.

THEORY OF KNOWLEDGE 2.1

The meaning and implications of maximum social welfare

We must be very careful when interpreting the meaning of *maximum social welfare*. The achievement of *MB* = *MC*, or maximum social surplus, deals with the *what/how much to produce* and *how to produce* questions, and therefore with how a society can realise allocative efficiency.

Even if it is assumed that allocative efficiency is achieved, in which case we have 'maximum social welfare', this tells us nothing about how output and income are, or should be, distributed. The achievement of allocative efficiency is, in fact, consistent with any possible distribution of output or income. This means that we can have an extremely unequal distribution of output and income, where one person in society gets all the output, or a highly equal distribution where everyone gets an equal share, and both of these situations can lead to allocative efficiency. This idea highlights the point that *when we refer to maximum social welfare, we are only talking about making the best possible use of scarce resources, while saying nothing about who gets the benefits of what the resources produce.* For any distribution of income and output, ranging from the most unequal to the most equal, it is possible to have a situation of maximum social surplus and therefore allocative efficiency.

The pursuit of efficiency, dealing with the *what/how much to produce* and *how to produce* questions, is based on social scientific investigation, and tries to discover the most effective ways to increase efficiency in resource allocation. Many economists would argue that this is a matter of positive economics (see Chapter 1 on the distinction between positive and normative economics). On the other hand, the *for whom to produce* question is a matter of normative economics. Positive economic thinking, based on the scientific method described in Chapter 1, is not intended to make judgements about what is fair or unfair, or equitable or inequitable. It is intended to deal with issues of things that 'are' or 'will be' under different conditions. Therefore, while it can tell us about what methods are most likely to lead to increases in efficiency, it is not intended to tell us about how income and output should be distributed. The issue of equity (introduced in Chapter 1) is a normative issue, because what is considered 'fair' is a matter of beliefs and value judgements about things that 'ought to be'. What share of total income, or what particular goods and services *ought* individuals to have? Is there a minimum income that people *ought* to have? If so, based on what kinds of equity principles *ought* distribution and redistribution to take place?

There are no 'right' or 'wrong' answers to these questions. The answers that are chosen are based on beliefs and value judgements about what is good for society and the people within it. According to the standard view, economists should be concerned with positive aspects of economics. Their positive thinking could tell us about the likely consequences of policies to change the distribution of income, but should not make recommendations about how the distribution of income ought to change.

We will continue with the topic in Theory of knowledge 6.1 in Chapter 6.

Thinking points

- What is the significance of language in conveying meaning; does the expression 'maximum social welfare' accurately reflect its actual meaning?
- If economists, as social scientists, cannot make recommendations about normative issues like the distribution of income, how are these decisions made? (Think about the political process, social values, tradition and history.)
- Based on your reading about economics in the press and listening to the news, do you think that economists in the real world make a clear distinction between positive and normative ideas?
- Do you agree with the principle that economists (and social scientists generally) should only be concerned with positive thinking (social scientific investigation) and should leave normative issues (about things that ought to be) to societal decision-making?

2.6 Critique of the maximising behaviour of consumers and producers (HL only)

LEARNING OBJECTIVES

After studying this section you will be able to:

- define all the terms appearing in **orange bold** in the text (AO1)
- evaluate the assumptions of consumer rationality, utility maximisation and perfect information (AO3)
- referring to the following points, discuss the limitations of standard consumer behaviour made by behavioural economists (AO3)
  - biases including rule of thumb, anchoring, framing, availability
  - bounded rationality
  - bounded self-control
  - bounded selfishness
  - imperfect information
- discuss policies inspired by behavioural economics (AO3)
  - choice architecture with respect to default, restricted and mandated choices
  - nudge theory
- distinguish between various objectives of businesses (AO3)
  - profit maximisation
  - alternative objectives including corporate social responsibility, market share, satisficing, growth

Rational economic decision-making

Standard economic theories and models are based on an important assumption, that of 'rational self-interest', or *rational economic decision-making*. This means that individuals are assumed to act in their best self-interest, trying to *maximise* (make as large as possible) the satisfaction they expect to receive from their economic decisions. It is assumed that consumers spend their money on purchases to maximise the satisfaction they get from buying different goods and services. Similarly, it is assumed that firms (or producers) try to maximise the profits they make from their businesses; workers try to secure the highest possible wage when they get a job; investors in the stock market try to get the highest possible returns on their investments, and so on.

As we will see in the discussion below, the assumption of rational economic decision-making and the principle of self-interested maximisation underlying economic theory has come to be seriously questioned.

Rational consumer choice

The standard theory of consumer behaviour is based on some assumptions of **rational consumer choice** which we discuss below.

Assumptions of rational consumer choice

Consumer rationality

According to *rational consumer choice* consumers make purchasing decisions according to their tastes and preferences which satisfy three assumptions.

1. The consumer is able to rank goods according to her preferences. This means that if confronted with two goods, or two groups of goods, A and B, the consumer can say with certainty that she prefers A to B, or she prefers B to A, or she is indifferent between A and B. (This is called the *completeness* assumption.)
2. Preferences among alternative choices are consistent. This means that if the consumer prefers A to B, and B to C, then she must prefer A to C. (This is called the *transitivity* assumption.)
3. The consumer always prefers more of a good to less. If there are two groups of goods where one group contains more of one good than the other group, the consumer will always prefer the group that has more of the one good. (This is called the *non-satiation* assumption.)

Perfect information

It is further assumed that consumers have at their disposal perfect information about all their alternatives, so that there is no uncertainty. In other words, when a consumer chooses good A over good B, she possesses all possible information about good A and good B, so that there are no doubts, questions or uncertainties about the goods being chosen. The consumer has knowledge of all possible products, product qualities and prices.

Utility maximisation

Utility, as we saw earlier in this chapter, refers to the satisfaction that consumers derive from consuming something. According to the theory of consumer behaviour, consumers *maximise their utility*, meaning they make it as large as possible, by buying the combination of goods and services that results in the greatest amount of utility for a given amount of money spent (the consumer's budget or income). This behaviour is based on the assumptions of rationality and perfect information that were presented above.

The concept of marginal utility, and the law of diminishing marginal utility, are fundamental building blocks of the theory of consumer behaviour that allow economists to show how consumers make their choices among alternatives in order to maximise utility.[5]

Behavioural economics: limitations of the assumptions of rational behaviour

Behavioural economics was introduced in Chapter 1, Section 1.5. It is a new branch of economics strongly influenced mainly by psychology, but also by sociology and neuroscience, based on the idea that human behaviour is far more complex than consumer rationality assumes. While the assumption of rationality has always been questioned on the grounds that people very often do not behave as rationally as is presumed, it is only since about the beginning of the 21st century that behavioural economics began to flourish into a distinct branch of economics.

Behavioural economics criticises consumer rationality and the idea of utility maximisation on the following grounds.

Biases

A **bias** (also known as *cognitive bias*) is a term from psychology that refers to systematic errors in thinking or evaluating. Biases are departures from normal standards of thought or judgment. Biases that affect consumer choices include the following:

- **Rules of thumb.** **Rules of thumb** are simple guidelines based on experience and common sense, simplifying complicated decisions that would have to be based on the complex consideration of every possible choice. Examples of rules of thumb are: one portion of salad is equal to two handfuls; a cup of filter coffee has 80 mg of caffeine; one serving of fruit is the size of a fist. These examples show that rather than make complicated measurements and calculations people often rely on simple rules to simplify decisions.
- **Anchoring.** **Anchoring** involves the use of irrelevant information to make decisions, which often occurs due to its being the first piece of information that the consumer happens to come across. For example, you find a pair of jeans you want that costs $150, and then you find a similar one for $100 that you buy thinking it's a bargain, but then you discover you could have gotten the same thing for $50 somewhere else.
- **Framing.** **Framing** deals with how choices are presented to decision-makers (framed). For example, consumers prefer beef described as 80% lean rather than 20% fat. Yet the rational consumer would be indifferent between the two as the information is exactly the same, it is only presented (framed) differently. In another example, consumers could be willing to pay a higher price for a faded pair of jeans in a boutique than for the identical pair of jeans in a discount store. In the first example the purchase is framed by language; in the second it is framed by the seller's environment.
- **Availability.** **Availability** refers to information that is most recently available, which people tend to rely on more heavily, though there is no reason to expect that this information is any more reliable than other information that was available at an earlier time. This may be due at least partly to the fact that people remember recent events or information more readily than older events or information. For example, if a consumer has recently received a lot of advertising information about the superiority of one brand over other brands, she may be more likely to select that brand, having been influenced by the recent advertising.

[5] It is beyond the scope of this book to explain how the law of diminishing marginal utility is used to explain rational consumer in order to maximise utility.

The presence of biases that influence consumer choices has led to alternative concepts and ideas put forward that may be more accurate descriptions of consumer behaviour:

Bounded rationality

Bounded rationality is an idea developed by Herbert Simon, who received the Nobel Prize in economics on 1971 for his work on economic decision-making. Simon argued that people do not have an unlimited capacity to process information and that searching for information needed to maximise utility is itself a costly process. He therefore suggested the concept of bounded rationality to describe consumer behaviour. Bounded rationality is the idea that consumers are rational only within limits, as consumer rationality is limited by consumers' insufficient information, the costliness of obtaining information, and the limitations of the human mind to process large amounts of information. According to Simon, rather than maximise, consumers *satisfice*, meaning they seek a satisfactory outcome rather than an optimal (or best) one.

Bounded self-control

Bounded self-control is related to bounded rationality (see above). It refers to the idea that people in reality exercise self-control only within limits. This means they often do not have the self-control that would be required of them to make rational decisions. People may at times eat too much, spend too much, save too little or work too little; all this is inconsistent with the assumption of rational behaviour.

Bounded selfishness

Another related concept is that of **bounded selfishness**, which is the idea that people are selfish only within limits, and that the assumption of self-interested behaviour underlying the maximisation principle cannot explain the numerous accounts of selfless behaviour and willingness to contribute to the public good even at the cost of reduced personal welfare (these points are further discussed in Theory of knowledge 2.2 below).

Imperfect information

The theory of consumer behaviour as we have seen is based on the assumption that consumers have access to all the information needed to make fully informed decisions, regarding prices, products, product quality. If they did not have access to such complete information, their decisions and choices would not be optimal, or best. However, it is abundantly clear that consumers do not and cannot possibly have such full access to information. This means they are unable to maximise utility as they make choices based on faulty and incomplete information.

THEORY OF KNOWLEDGE 2.2

Altruism, perceptions of fairness and self-interested behaviour of rational economic decision-makers

Economists have long been aware of the presence of altruism and acts based on perceptions of fairness in the behaviour of economic decision-makers. For example, when consumers make anonymous donations to charity or volunteer their services through many hours of work for the public good, they are not motivated by expectations of monetary gain. Firms, too, often act in the public interest by not firing workers in times of recession so as not to increase unemployment and cause personal hardship, even though this cuts into their profits.

Economists sometimes argue that there is no real conflict between such altruistic behaviour and behaviour based on self-interest, because people engaging in altruistic and charitable acts gain satisfaction from their contributions to society, and this works to increase their utility. Selflessness and caring for others are thus interpreted to be consistent with rational self-interest and with utility maximisation. Therefore in this view consumers are behaving rationally even when they behave altruistically.

A serious problem with this argument is that if the assumption of utility maximisation is interpreted so broadly that every human act is consistent with it, it loses its usefulness as a theory of consumer behaviour. To put it simply, if it explains everything, by being consistent with all human actions, it explains nothing. In the words of Herbert Simon, 'Economic theory has treated economic gain as the primary human motive.

CONTINUED

An empirically grounded theory would assign comparable weight to other motives, including altruism and the organisational identifications associated with it.'[6] In other words, altruism should be treated separately, and not be made consistent with the assumption of rational self-interest through an all-inclusive interpretation of utility.

Simon's thinking brings to mind the ideas of Adam Smith, discussed in Chapter 1, Section 1.5. In all likelihood, Smith would be extremely surprised and disappointed to discover that his contributions to economics regarding the invisible hand of markets have been taken out of context to mean that the best results for society arise because people behave *only on the basis of self-interest*. As noted in Chapter 1, in his early book, *The Theory of Moral Sentiments*, Smith argued that people's selfishness is restrained by 'moral sentiments', which are the need for self-respect and the respect of others, arising from empathy (which he refers to as 'sympathy') and the desire for honour and justice. Other economists of the classical period, among whom was David Ricardo, agreed with Smith that people's motivations are complex, involving a combination of self-interest and other motives like empathy and concern for other people. It was only much later, in the late 19th and early 20th centuries, that economists simplified and reduced human motivations solely to the idea of utility maximisation in the name of rational self-interest. This gave rise to the term *homo economicus* (from the Latin word *homo* meaning man, and the Greek word *economicus* meaning economic) or what translates into *economic man*, a mythical being who goes about life rationally maximising his self-interest. As Gary Becker, a noted Nobel Prize winner (1992) stated decades later, 'self-interest is assumed to dominate all other motives'.

Since Simon's time, economists and other scientists have carried out thousands of experiments that provide ample evidence of the presence of altruism in human beings. It is believed that altruism may derive from culture, or from psychological development and socialisation, or from innate biological traits. (See also Theory of knowledge 5.1 in Chapter 5 on sustainability and rational self-interest.)

Thinking points

Homo economicus, an assumed being who is forever rationally maximising utility as a consumer (or forever rationally maximising profit as a producer), is a fundamental building block of microeconomic theory. This mythical being came into existence at a time when economists were trying to imitate the methods of the sciences, especially physics. Just as physics assumes the existence of tiny particles (like atoms) that behave in a certain predictable way, so consumers and firms became the 'particles' of economics that behave in certain predictable ways. Although atoms are not visible with the naked eye, physicists have developed tools to detect their existence and describe their behaviour.

- What tools do economists use to make assumptions about the existence and behaviour of homo economicus?
- In what ways are physicists' tools the same or different from economists' tools? Do you think physicists' and economists' tools are reliable to the same extent?

Behavioural economics in action

Behavioural economics studies how individuals and groups of individuals make decisions. Instead of relying on assumptions about human behaviour, as in the theory of consumer behaviour, behavioural economics relies extensively on experiments that try to observe and understand how people behave and react in a variety of situations.

Nudge theory

The word *nudge* means poking someone gently to get attention. As an economics term, a **nudge** has come to mean a method designed to influence consumers' choices in a predictable way, without offering financial incentives or imposing sanctions, and without limiting choice. As an economics term the word *nudge* was coined by two American economists, Richard Thaler and Cass Sunstein

[6] Herbert Simon, Altruism and Economics, *Eastern Economic Journal* Vol. 18, No. 1 (Winter, 1992), pp. 73–83

in their now classic book of 2008, *Nudge: Improving Decisions about Health, Wealth and Happiness*. Nudges try to influence people to behave in socially desirable ways, but without imposing any constraints on their behaviour, and by maintaining freedom of choice. Richard Thaler received the Nobel Prize in Economics in 2017 in part for his work on behavioural economics.

While behavioural economics was initiated in the United States, it has taken a strong hold over policy in many countries in the world, with some countries establishing nudge units to pursue and support this approach to a variety of public policy areas. It is not only governments but also the private sector that have undertaken nudge initiatives.

Examples of nudges and their results include the following:

- In the United Kingdom, tax payments of late taxpayers increased by 15% when they were told by the UK revenue service that most people in their area had already paid their taxes. This worked because the late taxpayers were made to feel they were the exception to what was the norm in their community.
- In a Danish municipality, the streetlights turn red when solar panels are no longer sufficient to power the light, making residents more aware of their electricity consumption.
- Putting healthy foods in more visible and accessible positions in stores has been shown to increase sales of such foods. In addition, colourful footsteps leading consumers to the locations of healthy foods has also been effective.
- When non-payers of vehicle taxes received personalised messages rather than warning letters, the number of payers doubled; when the message included a photo of the car in question, the number of payers tripled.
- When taxpayers received notifications from tax authorities in white envelopes instead of brown with their names handwritten rather than typed, there was a 16% increase in people paying their tax.

These examples indicate that people respond favourably when *nudged* to perform actions that they might not otherwise be inclined to do.

Behavioural patterns such as these are taken as evidence that borrowing from psychology, sociology and neuroscience in an attempt to understand why human beings behave the way they do can have important results in changing behaviour to encourage achievement of socially desirable results.

Behavioural economics has been more successful in the area of finance, where it has already strongly influenced economic thinking. Other areas where it is making some headway are development, health, law and economics, public economics, organisational economics and wage determination.

Choice architecture

Choice architecture is the design of particular ways or environments in which people make choices; it is based on the idea that consumers make decisions in a particular context and that choices of decision-makers are influenced by *how options are presented to them*. Choice architects are individuals or organisations that arrange the context in which choices are made. The term *choice architecture* was also coined by Richard Thaler and Cass Sunstein in their book on nudges noted above. *Framing*, which was explained earlier, is a very important part of choice architecture, as it defines the context in which choices are presented to decision-makers, thus trying to influence their decisions.

Choice architecture offers different kinds of choices that are described below.

- **Default choice** is a choice that is made by default, which means doing the option that results when one does not do anything. People often make choices by default due to habit or lack of interest in taking a deliberate action, even if doing nothing may not the best choice for them. Sometimes they feel more comfortable not having to make a choice. Therefore one way of inducing people to follow a particular course of action is to provide it as a default choice. See Real world focus 2.3 below for an example of the role of default choice in organ donation.
- **Restricted choice** is a choice that is limited by the government or other authority. People everywhere are subject to countless restrictions of all kinds, ranging from speed limits, voting age, and recycling regulations to smoking prohibitions and social norms like shaking hands. It is argued that restrictions such as these are necessary because people have too many choices available to them, and in the absence of more and better information, or due to poor judgement, they often make poor choices. Therefore choice architecture can take advantage of restrictions to encourage people to make choices with socially desirable outcomes.

REAL WORLD FOCUS 2.3

Choice architecture, nudging and organ donation

In the area of organ donation, it is a challenge to increase the number of organ donors in many countries around the world. Many countries have a *default choice* for people to be either (a) donors, in which case they automatically become a donor if they do nothing, and would have to make a deliberate effort to register as a non-donor; or (b) non-donors, in which case they automatically become a non-donor and would have to register to become a donor. In countries where the default option is to be a donor, for example Austria, Belgium, France, Hungary, Poland, Portugal, and the United Kingdom, the participation rate in organ donation is very high, sometimes as much as 90% of the population or more. By contrast, in countries where the default option is to be a non-donor, for example Denmark, Germany, and the United States, the participation rate is much lower.



Figure 2.19: Organ donor card

Organ donation can also be used to illustrate *mandated choice*. In the state of Illinois in the United States, in order to get a driver's licence renewed it is mandatory (compulsory) to answer a question on whether one wishes to be an organ donor. The result has given rise to a 60% rate of signing up to be a donor, compared to only 38% for the US national average where mandated choice does not exist.

The default choice that makes people donors by default therefore appears to be effective. However, nudges may in some cases backfire, giving rise to opposite results than the desired ones. In the Netherlands, a 2016 law made being a donor the default choice, in the hope that the number of donors would increase. Yet in a drive that took place shortly after the bill was passed, the surprising result was that there were *six times as many non-donors as donors*. A likely explanation for such an unexpected result may be that the Dutch population rebelled against the nudger's intentions. There had been a lot of press coverage of the government's decision to change the default choice, and it appears the Dutch people did not want the government to make choices for them.

Sources: *Dee Gill, How to spot a nudge gone rogue, UCLA Anderson Review, School of Management, 12 September 2018*

Applying your skills

1. According to the experiences of many countries, the default choice for organ donation appears to be succeeding in increasing the number of donors. Can you think of other areas where this type of default choice could be applied?
2. What is the lesson about nudging that can be learned from the Dutch experience?
3. Some observers argue that nudging, in spite of its potential benefits, also carries a risk of manipulation of the public's opinion in subtle ways that may not always be detectable. What does the Dutch experience say about this possible risk? (See also Question 4 under *Inquiry and reflection* at the end of this chapter.)

- **Mandated choice** is a choice between alternatives that is made mandatory (compulsory) by the government or other authority. It can be thought of as required choice. It is a free choice, but it is compulsory to make that free choice. See Real world focus 2.3 for an example of the role of default choice in organ donation.

Note that default choice, restricted choice and mandated choice are different types of *nudges* within *choice architecture* that are intended to work toward influencing people's choices in a direction held to be socially desirable.

The response of mainstream economists to the critique of utility theory

In response to the criticisms of utility theory and rational behaviour, many mainstream economists would argue that in spite of its imperfections, the theory of consumer behaviour based on the assumption of rationality does a reasonably good job in explaining and predicting how consumers behave under many circumstances. There is evidence that consumers do respond rationally to monetary factors. For example, an increase in the gasoline or petrol tax will most likely lead to a lower quantity of petrol demanded, as predicted by the downward sloping demand curve based on consumer rationality. Examples abound of tests that have empirically shown (not refuted) the existence of downward sloping demand curves, which are based on the theory of utility maximisation. So even if consumers do not behave rationally all the time according to the rules of rationality, on average, overall, their behaviour is consistent with the predictions of utility theory.

Moreover, some economists would argue that choice architecture and the use of nudges is manipulative, as it attempts to exercise control and influence over consumer choices. The idea of free choice is therefore weakened and becomes an illusion of free choice rather than actual free choice. What may be worse is that it often affects consumers on a sub-conscious level, making them do things that they might not choose to do if they were fully aware of the full implications of their choices.

In effect, choice architecture takes advantage of the incomplete information available to the consumer, and selectively provides information (such as through framing and choice architecture) to encourage consumers to act in a particular way.

For these reasons it is argued that there should be transparency and open discussion of nudges used for economic policy.

Evaluating behavioural economics and economic policy

Potential advantages

- Behavioural economics may be a relatively simple and low-cost way to influence people's behaviour to act in socially desirable ways.
- It has been used successfully in a number of areas, suggesting that the methods of choice architecture and nudging may have numerous possible applications in areas that are as yet unexplored.
- It offers consumers and citizens, generally, freedom of choice without forcing them to do anything or preventing them from doing anything, hence without restricting their choices.
- It may be able to overcome the weaknesses of the theory of consumer behaviour, which is not always able to explain the inconsistencies and seeming irrationality of actual consumer behaviour.
- Policies are based on principles of psychology, such as framing, many of which have been previously tested over many years.
- The development of policies is based on trials, indicating the use of a flexible trial-and-error method of discovering policy measures that can work in achieving desired results.

Potential disadvantages

- The body of knowledge being developed is not based on any understanding of human behaviour, and is therefore unable to lead to a systematic and unifying theory of how consumers behave with general applicability.
- The resulting unsystematic approach (different policies for different situations) may not be valid over time or across different income groups, social groups or cultures; this reduces the applicability of the policies being developed over time and across social, economic and cultural groups.
- There may be risks of using psychological principles to manipulate consumers, in much the same ways that advertising and marketing have been using similar principles over many years in order to

manipulate consumer tastes and preferences for the purpose of convincing consumers to act in ways that are not necessarily in their best interests (such as advertising for sugary and fatty food products that are unhealthy).

- There may be risks that behavioural policies may be used as substitutes for necessary but politically costly economic policies, such as the imposition of taxes on socially harmful goods known as *demerit goods* or goods leading to *negative externalities* (see Chapter 5).
- Traditional economic policies (for example indirect taxes, subsidies, to be discussed in Chapters 4 and 5) may be more effective.
- It may be a new form of government regulation, only camouflaged under the guise of freedom of choice, and therefore more dangerous as people may not be aware of its existence and its effects on their choices.
- Choice architecture and nudging may successfully affect people's choices, but these choices may not be a reflection of their true preferences.

TEST YOUR UNDERSTANDING 2.9

1 a Discuss what it means to be 'a rational consumer' in economics.

b To what extent are you personally a rational consumer?

2 For each of the biases discussed in the text, explain why they could not occur according to the theory of consumer behaviour.

3 In the examples of nudges given in the text, identify the framing element in the nudge.

4 Distinguish between default, restricted and mandated choice. Explain how these can be used as nudges.

5 Distinguish between bounded rationality, bounded self-control and bounded selfishness. Provide an example for each of these. Explain why they are all inconsistent with rational consumer behaviour.

6 Discuss the potential advantages and disadvantages of behavioural economics.

Firm business objectives

Rational producer behaviour: profit maximisation

Standard economic theory of the firm assumes **rational producer behaviour** according to which firms are guided by the goal to maximise profit. Profit maximisation involves determining the level of output that the firm should produce to make profit as large as possible.

On a very general level, profit is equal to the total revenue earned by a firm minus the total costs incurred by the firm in the process of producing its output.

Suppose you have a pizza restaurant and you sell 400 pizzas a day at $7 each. You have revenues of 7 x 400 = $2800 per day. On the other hand the costs of buying the pizza ingredients, paying your workers and meeting all other expenses of running your restaurant amount to $2650 per day. Your profit per day is

Profit = revenues – costs = 2800 – 2650 = $150 per day.

The objective of profit maximisation is to make the difference between revenues and costs as large as possible so as to make profit as large as possible. We will come back to the topic of profit maximisation in Chapter 7 (at HL).

Alternative business objectives

Over the years, economists have developed many theories about firm behaviour. The following is a brief survey of some of the more important ones. The common theme of all of these is that profit maximisation may not be the overriding objective of firms, as in fact they may have other goals that may be more important.

Corporate social responsibility: ethical and environmental concerns

The self-interested behaviour of firms often leads to negative consequences for society. It is often the case that the well-being of firms is not consistent with the well-being of society. A prime example is the self-interested firm that pollutes the environment. (Such actions of firms will be examined in Chapter 5 under the topics of negative externalities and common pool resources.) In addition, firms can engage in actions that most consumers would consider to be ethically unacceptable, such as the practice in many developing countries of employing poorly paid children forced to work long hours, or employing labour that is forced to work under

unhealthy or dangerous conditions. These situations may arise in countries where there is widespread poverty, and government legislation protecting the rights of children and workers is either non-existent or poorly enforced.

However, many firms increasingly recognise that the pursuit of self-interest need not necessarily conflict with ethical and environmentally responsible behaviour. A negative image of the firm held by workers and buyers of the product can cut deeply into the firm's revenues and profits by lowering worker productivity and the firm's sales. Further, socially irresponsible firm behaviour may lead to government regulation of the firm intended to minimise the negative consequences of the firm's actions for society, whereas socially responsible behaviour could instead result in less government regulation. Therefore, firms face incentives to display **corporate social responsibility** by engaging in socially beneficial activities. These can take many forms, including:

- avoidance of polluting activities
- engaging in environmentally sound practices
- support for human rights, such as avoiding exploitation of child labour and labour in general in less developed countries, or avoiding investments in countries with politically oppressive regimes
- art and athletics sponsorships
- donations to charities.

Many of these practices are the result of increased consumer awareness of social and environmental issues, growing consumer concern over ethical and environmental aspects of business practices, and even consumer activism that results in boycotts of offending firms. One indication of the influence and concern of consumers is the rapidly growing interest in investments in companies (through stock markets) that meet certain social, ethical and ecological criteria.

Economists used to think that ethical and environmentally responsible behaviour of firms would reduce their profits. This was based on considering only the cost aspect of profits; for example, firms using cheap child labour face lower costs, and hence will make higher profits than firms avoiding such practices. Yet profits depend not only on costs, but also on revenues that will fall if consumers avoid buying the products of offending firms.

A number of studies have attempted to measure the effects of socially responsible behaviour on the profits of firms. Does ethical and environmentally responsible behaviour lower or increase firms' profits? The results of these studies have been inconclusive. The behaviour of firms themselves, however, suggests that they often do not want to risk consumer displeasure.

Market share

Market share refers to the percentage of total sales in a market that is earned by a single firm. For example, if a product has total sales within a country of $50 million, and a firm sells $10 million worth of that product, then that firm has a 20% market share. A high market share means that the firm is enjoying large sales of its product or products, and is an indication of the product's popularity among buyers.

A large market share means that it is likely that the firm is achieving *economies of scale* (falling costs per unit of output as the firm grows, to be studied at HL in Chapter 7). Economies of scale allow a firm to increase its profitability.

In general, market share is so important that large companies such as multinational corporations (large firms that have productive facilities in more than one country) often measure their performance in terms of their market share in specific countries. Market share is an important indicator of performance because it allows the firm to monitor how well it is doing in relation to its competitors.

A firm that succeeds in maintaining its market share over time means that it is likely earning revenues that are growing at the same rate as the overall market. A growing market share is a sign that the firm is doing well in relation to its competitors. It is likely that its revenues are growing faster than the revenues of competitor firms. A falling market share on the other hand means that the firm is not doing as well as its competitors in terms of sales of its products.

In order to increase its market share, the firm may try to lower its prices, or introduce new or innovative products into the market, or use advertising. Each of these strategies may have the effect of lowering profits. Lower prices may reduce revenues (though not necessarily, this will be explored in Chapter 3). The introduction of new products involves increased costs on research and development, while advertising also involves increased firm spending to pay for the advertising. Possible lower prices with higher costs will cut into profits. However, as long as the firm remains profitable, and it continues to earn a satisfactory level of profits, the benefits of maintaining or increasing market share are likely to make the lower profits worthwhile.

Growth maximisation

Another possible objective is maximisation of growth rather than profits.[7] **Growth maximisation** can be attractive for the following reasons:

- A growing firm can achieve economies of scale (see Chapter 7) and lower its average costs, thus increasing its profitability.
- As a firm grows it can diversify into production of different products and markets and reduce its dependence on a single product or market.
- A larger firm has greater market power and increased ability to influence prices, thus again potentially increasing its profitability.
- A larger firm reduces its risks because it may be less affected in an economic downturn (a recession) and is less likely to be taken over (bought) by another firm.
- The objective of growth maximisation reconciles the interests of both owners and managers, because both groups have much to gain from a growing firm. Other maximisation objectives may pit firm owners against firm managers; for example, profit maximisation is favoured by owners while revenue maximisation is favoured by managers.

Revenue maximisation

In one theory of firm behaviour, it is argued that the separation of firm management from firm ownership, which increasingly dominates business organisation, has meant that firms' objectives have changed. Whereas profit maximisation may be the dominant motive of the traditional owner-managed firm, firm managers who are hired by the owners to perform management tasks may be more interested in increasing sales and maximising the revenues that arise from larger quantities sold. This goal of firms is referred to as *revenue maximisation*.[8] Increasing sales and maximising revenues may be more useful to a firm than profit maximisation for the following reasons:

- Sales can be identified and measured more easily over the short term than profits, and increased sales targets can be used to motivate employees.
- Rewards for managers and employees are often linked to increased sales rather than increased profits.
- It is often assumed that revenue from more sales will increase more rapidly than costs; if this is the case, profit (= total revenues – total costs) will also increase.
- Increased sales give rise to a feeling of success, whereas declining sales create a feeling of failure.

(*Note that revenue maximisation, while important, is not in the syllabus.*)

Satisficing

Many of the above objectives assume that the firm tries to maximise some variable, whether it is profit, market share, growth or revenue. Herbert Simon, a Nobel Prize-winning economist (also mentioned earlier in connection with the idea of bounded rationality as a critique of consumer rationality), has argued that the large modern enterprise cannot be looked upon as a single entity with a single maximising objective; instead it is composed of many separate groups within the firm, each with its own objectives which may overlap or may conflict. This multiplicity of objectives does not allow the firm to pursue any kind of maximising behaviour. Firms therefore try to establish processes through which they can make compromises and reconcile conflicts to arrive at agreements, the result of which is the pursuit of many objectives that are placed in a hierarchy. This behaviour was termed **satisficing** by Simon. (Simon used this term to describe the behaviour of consumers as well.) It refers to the idea that firms try to achieve a satisfactory level of profits together with satisfactory results for many more objectives, rather than optimal or 'best' results for any one objective.

TEST YOUR UNDERSTANDING 2.10

Imagine you are the owner of a business. Discuss and justify any business goals you may have that are not directly concerned with profit maximisation.

[7] This is based on the work of R. Marris and others.
[8] The revenue-maximisation goal of firms was described by W. J. Baumol in 1959.

THEORY OF KNOWLEDGE 2.3

How important are the criticisms of profit maximisation as the firm's main goal?

Standard economic theory assumes that profit maximisation is the most important goal of firms. As we will see in Chapter 7, the study of firm behaviour is based very heavily on the assumption of profit maximisation. Yet this assumption is criticised for several reasons:

- The use of marginal concepts (for example *marginal cost*) in the theory is unrealistic; firms cannot easily identify marginal costs, and do not even try to do so; therefore, this theory does not accurately describe methods actually used by firms to determine price and output.
- The model is based on the assumption that firms have perfect information at their disposal, whereas in fact the information on which they base their decisions is highly fragmentary and uncertain; firms do not know what demand curves they face for their products and they do not know how competitor firms will behave in response to their actions.
- Short-run profit maximisation may be unrealistic; firms may prefer lower profits in the short run in exchange for larger profits over the long run.
- The factors determining demand and supply for products and resources are continuously changing, with demand and supply curves continuously shifting, so that any profit-maximising decisions regarding prices and output made today under current conditions may be irrelevant by the time the output is produced and ready for sale in the market.
- There is real-world evidence suggesting that firm behaviour may be motivated by a variety of objectives other than profit maximisation, which were discussed above.

Milton Friedman, an American Nobel Prize-winning economist, argued in a famous book[9] that it does not matter if the assumptions of a theory are unrealistic, as long as the theory has predictive powers. In fact, good theories are often based on unrealistic assumptions that do not accurately describe the real world, because the role of assumptions is to portray only the important aspects of a process that is modelled or theorised about, ignoring the irrelevant details.

Paul Samuelson, another American Nobel Prize-winning economist, fundamentally disagreed. Samuelson argued that the predictions of a theory can only be as empirically valid as the theory itself, and as the assumptions on which the theory rests. If the assumptions are unrealistic or invalid, then the theory and its predictions will similarly be invalid; it is not possible to have a theory with predictive powers if its assumptions are unrealistic. If the predictions of a theory are empirically valid, so is the theory and its assumptions. Logically, then, it is not possible to separate the predictions of a theory from the assumptions of the theory; they all stand or fall together.

Thinking points

- Remember that a theory tries to explain real-world events. Does it matter if a theory is based on unrealistic assumptions?
- As you read through Chapter 7, you may want to keep these issues in mind, as we will encounter further unrealistic assumptions in some market models discussed in that chapter (see also Theory of knowledge 7.1 about perfect competition).

[9] Milton Friedman (1953) 'The Methodology of Positive Economics' in Essays in Positive Economics, University of Chicago Press.

INQUIRY AND REFLECTION

The following questions will help you reflect on your learning and enhance your understanding of key topics in this chapter. They are suggestions for inquiries that you can undertake on your own or in groups in order to revise the learning objectives of this chapter.

1 Advertising makes great use of biases that influence consumers' way of thinking. As you watch TV, when a commercial comes on, try to see if the message being conveyed to the consumer makes use of biases in order to convince consumers to buy the product.

2 An area that has been examined extensively involves providing healthy lunches at schools and using nudges to encourage parents and students to prefer these. Investigate what kinds of nudges have been suggested, when and where they have been implemented, and what were the results. Try to design a programme where you could recommend nudges that you think may be effective in increasing demand for healthy school meals.

3 Was your decision to study IB economics a rational choice? Identify the conditions that would have to be satisfied for this choice to have been a rational one. If it was not rational, identify some biases that may have affected your choice.

4 Richard Thaler and Cass Sunstein, who became famous for their contributions to nudges. coined the phrase *libertarian paternalism* to describe nudges as a form of government intervention. The word *libertarian* (from the Latin word *libertas* which means freedom) refers to a philosophy where liberty or freedom and autonomy are the main principles, meaning that people should have freedom of choice and should be free of authority. The word *paternalism* (from the Greek word *pater* which means father) refers to actions that limit people's freedom or autonomy; such actions are often justified on the grounds that they are in the people's best interests. Consider the phrase *libertarian paternalism*, and determine whether these two concepts may be contradictory or whether they can be made consistent with each other as Thaler and Sunstein have tried to do through the policy of nudges.

EXAM STYLE QUESTIONS

You can find questions in the style of IB exams in the digital coursebook.



Chapter 3

Elasticities

BEFORE YOU START

- When the price of a good or service that you want rises, how likely is it that you will purchase it anyway? You might say it depends. What factors might influence your decision?
- If producers of a good or service want to increase their revenue, sometimes they increase their price. Does this strategy always work? Why or why not?
- If someone is poor and over time becomes richer, what kinds of changes might you expect to see in her or his purchasing habits? What if she or he is rich and over time becomes poorer?

The topics above relate to the concept of elasticity. **Elasticity** is a measure of the responsiveness of a variable to changes in price or any of the variable's determinants. In this chapter we will examine three kinds of elasticities:

- price elasticity of demand, which examines the responsiveness of quantity demanded to changes in price
- income elasticity of demand, which examines the responsiveness of demand to changes in income
- price elasticity of supply, which examines the responsiveness of quantity supplied to changes in price.

Each of these has a number of applications to important economic problems.

Photo: Group of people shopping

3.1 Price elasticity of demand (*PED*)

LEARNING OBJECTIVES

After studying this section you will be able to:

- define all the terms appearing in **orange bold** in the text (AO1)
- use the formula for price elasticity of demand (*PED*) to calculate *PED*, changes in price, changes in quantity and total revenue (AO4)
- identify the various degrees and range of values for *PED* (AO2)
- draw diagrams showing the range of values for *PED*, including (AO4)
  - relatively elastic and inelastic demand
  - constant values for perfectly elastic demand, perfectly inelastic demand and unitary *PED*
- explain and illustrate in a diagram changing *PED* along a straight-line downward sloping demand curve (HL only) (AO2, AO4)
- analyse the determinants of *PED* (AO2)
- explain the relationship between *PED* and total revenue (AO2)
- draw a diagram to show how total revenue changes in response to a price change depending on elastic or inelastic demand (AO4)
- apply *PED* to:
  - discuss its importance for firms and government decision-making (AO3)
  - analyse the reasons why primary commodities generally have a lower *PED* than manufactured products (HL only) (AO2)

Price elasticity of demand

Understanding price elasticity of demand (*PED*)

According to the law of demand, there is a negative relationship between price and quantity demanded: the higher the price, the lower the quantity demanded, and vice versa, *ceteris paribus*. We now want to know *by how much* quantity responds to change in price.

Price elasticity of demand (*PED*) is a measure of the responsiveness of the quantity of a good demanded to changes in its price. *PED* is calculated along a given demand curve. In general, if quantity demanded is highly responsive to a change in price, demand is referred to as being *price elastic*; if quantity demanded is not very responsive, demand is *price inelastic*.

The formula for *PED*

Suppose we are considering *PED* for good *X*. The formula used to measure its *PED* is:

$$\text{price elasticity of demand } (PED) = \frac{\text{percentage change in quantity demanded}}{\text{percentage change in price}}$$

If we abbreviate 'change in' by the Greek letter Δ, this formula can be rewritten as:

$$PED = \frac{\%\Delta Q}{\%\Delta P}$$

Simplifying, the above formula can be rewritten as:

$$\text{PED} = \frac{\frac{\Delta Q}{Q} \times 100}{\frac{\Delta P}{P} \times 100} = \frac{\frac{\Delta Q}{Q}}{\frac{\Delta P}{P}} \text{ where}$$

$\Delta Q = Q_{\text{final value}} - Q_{\text{initial value}}$

$Q = Q_{\text{initial value}}$

$\Delta P = P_{\text{final value}} - P_{\text{initial value}}$

$P = P_{\text{initial value}}$

The sign of *PED*

Since price and quantity demanded are negatively related, *PED* is a negative number. Any percentage increase in price (a positive denominator) gives rise to a percentage decrease in quantity demanded (a negative numerator), leading to a negative *PED*. Similarly, a percentage price decrease results in a percentage quantity increase. However, *the common practice is to drop the minus sign and consider PED as a positive number*. (In mathematics this is called taking the absolute value.) This is done to avoid confusion when making comparisons between different values of *PED*. Using positive numbers, we can say, for example, that a *PED* of 3 is larger than a *PED* of 2. (Had we been using the minus sign, −2 would be larger than −3.)

The use of percentages

Elasticity is measured in terms of percentage changes of P and Q for two reasons:

- We need a measure of responsiveness that is independent of units. It makes little sense to compare units of oranges with units of computers or cars. Also, we want to be able to compare elasticities across countries that have different currencies; an elasticity measured in terms of euros will not be comparable with an elasticity measured in yen or pounds. The use of percentages allows us to express elasticities in common terms.
- It is meaningless to think of changes in prices or quantities in absolute terms (for example, a $15 increase in price or a 20 unit decrease in quantity). A $15 price increase means something very different for a good whose original price is $100 than for a good whose original price is $5000. In the first case there is a 15% increase, and in the second there is a 0.3% increase. Using percentages allows us to put responsiveness into perspective.

The same arguments apply to all other elasticities we will consider.

Calculating *PED*, change in price and change in quantity

We can now use the formula above to calculate *PED*. Suppose consumers buy 6000 TVs when the price is $255 per unit, and they buy 5000 TVs when the price is $300.

$$PED = \frac{\frac{6000-5000}{5000}}{\frac{255-300}{300}} = \frac{\frac{1000}{5000}}{\frac{-45}{300}} = \frac{0.20}{-0.15} = -1.33$$

or 1.33 since we drop the minus sign. Therefore *PED* for TVs is 1.3.[1]

Note that elasticity is measured as a number, not as a percentage, and there are no units.

If we know *PED* and the percentage change in P, we can calculate the percentage change in Q. Suppose *PED* = 1.25 and price of good X *increases* by 12%. Calculate the percentage change in Q demanded:

$PED = 1.25 = \frac{\%\Delta Q}{-0.12} \Rightarrow \%\Delta Q = -0.12x1.25 = -0.15$ or −15%. Quantity *decreased* by 15%.

Similarly, if we know *PED* and the percentage change in Q demanded, we can calculate the percentage change in P.

[1] You may note that the value of this elasticity of demand depends on the choice of the initial price–quantity combination. In the calculation above, this was taken to be 300, 5000. If we had taken 255, 6000 as the initial price–quantity combination, we would get a *PED* value of 0.94. (You could calculate this as an exercise.) This difficulty can be overcome by use of the 'midpoint formula'. Note that you are not required to know the midpoint formula:

$$PED = \frac{\frac{\Delta Q_X}{\text{average } Q_X}}{\frac{\Delta P_X}{\text{average } P_X}}$$

In the previous example,

$$PED = \frac{\frac{1000}{5500}}{\frac{45}{277.5}} = 1.12, \text{ where } 5500 = \frac{(5000+6000)}{2} \text{ and } 277.5 = \frac{(255+300)}{2}$$

i.e. we use the average of the two Q_x values and the average of the two P_x values instead of the initial Q_x and initial P_x. Note that you are not required to know the midpoint formula.

Suppose $PED = 0.80$ and quantity of good Y demanded *falls* by 16%. Calculate the percentage change in P:

$$PED = 0.80 = \frac{-0.16}{\%\Delta P} = \Rightarrow 0.80 \times \%\Delta P = -0.16$$

$\Rightarrow \%\Delta P = 0.20$ or 20%. Price *increased* by 20%.

You may note that here we have used negative signs in the calculations to show decreases, however this is not necessary since the final elasticity value is taken as a positive number (absolute value).

Total revenue calculations (referred to in the learning objectives) will be presented below.

TEST YOUR UNDERSTANDING 3.1

1 a Explain the meaning of price elasticity of demand.

b Outline why we say it measures responsiveness of quantity along a given demand curve.

2 State why we treat *PED* as if it were positive, even though it is usually negative.

3 It is observed that when the price of pizzas is $16 per pizza, 100 pizzas are sold; when the price falls to $12 per pizza, 120 pizzas are sold. Calculate price elasticity of demand.

4 A 10% increase in the price of a particular good gives rise to an 8% decrease in quantity bought. Calculate the price elasticity of demand.

5 The *PED* for good X is 0.8. If the price of good X increased by 15%, calculate the percentage decrease in quantity demanded.

6 The *PED* of good Y is 1.5. If quantity of good Y increases by 30%, calculate the percentage decrease in the price of good Y.

The range of values for *PED*

The value of *PED* involves a comparison of two numbers: the percentage change in quantity demanded (the numerator in the *PED* formula) and the percentage change in price (the denominator). This comparison yields several possible values and range of values for *PED*. These are illustrated in Figure 3.1 and summarised in Table 3.1.

- **Demand is price inelastic when $PED < 1$ (but greater than zero).** The percentage change in quantity demanded is smaller than the percentage change in price, so the value of *PED* is less than one; quantity demanded is relatively unresponsive to changes in price, and demand is **price inelastic**. Figure 3.1(a) illustrates price inelastic demand: the percentage change in quantity demanded (a 5% decrease) is smaller than the percentage change in price (a 10% increase), therefore *PED* is less than one.
- **Demand is price elastic when $PED > 1$ (but less than infinity).** The percentage change in quantity demanded is larger than the percentage change in price, so the value of *PED* is greater than one; quantity demanded is relatively responsive to price changes, and demand is **price elastic**. In Figure 3.1(b) the percentage change in quantity demanded (−10%) is larger than the percentage change in price (5%), therefore *PED* is greater than one.

In addition, there are three special cases where PED is constant (unchanging) along the full length of the demand curve:

- **Demand is unit elastic when $PED = 1$.** The percentage change in quantity demanded is equal to the percentage change in price, so *PED* is equal to one; demand is then *unit elastic*; there is **unitary *PED***. Figure 3.1(c) shows a unit elastic demand curve, where the percentage change in quantity demanded (−5%) is equal to the percentage change in price (5%).
- **Demand is perfectly inelastic when $PED = 0$.** The percentage change in quantity demanded is zero; there is no change in quantity demanded, which remains constant at Q_1 no matter what happens to price; *PED* is then equal to zero and demand is **perfectly inelastic**. For example, a heroin addict's quantity of heroin demanded is unresponsive to changes in the price of heroin. Figure 3.1(d) shows that a perfectly inelastic demand curve is vertical.
- **Demand is perfectly elastic when *PED* = infinity.** When a change in price results in an infinitely large response in quantity demanded, demand is **perfectly elastic**. As shown in Figure 3.1(e) the perfectly elastic demand curve is horizontal. At price P_1, consumers will buy any quantity that is available. If price falls, buyers will buy all they can (an infinitely large response); if there is an increase in price, quantity demanded drops to zero. This apparently strange kind of demand will be considered in Chapter 7 (at HL).

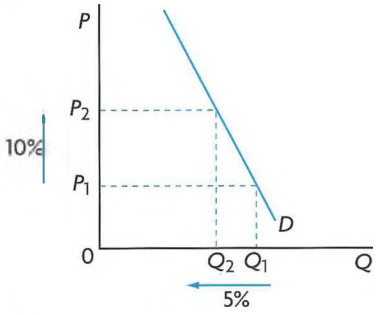
The numerical value of *PED* can therefore vary from zero to infinity. In general, the larger the value of *PED*, the greater the responsiveness of quantity demanded. While the cases of unit elastic, perfectly inelastic and perfectly elastic demand are rarely encountered in practice, they have important applications in economic theory.

Value of *PED*	Classification	Interpretation
Frequently encountered cases		
$0 < PED < 1$ (greater than zero and less than one)	price inelastic demand	quantity demanded is relatively unresponsive to price
$1 < PED < \infty$ (greater than 1 and less than infinity)	price elastic demand	quantity demanded is relatively responsive to price
Special cases: constant PED along the length of the demand curve		
$PED = 1$	unit elastic demand	percentage change in quantity demanded equals percentage change in price
$PED = 0$	perfectly inelastic demand	quantity demanded is completely unresponsive to price
$PED = \infty$	perfectly elastic demand	quantity demanded is infinitely responsive to price

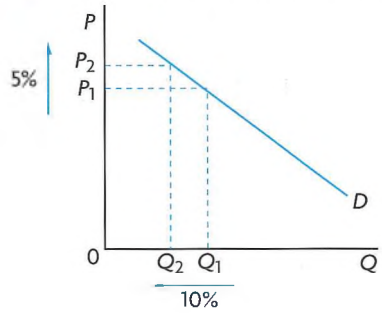
Table 3.1: Characteristics of price elasticity of demand

Frequently encountered cases

a Price inelastic demand: $0 < PED < 1$

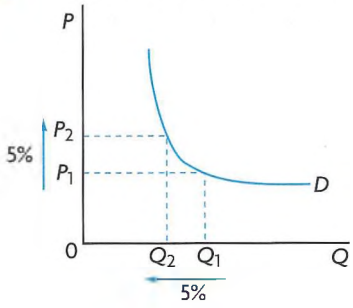


b Price elastic demand: $1 < PED < \infty$

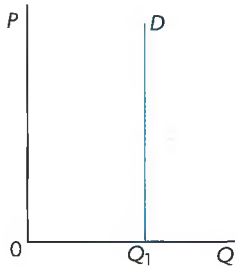


Special cases

c Unit elastic demand: $PED = 1$



d Perfectly inelastic demand: $PED = 0$



e Pefectly elastic demand: $PED = \infty$

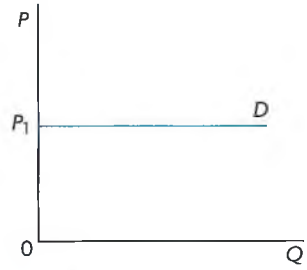


Figure 3.1: Demand curves and *PED*

PED and the steepness of the demand curve

The variety of demand curves and their *PED*s in Figure 3.1 suggest that the flatter the demand curve, the more elastic the demand (the higher the *PED*); the steeper the demand curve, the less elastic the demand (the lower the *PED*).

However, it is not always accurate to conclude that demand is more elastic or less elastic in different demand curves simply by comparing their steepness. The reason is that demand curves drawn on different scales are not comparable. Figure 3.2 shows two identical demand curves with different scales on the horizontal axis. It would be incorrect to conclude that the steeper demand curve has a less elastic demand.

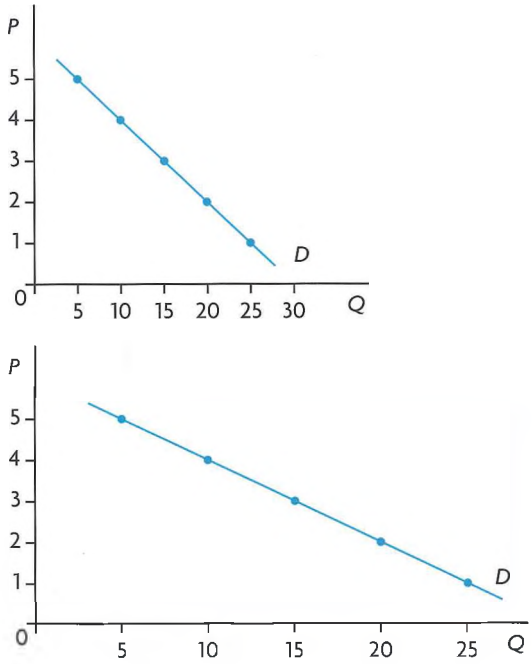


Figure 3.2: Two identical demand curves

So when is it correct to compare *PED*s of demand curves by referring to their steepness? *This can be done when* the demand curves intersect at some point, such as demand curves D_1 and D_2 in Figure 3.3. In this figure, for any price, D_1 is flatter and more elastic than D_2. For example, if price falls from P_1 to P_2, the resulting percentage change in quantity will be larger for D_1 (increase from Q_1 to Q_3) than for D_2 (increase from Q_1 to Q_2). In general, when demand curves intersect, then for any given price, the flatter the demand curve, the more elastic is the demand. This generalisation holds for comparisons between two demand curves at a particular price.

We often use the relative steepness of demand curves to be an indication of *PED*, so that comparing two demand curves, the one that is flatter is said to be more elastic while the one that is steeper is said to be more inelastic. (This is done on the assumption that if they were drawn on the same diagram they would intersect at some point.)

Intersecting demand curves and PEDs

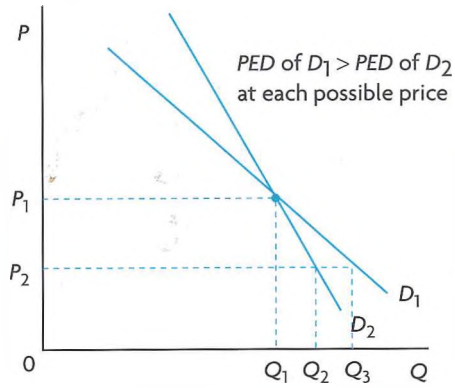


Figure 3.3: Demand curves and PEDs

TEST YOUR UNDERSTANDING 3.2

1 Specify the value for each of the following *PED*s and show, using diagrams, the shape of the demand curve that corresponds to each one:
   - a perfectly elastic demand,
   - b unitary *PED* (unit elastic demand) and
   - c perfectly inelastic demand.

2 Provide examples of goods likely to have demand that is
   - a elastic, and
   - b inelastic.

3 Identify which price elasticity of demand values or range of values we see most frequently in the real world.

4 Assuming two demand curves intersect, explain which of the two will be relatively more elastic for a given price.

Changing *PED* and the straight-line demand curve (HL only)

Why *PED* varies

Along any *downward-sloping, straight-line demand curve*, the *PED* varies (changes) as we move along the curve. This applies to all demand curves of the types shown in Figure 3.1(a) and (b). It excludes unit elastic, perfectly inelastic and perfectly elastic demand curves (where *PED* is constant). We can see in Figure 3.4 that when price is low and quantity is high, demand is inelastic; as we move up the demand curve towards higher prices and lower quantities, demand becomes more and more elastic. The figure shows the *PED* values along different parts of the demand curve.

The reason behind the changing *PED* along a straight-line demand curve has to do with how *PED* is calculated. At high prices and low quantities, the percentage change in Q is relatively large (since the denominator of $\Delta Q/Q$ is small), while the percentage change in P is relatively small (because the denominator of $\Delta P/P$ is large). Therefore, the value of *PED*, given by a large percentage change in Q divided by a small percentage change in P results in a large *PED* (elastic demand). At low prices and high quantities, the opposite holds. The value of *PED* is given by a low percentage change in Q divided by a high percentage change in P, resulting in a low *PED* (inelastic demand).

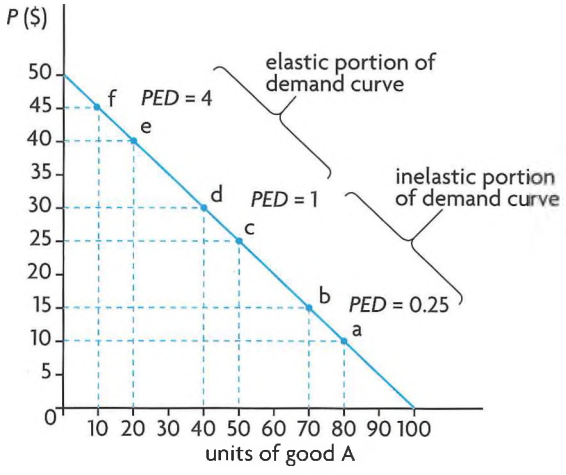


Figure 3.4: Variability of *PED* along a straight-line demand curve

On any downward-sloping, straight-line demand curve, demand is price-elastic at high prices and low quantities, and price-inelastic at low price and large quantities. At the midpoint of the demand curve, there is unit elastic demand.

Therefore, the terms 'elastic' and 'inelastic' should not be used to refer to an entire demand curve (with the exception of the three special cases where *PED* is constant), but only to a portion of the demand curve corresponding to a particular price or price range.

Why *PED* varies along a straight-line demand curve from a different perspective

A far simpler explanation of the varying *PED* along a straight-line demand curve is the following.

Note that the slope of demand curve[2] $= \frac{\Delta P}{\Delta Q}$, while

$$PED = \frac{\frac{\Delta Q}{Q}}{\frac{\Delta P}{P}} = \frac{\Delta Q}{\Delta P} \times \frac{P}{Q} = \frac{1}{slope} \times \frac{P}{Q}$$

Now we know that the slope along a straight line is always constant, therefore the inverse of the slope = 1/slope is also constant.

Therefore *PED* is equal to a constant number multiplied by P/Q. As we move down the demand curve, we know that P gets smaller and smaller, while Q gets bigger and bigger, therefore P/Q becomes continuously smaller. This means that as we move down the demand curve, *PED*, which is equal to a constant number times an ever decreasing number, must be getting smaller and smaller. We thus have a simple explanation of why *PED* continuously decreases as we move down the demand curve.

[2] It may be noted that strictly speaking the slope of the demand curve is $\Delta Q/\Delta P$. The reason is that mathematically the slope is defined as the change in the dependent variable divided by the change in the independent variable. In the case of demand, Q is the dependent variable and P is the independent variable. However because of the practice of reversing the P and Q axes, and putting P on the vertical axis, we may take the slope as being $\Delta P/\Delta Q$. In any case this does not make any difference to the point of the argument in the text because both $\Delta Q/\Delta P$ and $\Delta P/\Delta Q$ are constant along a straight line.

TEST YOUR UNDERSTANDING 3.3

1 Using the information in Figure 3.4, calculate *PED* between:

- a points a and b, where price increases from $10 to $15;
- b points c and d, where price increases from $25 to $30; and
- c points e and f, where price increases from $40 to $45.
- d State the general principle about values of the *PED* along the straight-line demand curve that your calculations show.

Determinants of price elasticity of demand

We will now consider the factors that determine whether the demand for a good is elastic or inelastic.

Number and closeness of substitutes

The more substitutes a good (or service) has, the more elastic is its demand. If the price of a good with many substitutes increases, consumers can switch to other substitute products, therefore resulting in a relatively large drop (large responsiveness) in quantity demanded. For example, there are many brands of toothpaste, which are close substitutes for each other. An increase in the price of one, with the prices of others constant will lead consumers to switch to the others; hence demand for a specific toothpaste brand is price elastic. If a good or service has few or no substitutes, then an increase in price will bring forth a relatively small drop in quantity demanded. An increase in the price of petrol (gasoline) is likely to lead to a relatively small decrease in quantity demanded, because there are no close substitutes; therefore, demand for petrol is price inelastic.

Also important is the closeness of substitutes. For example, Coca-Cola® and Pepsi® are much closer substitutes than Coca-Cola and orange juice; we say that Coca-Cola and Pepsi have greater *substitutability*. The closer two substitutes are to each other, the greater the responsiveness of quantity demanded to a change in the price of the substitute, hence the greater the *PED*, because it is easier for the consumer to switch from one product to the other.

A factor that affects the number of substitutes is whether the good is defined broadly or narrowly. For example, *fruit* is a broad definition in comparison to *specific fruits* like oranges, apples and pears, which are narrowly defined. If we had considered fruit in relation to food, *food* is broad and *fruit* is narrow. The point here is that the narrower the definition of a good, the greater the number of close substitutes and the more elastic the demand. The demand for apples is more elastic than the demand for fruit; demand for fruit is more elastic than the demand for food. Similarly, a Honda has a higher *PED* than all cars considered together.

Necessities versus luxuries

Necessities are goods or services we consider to be essential or necessary in our lives; we cannot do without them. **Luxuries** are not necessary or essential. The demand for necessities is less elastic than the demand for luxuries. For example, the demand for medications tends to be very inelastic because people's health or life depend on them; therefore, quantity demanded is not very responsive to changes in price. The demand for food is also inelastic, because people cannot live without it. On the other hand, the demand for diamond rings is elastic as most people view them as luxuries. In general, the more necessary a good, the less elastic the demand.

A special case of necessity is a consumer's addiction to a good. The greater the degree of addiction to a substance (alcohol, cigarettes, and so on), the more inelastic is the demand. A price increase will not bring forth a significant reduction in quantity demanded if one is severely addicted.

Length of time

The longer the time period in which a consumer makes a purchasing decision, the more elastic the demand. As time goes by, consumers have the opportunity to consider whether they really want the good, and to get information on the availability of alternatives to the good in question. For example, if there is an increase in the price of heating oil, consumers can do little to switch to other forms of heating in a short period of time, and therefore demand for heating oil tends to be inelastic over short periods. But as time goes by, they can switch to other heating systems, such as gas, or they can install better insulation, and demand for heating oil becomes more elastic.

Proportion of income spent on a good

The larger the proportion of one's income needed to buy a good, the more elastic the demand. An item such as a pen takes up a very small proportion of one's income, whereas a car takes up a much larger proportion. For the same percentage increase in the price of pens and in the price of cars, the response in quantity demanded is likely to be greater in the case of the car than in the case of pens.

TEST YOUR UNDERSTANDING 3.4

1 Identify and explain the determinants of price elasticity of demand.

2 State in which case demand is likely to be more elastic in each of the following pairs of goods, and why:

- a chocolate or Cadbury's chocolate
- b orange juice or water
- c cigarettes or sweets

CONTINUED

- d a notepad or a computer
- e heating oil in one week or in one year
- f bread or meat.

Applications of price elasticity of demand

Price elasticity of demand is a very important concept in economics, with numerous applications. Some of these will be considered below; others will be studied in later chapters.

PED and total revenue

PED and the effects of price changes on total revenue

Total revenue (TR) is the amount of money received by firms when they sell a good (or service), and is equal to the price (P) of the good times the quantity (Q) of the good sold. Therefore, $TR = P \times Q$.

REAL WORLD FOCUS 3.1

What happens when demand is highly price inelastic?

A girl sells lemonade at a stand for 50 cents (= $0.50) a cup. On a very hot day, the lemonade becomes even more popular, and the girl realises she can raise her price a little and still sell all her lemonade. One afternoon, a diabetic boy comes along asking for lemonade with extra sugar because his blood sugar has fallen to dangerously low levels. The girl sees an opportunity and increases her price by 500%. The boy doesn't have enough money, but she tells him she will give him the lemonade right away provided he promises to run home afterward, get the money and return to pay her the full price. Having no choice, the boy agrees.



Figure 3.5: Little girl selling lemonade

Applying your skills

1 a Describe the boy's price elasticity of demand for sweet lemonade at that particular moment.

b Identify what determinant of *PED* accounts for this.

2 What would have happened to the quantity of lemonade demanded if the other children were faced with a 500% increase in its price? Explain in terms of their price elasticity of demand for lemonade.

Source: *Adapted from Teymour Semnani, 'Free markets don't always do the right thing regarding health care' in The Deseret News, 15 November 2009.*

We are interested in examining what will happen to the firm's total revenue (*TR*) when there is a change in the price of the good it produces and sells. We know that an increase in *P* leads to a decrease in *Q* demanded and vice versa. What can we say about the resulting change in total revenue? Will it increase, decrease or stay the same? The change will depend on price elasticity of demand of the good. We have the following three possibilities:

Demand is elastic (*PED* > 1)

When demand is elastic, an increase in price causes a fall in total revenue, while a decrease in price causes a rise in total revenue. To see why, consider that if demand is elastic, so *PED*>1, an increase in price results in a proportionately larger decrease in quantity demanded. For example, if price rises by 10% quantity demanded will fall by more than 10%. The decrease in quantity has a bigger impact on total revenue than the increase in price; therefore, total revenue falls. If there is a price decrease, a 10% price fall results in a larger than 10% increase in quantity demanded, and total revenue increases.

When demand is elastic, an increase in price causes a fall in total revenue, while a decrease in price causes a rise in total revenue.

Demand is inelastic (*PED* < 1)

When demand is inelastic, an increase in price causes an increase in total revenue, while a decrease in price causes a fall in total revenue. Since *PED* < 1, an increase in price causes a proportionately smaller decrease in quantity demanded. For example, a 10% price increase produces a smaller than 10% decrease in quantity demanded, and total revenue rises. If price falls, a percentage price decrease gives rise to a smaller percentage increase in quantity demanded and total revenue falls. In both cases, the effect on total revenue of the change in price is larger than the effect of the change in quantity.

When demand is inelastic, an increase in price causes an increase in total revenue, while a decrease in price causes a fall in total revenue.

Demand is unit elastic (unitary *PED*; *PED* = 1)

When demand is unit elastic, the percentage change in quantity is equal to the percentage change in price, and total revenue remains constant.

When demand is unit elastic, a change in price does not cause any change in total revenue.

Using diagrams to illustrate *PED* and the effects of price changes on total revenue

Figure 3.6 shows three demand curves. The first shows a price range where *PED*>1, the second a price range where *PED*<1 and the third unitary *PED* throughout its range. In all three diagrams total revenue or *TR* is shown by the rectangles represented by $P \times Q$.

Elastic demand: as P increases TR falls: P and TR change in opposite directions

In Figure 3.6(a), where demand is elastic, at the initial price and quantity, P_1 and Q_1, total revenue is given by the sum of the rectangles A and B. When price increases to P_2 and quantity drops to Q_2, total revenue is given by the sum of the rectangles A and C. Due to the price increase rectangle B was lost and the rectangle C was gained. Since the loss (B) is larger than the gain (C), total revenue fell.

We can use the same diagram to explore a price decrease when *PED* > 1, simply by assuming that the initial price and quantity are P_2 and Q_2; price then falls to P_1 while quantity increases to Q_1. The gain in *TR* is given by rectangle B, which is greater than the loss shown by rectangle C, thus total revenue increases.

Inelastic demand: as P increases TR increases: P and TR change in same direction

These results can be seen in Figure 3.6(b). With a price increase, total revenue gained (rectangle C) is larger than total revenue lost (rectangle B); therefore, total revenue increases. If price falls from P_2 to P_1, the gain in total revenue (rectangle B) is smaller than the loss (rectangle C) and total revenue falls.

Unitary *PED*: any change in P leaves TR unchanged

In Figure 3.6(c), as price and quantity change, the gain in total revenue is exactly matched by the loss, and total revenue remains unchanged.

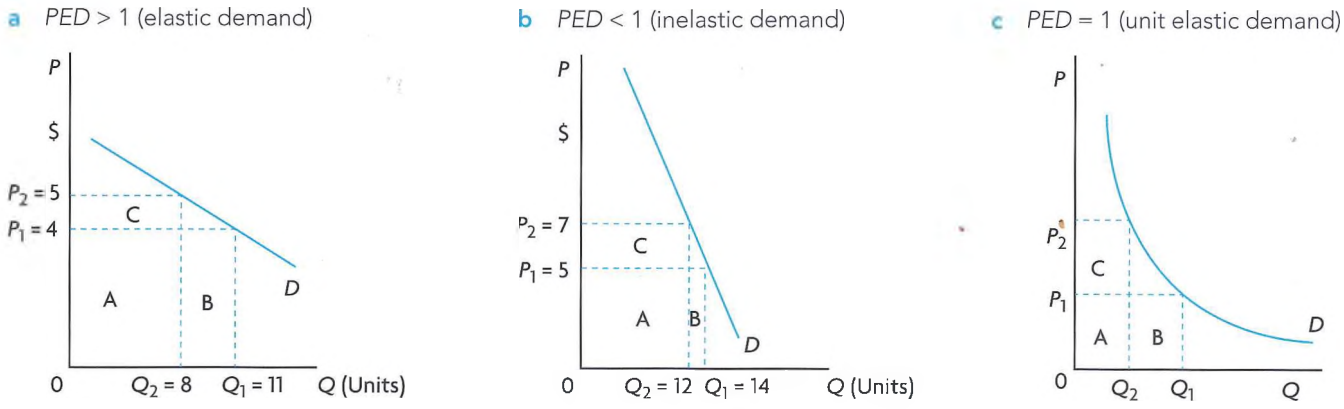


Figure 3.6: PED and total revenue

Calculating total revenue at different points on the demand curve

We can use the information in Figure 3.6 to calculate total revenue as price changes. In Figure 3.6(a) where demand is elastic, initially $TR = P_1 \times Q_1 = 4 \times 11 = \$44$. After the price increase $TR = P_2 \times Q_2 = 5 \times 8 = \$40$. Therefore the increase in price caused a fall in total revenue, confirming the point that when demand is elastic P and TR change in opposite directions.

In Figure 3.6(b) demand is inelastic. Initially $TR = P_1 \times Q_1 = 5 \times 14 = \$70$. After the price increase $TR = P_2 \times Q_2 = 7 \times 12 = \$84$. The price increase here led to an increase in total revenue confirming the point that when demand is inelastic price and total revenue change in the same direction.

PED and firm pricing decisions

The above discussion shows that businesses must take *PED* into account when considering changes in the price of their product. If a business wants to increase total revenue, it must drop its price if demand is elastic, or increase its price if demand is inelastic. If demand is unit elastic, the firm is unable to change its total revenue by changing its price.

A firm's total revenue should not be confused with *profit*. Profit is total revenue minus total costs.

PED and indirect taxes

Governments often impose taxes on specific goods. Such taxes are a type of *indirect tax* (to be discussed in Chapter 4). If governments are interested in increasing their tax revenues, they must consider the *PED* of the goods to be taxed for the following reason.

> The lower the price elasticity of demand for the taxed good, the greater the government tax revenues.

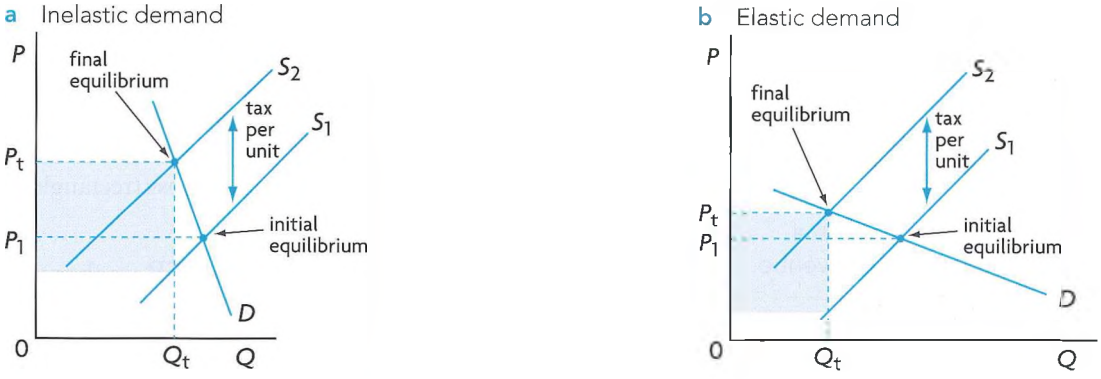


Figure 3.7: PED, indirect taxes and government tax revenue

This can be seen in Figure 3.7, showing the case of inelastic demand in part (a) and elastic demand in part (b).[3] When a tax is imposed on a good, it has the effect of shifting the supply curve upward. The reason is that for every level of output the firm is willing and able to supply to the market, it must receive a price that is higher than the original price by the amount of the tax. (This is equivalent to a leftward shift of the supply curve; for an explanation see 'Quantitative techniques' chapter in the digital coursebook.) The curve shifts from S_1 to S_2 so that the vertical distance between S_1 and S_2 is equal to the amount of the tax per unit of output. The new, after-tax equilibrium occurs at price P_t and quantity Q_t, determined by the intersection of the demand curve, D, and the new supply curve, S_2. The shaded area represents the government's tax revenue, obtained by multiplying the amount of tax per unit times the number of units, or quantity Q_t. A comparison of the two figures indicates that tax revenue is larger when demand is inelastic. This result follows from the principle that when demand is inelastic ($0 < PED < 1$), an increase in price (here due to the increase in the tax) leads to a proportionately smaller decrease in quantity demanded, and hence to an increase in total revenue (i.e. tax revenue). Indirect taxes are therefore usually imposed on goods like cigarettes and petrol (gasoline), which have a low *PED*.

TEST YOUR UNDERSTANDING 3.5

1 Explain and show, using diagrams, how total revenue will change if

- a price increases and demand is elastic,
- b price decreases and demand is inelastic,
- c price increases and demand is perfectly inelastic,
- d price increases and demand is inelastic,
- e price decreases and demand has unit elasticity, and
- f price decreases and demand is elastic.

2 Using diagrams, discuss how a firm's knowledge of price elasticity of demand for its product can help it in its pricing decisions.

3 Refer to Figure 3.2 in this chapter.

- a Calculate the change in total revenue when price increases from \$4 to \$5.
- b Calculate the change in total revenue when price increases from \$1 to \$2.
- c Given your results in (a) and (b), comment on the size of the *PED* for a price change from \$4 to \$5, and the size of the *PED* for a price change from \$1 to \$2.
- d Calculate *PED*s for the two price changes to confirm your answers.

4 (HL only) Referring to Figure 3.4 calculate the change in total revenue that results when:

- a price increases from \$10 to \$15, and
- b price increases from \$40 to \$45.
- c Noting when total revenue increased and when it decreased, explain what your calculations show about *PED* along a straight-line demand curve.

5 The government would like to levy indirect taxes on certain goods to raise tax revenue. Using diagrams, explain how price elasticity of demand can help it decide which products it should tax.

[3] Note we are assuming that the two demand curves are drawn on the same scale, and that if they were drawn in the same diagram they would intersect, therefore the *PED*s are comparable.

PED in relation to primary commodities and manufactured products (HL only)

Why many primary commodities have a lower *PED* compared with the *PED* of manufactured products

Primary commodities are goods arising directly from the use of natural resources, or the factor of production 'land' (see Chapter 1). Primary commodities therefore include agricultural, fishing and forestry products, as well as products of extractive industries (oil, coal, minerals, and so on). Agricultural products include food, as well as other, non-edible commodities (such as cotton and rubber).

Many primary commodities have a low *PED*, which is usually lower than the *PED* of manufactured products (as well as services). **Manufactured products** are goods produced by labour usually working together with capital as well as raw materials, such as for example cars, computers and televisions. Food has a highly price inelastic demand, because it is a necessity and it has no substitutes. The same applies to a variety of other primary products (such as oil and minerals). In the case of food, in developed countries the *PED* is estimated to be between 0.20 and 0.25. By contrast, the demand for manufactured products tends to be more price elastic, because these products, though they may be necessities (in some cases), usually do have substitutes. Therefore, given a price change, quantity demanded is generally more responsive in the case of manufactured products compared with primary commodities. (Note, however, that there are exceptions. For example, medications are manufactured products, yet their demand tends to be inelastic because they are necessities and have no substitutes.)

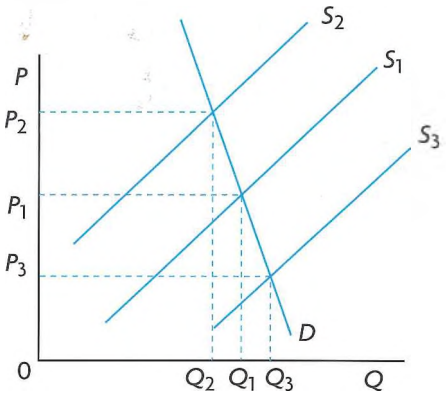
> Many primary commodities have a relatively low *PED* (price inelastic demand) because they are necessities and have no substitutes (for example, food and oil). The *PED* of manufactured products is relatively high (price elastic demand) because they usually have substitutes.

Consequences of a low *PED* for primary commodities (Supplementary material)

Low price elasticity of demand, together with fluctuations in supply over short periods of time, creates serious problems for primary commodity producers, because they result in large fluctuations in primary commodity prices, and these also affect producers' incomes.

Consider the diagrams in Figure 3.8. Part (a) shows relatively inelastic demand (such as for primary commodities) and part (b) shows relatively elastic demand (such as for manufactured products).

a Primary commodities: supply shifts with inelastic demand



b Manufactured products: supply shifts with elastic demand

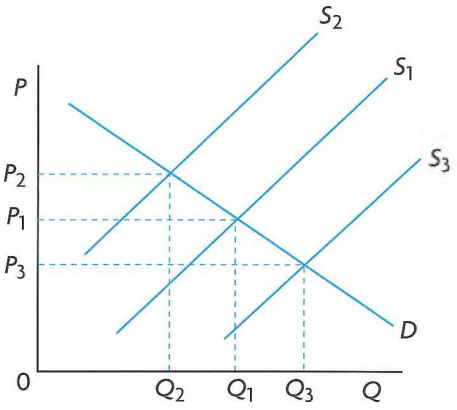


Figure 3.8: Price fluctuations are larger for primary commodities because of low PED

Both diagrams show the effects on price and quantity when there is a decrease in supply (from S_1 to S_2) and when there is an increase in supply (from S_1 to S_3). A comparison of the two diagrams reveals that shifts in the supply curve result in large price fluctuations when demand is inelastic, and much milder ones when demand is elastic. Large price fluctuations over short periods of time are referred to as *price volatility*. (Volatility means instability or high variability.)

Figure 3.8 explains why, in the real world, prices of primary commodities can be highly volatile, whereas they tend to be much less so in the manufacturing and services sectors.

Two results follow from this:

- As primary commodity prices fluctuate widely, so do producers' incomes, which depend on the revenues (price × quantity) producers receive from selling their output.
- In view of the relationship between *PED* and total revenue (see above), a fall in the supply of a primary commodity with inelastic demand (from S_1 to S_2 in Figure 3.8(a)) leads to an increase in total revenue of producers because the percentage increase in price is larger than the percentage decrease in quantity. An increase in supply leads to lower revenues (the percentage decrease in price is larger than the percentage increase in quantity).

These points lead to some unexpected conclusions. They show that a poor crop in agriculture, say due to poor weather conditions, which results in a fall in supply (S_2 in Figure 3.8(a)), leads to higher prices and higher total revenue for farmers. A good crop resulting in a supply increase, or S_3, leads to lower prices and lower farmers' revenues. We come, therefore, to the ironic conclusion that a poor crop may be good for farmers while a good crop may be bad for them.

If supply of agricultural products were relatively stable, the problem would be less serious as agricultural product prices would also be more stable. However, agricultural production depends on many factors beyond the farmer's control, such as drought, pests, floods, frost and other such natural disasters, as well as exceptionally good weather conditions, which occur over short periods of time. These cause frequent and large supply changes (supply curve shifts).

The problem of unstable farmer revenues is an important reason behind government intervention to support farmer incomes, which we will study in Chapter 4. The implications of unstable primary product prices for farmer revenues and the economy will be explored in Chapter 19 .

TEST YOUR UNDERSTANDING 3.6

1 Suppose flooding destroys a substantial portion of this season's crop. Using diagrams, explain what is likely to happen to farmers' revenues, assuming the demand for the product they produce is inelastic.

2 a Using examples, explain why many primary commodities have a relatively low *PED* while many manufactured products have a relatively high *PED*.

b Use the concept of *PED* and diagrams to explain why agricultural product prices tend to fluctuate more (are more volatile) compared with manufactured product prices over the short term. (Optional)

3.2 Income elasticity of demand (*YED*)

LEARNING OBJECTIVES

After studying this section you will be able to:

- define all the terms appearing in **orange bold** in the text (AO1)
- use the formula for income elasticity of demand (*YED*) to calculate *YED*, change in income and quantity demanded (AO4)
- draw an Engel curve diagram to show income elastic, income inelastic and inferior goods (AO4)
- depending on the sign of *YED*, distinguish between normal and inferior goods (AO2)
- depending on the value of *YED* (less than one or greater than one), distinguish between necessities, services and luxury goods (AO2)
- apply *YED* to (HL only) (AO3)
  - discuss its relevance to firms
  - discuss changes in the sectoral structure of an economy

Income elasticity of demand

Understanding income elasticity of demand

Consumer income is an important factor influencing demand for a good and the position of the demand curve. We have already encountered the role of income in Chapter 2, where we saw that income is a factor that causes demand curve shifts (see Chapter 2).

Income elasticity of demand (*YED*) is a measure of the responsiveness of demand to changes in income, and involves demand curve shifts. It provides information on the direction of change of demand given a change in income (increase or decrease) and the size of the change (size of demand curve shifts).

Calculating *YED*, change in price and change in quantity

The formula for *YED* has the same basic form as the other elasticity formulae, and shows the relationship between the percentage change in quantity demanded of a good, *X*, and the percentage change in income, which we abbreviate as *Y*:

income elasticity of demand (*YED*)

$$= \frac{\text{percentage change in quantity demanded of good}}{\text{percentage change in income}} \times$$

$$YED = \frac{\%\Delta Q}{\%\Delta P}$$

Which can be rewritten as:

$$\frac{\frac{\Delta Q}{Q} \times 100}{\frac{\Delta Y}{Y} \times 100} = \frac{\frac{\Delta Q}{Q}}{\frac{\Delta Y}{Y}} \text{ where}$$

$\Delta Q = Q_{\text{final value}} - Q_{\text{initial value}}$

$Q = Q_{\text{initial value}}$

$\Delta Y = Y_{\text{final value}} - Y_{\text{initial value}}$

$P = Y_{\text{initial value}}$

Suppose your income increases from $800 per month to $1000 per month, and your purchases of clothes increase from $100 to $140 per month. What is your income elasticity of demand for clothes?

$$YED = \frac{\frac{40}{100}}{\frac{200}{800}} = \frac{0.40}{0.25} = +1.6$$

Your income elasticity demand for clothes is +1.6.

If we know *YED* and the percentage change in quantity demanded, it is simple to calculate the percentage change in income. Suppose *YED* = 0.75 and quantity demanded of a *normal* good has increased by 15%. The percentage change in income is found by taking

$$YED = 0.75 = \frac{0.15}{\%\Delta Y} \Rightarrow 0.75x\%\Delta Y = 0.15 \Rightarrow \%\Delta Y$$

= 0.20 or 20%.

Income increased by 20%.

Similarly if we know *YED* and the percentage change in income, we can calculate the percentage change in quantity demanded. Suppose *YED* = 1.25 and income increases by 20%. The percentage change in Q is found by:

$$YED = 1.25 = \frac{\%\Delta Q}{0.20} \Rightarrow \%\Delta Q = 1.25x0.20 = 0.25 \text{ or } 25\%$$

Quantity demanded increased by 25%.

Interpreting income elasticity of demand

Income elasticity of demand provides two kinds of information:

- the sign of *YED*: positive or negative
- the numerical value of *YED*: whether it is greater or smaller than one (assuming it is positive).

The sign of income elasticity of demand: normal or inferior goods

The sign of *YED* tells us whether a good is normal or inferior:

- *YED* > 0. A positive income elasticity of demand indicates that the good in question is *normal*; demand for the good and income change in the same direction (both increase or both decrease). Most goods are normal goods (see Chapter 2).
- *YED* < 0. A negative income elasticity of demand indicates that the good is *inferior*: demand for the good and income move in opposite directions (as one increases the other decreases). Examples include bus rides, used clothes and used cars; in these cases, as income increases, the demand for these goods fall as consumers switch to consumption of normal goods (new cars, new clothes and so on; see Chapter 2).

The difference between normal and inferior goods can be seen in Figure 3.9, showing a demand curve, D_1, and shifts of the curve that occur in response to increases in income. As income increases, the demand curve shifts rightward from D_1 to D_3 or D_4 when goods are normal ($YED > 0$), but shifts leftward to D_2 when goods are inferior good ($YED < 0$).

The numerical value of income elasticity of demand: necessities, luxuries and services

Here we are making a distinction between goods that have a *YED* that is positive and less than or greater than one:

- *YED* < 1: *Necessities*. If a good has a *YED* that is positive but less than one, it has **income inelastic demand**: a percentage increase in income produces a smaller percentage increase in quantity demanded. Necessities are income inelastic goods.
- *YED* > 1: *Luxuries* and *services*. If a good has a *YED* that is greater than one, it has **income elastic demand**: a percentage increase in income produces a larger percentage increase in quantity demanded. Luxuries and services are income elastic.

Necessities, such as food, clothing and housing, tend to have a *YED* that is positive but less than one; they are normal goods that are income inelastic. In the case of food, as income increases, people buy more food but the amount of income spent on food increases more slowly than income. In developed countries, *YED* for food is about 0.15 to 0.2. This means that a 1% increase in income produces a 0.15% to 0.2% increase in spending on food; or a 10% increase in income results in a 1.5% to 2% increase in spending on food. By contrast, luxuries, such as jewellery and expensive cars, as well as many services, such as travel to other countries, private education and eating in restaurants are income elastic: as income increases, the amount of income spent on such goods increases faster than income (the denominator in the *YED* formula is smaller than the numerator).

What is a necessity and what is a luxury depends on income levels. For people with extremely low incomes, even food and certainly clothing can be luxuries. As income increases, certain items that used to be luxuries become necessities. For example, items like Coca-Cola® and coffee for many poor people in less developed countries are luxuries, whereas for consumers in developed countries they have become necessities. Income elasticity of demand for particular items therefore varies widely depending on income levels. While *YED* for food is about 0.15–0.20 in more developed countries, it is about 0.8 in poor countries. For an increase in income of 10%, spending on food increases by only 1.5%–2% in rich countries and by 8% in poor countries.

In Figure 3.9, we see that in the case of necessities, an increase in income will produce a relatively small rightward shift in the demand curve; in the case of luxuries and services, the rightward shift will be larger.

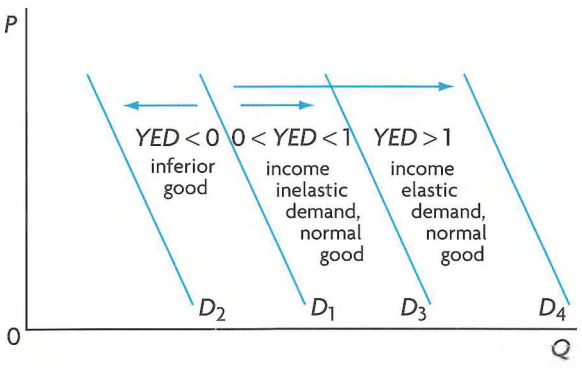


Figure 3.9: Demand curve shifts in response to increases in income for different *YED*s

TEST YOUR UNDERSTANDING 3.7

1 Explain the meaning of income elasticity of demand. Why do we say it involves *a shifting demand curve*?

2 Explain the difference between normal and inferior goods and provide examples of each.

3 Your income increases from £1000 a month to £1200 a month. As a result, you increase your purchases of pizzas from 8 to 12 per month, and you decrease your purchases of cheese sandwiches from 15 to 10 per month.

   a Calculate your income elasticity of demand for pizzas and for cheese sandwiches.

   b What kind of goods are pizzas and cheese sandwiches for you?

   c Show using diagrams the effects of your increase in income on your demand for pizzas and cheese sandwiches.

4 A 15% increase in income leads to a 10% increase in demand for good A and 20% increase in demand for good B.

   a Identify which of the two goods is income elastic and which is income inelastic.

   b Identify which of the two goods is likely to be a necessity good and which a luxury good.

The Engel curve

The Engel curve is a far more accurate way to illustrate *YED* than demand curve shifts. It is named after Ernst Engel, a German statistician and economist who lived in the 19th century, and who was the first to study the relationship between consumer income and demand for a product. An Engel curve is shown in Figure 3.10, where the vertical axis measures the income of a consumer per week, the horizontal axis measures the quantity of hot dogs she or he buys each week, while the solid line shows the curve itself. We can see straight away that hot dogs are a normal good from point A to point C, since as income increases from $100 to $250, the quantity increases from 4 hot dogs to 9 hot dogs. As income increases further to $350 the quantity remains constant at 9 hot dogs, but as income increases even more the quantity falls from 9 to 8 hot dogs. We can see therefore that when income goes above $350, hot dogs become an inferior good for this consumer.

With income on the vertical axis and quantity on the horizontal axis of an **Engel curve** diagram, we can see the following:

- YED>0 in the upward sloping part of the curve showing quantity and income both increasing, which indicates the good is normal
- YED<0 in the downward sloping part showing quantity decreasing as income increases, which indicates the good is inferior.

The Engel curve can also show whether a good is income inelastic or income elastic. In Figure 3.10, for very low incomes less than $150, hot dogs are a luxury as $YED>1$. At the higher income levels between $250 and $350, hot dogs have become a necessity as $YED<1$.

There is a simple rule that allows us to distinguish between a luxury and a necessity on the upward sloping part of the Engel curve.

Imagine each segment of the Engel curve extending backward to touch either the vertical axis or the horizontal axis, as shown by the dotted lines:

- *YED*>1 if the line touches the vertical axis, as with the line AB, so that it is a luxury or service
- *YED*<1 if the line touches the horizontal axis, as with line BC, so that it is a necessity.

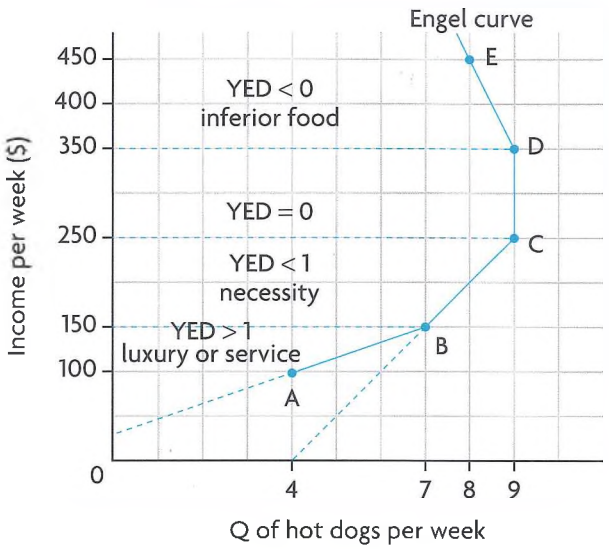


Figure 3.10: The Engel curve showing different YEDs

(The mathematically inclined student can easily see why this is so. As we know

$$YED = \frac{\frac{\Delta Q}{Q}}{\frac{\Delta Q}{Y}} = \frac{\Delta Q}{\Delta Y} \times \frac{Y}{Q} = \frac{\frac{Y}{Q}}{\frac{\Delta Y}{\Delta Q}}$$

Imagine now a line joining the origin with point B on the Engel curve. The slope of this line is Y/Q. Note also that the slope of the Engel curve is ΔY/ ΔQ. Therefore *YED* has been written as the ratio of the two slopes: slope of OB/slope of Engel curve. Now note that OB is steeper than AB, therefore the slope of OB>the slope of the Engel curve from A to B, and so *YED*>1 up to point B. On the other hand, OB is flatter than BC, therefore the slope of OB<the slope of the Engel curve from B to C, and so *YED*<1 from B to C.)

The Engel curve shows a continuum: at very low incomes a good may be a luxury; as income increases it becomes a necessity and finally at high income levels it becomes inferior.

The information in Figure 3.10 can be used to calculate *YED*s in order to confirm the above points. This will be given to you as an exercise.

TEST YOUR UNDERSTANDING 3.8

1 Draw an Engel curve for a typical good and explain how it changes from income elastic to income inelastic to inferior as income increases.

2 Describe why income elasticity of demand for food has been estimated to be about 0.15 to 0.2 in more developed countries and about 0.8 in less developed countries.

3 Using the information in Figure 3.10, calculate *YED* for an increase in income

- a from $100 to $150,
- b $150 to $250,
- c $250 to $350, and
- d $350 to $450.
- e Explain what your results for these *YED*s tell you about the nature of the good for the various income levels.

Applications of income elasticity of demand (HL only)

YED and producers: the rate of expansion of industries

Over time, as countries experience economic growth, society's income increases. Increasing income means a growing demand for goods and services. Suppose that total income in an economy grows at an average rate of about 3% per year. If goods and services have income elastic demand (*YED* > 1), this means that demand for these goods and services grows at a higher rate than 3%. Examples usually include restaurants, movies, health care and foreign travel. Other goods and services have income inelastic demand (*YED* < 1), meaning that the demand for these grows at a rate of less than 3%. Examples include food, clothing and furniture. The first group (with the elastic demand) includes goods and services produced by industries that grow and expand faster than total income in the economy, while the second group includes goods or services produced by industries growing more slowly than total income.

The higher the *YED* for a good or service, the greater the expansion of its market is likely to be in the future;

the lower the *YED*, the smaller the expansion. Producers interested in producing in an expanding market may therefore want to know *YED*s of various goods and services.

In contrast to periods of economic growth, if an economy is experiencing a recession (falling output and incomes, see Chapter 8), goods and services with high *YED*s ($YED>1$) are the hardest hit, experiencing the largest declines in sales. Products with low *YED*s ($YED<1$) can avoid large falls in sales, while inferior goods ($YED<0$) can even experience increases in sales.

YED and the sectoral structure of the economy

The implications of differing *YED*s for the economy follow from what happens to particular industries in the economy as income grows, discussed above.

Every economy has three sectors (or parts): the primary sector including primary products (agriculture, forestry, fishing and extractive industries), the manufacturing sector and the services sector (including entertainment, travel, banking, insurance, health care, education, and so on). With economic growth, the relative size of the three sectors usually changes over time, and these changes can be explained in terms of income elasticity of demand.

Agriculture, the main part of the primary sector, produces food, which as noted above has a *YED* that is positive but less than one (it is income inelastic). As society's income grows over time, the demand for agricultural output grows more slowly than the growth in income. Other primary products also have a low income elasticity of demand. For example, cotton and rubber have synthetic substitutes, so as income increases a relatively larger proportion of it is spent on the synthetic materials, while a relatively lower fraction goes towards cotton and rubber. By contrast, manufactured products (cars, televisions, computers, and so on) have a *YED* that is usually greater than one (income elastic), so that as society's income grows, the demand for these products grows faster than income. Many services have even higher *YED*s, so the percentage increase in the demand for these is much larger.

Therefore, over time, the share of agricultural output in total output in the economy shrinks, while the share of manufactured output grows. With continued growth, the services sector expands at the expense of both agriculture and manufacturing.

REAL WORLD FOCUS 3.2

Income elasticity of demand and growth in demand for food products

According to the World Bank (an international organisation that lends to developing countries, see Chapter 20), the global rate of growth in demand for agricultural commodities like rice and wheat will fall from an average of 2.8% per year in the period 2010–2016 to an average of 1.8% in the period 2017–2018. The reason for this expected trend is that as consumers' incomes increase, consumers switch from commodities like rice and wheat to other foods that have a high protein and fat content.

Applying your skills

1. Explain the meaning of income elastic demand and income inelastic demand.
2. Based on the information in the text, outline your conclusion about the *YED* of commodities like rice and wheat, in comparison with the *YED* of foods with a high protein and fat content. In other words, which of the two is likely to be greater?
3. Explain why goods and services with highly income elastic demand stand to gain the most from rising consumer incomes.

Source: *Manas Chakravarty 'World Bank says long term growth in commodity consumption to weaken', 13 June 2018, Livemint www.livemint.com*

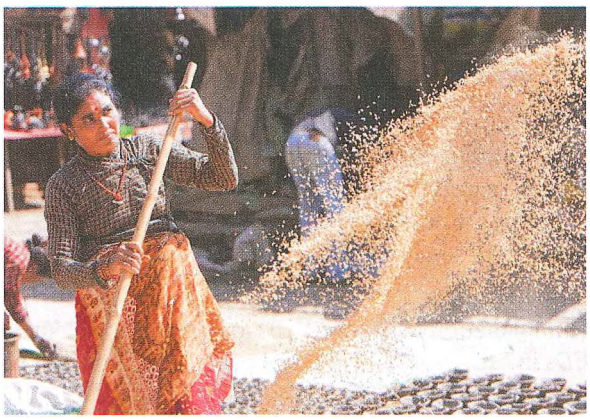


Figure 3.11: A Nepali woman winnowing rice, which is the major crop in Nepal

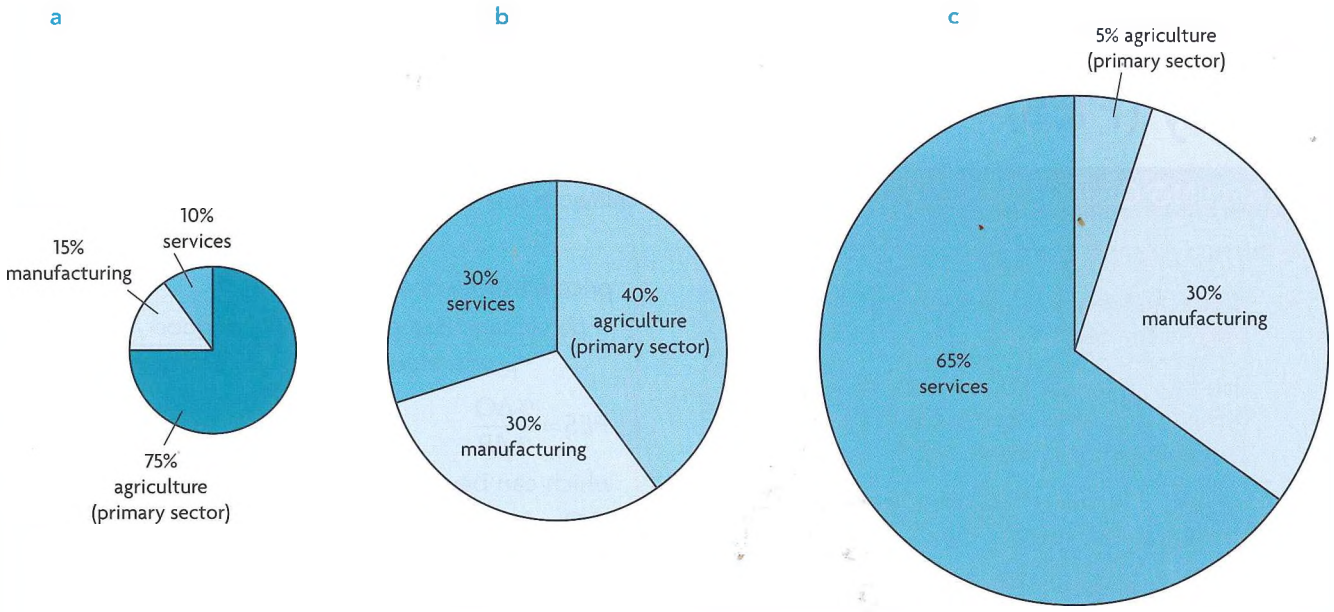


Figure 3.12: Changing relative shares (as percentage of total output) of primary, manufacturing and services sectors for a hypothetical economy as it grows

In Figure 3.12, this is shown for a hypothetical growing economy in the changes of relative sector sizes from parts (a) to (b) to (c).

Economically less developed countries usually have a large primary sector due to the importance of agriculture and extractive activities, while manufacturing and services are far less important. The developed countries of today were in a similar position many decades ago. The historical experience of both more and less developed countries shows that with economic growth, the primary sector becomes less and less important, and is partly replaced by manufacturing and services. As the economy grows further, the relative importance of the primary sector continues to shrink, and manufacturing becomes increasingly replaced by services. Thus, while less developed countries are usually dominated by the primary sector, more developed countries are dominated by services. In the developed world today, among the industries experiencing the fastest growth are services, including education, health care, travel and financial services.

Note that if total output is increasing over time, a falling share for a particular sector (such as the primary sector) does not necessarily mean that primary sector output is falling. Most likely it means that this sector's output is growing, but more slowly than total output. An increasing share for a sector means that its output is growing more rapidly than total output.

TEST YOUR UNDERSTANDING 3.9

1 Discuss why firms would be interested in knowing the *YED* of various goods and services.

2 Explain a likely reason behind the observed rapid growth in certain service industries, including health care, education and financial services, compared with other industries such as food (in the primary sector) and furniture (in the secondary sector).

3 Discuss the role of *YED* in the observed pattern of change in the relative shares of the primary, secondary and tertiary sectors in the economy as a country grows and develops.

3.3 Price elasticity of supply (*PES*)

LEARNING OBJECTIVES

After studying this section you will be able to:

- define all the terms appearing in **orange bold** in the text (AO1)
- use the formula for price elasticity of supply (*PES*) to calculate *PES*, changes in price and changes in quantity (AO4)
- identify the various degrees and range of values for *PES* (AO2)
- draw diagrams showing the range of values for *PES*, including relatively elastic and inelastic supply; and constant values for perfectly elastic supply, perfectly inelastic supply and unitary *PES* (AO4)
- analyse the determinants of *PES* (AO2)
- apply *PES* to analyse the reasons why primary commodities generally have a lower *PES* than manufactured products (HL only) (AO2)

Price elasticity of supply

Understanding price elasticity of supply

Until now, we have studied two demand elasticities, both of which involve consumer responses. We now turn to examine price elasticity of supply, which concerns firm (business) responses to changes in price. According to the law of supply, there is a positive relationship between price and quantity supplied: when price increases, quantity supplied increases and vice versa, *ceteris paribus*. But by how much does quantity supplied change?

Price elasticity of supply (*PES*) is a measure of the responsiveness of the quantity of a good supplied to changes in its price. *PES* is calculated along a given supply curve. In general, if there is a relatively large responsiveness of quantity supplied, supply is referred to as being *elastic*; if there is a relatively small responsiveness, supply is *inelastic*.

Calculating *PES*

The formula for price elasticity of supply (*PES*) follows the same general form of elasticity formulae, only now we consider the relationship between the percentage change in the price of a good, *X*, and the percentage change in quantity of *X* supplied:

price elasticity of supply (*PES*)

$$= \frac{\text{percentage change in quantity of good } X \text{ supplied}}{\text{percentage change in price of good } X}$$

$$PES = \frac{\%\Delta Q}{\%\Delta P}$$

which can be rewritten as

$$\frac{\frac{\Delta Q}{Q} \times 100}{\frac{\Delta P}{P} \times 100} = \frac{\frac{\Delta Q}{Q}}{\frac{\Delta P}{P}} \text{ where}$$

$$\Delta Q = Q_{\text{final value}} - Q_{\text{initial value}}$$

$$Q = Q_{\text{initial value}}$$

$$\Delta P = P_{\text{final value}} - P_{\text{initial value}}$$

$$P = P_{\text{initial value}}$$

Calculating PES, change in price and change in quantity

Suppose the price of strawberries increases from €3 per kg to €3.50 per kg, and the quantity of strawberries supplied increases from 1000 to 1100 tonnes per season. Calculate *PES* for strawberries.

$$PES = \frac{\frac{100}{1000}}{\frac{0.50}{3.00}} = \frac{0.10}{0.17} = +0.59$$

Price elasticity of supply for strawberries is +0.59.

If you are given *PES* and the percentage change in quantity supplied, you can find the percentage change in price by solving for this based on the formula above, in exactly the same way as in the case of *PED*. Similarly, you can solve for the percentage change in quantity supplied if you are given *PES* and percentage change in price.

Interpreting price elasticity of supply

The range of values for *PES*

Price elasticity of supply ranges in value from zero to infinity. Because of the positive relationship between price and quantity supplied, *PES* is positive.

The value of *PES* involves a comparison of the percentage change in quantity supplied (the numerator in the formula for *PES*) with the percentage change in price (the denominator). This comparison yields the following possible values and range of values of *PES*, which are illustrated in Figure 3.13 and summarised in Table 3.2:

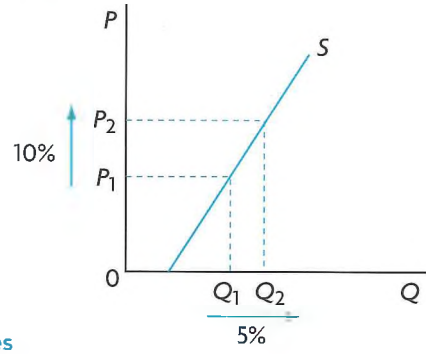
- **Supply is price inelastic when *PES*<1.** The percentage change in quantity supplied is smaller than the percentage change in price, so the value of *PES* is less than one; quantity supplied is relatively unresponsive to changes in price, and supply is *price inelastic* or *inelastic*. Figure 3.13(a) shows an inelastic supply curve (*PES*<1), where a 10% price increase leads to a 5% increase in quantity supplied. When *PES*<1, the supply curve extends upward and to the right from the horizontal axis; its end-point cuts the horizontal axis.[4]
- **Supply is price elastic when *PES*>1.** The percentage change in quantity supplied is larger than the percentage change in price, so the value of the *PES* is greater than one; quantity supplied is relatively responsive to price changes, and supply is *price elastic* or *elastic*. Figure 3.13(b) shows an elastic supply curve (*PES*>1) where the percentage increase in price (10%) is smaller than the percentage increase in quantity (15%). When *PES*>1, the supply curve extends upward and to the right from the vertical axis; its end-point cuts the vertical axis.[5]

In addition, there are three special cases of constant PES along the length of the supply curve:

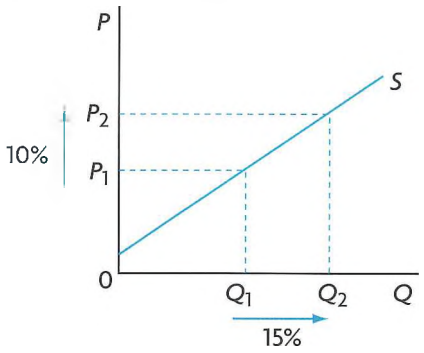
- **Supply is unit elastic when PES = 1.** The percentage change in quantity supplied is equal to the percentage change in price, so *PES* is equal to one; supply is *unit elastic;* there is **unitary *PES*.** In Figure 3.13(c), all three supply curves shown

Frequently encountered cases

a Price inelastic supply: *PES* < 1

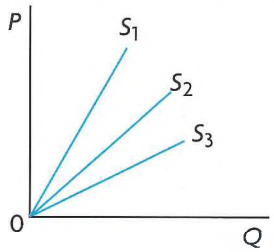


b Price elastic supply: *PES* > 1

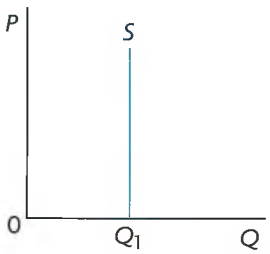


Special cases

c Unit elastic supply: *PES* = 1



d Perfectly inelastic supply: *PES* = 0



e Perfectly elastic supply: *PES* = ∞

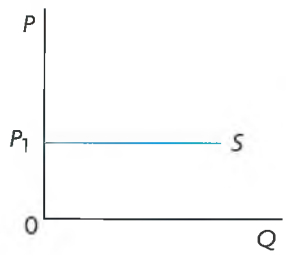


Figure 3.13: Supply curves and *PES*

[4] The reasoning here is exactly the same as in the case of the Engel curve where *YED*<1 if the extension of the Engel curve touches the horizontal axis.

[5] Here, too, the logic is the same as in the Engel curve where *YED*>1 if the Engel curve touches the vertical axis.

are unit elastic supply curves, i.e. for all three, $PES = 1$. Any supply curve that passes through the origin has a PES equal to unity. The reason for this is that along any straight line that passes through the origin, between any two points on the line the percentage change in the vertical axis (the price) is equal to the percentage change in the horizontal axis (the quantity). Therefore, for lines that pass through the origin, it is important not to confuse the steepness of the curve with the elasticity of the curve.

- **Supply is perfectly inelastic when PES = 0.** The percentage change in quantity supplied is zero; there is no change in quantity supplied no matter what happens to price; PES is equal to zero and supply is said to be **perfectly inelastic**. In Figure 3.13(d), the supply curve is vertical at the point of fixed quantity supplied, Q_1. This is the same as the supply curve shown in Figure 2.7 in Chapter 2. Examples of a vertical supply curve include the supply of fish at the moment when fishing boats return from sea; the season's entire harvest of fresh produce brought to market; the supply of Picasso paintings.
- **Supply is perfectly elastic when PES = ∞.** The percentage change in quantity supplied is infinite; any change in price leads to an infinitely large response in quantity supplied; supply in this case is called **perfectly elastic**, and is shown in Figure 3.13(e) as a horizontal line. (We will encounter such a supply curve in Chapter 14.)

Price elasticities of supply most commonly encountered in the real world are those representing elastic or inelastic supply, with perfectly elastic, perfectly inelastic and unit elastic supply being special cases. Note that only when two supply curves intersect (when they share a price and quantity combination) is it possible to make comparisons of price elasticities of supply by reference to the steepness of the curves. (We have the same condition for making comparisons of PEDs in the case of demand curves as explained earlier). In the case of intersecting supply curves, the flatter the supply curve, the more elastic it is at any given price. For example, in Figure 3.14, at any one particular price level, S_3 is more elastic than S_2, which is more elastic than S_1.

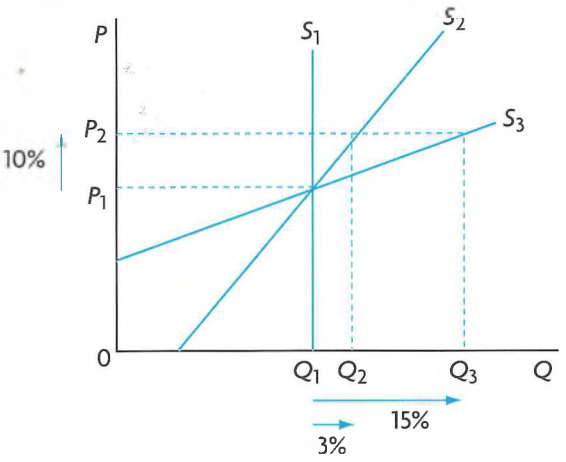


Figure 3.14: The length of time and *PES*

Value of PES	Classification	Interpretation
Frequently encountered cases		
$0 < PES < 1$ (greater than zero and less than one)	inelastic supply	quantity supplied is relatively unresponsive to price
$1 < PES < \infty$ (greater than one and less than infinity)	elastic supply	quantity supplied is relatively responsive to price
Special cases		
$PES = 1$	unit elastic supply	percentage change in quantity supplied equals percentage change in price
$PES = 0$	perfectly inelastic supply	quantity supplied is completely unresponsive to price
$PES = \infty$	perfectly elastic supply	quantity supplied is infinitely responsive to price

Table 3.2: Characteristics of price elasticity of supply

Determinants of price elasticity of supply

We will now consider the factors that determine whether the supply for a good is elastic or inelastic.

Length of time

An important factor determining *PES* is the amount of time firms have to adjust their inputs (resources) and the quantity supplied in response to changes in price. Over a very short time, the firm may be unable to increase or decrease any of its inputs to change the quantity it produces. In this case, supply is highly inelastic, and may even be perfectly inelastic ($PES = 0$). In Figure 3.14, this is represented by S_1. For example, a fishing boat upon its return from a fishing trip has only so many fish to supply in the market. Even if the price of fish rises, there can be no response in quantity supplied. As the length of time that firms have increases, the responsiveness of quantity supplied to price changes begins to rise, and *PES* increases.

In Figure 3.14, the supply curve S_2 corresponds to a time period when the fishing boat can be taken out to sea more often, and more labour can be hired to fish, so as price increases to P_2, quantity supplied increases to Q_2 (the 10% price increase from P_1 to P_2 leads to a 3% increase in quantity supplied, indicating inelastic supply, as $PES < 1$). If an even longer time period goes by, the ability of firms to respond to price changes becomes much greater. The owner of the fishing boat can now not only hire more labour but can also buy more fishing boats, thus greatly increasing the amount of fish that can be supplied. This is shown by the supply curve S_3, for which the price P_2 gives rise to the much larger quantity Q_3 (the 10% price increase from P_1 to P_2 leads to a 15% increase in quantity supplied, indicating elastic supply, as $PES > 1$). Therefore, the larger amount of time firms have to adjust their inputs increases, the larger the *PES*.

Mobility of factors of production

The more easily and quickly resources can be shifted out of one line of production and into another (where price is increasing), the greater the responsiveness of quantity supplied to changes in price, and hence the greater the *PES*. For example, a farm worker can move more easily from strawberry cultivation to corn cultivation than the same farm worker can move to car production.

Spare (unused) capacity of firms

Sometimes firms may have capacity to produce that is not being used (for example, factories or equipment may be idle for some hours each day). If this occurs, it is relatively easy for a firm to respond with increased output to a price rise. But if the firm's capacity is fully used, it will be more difficult to respond to a price rise. The greater the spare (unused) capacity, the higher the *PES* (the more elastic the supply).

Ability to store stocks

Some firms store stocks of output they produce but do not sell right away. Firms that have an ability to store stocks are likely to have a higher *PES* for their products than firms that cannot store stocks.

Rate at which costs increase

If the costs of producing extra output increase rapidly, then supply will be inelastic, as firms will have difficulty expanding their output since they are unlikely to want to incur large costs. On the other hand, if the costs of producing more output rise slowly, it will be easier for firms to expand their output so supply will be elastic. For example, if the price of fertiliser is rising rapidly, thus raising the farm's costs of production, the farmer will find it more difficult to expand output quickly, therefore *PES* is likely to be lower than if the price of fertiliser were stable.

TEST YOUR UNDERSTANDING 3.10

1 a Explain the meaning of price elasticity of supply.

b Why do we say it measures responsiveness of quantity *along a given supply curve*?

2 Identify the value or range of values for each of the following *PES*s, and show, using diagrams, the shape of the supply curve that corresponds to each one:

a perfectly elastic supply

b unit elastic supply

c perfectly inelastic supply.

CONTINUED

3 a Identify the price elasticity of supply values or range of values that we see most frequently in the real world.

b Compare these by drawing supply curves in a single diagram.

4 Using examples, explain the determinants of *PES*.

5 Suppose that in response to an increase in the price of good *X* from $10 to $15 per unit, the quantity of good *X* produced

a does not respond at all during the first week,

b increases from 10000 units to 12000 units over five months, and

c increases from 10000 to 18000 units over two years. Calculate *PES* for each of these three time periods and identify when it is price elastic, price inelastic or perfectly inelastic.

6 a Explain what factors can account for the difference in the size of the three elasticities of question 5.

b Draw a supply curve that is likely to correspond to each of the three elasticities in a single diagram.

Applications of price elasticity of supply (HL only)

PES in relation to primary commodities and manufactured products

Why many primary commodities have a lower PES compared with the PES of manufactured products

In general, primary commodities usually have a lower *PES* than manufactured products. The main reason is the time needed for quantity supplied to respond to price changes. In the case of agriculture, it takes a long time for resources to be shifted in and out of agriculture. Farmers need at least a planting season to be able to respond to higher prices. In most areas there is a limited amount of new land that can be brought into cultivation. In some regions of the world land appropriate for agriculture is shrinking due to environmental destruction (caused by over-farming that depletes the soil of minerals needed by crops). Under such conditions, what is needed is an increase in output per unit of land cultivated (crop yields), but this requires technological change in agriculture, involving new seeds or other inputs that are more productive, and takes a great deal of time. Also needed are more and better irrigation systems, although many countries face a growing water shortage. All these factors explain why a long time is needed for the quantity of an agricultural commodity to respond to increases in price.

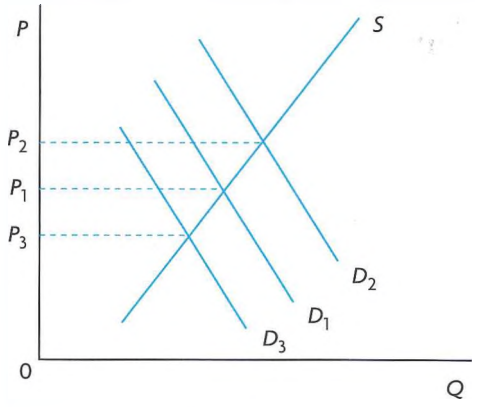
In the case of other primary products, such as oil, natural gas and minerals, time is needed to make the necessary investments and to begin production. Because of the costs involved, firms do not respond quickly to price increases, and wait for a serious shortage (excess demand) in the commodity to arise before they take actions to increase production.

Consequences of a low PES for primary commodities (Supplementary material)

Earlier, in our discussion of price elasticity of demand (*PED*), we saw that price inelastic demand for primary products is an important factor contributing to short-term price and revenue instability for producers such as farmers. Inelastic supply of agricultural and other primary products also contributes to price and income instability for primary product producers.

Figure 3.15 shows a fluctuating demand curve: in part (a) it interacts with inelastic supply, which is typical in the case of primary products, and in part (b) with elastic supply, more typical of manufactured products. Clearly, price fluctuations are larger in the case of inelastic supply. Large price fluctuations mean large revenue fluctuations, or unstable revenue for producers of primary commodities. We will come back to the implications of unstable prices and revenues for producers and for the economy in Chapters 4 and 19.

a Primary commodities: demand shifts with inelastic supply



b Manufactured products: demand shifts with elastic supply

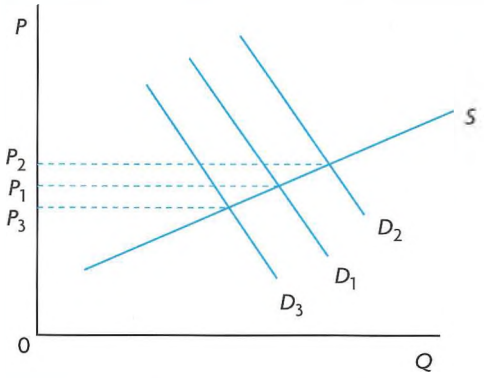


Figure 3.15: Price fluctuations are larger for primary commodities because of low *PES*

Short-term and long-term price elasticities of supply

It was noted above that agricultural products (and other primary commodities) usually have lower price elasticities of supply than manufactured products because they need more time to respond to price changes. This suggests that over longer periods of time the *PES* of agricultural products is larger.

Table 3.3 shows that this is in fact the case. The longer the time producers have to make the necessary adjustments, the greater the responsiveness of quantity supplied to price changes (see Figure 3.14).

Commodity	Short-term *PES*	Long-term *PES*
Cabbage	0.36	1.20
Carrots	0.14	1.00
Cucumbers	0.29	2.20
Onions	0.34	1.00
Green peas	0.31	4.40
Tomatoes	0.16	0.90
Cauliflower	0.14	1.10
Celery	0.14	0.95

Table 3.3: Short-run and long-run *PES* for selected agricultural commodities

TEST YOUR UNDERSTANDING 3.11

1 a Explain why the *PES* for many primary commodities is relatively low and for many manufactured products is relatively higher.

b Use the concept of *PES* to explain why agricultural product prices are volatile over the short term. (Optional)

Summary of *PED*, *YED* and *PES*

Table 3.4 provides a summary of key characteristics of all the elasticities considered in this chapter.

Elasticity	Possible values	Description	Examples
Price elasticity of demand $PED = \frac{\%\Delta Q}{\%\Delta P}$	$PED = 0$	perfectly inelastic	concept used in economic theory
	$0<PED<1$	price inelastic	gasoline, cigarettes, food
	$PED = 1$	unit elastic	concept used in economic theory
	$PED>1$	price elastic	yachts, expensive holidays
	$PED = \infty$	perfectly elastic	concept used in economic theory

Elasticity	Possible values	Description	Examples
Income elasticity of demand $YED = \frac{\%\Delta Q}{\%\Delta Y}$	$YED > 0$	normal good	new cars, new clothes
	$YED < 0$	inferior good	used cars, used clothes
	$YED > 1$	income elastic, luxury	expensive cars and clothes, many services
	$YED < 1$	income inelastic, necessity	food, medicines
Price elasticity of supply $PES = \frac{\%\Delta Q}{\%\Delta P}$	$PES = 0$	perfectly inelastic	concept used in economic theory
	$PES < 1$	price inelastic	oil and gasoline, some agricultural products
	$PES = 1$	unit elastic	concept used in economic theory
	$PES > 1$	price elastic	any good that can be produced quickly
	$PES = \infty$	perfectly elastic	concept used in economic theory

Table 3.4: Elasticity concepts: a summary

INQUIRY AND REFLECTION

The following questions will help you reflect on your learning and enhance your understanding of key topics in this chapter. They are suggestions for inquiries that you can undertake on your own or in groups in order to revise the learning objectives of this chapter.

1 Research and find products that are subject to indirect taxes in the country you live in. Do you think they are likely to be price elastic or price inelastic? Is the government likely to have high or low revenues arising from the indirect taxes?

2 Think of some goods that are likely to have changing *YED*s on account of their being luxuries, necessities and inferior goods depending on the income level.

3 Consider a good with a *PES* that ranges from zero, positive but less than one, and greater than one depending on the time period and explain why *PES* varies. Try to consider as many determinants of supply to explain your answer.

EXAM STYLE QUESTIONS

You can find questions in the style of IB exams in the digital coursebook.



Real world issue 2: When do markets fail to meet important economic objectives, and what can government intervention offer?

CONCEPTUAL UNDERSTANDINGS

1 The market is often unable to achieve socially desirable outcomes, resulting in reduced ***efficiency***, environmental ***sustainability*** and ***equity***.

2 The issues arising from the inability of the market to achieve socially desirable outcomes include

- **a** market failure, where the market fails to achieve allocative ***efficiency*** thus resulting in welfare loss
- **b** overuse of resources, that challenges environmental ***sustainability***
- **c** inequity, which results in inequalities.

3 There are numerous policy tools that governments have at their disposal to deal with the above issues; government ***intervention*** works to change market outcomes, and has varying degrees of effectiveness in a variety of real-world situations.

We have seen in previous chapters that markets are a mechanism for allocating resources and distributing (rationing) output among users. But markets sometimes do not give rise to socially desirable results. Governments, therefore, intervene in markets in order to deal with such issues as pollution, destruction of the environment, provision of more schools and health services, prevention of monopolies and greater fairness in how income is distributed.

Deforestation in the Amazon



Chapter 4

Government intervention in microeconomics

BEFORE YOU START

- Sometimes governments tax goods and services. What might be some reasons for doing so?
- Sometimes governments give money to (subsidise) certain producers. Why do you think government would do this?
- Besides taxing and subsidising, can you think of other ways that governments intervene in markets?

This chapter will examine the following types of government intervention in markets:

- price controls:
  - price ceilings
  - price floors
- indirect taxes
- subsidies.

We will see what effects these policies have on markets and we will evaluate their effects on stakeholders.

Photo: Berlin, Germany, January 2019. Farmers protest against distribution of EU subsidies: large farms receive too much

4.1 Government intervention in markets

LEARNING OBJECTIVES

After studying this section you will be able to:

- define all the terms appearing in **orange bold** in the text (AO1)
- outline key reasons why governments intervene in markets (AO2)

Why governments intervene in markets

In Chapter 1, we saw that an important justification for government intervention is to compensate for the inability of markets to carry out all socially desirable economic activities effectively. This section will provide a review of the main reasons for government intervention at the microeconomic level. Intervention at the macroeconomic level will be studied in Chapter 13.

Earn revenue for the government

Governments earn revenues from *indirect taxes*, which are taxes on goods and services. Note that these kinds of taxes are different from income taxes, which you will learn about in Chapter 12. In Chapter 3 we saw that the lower the price elasticity of demand for a good, the greater the amount of tax revenue earned. This explains why indirect taxes are often imposed on goods that have a price inelastic demand, meaning $0<PED<1$, such as cigarettes, alcohol and petrol/gasoline.

Provide support to firms

Governments provide support to firms for several reasons. Small firms that have just been set up may require financial assistance to become well established so they can compete with larger firms. Support may also be offered to firms in an industry whose growth the government would like to encourage, such as industries that produce environmentally-friendly forms of energy (for example, wind power and solar power). In these two cases, the government may offer *subsidies* or *price floors*, both of which we will study in this chapter. Further, the government may want to protect domestic firms from foreign competition arising from imports, in which case it may offer trade protection measures such as *tariffs* and *quotas*, to be studied in Chapter 14.

Provide support to households on low incomes

Households often do not earn enough income to be able to provide all necessities to meet their basic needs (food, shelter, clothing). There are several methods the government can use to support such households, including *subsidies*, *price ceilings* and *direct provision of services*, such as the provision of free education and free health care, all of which we will study in this chapter. Also, it provides *transfer payments*, which include such payments as unemployment benefits, child benefits, maternity benefits and many more; these will be considered in Chapters 12 and 20.

Influence the levels of production of firms

When the government provides support to firms through any of the methods mentioned above (subsidies, price floors or trade protection measures), one of the consequences is to also to increase the firms' level of production. On the other hand, *indirect taxes* have the opposite effect; they work to decrease firms' level of output.

Influence levels of consumption of households/consumers

In some cases, the government would like to influence consumers to consume greater quantities of goods and services that are held to be beneficial to them (known as *merit goods*) or to reduce consumption of goods and services held to be harmful (*demerit goods*). Examples of merit goods are education and health care; examples of demerit goods are cigarettes and fatty foods. Merit and demerit goods will be explained in Chapters 5 and 6. To increase consumption levels, the government can use *subsidies* and *direct provision of services* (to be discussed in this chapter), or *nudges* (see also Chapter 2), or *command and control methods*, such as compulsory education up to a certain age, also to be discussed in this chapter. To reduce consumption levels, the government can use *indirect taxes* which raise the price of the good or service, thus lowering the quantity demanded, or *nudges*, or *command and control methods* like prohibiting smoking in public places.

Note that **command and control** refers to government laws and regulations that must be followed. It refers to the command approach to decision-making explained in Chapter 1.

Correct market failure

Market failure refers to the failure of the market to achieve allocative efficiency (introduced briefly in Chapter 2 and to be discussed at length in Chapters 5–7). When market failure occurs, it means that the market produces quantities of a good or service that are too large or too small in relation to what society mostly prefers. In addition, it may mean that certain goods that are socially desirable are not produced at all by the market. When market failure occurs, government intervention is required in order to try to correct it, so that the economy will come closer to achieving socially desirable results. Policies the government can use to deal with issues like this include *indirect taxes*, *subsidies*, *nudges*, *direct provision of services* and *command and control methods*. All these will be considered in Chapters 5–7.

Promote equity (equality)

The market system as a rule does not achieve an equitable (or fair) distribution of income and wealth. As we saw in Chapter 1, economists usually interpret equity to mean equality. Most societies consider that the market system results in income and wealth distributions that are too unequal, meaning that a relatively small proportion of the population receives too large a share of income, and the reverse, where a large proportion of the population receives only a relatively small share of income. Governments therefore undertake to redistribute income. Some policies that we will consider in this chapter include *price ceilings and subsidies*. Other important policies will be considered in Chapters 12 and 20.

What forms does government intervention in markets take at the microeconomic level?

The following forms of government intervention are undertaken at the microeconomic level. When we say that they operate on the microeconomic level, what we mean is that *each of these works to influence demand or supply for a good or service, thus affecting market outcomes*. The main forms of these interventions include the following:

- price controls:
  - price ceilings
  - price floors
- indirect taxes
- subsidies
- direct provision of services
- command and control regulation and legislation
- consumer nudges (explained in Chapter 2 at HL).

This chapter will consider price controls, indirect taxes and subsidies.

4.2 Price controls

LEARNING OBJECTIVES

After studying this section you will be able to:

- define all the terms appearing in **orange bold** in the text (AO1)
- explain the consequences of price ceilings and price floors on markets and stakeholders (AO2)
- draw diagrams to illustrate the effects of price ceilings and price floors on markets and stakeholders (AO4)
- evaluate the effects of price ceilings and price floors on markets and stakeholders (AO3)
- calculate the effects of price ceilings and price floors on markets and stakeholders (HL only) (AO4)

Introduction to price controls

The first type of intervention we will consider involves price controls.

Price controls refer to the setting of minimum or maximum prices by the government (or private organisations) so that prices are unable to adjust to their equilibrium level determined by demand and supply. Price controls result in market disequilibrium, and therefore in shortages (excess demand) or surpluses (excess supply).

Price controls differ in a fundamental way from other types of government intervention that we will study below (indirect taxes and subsidies). When a tax is imposed or a subsidy granted, the market *settles at a new equilibrium*, where there is a balance of demand with the new supply. Price controls differ because, once they are imposed, they do not allow a new equilibrium to be established, and instead force a situation where there is *persisting market disequilibrium*.

Market disequilibrium means that the market is prevented from reaching a market-clearing price, and there emerge shortages (excess demand) or surpluses (excess supply) (see Chapter 2).

In the discussion that follows, it is important to bear in mind that the term *surplus* has two different meanings. In one sense it refers to excess supply resulting when quantity supplied is greater than quantity demanded, as we discussed in Chapter 2. In the second sense it refers to the benefits that consumers or producers receive from buying or selling (see Chapter 2). This is not as confusing as it may sound, because *surplus* in the second sense is referred to as 'consumer surplus', or 'producer surplus' or 'social surplus' (or 'community surplus').

Price ceilings: setting a legal maximum price

What is a price ceiling?

A government may in some situations set a legal *maximum price* for a particular good; this is called a *price ceiling*. It means that the price that can be legally charged by sellers of the good must not be higher than the legal maximum price. Price ceilings are usually set in order to make certain goods more affordable to people on low incomes.

Figure 4.1 shows how this works. The equilibrium price is P_e, determined by the forces of demand and supply. The price ceiling, P_c, is set by the government at a level below the equilibrium price, leading to a shortage (excess demand), since quantity demanded, Q_d, is greater than quantity supplied, Q_s. If the market were free, the forces of demand and supply would force price up to P_e. However, now this cannot happen, because the price hits the legally set price ceiling.

Note that to have an effect, the price ceiling must be *below* the equilibrium price. If it were higher than the equilibrium price, the market would achieve equilibrium, and the price ceiling would have no effect.

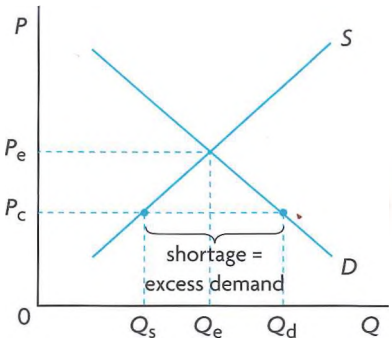


Figure 4.1: Price ceiling (maximum price) and market outcomes

A **price ceiling** is a **maximum price** set *below* the equilibrium price, in order to make goods more affordable to people on low incomes.

Consequences of price ceilings

By imposing a price that is below the equilibrium price, a price ceiling results in a lower quantity supplied and sold than at the equilibrium price. This is shown in Figure 4.1, where the price ceiling, P_c, gives rise to quantity, Q_s, that firms supply, which is less than the equilibrium quantity, Q_e, that suppliers would supply at price P_e.

In addition, the price ceiling, P_c, gives rise to a larger quantity demanded than at the equilibrium price: the quantity consumers want to buy at price P_c is given by Q_d, which is greater than quantity, Q_e, that they would buy at price P_e.

A price ceiling does not allow the market to clear; it creates a situation of disequilibrium where there is a shortage (excess demand).

Consequences for markets

Shortages

A price ceiling, P_c, set below the equilibrium price of a good creates a shortage. At P_c, not all interested buyers who are willing and able to buy the good are able to do so because there is not enough of the good being supplied. In Figure 4.1, the shortage is equal to $Q_d - Q_s$.

Non-price rationing

The term *rationing* refers to a method of dividing up something among possible users (see Chapter 2). In a free market, this is achieved by the price system: those

who are willing and able to pay for a good will do so, and the good is rationed among users according to who buys it; this is called *price rationing*. However, once a shortage arises due to a price ceiling, the price mechanism no longer achieves its rationing function. Some demanders willing and able to buy the good at P_c in Figure 4.1 will go unsatisfied. How will the quantity Q_s be distributed among all interested buyers? This can only be done through *non-price rationing* methods, which include:

- waiting in line and the first-come-first-served principle: those who come first will buy the good
- the distribution of coupons to all interested buyers, so that they can purchase a fixed amount of the good in a given time period
- favouritism: the sellers can sell the good to their preferred customers.

Underground (or parallel) markets

Underground (or *parallel*) *markets* involve buying/selling transactions that are unrecorded, and are usually illegal. In the case of price ceilings, they are a special kind of price rationing. They involve buying a good at the maximum legal price, and then illegally reselling it at a price above the legal maximum. Underground markets can arise when there are dissatisfied people who have not succeeded in buying the good because there was not enough of it, and are willing to pay more than the ceiling price to get it. If there were no shortage, the price of the good would be at its equilibrium price, and no one would be interested in paying a higher than equilibrium price for it. Underground markets are inequitable, and frustrate the objective sought by the price ceiling, which is to set a maximum price.

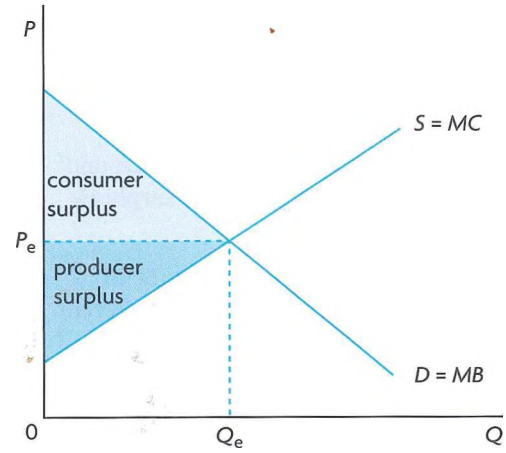
Underallocation of resources to the good and allocative inefficiency

The price ceiling, being lower than the equilibrium price, results in a smaller quantity supplied; in Figure 4.1 $Q_s < Q_e$, not enough resources are allocated to the production of the good, resulting in underproduction relative to the social optimum (or 'best'). Society is worse off due to underallocation of resources and allocative inefficiency.

Negative welfare impacts

Allocative efficiency, and the conditions of maximum social surplus and $MC = MB$ were explained in Chapter 2. In a competitive free market equilibrium, shown in Figure 4.2(a), consumer surplus appears as the shaded area above price P_e and under the demand curve up to quantity Q_e; producer surplus is the shaded area above the supply curve and under price P_e up to Q_e. At the competitive free market equilibrium, the sum of consumer and producer surplus, or social surplus, is maximum, and $MB = MC$, indicating that allocative efficiency is achieved.

a Consumer and producer surplus in a competitive free market: maximum social surplus



b Welfare impacts of a price ceiling (maximum price)

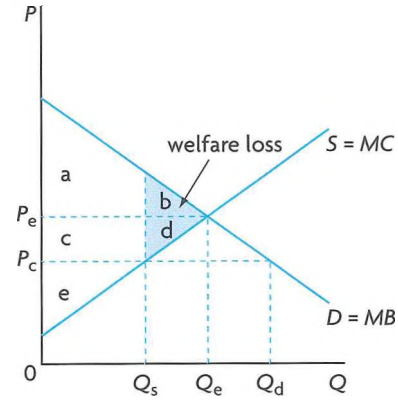


Figure 4.2: Effects of a price ceiling (maximum price) on consumer and producer surplus

In Figure 4.2(b), if there were no price control, the market would determine price P_e and quantity Q_e at equilibrium. Consumer surplus would be equal to areas a + b. Producer surplus , would be equal to areas c + d + e. Consumer plus producer surplus would be maximum, and equal to a + b + c + d + e. Also, $MB = MC$, and there would be allocative efficiency.

If a price ceiling, P_c, is imposed, only the quantity Q_s is produced and consumed. Consumer surplus is now the area under the demand curve and above P_c, but only up to Q_s, since that is all that is consumed. Therefore, consumer surplus becomes a + c. Producer surplus is the area above the supply curve and below P_c, also only

up to Q_s since that is all that is produced. Producer surplus therefore falls to area e. Total social surplus after the price ceiling is a + c + e. Comparing with total social surplus before the price ceiling, we see that the shaded areas b and d have been lost and represent *welfare loss*. Welfare loss represents benefits that are lost to society because of resource misallocation.

Welfare loss (also known as *deadweight loss*) represents social surplus or welfare benefits that are lost to society because resources are not allocated efficiently.

We can see there is allocative inefficiency also because $MB > MC$ at the point of production, Q_s: the benefit consumers receive from the last unit of the good they buy is greater than the marginal cost of producing it. Therefore, society is not getting enough of the good, as there is an underallocation of resources to its production.

A *price ceiling* creates a welfare loss, indicating that the price ceiling introduces allocative inefficiency due to an underallocation of resources to the production of the good. This is shown by $Q_s < Q_e$. Also, $MB > MC$, indicating that society is not getting enough of the good.

Consequences for stakeholders

Stakeholders are individuals or groups of individuals who have an interest in something and are affected by it.

Consumers

Consumers partly gain and partly lose. They gain area c from producers but they lose area b (see Figure 4.2(b)). This is because those consumers who are able to buy the good at the lower price are better off. However, some consumers remain unsatisfied since they don't get to buy the good at all, because at the ceiling price there is not enough of the good to satisfy all demanders.

Producers

Producers are worse off, because with the price ceiling they sell a smaller quantity of the good at a lower price; therefore, their revenues drop from $P_e \times Q_e$ to $P_c \times Q_s$. This is clear also from their loss of some producer surplus, area c (which is transferred to consumers), as well as area d (welfare loss) in Figure 4.2(b).

Workers

The fall in output (from Q_e to Q_s) means that some workers are likely to be fired, resulting in unemployment; clearly these workers will be worse off.

Government

There will be no gains or losses for the government budget, yet the government may gain in political popularity among the consumers who are better off due to the price ceiling.

The examples of rent controls and food price controls

Price ceilings are for the most part set in order to make certain goods considered to be necessities more affordable to low-income earners.

Rent controls

Rent controls consist of a maximum legal rent on housing, which is below the market-determined level of rent (the price of rental housing). It is undertaken by governments in some cities around the world to make housing more affordable to low-income earners. Consequences of rent controls include:

- housing becomes more affordable to low-income earners
- a shortage of housing, as the quantity of housing demanded at the legally maximum rent is greater than the quantity available
- long waiting lists of interested tenants waiting for their turn to secure an apartment/flat
- a market for rented units where tenants sublet their apartments at rents above the legal maximum (an underground market)
- run-down and poorly maintained rental housing because it is unprofitable for landlords to maintain or renovate their rental units since low rents result in low revenues.

Food price controls

Some governments use food price controls as a method to make food more affordable to low-income earners, especially during times when food prices are rising rapidly. The results of food price controls follow the same patterns as discussed above: lower food prices and greater affordability; food shortages as quantity demanded is greater than quantity supplied; non-price rationing methods (such as queues) to deal with the shortages; development of underground markets; falling farmer incomes due to lower revenues; more unemployment in the agricultural sector; misallocation of resources; possible greater popularity for the government among consumers who benefit.

REAL WORLD FOCUS 4.1

Price controls in Vietnam

Some years ago, due to high rates of inflation (a rising general price level), the Vietnamese government considered the imposition of price controls. These consisted of price ceilings on numerous products, including chemical fertilisers, salt, milk powder, rice, sugar, animal feeds, coal, cement, paper, textbooks and many more. If it introduced these measures, the pricing rules would apply not just to domestic government-owned businesses but also private firms and foreign-owned businesses. It was feared that Vietnam might be moving away from its freer market orientation of recent years and back toward the ways of a command economy. Foreign diplomats warned the government that price controls would damage business confidence in the country.

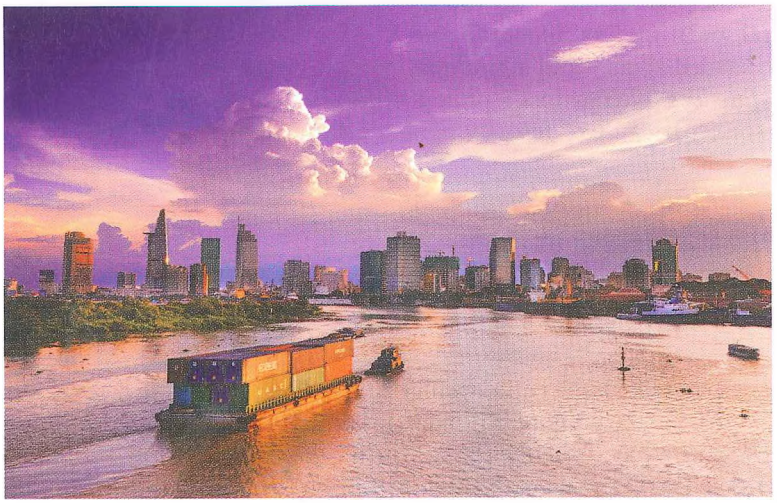


Figure 4.3: Vietnam. Transport on the Saigon River

Applying your skills

1 Outline what it means to move toward the ways of a 'command economy'.

2 Discuss the consequences for the economy and stakeholders that might have arisen if the government imposed the price controls.

3 Why do you think that price controls could damage business confidence?

Source: *Adapted from The Economist Intelligence Unit, 'Vietnam economy: reform roll-back?' in ViewsWire News Analysis, 21 September 2010.*

TEST YOUR UNDERSTANDING 4.1

1 Define a price ceiling and, providing examples, outline some reasons why governments impose them.

2 Using a diagram, explain why price controls lead to disequilibrium market outcomes.

3 Draw a diagram illustrating a price ceiling, and analyse its effects on market outcomes (price, quantity demanded, quantity supplied, market disequilibrium) and consequences for the economy (shortages, non-price rationing, allocative inefficiency, welfare loss).

4 a Explain the difference between price rationing and non-price rationing.

b Describe the circumstances under which non-price rationing arises.

c Identify some forms of non-price rationing.

d Outline why underground markets are a form of price rationing.

5 a Draw a diagram showing producer and consumer surplus in a free market competitive equilibrium.

CONTINUED

b Assuming a price ceiling is imposed in this market, draw a new diagram showing the new consumer surplus, producer surplus and welfare loss.

c Comparing your diagrams for parts (a) and (b), what can you conclude about consumer surplus, producer surplus and welfare loss?

d Describe the relationship between marginal benefits and marginal costs at the new equilibrium. Outline what this reveals about allocative efficiency (or inefficiency).

6 Examine the consequences of price ceilings for different stakeholders in the case of:

a rent controls, and

b food price controls.

Calculating the effects of price ceilings on stakeholders and welfare (HL only)

Figure 4.4 provides us with a numerical example of a price ceiling. At equilibrium, price is equal to £8 and quantity demanded and supplied is 20000 units of the good per week. When a price ceiling is imposed at P_c = £5.00 per unit, quantity demanded becomes Q_d = 30000 units, and quantity supplied Q_s = 10000 units.

Shortage (excess demand)

The shortage, or excess demand, is equal to $Q_d - Q_s$, which in this case is 30000 – 10000 = 20000 units per week.

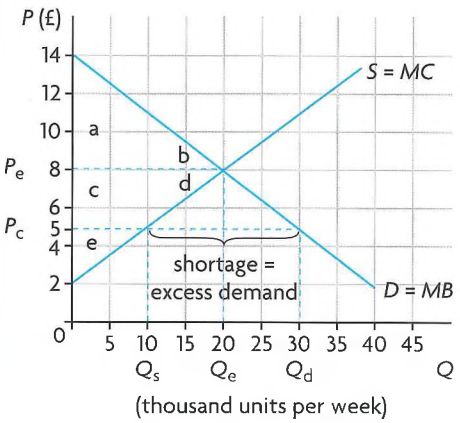


Figure 4.4: Calculating effects of a price ceiling

Change in consumer expenditure

Consumer expenditure is given by the price per unit of the good times the number of units purchased.

At equilibrium, prior to the price ceiling, consumers spend $P_e \times Q_e$ = £8 × 20000 units = £160000. After the price ceiling is imposed, consumers spend $P_c \times Q_s$ = £5.00 × 10000 units = £50000. The change is therefore £160000 – £50000 = £110000, meaning that consumers now spend £110000 less than at equilibrium.

Change in producer (firm) revenue

Firm revenue is the same as consumer expenditure both before and after the imposition of the price ceiling. This is because revenue is equal to price per unit times quantity of units sold, and both the price (P_c) and the quantity (Q_s) are the same for both consumers and producers. Therefore, before the price ceiling is imposed, firm revenue is £160000, and after the price ceiling, firm revenue is reduced to £50000. Therefore, firm revenues fall by the amount £110000.

Note that the change in consumer expenditure and the change in firm revenue are the same, because price and quantity were the same for consumers and firms both before and after the price ceiling was imposed.

Change in consumer and producer surplus and welfare loss

In Chapter 2, we saw how consumer and producer surplus are calculated. You may remember that at market equilibrium consumer surplus is half the area of the rectangle whose one side equals the P intercept of the demand curve minus the price paid by consumers, and whose other side equals the number of units purchased:

$$\text{Consumer surplus} = \frac{(P \text{ intercept of D curve minus } P \text{ of consumers}) \times Q \text{ purchased}}{2}$$

Therefore, using the information in Figure 4.4, consumer surplus before the price ceiling is:

$$\text{Initial consumer surplus} = \frac{(14-8)\times 20\,000}{2} = \frac{6\times 20\,000}{2}$$

$$= £60\,000$$

Now notice that after the price ceiling is imposed, consumer surplus no longer has the area of a triangle. It consists of a trapezium composed of areas a + c. The formula for a trapezium was also presented in Chapter 2. *It is the sum of the two parallel sides times the distance between them divided by 2*. Therefore:

$$\text{Final consumer surplus} = \frac{[(14-5)+(11-5)]\times 10\,000}{2}$$

$$= \frac{(9+6)\times 10\,000}{2} = \frac{150\,000}{2} = £75\,000$$

Therefore consumer surplus has increased by £75 000 – £60 000 = £15 000.

Producer surplus at market equilibrium is half the area of the rectangle whose one side equals the price received by producers minus the P intercept of the *initial* supply curve, S_1 and whose other side equals the number of units sold:

$$\text{Producer surplus} = \frac{(P \text{ of producers minus } P \text{ intercept of } S \text{ curve})\times Q \text{ sold}}{2}$$

Therefore producer surplus before the price ceiling is:

$$\text{Initial producer surplus} = \frac{(8-2)\times 20\,000}{2} = \frac{6\times 20\,000}{2}$$

$$= £60\,000$$

Notice that after the price ceiling is imposed producer surplus is the area of triangle e therefore we can apply the triangle formula:

$$\text{Final producer surplus} = \frac{(5-2)\times 10\,000}{2} = \frac{3\times 10\,000}{2}$$

$$= £15\,000$$

Producer surplus has decreased by £60 000 – £15 000 = £45 000.

The welfare loss is given by the area of triangles a plus b:

$$\text{Welfare loss} = \frac{(11-5)\times(20\,000-10\,000)}{2} = \frac{6\times 10\,000}{2}$$

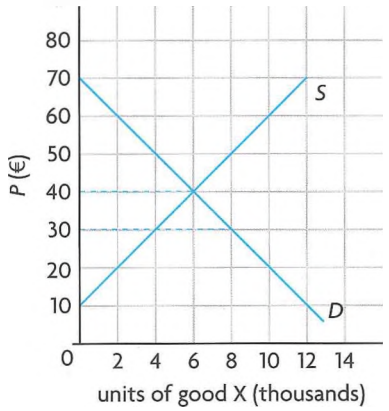
$$= £30\,000$$

TEST YOUR UNDERSTANDING 4.2

1 In the example of the market illustrated in Figure 4.4, comment on the effect of a price ceiling set at £10.

2 The diagram below shows a price ceiling of €30 that has been set for good X. Calculate:

- a the shortage (excess demand),
- b the change in consumer expenditure,
- c the change in producer revenue,
- d the change in consumer surplus,
- e the change in producer surplus, and
- f welfare loss.



Price floors: setting a legal minimum price

What is a price floor?

A legally set *minimum price* is called a *price floor*. The price that can be legally charged by sellers of the good must not be lower than the price floor, or minimum price. In Figure 4.5, a price floor, P_f, is set above the equilibrium price, P_e. At P_e, consumers are willing and able to buy Q_d of the good, but firms are willing and able to supply Q_s of the good. Therefore, a surplus, or excess supply, equal to the difference between Q_s and Q_d, arises. If the market were free, the forces of demand and supply would force the price down to P_e. However, now this cannot happen.

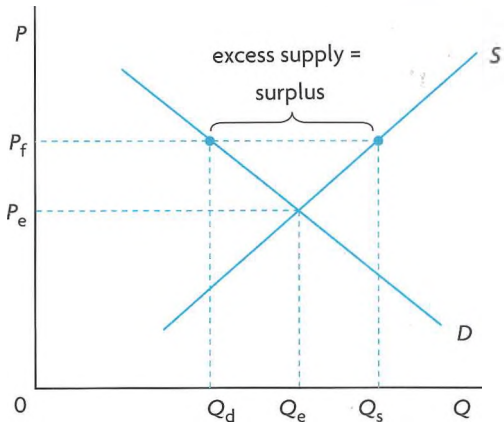


Figure 4.5: Price floor (minimum price) and market outcomes

Note that to have an effect, the price floor must be *above* the equilibrium price. If it were below the equilibrium price, the market would achieve equilibrium and the price floor would have no effect.

Why governments impose price floors

Price floors are commonly used for two reasons: (a) to provide income support for farmers by offering them prices for their products that are above market-determined prices; and (b) to protect low-skilled, low-wage workers by offering them a wage (the minimum wage) that is above the level determined in the market. Note that the first of these involves price control in product markets, while the second concerns price control in a resource market. While market outcomes are similar, each type of price control has different consequences for the economy and stakeholders. We will therefore consider each one separately.

A **price floor** is a **minimum price** set *below* the equilibrium price, in order to provide income support to farmers or to increase the wages of low-skilled workers.

To avoid getting confused about which is which, note that the position of a price floor and a price ceiling in relation to the equilibrium price is always *the opposite* of the floor and ceiling of a room. The *price floor is above* and the *price ceiling is below*.

Consequences of price floors

Consequences for markets

Farmers' incomes in many countries, resulting from the sale of their products in free markets, are often unstable or too low. Some important reasons for both instability and low incomes were considered in Chapter 3. Unstable incomes arise from unstable agricultural product prices, which are due to low price elasticities of demand and low price elasticities of supply for agricultural products. Low farmer incomes may arise from low income elasticities of demand.

One method governments use to support farmers' incomes is to set price floors for certain agricultural products, the objective being to raise the price above their equilibrium market price. The consequences are explained below.

Surpluses

Figure 4.6 illustrates the market for an agricultural product with a price floor, P_f, set above the equilibrium price, P_e. The price floor results in a larger quantity supplied, Q_s, than the quantity supplied at market equilibrium, Q_e. In addition, the price floor, P_f, leads to a smaller quantity demanded and purchased than at the equilibrium price: the quantity consumers want to buy at P_f is Q_d, which is smaller than the quantity Q_e that they bought at price P_e.

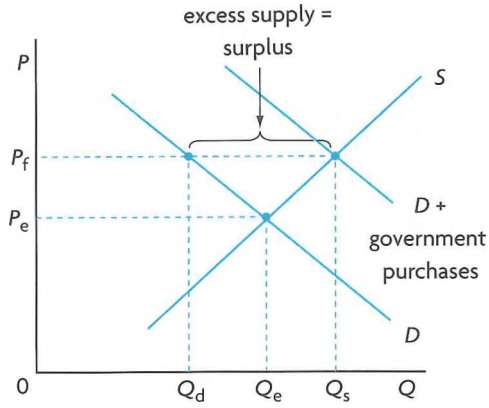


Figure 4.6: An agricultural product market with price floor and government purchases of the surplus

A price floor results in disequilibrium where there is a surplus (excess supply). A common practice is for the government to buy the excess supply, and this causes the demand curve for the product to shift to the right to the new demand curve '*D* plus government purchases'. By buying up the excess supply, the government is able to maintain the price floor at P_f.

It should be noted that if the government did not buy the surplus, the price would fall back to its equilibrium level. The reason is that farmers would have excess output with no buyers, and so would have to lower the price in order to be able to sell all the surplus.

Government measures to dispose of surpluses

The government must make a decision about what to do with the surplus (excess supply) it purchases. One option is to store it, giving rise to additional costs for storage above the costs of the purchase. Another method is to export the surplus (sell it abroad); this often requires granting a subsidy (this is money given to producers, to be discussed later in this chapter) to lower the price of the good since foreign countries would not want to buy it at the high price. Clearly, subsidies involve additional costs for the government. In general, any course chosen by the government to get rid of the surpluses is problematic.

Firm inefficiency

Higher than equilibrium product prices can lead to inefficient production; inefficient firms with high costs of production do not face incentives to cut costs by using more efficient production methods because the high price offers them protection against lower-cost competitors. This leads to inefficiency.

Overallocation of resources to the production of the good and allocative inefficiency

Too many resources are allocated to the production of the good, resulting in a larger than optimum (or 'best') quantity produced. Whereas the optimum quantity is Q_e, actually Q_s is produced.

Negative welfare impacts

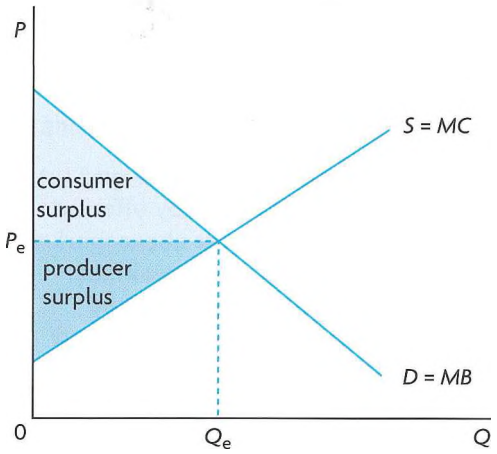
In Figure 4.7(a), price P_e and quantity Q_e represent market equilibrium with no price floor, and where social surplus is maximum. In part (b), when there is no price floor, consumer surplus is given by a + b + c and producer surplus by d + e. Also, $MB = MC$.

After a price floor, P_f, is imposed, consumer surplus becomes the area under the demand curve and above P_f, up to the quantity consumers buy, Q_d, and so falls to a. Producer surplus becomes the area above the supply curve and below P_f, up to the quantity produced, Q_s, and so becomes d + e + b + c + f. This means that the sum of consumer plus producer surplus *increases* by the area f after the price floor is imposed. This happens because producers gain the area b and c lost by consumers, and in addition gain f.

On the other hand, government spending to buy the excess supply is equal to the price paid per unit, P_f, times the surplus quantity it purchases: $P_f \times (Q_s - Q_d)$, corresponding to the rectangle outlined in bold. Since government spending is financed out of taxes with alternative uses (opportunity costs), government spending to maintain the price floor involves losses for society.

Therefore, there is a gain in surplus of f and a loss equal to the rectangle shown in bold. Subtracting the loss from the gain we are left with the blue shaded area, which is welfare loss, representing loss of benefits due to allocative inefficiency caused by overallocation of resources to the production of the good. This is also shown by $MB < MC$ at the point of production, Q_s, indicating that society would be better off if less of the good were produced.

a Consumer and producer surplus in a competitive free market: maximum social surplus



b Welfare impacts of a price floor (minimum price)

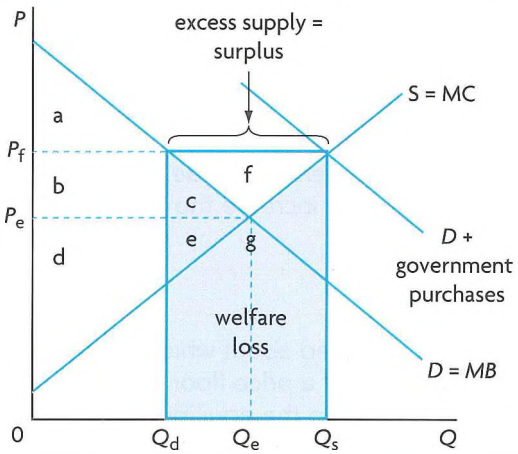


Figure 4.7: Welfare impacts of a price floor (minimum price) for agricultural products and government purchases of the surplus

Another way to see that the blue shaded area is the welfare loss is to note simply that the price floor results in a gain of f and a loss of c + e + f + g, or the bold rectangle of government spending. And so the net loss is simply that part of government spending that is not gained. In other words the net change is f − (c + e + f + g) = − (c + e + g).

A price floor creates welfare loss, indicating that the price floor introduces allocative inefficiency due to an overallocation of resources to the production of the good, shown by $Q_s > Q_e$. Also $MB < MC$, indicating that society is getting too much of the good.

Consequences for stakeholders

Consumers

Consumers are worse off, as they must now pay a higher price for the good ($P_f > P_e$), while they buy a smaller quantity of it ($Q_d < Q_e$). This is clear also from their loss of some consumer surplus.

Producers

Producers gain as they receive a higher price and produce a larger quantity, and since the government buys up the surplus, they increase their revenues from $P_e \times Q_e$ to $P_f \times Q_s$. Remember, this is the main rationale of agricultural price floors. Also, producers become protected against low-cost competition and do not face as strong incentives to become efficient producers; they are therefore less likely to go out of business if they are producing inefficiently (with higher costs).

Workers

Workers are likely to gain as employment increases on account of greater production of the good.

Government

When the government buys the excess supply, this is a burden on its budget, resulting in less government funds to spend on other desirable activities in the economy. The costs to the government are paid for out of taxes (and therefore by taxpayers). In addition, there are further costs of storing the surplus or subsidising it for export (sale to other countries).

Stakeholders in other countries

The European Union, the United States and many other more developed countries rely on price floors for agricultural products to support their farmers.

The surpluses are sometimes exported (sold to other countries), leading to lower world prices due to the extra supply made available in world markets. Countries that do not have price supports are forced to sell their agricultural products at low world prices. The low prices in these countries signal to local farmers that they should cut back on their production, resulting in an underallocation of resources to these products. These events often work against the interests of less developed countries (this topic will be discussed in Chapter 19).

Overall, a global misallocation of resources can result in a waste of resources, as price floors cause high-cost producers to produce more and low-cost producers to produce less than the social optimum.

TEST YOUR UNDERSTANDING 4.3

1 Define a price floor, and providing examples, explain some reasons why governments impose them.

2 Draw a diagram illustrating a price floor that is imposed in a product market, and analyse its effects on market outcomes (price, quantity demanded, quantity supplied, market disequilibrium, excess supply, firm inefficiency, allocative inefficiency, welfare loss).

3 Identify some measures governments can take to dispose of surpluses that result from the imposition of a price floor in an agricultural product market. Comment on the problems associated with these measures.

4 Assuming a price floor is imposed in a market for an agricultural product, and that the government purchases the entire excess supply that results in order to maintain the price:

   a draw a diagram illustrating welfare loss, and

   b comment on the relationship between marginal benefits and marginal costs in the new equilibrium and what it reveals about allocative efficiency (or inefficiency).

5 Examine the consequences for different stakeholders of a price floor for an agricultural product whose excess supply is purchased by the government.

Calculating effects of price floors on stakeholders and social welfare (HL only)

Figure 4.8 provides a numerical example of a price floor on an agricultural product. At equilibrium, price is equal to £20 and quantity is equal to 60000 kg per week. When a price floor is imposed at P_f = £25, quantity demanded is Q_d = 40000 kg per week and quantity supplied is Q_s = 80000 kg per week.

Surplus (excess supply)

The surplus, or excess supply is equal to $Q_s - Q_d$, which in this case is 80000 – 40000 = 40000 kg per week.

Change in consumer expenditure

Consumer expenditure is given by the price per kg of the good times the number of kg purchased per week. At equilibrium, before the price floor, consumers spend $P_e \times Q_e$ = £20 × 60000 kg = £1.2 million per week. After the price floor is imposed, consumers spend $P_f \times Q_d$ = £25 × 40000 kg = £1 million per week. Therefore, consumers spend £200000 less on the good per week.

Change in producer revenue

Before the price floor, producer revenue is the same as consumer expenditure, since revenue is equal to price per kg times quantity sold, and both the price (P_e) and the quantity (Q_e) are the same for consumers and producers. Therefore producer revenue before the price floor is £1.2 million per week. Once the price floor is imposed and the government purchases the surplus (excess supply), firms receive revenues of $P_f \times Q_s$, and so producer revenue increases to £25 × 80000 kg = £2 million per week. Therefore, the change is £800000, or additional producer revenue of this amount per week.

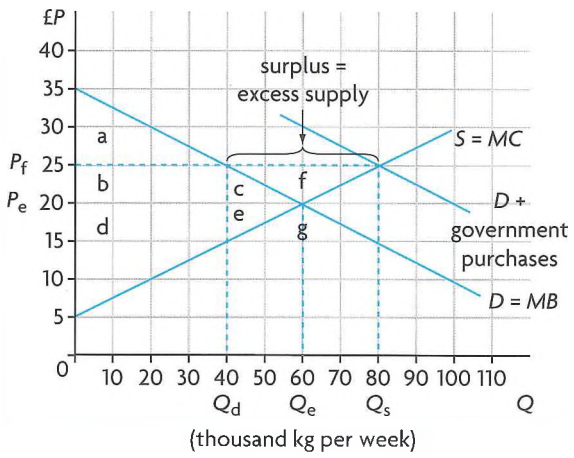


Figure 4.8: Calculating effects of a price floor on an agricultural product with government purchases of the surplus

Government expenditure

In order to purchase the excess supply of the agricultural product, the government spends an amount equal to the price of the good at the price floor times the number of kg purchased, or $P_f \times (Q_s - Q_d)$ = £25 × 40000 = £1 million.

Note that government expenditure (£1 million) is equal to total producer revenue (£2 million) minus total consumer expenditure (£1 million) per week.

Change in consumer and producer surplus and welfare loss

Notice that all areas of consumer and producer surplus, both before and after the price floor, are the areas of triangles.

In this case, as we know, the formula for calculating consumer surplus is:

$$\text{Consumer surplus} = \frac{(P \text{ intercept of D curve minus } P \text{ of consumers}) \times Q \text{ purchased}}{2}$$

Therefore, using the information in Figure 4.8, consumer surplus before the price floor is:

$$\text{Initial consumer surplus } \frac{(35-20)\times 60\,000}{2} = £450\,000$$

After the price floor is imposed consumer surplus is:

$$\text{Final consumer surplus} = \frac{(35-25)\times 40\,000}{2} = \frac{10\times 40\,000}{2}$$

= £200000

Therefore, consumer surplus *decreased* by £450000 – £200000 = £250000.

Producer surplus is given by:

$$\text{Producer surplus} = \frac{(P \text{ of producers minus } P \text{ intercept of } S \text{ curve}) \times Q \text{ sold}}{2}$$

Before the price floor producer surplus is:

$$\text{Initial producer surplus} = \frac{(20-5)\times 60\,000}{2} = \frac{15\times 60\,000}{2}$$

= £450000

After the price floor is imposed producer surplus is:

$$\text{Final producer surplus} = \frac{(25-5)\times 80\,000}{2} = \frac{20\times 80\,000}{2}$$

= £800000

Therefore, producer surplus *increased* by £800000 – £450000 = £350000.

The welfare loss is given by the shaded area in Figure 4.7, which according to Figure 4.8 is the rectangle of government spending minus area f.

We have found above that government spending on the subsidy is £1 million.

$$\text{Area f} = \frac{(25-20)\times(80\,000-40\,000)}{2} = \frac{(5\times 40\,000)}{2}$$
$$= £100\,000$$

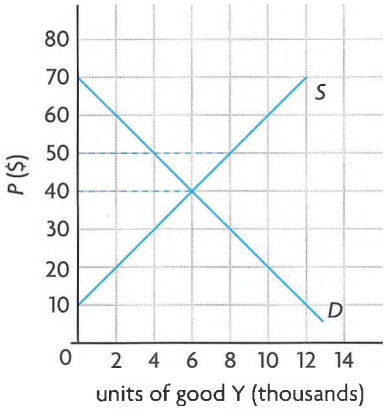
Therefore, welfare loss = £1 million – £100 000 = £900 000.

TEST YOUR UNDERSTANDING 4.4

1 In the example of the market illustrated in Figure 4.8, what would be the effect of a price floor set at £15?

2 The diagram below shows a price floor of $50 that has been set for good Y. Calculate:

- a the surplus (excess supply),
- b the change in consumer expenditure,
- c the change in producer revenue,
- d the change in consumer surplus,
- e the change in producer surplus, and
- f welfare loss.



Minimum wages

Impacts of minimum wages on market outcomes

Many countries around the world have **minimum wage** laws that determine the minimum price of labour (the wage rate) that an employer (a firm) must pay. The objective is to guarantee an adequate income to low-income workers, who tend to be mostly unskilled. (The market-determined wages of skilled workers are usually higher than the minimum wage.) Figure 4.9 shows the market for labour. The demand for labour curve shows the quantity of labour that firms are willing and able to hire at each wage, and the supply of labour curve shows the quantity of labour that workers supply at each wage. Supply and demand determine the equilibrium 'price' of labour, which is the wage, W_e, where the quantity of labour demanded is equal to the quantity of labour supplied, Q_e.

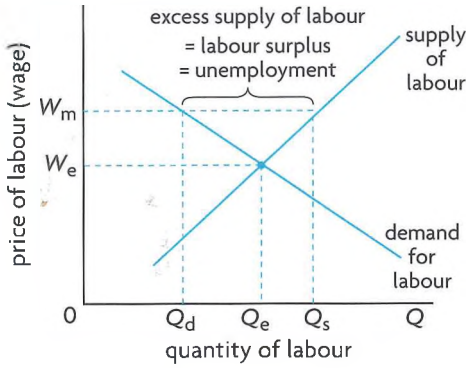


Figure 4.9: Labour market with minimum wage (price floor)

The minimum wage, W_m, lies above the equilibrium wage, W_e. Therefore, at W_m, the quantity of labour supplied, Q_s, is larger than the quantity of labour supplied when the labour market is in equilibrium (Q_e). The quantity of labour demanded, Q_d, is less than the quantity demanded at equilibrium, Q_e. There results a surplus of labour in the market equal to the difference between Q_s and Q_d. The labour market does not clear when there is a minimum wage.

Consequences of minimum wages for the economy

- ***Labour surplus (excess supply) and unemployment***

The imposition of a minimum wage in the labour market creates a surplus of labour equal to $Q_s - Q_d$ in Figure 4.9, which is unemployment, as it corresponds to people who would like to work but are not employed. The unemployment is due partly to the decrease in quantity of labour demanded by firms (the difference between Q_e and Q_d) and partly to an increase in the quantity of labour supplied (the difference between Q_s and Q_e). This occurs because the higher wage makes work more attractive, causing a movement up the labour supply curve. This unemployment is likely to involve unskilled workers.

- ***Illegal workers at wages below the minimum wage***

Illegal employment of some workers at wages below the legal minimum may result; this often involves illegal

immigrants who may be willing to supply their labour at very low wages.

- ***Misallocation of labour resources***

The minimum wage affects the allocation of *labour resources*, as it prevents the market from establishing a market-clearing price of labour. In Chapter 2, we saw how the wage acts as a signal and incentive to workers (the suppliers of labour) and firms (the demanders of labour) to determine the optimal allocation of labour resources. The imposition of a minimum wage changes these signals and incentives for unskilled labour, whose wage is affected by the price floor. Therefore, industries that rely heavily on unskilled workers are more likely to be affected, and will hire less unskilled labour.

- ***Misallocation in product markets***

Firms relying heavily on unskilled workers experience an increase in their costs of production, leading to a leftward shift in their *product* supply curve (see Chapter 2), resulting in smaller quantities of output produced. Therefore, the misallocation of labour resources leads also to misallocation in product markets.

Consequences of minimum wages for various stakeholders

- ***Firms (employers of labour)***

Firms are worse off as they face higher costs of production due to the higher labour costs.

- ***Workers (suppliers of labour)***

The impacts on workers are mixed. Some gain because they receive a higher wage than previously ($W_m > W_e$), but some lose because they lose their job. Note that the workers who lose their job are those represented by $Q_e - Q_d$. This is not the full amount of unemployment created by the minimum wage, because the minimum wage leads to *additional* unemployment of $Q_s - Q_e$, since more workers supply their labour in the market when the wage increases.

- ***Consumers***

Consumers are negatively affected, because the increase in labour costs leads to a decrease in supply of products (a leftward shift in firm supply curves) causing higher product prices and lower quantities.

Price floors and minimum wages in the real world

Economists agree that price floors for agricultural products lead to surpluses (excess supplies) and are highly inefficient for the reasons discussed above. Yet they continue to be used in many countries because of strong political pressures exerted by farmers who claim to need these for income support.

The effects of minimum wages, on the other hand, are controversial, as it is questionable whether they lead to unemployment to the extent that economic theory predicts. There is agreement that if a minimum wage is set at a high level relative to the free market equilibrium wage, it is likely to create some unemployment. Yet a large and growing number of studies show that a minimum wage may have no effect or even a positive effect on *total* employment. Some firms respond to the minimum wage by maintaining the same number of workers but cutting non-wage benefits (such as paid holidays or sick leave). Also, it is possible that labour productivity (defined as the amount of output produced per worker) may increase due to the minimum wage, as workers feel motivated to work harder, with the result that some firms hire more unskilled labour in response to minimum wages.

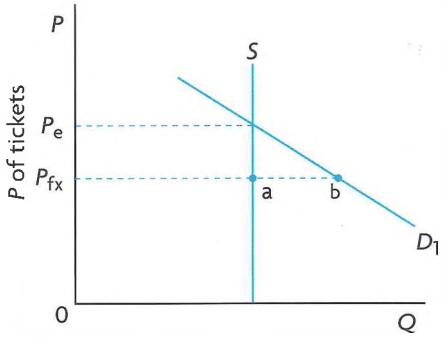
While the effects of minimum wages remain controversial, there is generally strong political support for their continued use on the grounds of greater equity in income distribution. In fact, the debate is not about whether or not there should be minimum wages, but rather what their level should be.

Setting fixed prices

Sometimes prices may be fixed at a particular level, such as with ticket prices for theatres, movies and sports events, where prices are usually fixed ahead of time by the organising body (which may be private or public), and so cannot increase or decrease according to supply and demand.

Figure 4.10 shows the market for tickets for a sports event. The supply curve is vertical because there is a fixed supply of tickets (due to a fixed number of seats; see Chapter 2). The ticket price is fixed at P_{fx} by the organising body. Figure 4.10(a) illustrates an event for which there is large demand, given by D_1. If the price could respond to market forces, it would rise to P_e, but since it is fixed at P_{fx} a shortage of tickets arises equal to the horizontal difference between points a and b. Figure 4.10(b) illustrates an event for which there is low demand, given by D_2. Here, the equilibrium price would have been P_e, however price is fixed at the higher level P_{fx}, resulting in a surplus of tickets equal to the horizontal difference between points c and d.

a Price fixing resulting in a shortage



b Price fixing resulting in a surplus

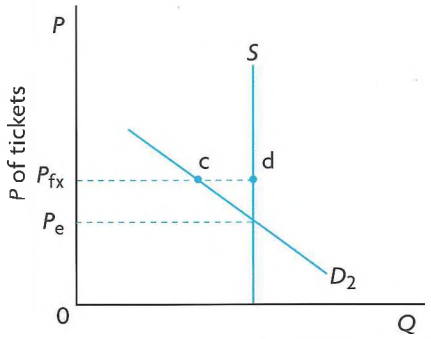


Figure 4.10: Price fixing and surpluses and shortages

TEST YOUR UNDERSTANDING 4.5

1 Define a minimum wage. Suggest reasons why many governments around the world impose them.

2 Draw a diagram illustrating the imposition of a minimum wage, and analyse its effects on market outcomes (the wage, quantity of labour demanded and supplied, market disequilibrium) and consequences for the labour market (unemployment of labour, illegal work, resource misallocation, welfare loss).

3 Suggest why, in practice, minimum wages may not lead to increased unemployment if they are not set too high.

4 Using diagrams show how excess demand or excess supply of tickets results when ticket prices are set at a level that is

a lower than equilibrium, and

b higher than equilibrium.

4.3 Indirect taxes

LEARNING OBJECTIVES

After studying this section you will be able to:

- define all the terms appearing in **orange bold** in the text (AO1)
- explain the consequences of indirect taxes on markets and stakeholders (AO2)
- draw diagrams to illustrate the effects of indirect taxes on markets and stakeholders (AO4)
- evaluate the effects of indirect taxes on markets and stakeholders (AO3)
- calculate the effects of indirect taxes on markets and stakeholders (HL only) (AO4)

Introduction to indirect taxes

The meaning of indirect taxes

Indirect taxes are imposed on spending to buy goods and services. They are paid partly by consumers, but are paid to the government by producers (firms), and for this reason are called 'indirect'. There are two types of indirect taxes:

- *excise taxes*, imposed on particular goods and services, such as petrol (gasoline), cigarettes and alcohol
- *taxes on spending on all (or most) goods and services*, such as *general sales taxes* (used in the United States) and *value added taxes* (used in the European Union, Canada and many other countries).

Indirect taxes differ from **direct taxes**, involving payment of the tax by the taxpayers directly to the government (see Chapter 12).

In this chapter, we will study excise taxes.

Indirect taxes and the allocation of resources

Taxes have the effect of changing the allocation of resources. In Chapter 2 we learned that prices act as signals and incentives, which determine the pattern of resource allocation. Since indirect (excise) taxes

are imposed on particular goods, they increase the price paid by consumers, causing consumers to reduce their spending on taxed goods. Excise taxes also lower the price received by producers, causing them to produce less. Therefore, by changing price signals and incentives, excise taxes affect the allocation of resources.

The interesting question is whether indirect taxes work to reduce or to increase allocative efficiency. The answer depends on the degree of allocative efficiency in the economy before the tax is imposed. If an economy begins with an efficient allocation of resources, the tax creates allocative inefficiency and a welfare loss. We will see how this happens below. In an economy with an inefficient resource allocation, indirect taxes potentially have the effect of improving resource allocation if they are designed to remove the source of allocative inefficiency. This will be studied in Chapter 5.

Why governments impose indirect taxes

Governments impose excise taxes for several reasons:

- **Indirect taxes are a source of government revenue.** Governments collect revenues from indirect taxes. As we know from Chapter 3, the lower the price elasticity of demand, the greater the government revenue generated.
- **Indirect taxes are a method to discourage consumption of goods that are harmful for the individual.** The consumption of certain goods is considered harmful for the individual (for example, cigarette smoking, excess alcohol consumption or gambling). Taxing these goods is likely to reduce their consumption. However, the extent to which these taxes are successful in reducing consumption depends on the price elasticity of demand; if it is low, an indirect tax will likely result in only a relatively small decrease in quantity demanded (see Chapter 5).
- **Indirect taxes can be used to redistribute income.** Some excise taxes focus on **luxury goods** (expensive cars, boats, furs, jewellery, and so on). The objective is to tax goods that can only be afforded by high-income earners. Payment of a tax on the purchase of these goods reduces after-tax income, thus narrowing differences with the incomes of lower-income earners.
- **Indirect taxes are a method to improve the allocation of resources (reduce allocative inefficiencies) by correcting negative externalities.** If there are market imperfections (in the form of negative externalities), preventing the achievement of allocative efficiency, indirect taxes can be used to try to improve the allocation of resources. This topic will be discussed in Chapter 5.

In this chapter, we assume that the economy begins with allocative efficiency in order to see how the introduction of indirect taxes leads to allocative inefficiency.

Indirect taxes: impacts on market outcomes and consequences for stakeholders

Distinguishing between specific and *ad valorem* taxes

Indirect, excise taxes can be:

- *specific taxes*, a fixed amount of tax per unit of the good or service sold; for example, €5 per packet of cigarettes
- *ad valorem taxes*, a fixed percentage of the price of the good or service; in this case, the amount of tax increases as the price of the good or service increases.

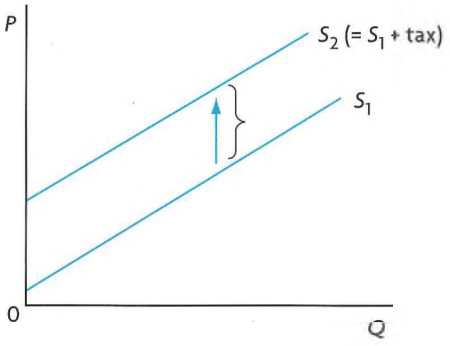
In our study of indirect taxes, we will consider only specific taxes, in other words a specific amount of tax for each unit of the good sold.

When a tax is imposed on a good or service, it is paid to the government by the firm. This means that for every level of output the firm is willing and able to supply to the market, it must receive a price that is higher than the original price by the amount of the tax. This involves a shift of the supply curve upward by the amount of the tax, and is shown in Figure 4.11(a). The tax causes a parallel upward shift, because the tax is a fixed amount for each unit of output. Therefore, in Figure 4.11(a) S_2 is parallel to S_1 (Note that this is equivalent to a leftward shift of the supply curve, meaning that for each price, the firm is willing to supply less output; this equivalence is explained in 'Quantitative techniques' chapter in the digital coursebook).

Illustrating and analysing impacts of an indirect tax on market outcomes

The impacts of specific taxes on market outcomes are shown in Figure 4.11(b). The supply curves are the same as in Figure 4.11(a); a demand curve has also been added. The pre-tax equilibrium is determined by the intersection of the demand curve D and the supply curve S_1, so the price paid by consumers and received by producers is P^* and quantity demanded and supplied is Q^*. If the government imposes a specific tax on the good, the supply curve shifts upwards to $S_2 = S_1 + \text{tax}$. The demand curve remains constant at D since demand is not affected. The new market equilibrium is determined by the demand curve D and the new supply curve S_2, so the price paid by consumers increases to P_c, and the quantity purchased falls to Q_t. The amount of tax per unit of output is shown on the vertical axis by $P_c - P_p$, or the vertical difference between the two supply curves. Whereas producers receive from consumers P_c per unit, they must pay the government $P_c - P_p$ per unit (tax per unit). Therefore, P_p is the final price received by producers after payment of the tax.

a How the supply curve shifts



b Market outcomes due to an indirect tax

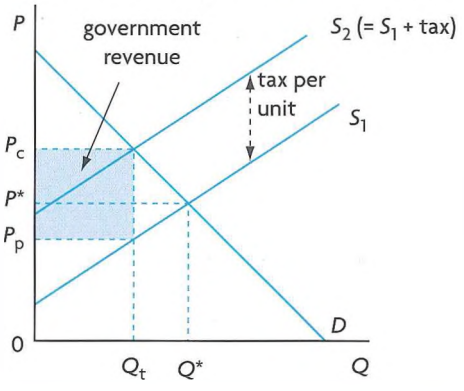


Figure 4.11: Supply curve shifts due to an indirect tax

The tax is said to 'drive a wedge' between the price P_c paid by consumers and the price P_p received by producers.

The market outcomes due to the tax are the following:

- equilibrium quantity produced and consumed falls from Q^* to Q_t
- equilibrium price increases from P^* to P_c, which is the price paid by consumers
- consumer expenditure on the good is given by the price of the good per unit times the quantity of units bought; it therefore changes from $P^* \times Q^*$ to $P_c \times Q_t$
- price received by the firm falls from P^* to P_p, which is $P_p = P_c - \text{tax per unit}$
- the firm's revenue falls from $P^* \times Q^*$ to $P_p \times Q_t$
- the government receives tax revenue, given by $(P_c - P_p) \times Q_t$, or the amount of tax per unit times the number of units sold; this is the shaded area in Figure 4.12
- there is an underallocation of resources to the production of the good: Q_t is less than the free market quantity, Q^*.

Consequences of indirect taxes for various stakeholders

Consumers

Consumers are affected in two ways: by the increase in the price of the good (from P^* to P_c, shown in Figure 4.11(b)) and by the decrease in the quantity they buy (from Q^* to Q_t). Both these changes make them worse off, as they are now receiving less of the good and paying more for it.

Producers (firms)

Producers are affected in two ways: by the fall in the price they receive (from P^* to P_p), and by the fall in the quantity of output they sell (from Q^* to Q_t). These effects translate into a fall in their revenues, from $P^* \times Q^*$ before the tax to $P_p \times Q_t$. Firms are therefore worse off as a result of the tax.

The government

The government is the only stakeholder that gains, as it now has revenue equal to $(P_c - P_p) \times Q_t$ in Figure 4.11(b). This is positive for the government budget.

Workers

A lower amount of output, from Q^* to Q_t, means that fewer workers are needed to produce it; therefore, the tax may lead to some unemployment. Workers are worse off if they become unemployed.

Society as a whole: consumer and producer surplus and welfare loss

Society is worse off as a result of the tax, because there is an underallocation of resources to the production of the good ($Q_t < Q^*$). What happens to social surplus after the imposition of the tax? We can see this in Figure 4.12. Part (a) shows the maximum consumer plus producer surplus in a competitive free market that we are familiar with. The effects of the indirect tax can be seen in part (b), where consumer surplus becomes the shaded area under the demand curve and above P_c up to Q_t. Producer surplus becomes the shaded area above the supply curve S_1 and below P_p up to Q_t. A portion of consumer surplus became government tax revenue, and another portion was lost as triangle a. A portion of producer surplus also became government tax revenue, and another portion was lost as triangle b. Total government revenue can be seen in Figure 4.12(b).

Note that when identifying producer surplus after the imposition of an indirect tax, we always refer to the *initial supply curve*, which is S_1 in Figure 4.12b.

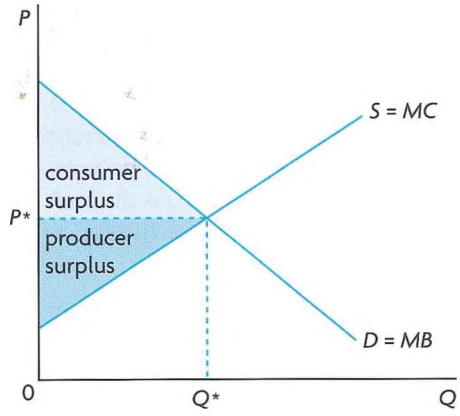
The consumer and producer surplus that is transformed into government tax revenue comes back to society in the form of government spending from the tax revenues. Therefore, the after-tax social surplus in Figure 4.12(b) is equal to after-tax consumer and producer surplus plus government revenue. However, after-tax social surplus is less than pre-tax social surplus by the amount of triangles a + b. The areas a + b represent social surplus that is completely lost, and is welfare loss.

In this case, welfare loss appears because the tax causes a smaller than optimum quantity to be produced: $Q_t < Q^*$. The tax has caused underproduction of the good relative to what is socially desirable, and an underallocation of resources, or allocative inefficiency.

Note that at the new point of production, Q_t, $MB > MC$, meaning that the benefits consumers receive from the last unit of the good they buy are greater than the marginal cost of producing it. Consumers would be better off if more of the good were produced (for an explanation see Chapter 2).

The imposition of an indirect tax results in reduced consumer and producer surplus, part of which is transformed into government revenue, and part of which is a welfare loss. The welfare loss in this case is the result of underallocation of resources to the production of the good (underproduction). This is also indicated by $MB > MC$: too little of the good is produced and consumed relative to the social optimum.

a Consumer and producer surplus in a competitive free market: maximum social surplus



b Consumer and producer surplus with an indirect tax: welfare loss

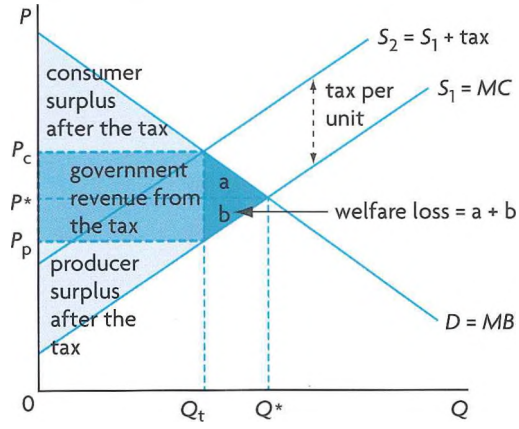


Figure 4.12: Effects of indirect taxes on consumer and producer surplus

TEST YOUR UNDERSTANDING 4.6

1 Describe what 'indirect' means in reference to 'indirect taxes'.

2 State some reasons why governments impose indirect taxes.

3 Outline how indirect taxes affect the allocation of resources.

4 Explain why in contrast to price controls, an indirect tax does not result in disequilibrium.

5 The government is considering imposing a €0.50 tax per litre of petrol (gasoline).

a Draw a diagram for the gasoline market before the imposition of the tax, showing the price paid by consumers, the price received by producers and the quantity of petrol (gasoline) that is bought/sold.

b Draw a diagram for the petrol (gasoline) market after the imposition of the tax, showing the price paid by consumers, the price received by producers and the quantity of petrol (gasoline) bought/sold.

6 For question (5):

a analyse the impacts on the market of the tax on petrol (gasoline), and

b discuss the consequences for stakeholders.

7 Using a diagram, show the effects of an indirect tax on consumer and producer surplus, as well as welfare loss. Explain why welfare loss arises as a result of a misallocation of resources.

Calculating the effects of indirect taxes on stakeholders and social welfare (HL only)

Suppose the government imposes an indirect (excise) tax on stromples of €6 per unit. This means that the supply curve will shift upward by €6 for each level of output Q.

How to graph the new supply curve, S_2

In Figure 4.13, the new supply curve, $S_2 = S_1$ + tax, lies €6 above the initial supply curve, S_1. We can count €6 upward along the vertical axis from the P intercept of S_1, and then draw a line parallel to S_1 from this new P intercept. This is gives us the new supply curve, S_2. For any Q, the vertical difference between the two supply curves is €6, which is the tax per unit of output.

How to find the new price paid by consumers, the price received by producers and the quantity bought and sold, following the imposition of the tax

After the tax is imposed, the demand curve D and the new supply curve, S_2, determine a new equilibrium price, which is P_c or the price paid by consumers, a new equilibrium quantity, Q_t, and P_p or the price received by producers ($P_p = P_c$ – tax per unit) (see also Figure 4.11(b)).

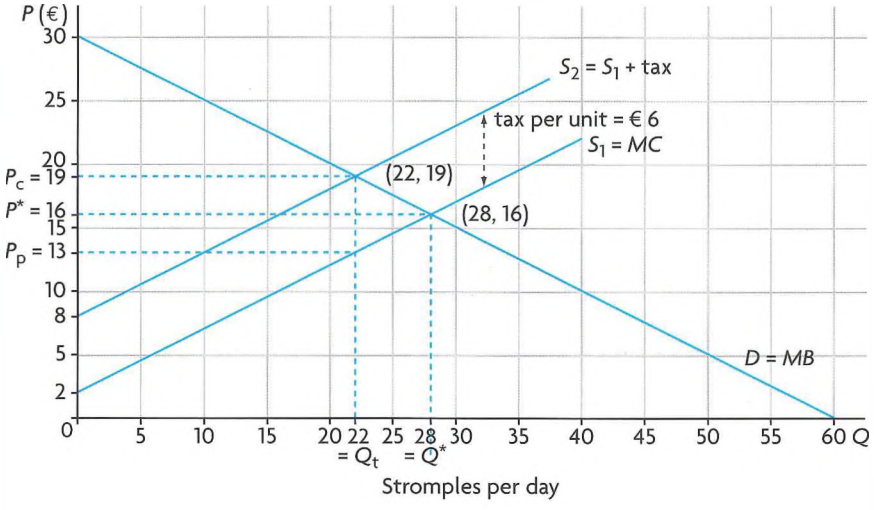


Figure 4.13: Demand and supply with indirect taxes

Therefore, at the new after-tax equilibrium, the price paid by consumers is P_c = €19, the equilibrium quantity of stromples demanded and supplied is 22 units per day, i.e. Q_t = 22, and the price received by producers is P_p = P_c – tax per unit = €19 – €6 = €13. These results are shown in Figure 4.13.

We have found that the price paid by consumers has increased from €16 to €19, the price received by producers has fallen from €16 to €13, and the quantity produced and consumed has fallen from 28 units to 22 units.

We now want to use this price and quantity information, together with the graph in Figure 4.13, to calculate the following: consumer expenditure, producer revenue, government revenue, consumer surplus and producer surplus.

Consumer expenditure

Consumer expenditure is given by the price paid per unit of stromples times the number of stromples purchased. Therefore, before the tax, consumers spent P^* × Q* = €16 × 28 units = €448 per day; after the tax was imposed, consumers spent P_c × Q_t = €19 × 22 units = €418 per day. Therefore consumer expenditure fell by €30 per day (=€448 – €418).

Producer revenue

Producer revenue is given by the price received per unit of stromples times the number of stromples sold. Therefore, before the tax, producer revenue was P^*×Q* = €16 × 28 units = €448 per day, which is the same as what consumers spent; firm revenue was exactly equal to consumer expenditure. After the tax was imposed, firm revenue fell to P_p × Q_t = €13 × 22 units = €286 per day. Producer revenue fell by €162 per day (= €448 – €286). Firm revenue is now less than consumer expenditure.

Government revenue

Government revenue can be calculated in two ways:

a It is equal to tax per unit (P_c – P_p) times the number of units sold, Q_t, and is therefore €6 × 22 stromples = €132.

b It is also equal to the difference between consumer expenditure and producer revenue after the tax: €418 – €286 = €132.

Calculating the effects of indirect taxes on consumer surplus, producer surplus and welfare loss

As we have seen above, consumer and producer surplus both before and after the indirect tax are the areas of triangles, therefore we use the familiar triangle formula to calculate them.

Figure 4.14 is the same as Figure 4.12, but also shows P and Q values, which are the same values as in Figure 4.13. In part (a), consumer surplus is the shaded area under the demand curve and above P^* = 16, up to Q^* = 28. In part (b), it is the shaded area under the demand curve and above P_c = 19, up to Q_t = 22.

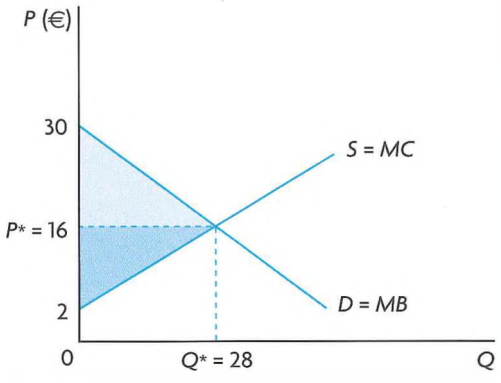
As you may recall, consumer surplus is:

$$\text{Consumer surplus} = \frac{(P \text{ intercept of D curve minus } P \text{ of consumers}) \times Q \text{ purchased}}{2}$$

Therefore, consumer surplus before the tax is:

$$\frac{(30 - P^*) \times Q^*}{2} = \frac{(30-16)\times 28}{2} = \frac{14 \times 28}{2} = \frac{392}{2} = €196$$

a Pre-tax equilibrium: maximum social surplus



b Post-tax equilibrium: welfare loss

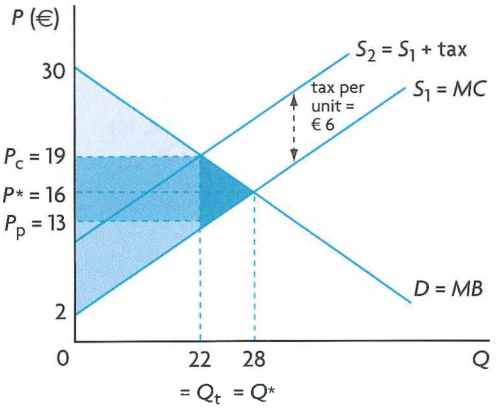


Figure 4.14: Calculating consumer and producer surplus before and after an indirect tax

Consumer surplus after the tax is:

$$\frac{(30-P_c)\times Q_t}{2}=\frac{(30-19)\times 22}{2}=\frac{11\times 22}{2}=\frac{242}{2}=€121$$

In the case of producer surplus, you may recall that it is:

Producer surplus =

$$\frac{(P \text{ of producers minus } P \text{ intercept of } S_1 \text{ curve})\times Q \text{ sold}}{2}$$

In Figure 4.14(a), producer surplus is the area above the supply curve S and below $P^* = 16$, up to $Q^* = 28$. In Figure 4.14(b), it is the area above the supply curve S_1 and below $P_p = 13$, up to $Q_t = 22$.

To calculate producer surplus, we can think of it as half the area of the rectangle whose one side equals the price received by producers minus the P intercept of the initial supply curve, S_1, and whose other side equals the number of units sold:

Producer surplus =

$$\frac{(P \text{ of producers minus } P \text{ intercept of } S_1 \text{ curve})\times Q \text{ sold}}{2}$$

Therefore, producer surplus before the tax is:

$$\frac{(P^*-2)^*Q^*}{2}=\frac{(16-2)\times 28}{2}=\frac{14\times 28}{2}=\frac{392}{2}=€196$$

Producer surplus after the tax is:

$$\frac{(P_p-2)\times Q_t}{2}=\frac{(13-2)\times 22}{2}=\frac{11\times 22}{2}=\frac{242}{2}=€121$$

Note that to calculate producer surplus after the tax has been imposed, we use the *initial supply curve*, S_1.

(You may have noticed that consumer and producer surplus are equal to each other, both before and after the tax; *this is coincidental, as they need not be equal to each other*.)

The welfare loss can be found by taking the pre-tax sum of consumer and producer surplus (total social surplus), and subtracting from that the post-tax sum of benefits (post-tax consumer surplus, producer surplus and tax revenue): 196 + 196 – (121 + 121 + 132) = 18.

This is also equal to the area of the triangle:

$$\frac{(P_c-P_p)(Q^*-Q_t)}{2}=\frac{(19-13)(28-22)}{2}=\frac{6\times 6}{2}=€18$$

TEST YOUR UNDERSTANDING 4.7

1 Using the concept of allocative efficiency explain why an indirect tax creates welfare loss.

2 In the market for good zeta, the *P* intercept of the demand curve is at the point where Q = 0 and P = 7, and the *P* intercept of the supply curve is at the point where Q = 0 and P = 1. The point of intersection of the demand curve and the supply curve at free market equilibrium is at the point where Q = 6 and P = 4.

- a Draw the demand and supply curves, and identify the equilibrium price and quantity.
- b Suppose that price is measured in $, and quantity of zeta in tonnes per day, and that a tax of $2 per tonne is imposed; draw the new supply curve and identify the price paid by consumers, the price received by producers and the new equilibrium quantity.
- c Explain why the increase in price paid by consumers is smaller than the amount of tax per unit.
- d Using your results, calculate the change in consumer expenditure, the change in firm revenue, government revenue, the change in consumer surplus, the change in producer surplus and welfare loss.
- e Identify, in your diagram, the areas that correspond to government revenue, welfare loss and after-tax consumer and producer surplus.
- f Outline how the relationship between marginal benefit and marginal cost at the new (after-tax) equilibrium relates to allocative efficiency (or inefficiency).

Tax incidence and price elasticities of demand and supply (Supplementary material)

When a good is taxed, part of the tax is paid by consumers and part by producers; therefore the tax burden is shared between the two. If you are interested in seeing how the tax is shared you can read about it in the digital coursebook as Supplementary material.

4.4 Subsidies

LEARNING OBJECTIVES

After studying this section you will be able to:

- define all the terms appearing in **orange bold** in the text (AO1)
- explain the consequences of subsidies on markets and stakeholders (AO2)
- draw diagrams to illustrate the effects of subsidies on markets and stakeholders (AO4)
- evaluate the effects of subsidies on markets and stakeholders (AO3)
- calculate the effects of subsidies on markets and stakeholders (HL only) (AO4)

Introduction to subsidies

The meaning of subsidies

A **subsidy**, in a general sense, refers to assistance by the government to individuals or groups of individuals, such as firms, consumers, industries or sectors of an economy. Subsidies may take the form of direct cash payments or other forms of assistance such as low-interest or interest-free loans (for example, to students, to low-income consumers for the purchase of goods and services such as housing, or to firms needing assistance), the provision of goods and services by the government at below-market prices; tax relief (i.e. paying lower or no taxes); and others.

In this section, we will consider only subsidies consisting of payments by the government to firms. Such payments are usually a fixed amount per unit of output, and are therefore *specific subsidies*.

Subsidies and the allocation of resources

Subsidies, like taxes, have the effect of changing the allocation of resources because they affect relative prices, thus changing the signals and incentives prices convey. A subsidy granted to a firm (or group of firms) has the effect of increasing the price received by producers, causing them to produce more, and lowering the price paid by consumers, causing them to buy more. Therefore, the allocation of resources changes and results in greater production and consumption than in the free market.

As with indirect taxes, we are interested in seeing whether the granting of a subsidy improves or worsens the allocation of resources. Here, too, the answer depends on the degree of allocative efficiency in the market before the subsidy. In an economy where resources are allocated efficiently, a subsidy introduces allocative inefficiency and welfare losses. This will be the topic of this section. But if the economy begins with allocative inefficiency (due to market imperfections), then a subsidy can work to improve the allocation of resources if it is designed to correct the source of the inefficiency. This will be examined in Chapter 6.

Why governments grant subsidies

There are several reasons why governments grant subsidies to firms:

- **Subsidies can be used to increase revenues (and hence incomes) of producers.** Subsidies have the effect of increasing the revenues of producers. Therefore, governments often grant subsidies to particular producers whose revenues (and therefore incomes) they would like to support.
- **Subsidies can be used to make certain goods (necessities) affordable to low-income consumers.** Subsidies have the effect of lowering the price of the good that is paid by consumers, thus making the good more affordable. For example, a government may wish to make a food staple (such as bread or rice) more affordable to low-income earners. It can do so by granting a subsidy to producers of the good.
- **Subsidies can be used to encourage production and consumption of particular goods and services that are believed to be desirable for consumers.** A subsidy has the effect of increasing the quantity of a good produced and consumed. If a government wishes to encourage consumption of a good because it is considered to be desirable (for example, education, vaccinations), it can use a subsidy to achieve this.
- **Subsidies can be used to support the growth of particular industries in an economy.** Since subsidies have the effect of increasing the quantity of output produced, if granted to firms in a particular industry, they support the growth of that industry. For example, subsidies to the solar industry are intended to promote the growth of solar power.

Other examples include chemicals, textiles, steel, fossil fuels and many more.

- **Subsidies can be used to encourage exports of particular goods.** Since subsidies lower the price paid by consumers, they are sometimes granted on goods that are exported (sold to other countries), since lower export prices increase the quantity of exports (see Chapter 14).
- **Subsidies are a method to improve the allocation of resources (reduce allocative inefficiencies) by correcting positive externalities.** It was noted above that market imperfections prevent the achievement of allocative efficiency; in some cases (such as when there are positive externalities), it may be possible to use subsidies to improve allocative efficiency (see Chapter 6).

Subsidies are a controversial topic in economics because they are very extensive and are often designed to achieve certain objectives that may not be consistent with other important objectives. For example, many countries grant subsidies to fossil fuels, which run contrary to objectives of sustainable development and which also contradict the objectives of other subsidies intended to support the growth of alternative energy. Fossil-fuel subsidies are known as 'perverse subsidies'. Subsidies for agriculture and exports are also highly controversial (see Chapter 14).

Subsidies: impacts on market outcomes and consequences for stakeholders

Illustrating and analysing impacts of subsidies on market outcomes

In Figure 4.15, the initial, pre-subsidy equilibrium is determined by the intersection of the demand curve D and the supply curve S_1, giving rise to equilibrium price P^* paid by consumers and received by producers, and equilibrium quantity Q^*. Now the government grants a subsidy consisting of a payment to the firm of a fixed amount for each unit of output sold. This means that for each unit of output the firm is willing and able to produce, it receives a lower price than the original by the amount of the subsidy; this produces a downward, parallel shift of the supply curve by the amount of the subsidy, to the new curve $S_2 = S_1 - \text{subsidy}$. (The reason we subtract the subsidy is that it works to *decrease the firms' costs of production*, thus causing a downward shift of the S curve. It is the exact opposite of the indirect tax which is added to the S curve to cause an upward shift. (See 'Quantitative techniques' chapter in the digital coursebook). The demand curve remains constant at D since demand is not affected. The demand curve and new supply curve S_2 determine a new equilibrium, where price is P_c (the price paid by consumers) and the quantity produced and sold increases to Q_{sb}. Since the vertical difference between the two supply curves represents the subsidy per unit of output, the firm receives price P_p, which is equal to the price paid by the consumer, P_c, plus the subsidy per unit of output.

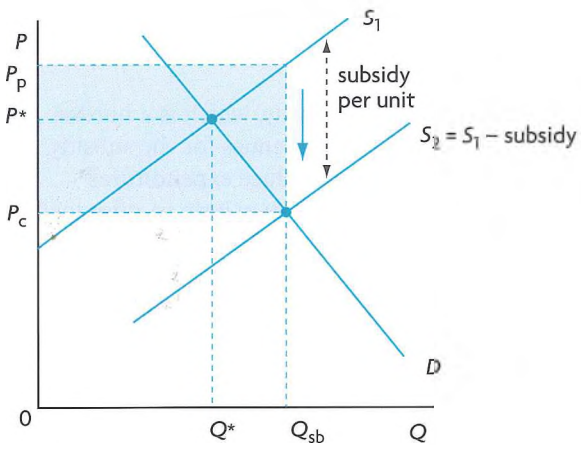


Figure 4.15: Impacts of subsidies on market outcomes

The market outcomes due to the subsidy are the following:

- equilibrium quantity produced and consumed increases from Q^* to Q_{sb}
- the equilibrium price falls from P^* to P_c; this is the price paid by consumers
- the price received by producers increases from P^* to P_p
- the amount of the subsidy is given by $(P_p - P_c) \times Q_{sb}$, or the amount of subsidy per unit multiplied by the number of units sold; this is the entire shaded area, and represents government spending to provide the subsidy
- there is an overallocation of resources to the production of the good: Q_{sb} is greater than the free market quantity, Q^*.

Consequences of subsidies for various stakeholders

Consumers

Consumers are affected by the fall in price of the good from P^* to P_c (Figure 4.15) and the increase in quantity

purchased (from Q^* to Q_{sb}). Both these changes make them better off.

Producers

Producers are also better off, because they receive a higher price ($P_p > P^*$) and produce a larger quantity ($Q_{sb} > Q^*$), seen in Figure 4.15. The price and quantity effects translate into an increase in revenues. Before the granting of the subsidy, firms had revenues of $P^* \times Q^*$. Following the subsidy, firm revenues increase to $P_p \times Q_{sb}$.

The government

The government pays the subsidy, which is a burden on its budget. To obtain the revenues for the subsidy, the government may have to reduce expenditures elsewhere in the economy, or it may have to raise taxes, or it may have to run a budget deficit (government expenditures greater than tax revenues). Whatever the case, the impact on the government's budget is negative.

Workers

As output expands from Q^* to Q_{sb}, firms are likely to hire more workers to produce the extra output, therefore workers who find new jobs are better off.

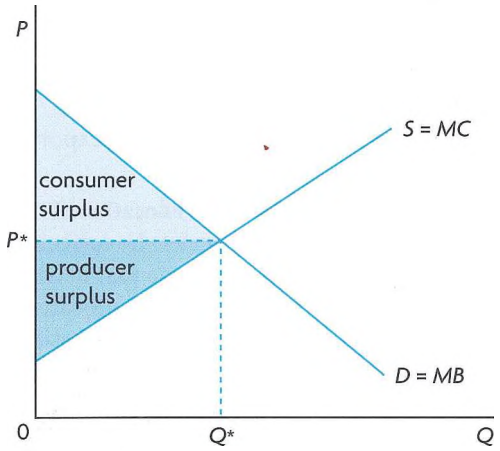
Society as a whole: consumer and producer surplus, and welfare loss

Society as a whole is worse off because there is an overallocation of resources to the production of the good; $Q_{sb} > Q^*$. In addition, society is worse off because the higher price received by producers protects relatively inefficient ones, allowing them to continue to produce.

Figure 4.16 shows consumer and producer surplus before and after the subsidy. In part (a), at the free market equilibrium before the subsidy, social (consumer plus producer) surplus is maximum and $MB = MC$, indicating the achievement of allocative efficiency.

After the granting of the subsidy, both consumer surplus and producer surplus increase. In Figure 4.16(b), consumer surplus is now the area under the demand curve and above price P_c, up to output Q_{sb}. The *gain* in consumer surplus is shown by the shaded area labelled 'gain in consumer surplus'. Producer surplus becomes the area above the supply curve S_1 and below the price P_p, up to output Q_{sb}. The *gain* in producer surplus is shown by the shaded area labelled 'gain in producer surplus'.

a Consumer and producer surplus in a competitive free market: maximum social surplus



b Consumer and producer surplus with a subsidy: welfare loss

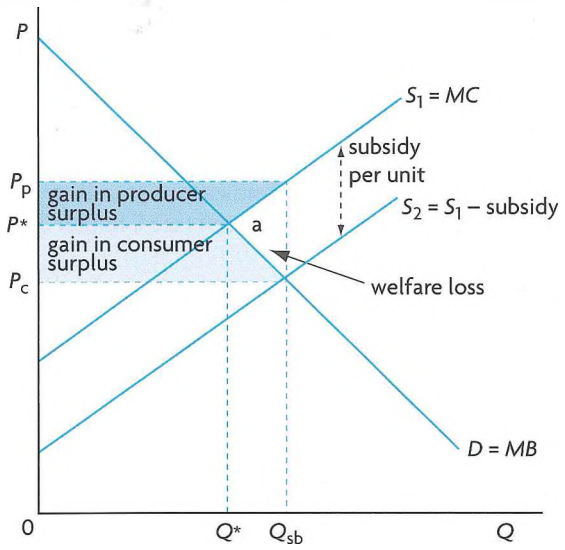


Figure 4.16: Effects of subsidies on consumer and producer surplus

Note that when identifying producer surplus after the granting of a subsidy, we always refer to the *initial supply curve*, which is S_1 in Figure 4.16.

At the same time that producers and consumers gain, the government loses because of the negative effects on its budget. The subsidy is paid for by taxes that have an opportunity cost (alternative uses that are sacrificed). As we know, government expenditure to provide the subsidy is $(P_p - P_c) \times Q_{sb}$. *This is*

exactly equal to the gains in consumer and producer surplus plus the triangle a. Therefore, the social losses due to government spending are greater than the gains in consumer and producer surplus by the amount a. The area a is welfare loss representing lost benefits for society, caused by a larger than optimum quantity produced: $Q_{sb} > Q^*$. The subsidy has caused overproduction relative to what is socially desirable, and an overallocation of resources, or allocative inefficiency.

We can also see in Figure 4.16(b) that at Q_{sb}, $MB < MC$, meaning that the benefit consumers receive from the last unit of the good they buy is less than the marginal cost of producing it. Therefore, society would be better off if less of the good were produced.

The granting of a subsidy results in greater consumer and producer surplus; however, society loses due to government spending on the subsidy. Since the loss from government spending is greater than the gain in consumer and producer surplus, welfare loss results, reflecting allocative inefficiency, which in this case is due to overallocation of resources to the production of the good (overproduction). This is also illustrated by $MB < MC$: too much of the good is being produced and consumed relative to the social optimum.

Foreign producers

If the subsidy is granted on exports (goods sold to other countries), it lowers price and increases the quantity of exports. While this is positive for domestic producers, it is negative for the producers of other countries who may be unable to compete with the lower price of the subsidised goods. (This topic will be discussed in Chapters 14 and 18.)

TEST YOUR UNDERSTANDING 4.8

1 Define subsidies and providing examples, outline some reasons why governments grant them.

2 The government is considering granting a €0.50 subsidy per kilogram of cheese.

   a Draw a diagram for the cheese market before the granting of the subsidy, showing the price paid by consumers, the price received by producers and the quantity of cheese that is bought/sold.

   b Using the same diagram, show what happens in the cheese market after the granting of the subsidy, indicating the price paid by consumers, the price received by producers, the quantity of cheese bought/sold and government spending on the subsidy.

3 Considering question 2 above:

   a analyse the impacts on the market of the subsidy, and

   b discuss the consequences for stakeholders.

4 Using a diagram, show the increase in consumer and producer surplus that results from the granting of a subsidy. Explain why welfare loss arises even though both consumer and producer surplus increase. Show the welfare loss in your diagram.

REAL WORLD FOCUS 4.2

Farm subsidies in the United States

Farmers in the United States have been receiving subsidies for certain agricultural products (corn, wheat, soybeans, cotton, rice and others) since the Great Depression of the 1930s. In 1996, a law was passed to end subsidies and create a free market in agriculture. However, the free market in agriculture never materialised. Support to farmers costs tax payers over $20 billion a year. This support is justified by the common belief that the government is helping small farmers survive. Yet according to the US Department of Agriculture (the agriculture ministry), the largest and wealthiest farmers receive the bulk of

CONTINUED

farm subsidies. The reason is that subsidies are paid according to the amount of crop produced. Smaller farmers receive very small amounts, while farmers who cultivate fruits and vegetables receive no subsidies at all. In fact, many billionaires (who are the owners of very large agricultural companies) receive farm subsidies. According to the Environmental Working Group, 50 people in the Forbes list of 400 wealthiest Americans received subsidies in the period 1995 to 2014. Since 2008, the top ten recipients of subsidies *each* received an average of $18.2 million.

Source: *Adapted from 'Farm subsidies no illogical entitlements – next senator must push for overhaul of ag policy' in Lexington Herald-Leader, 11 July 2010. Chris Edwards, 'Reforming federal farm policies', Tax and budget bulletin no 82, Cato institute, 12 April 2018 'Mapping the US farm subsidy $1million club,' Adam Andrzejewski, Forbes, 14 August 2018*



Figure 4.17: Washington, D.C., USA. The National Farmers Union urges Congress to pass a farm bill

Applying your skills

1 Discuss the consequences of farm subsidies for agricultural markets and stakeholders.

2 Suggest reasons why the government continues to grant agricultural subsidies even though they are costly and unfair.

Calculating the effects of subsidies on market outcomes and social welfare (HL only)

If the government grants a subsidy on stomfles of $4 per kg, the supply curve shifts downward by $4 for each kg of output Q.

How to graph the new supply curve, S_2

In Figure 4.18, the new supply curve, $S_2 = S_1 -$ subsidy, lies $4 below the initial supply curve, S_1. We can count $4 downward along the vertical axis from the P intercept of S_1, and then draw a line parallel to S_1 from this new P intercept. This gives us the new supply curve, S_2. For any Q, the vertical difference between the two supply curves is $4, which is the subsidy per unit of output.

Therefore, at the new equilibrium, the price paid by consumers is P_c = $18, the equilibrium quantity is Q_{sb} = 24 kg, and the price received by producers is $P_p = P_c$ + subsidy per unit = $18 + $4 = $22. These results are shown in Figure 4.18.

We have found that the price paid by consumers has fallen from $20 to $18 per kg, the price received by producers has increased from $20 to $22 per kg, and the quantity produced and consumed has increased from 20 kg to 24 kg.

We will now use this price and quantity information together with the graph in Figure 4.18 to calculate consumer expenditure, producer revenue, government expenditure, consumer surplus, producer surplus and welfare loss.

Consumer expenditure

Consumer expenditure equals the price paid per kg of stomfles times the number of kg purchased. Therefore, before the subsidy, consumers spent $P^* \times Q^*$ = $20 × 20 kg = $400 per day; after the subsidy, consumers spent $P_c \times Q_{sb}$ = $18 × 24 kg = $432 per day. Therefore consumer expenditure increased by $32 per day (= $432 – $400).

Producer revenue

Producer revenue is given by the price received per kg of stomfles times the number of kg sold. Therefore, before

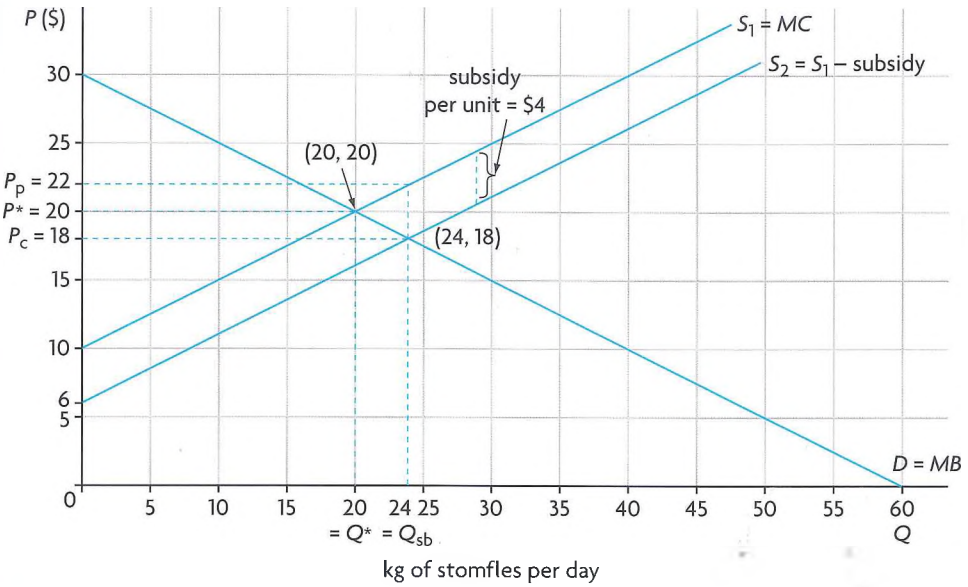


Figure 4.18: Demand and supply with subsidies

the subsidy was granted, producer revenue was $P^* \times Q^*$ = \$20 × 20 kg = \$400 per day, which is exactly the same as what consumers spent; firm revenue was exactly equal to consumer expenditure. After the subsidy was granted, producer revenue increased to $P_p \times Q_{sb}$ = \$22 × 24 kg = \$528 per day. Producer revenue increased by \$128 per day (= \$528 – \$400). Note that firm revenue is now more than consumer expenditure.

Government expenditure

Government expenditure on the subsidy can be calculated in two ways:

a It is equal to the subsidy per kg ($P_p - P_c$) times the number of kg sold (Q_{sb}), and is therefore \$4 × 24 kg = \$96 per day.

b It is also equal to the difference between producer revenue and consumer expenditure after the subsidy: \$528 – \$432 = \$96 per day.

Calculating the effects of subsidies on consumer and producer surplus and welfare loss

Consumer and producer surplus both before and after the subsidy are the areas of triangles, therefore we use the familiar triangle formula to calculate them.

Examining Figure 4.18, we can see that consumer surplus at the free market equilibrium (before the subsidy) is represented by the area under the demand curve and above price P^* = 20 up to quantity Q^* = 20.

After the subsidy, it is the area under the demand curve and above price P_c = 18 up to quantity Q_{sb} = 24.

Consumer surplus =

$$\frac{(P \text{ intercept of D curve minus } P \text{ of consumers}) \times Q \text{ purchased}}{2}$$

Therefore, consumer surplus before the subsidy is:

$$\frac{(30 - P^*) \times Q^*}{2} = \frac{(30 - 20) \times 20}{2} = \frac{10 \times 20}{2} = \frac{200}{2} = \$100$$

Consumer surplus after the subsidy is:

$$\frac{(30 - P_c) \times Q_{sb}}{2} = \frac{(30 - 18) \times 24}{2} = \frac{12 \times 24}{2} = \frac{288}{2} = \$144$$

Therefore, consumer surplus increased by \$44 (= \$144 – \$100).

Producer surplus before the subsidy in Figure 4.18, is the area above the supply curve S_1 and below P^* = 20 up to Q^* = 20. After the subsidy, it is the area above the supply curve S_1 and below P_p = 22 up to Q_{sb} = 24.

Producer surplus is also calculated using the same method as with indirect taxes:

Producer surplus =

$$\frac{(P \text{ of producers minus } P \text{ intercept of } S_1 \text{ curve}) \times Q \text{ sold}}{2}$$

Therefore, producer surplus before the subsidy is:

$$\frac{(P^*-10)\times Q^*}{2}=\frac{(20-10)\times 20}{2}=\frac{10\times 20}{2}=\frac{200}{2}=\$100$$

Producer surplus after the subsidy is:

$$\frac{(P_p-10)\times Q_{sb}}{2}=\frac{(22-10)\times 24}{2}=\frac{12\times 24}{2}=\frac{288}{2}=\$144$$

Therefore producer surplus increased by \$44 (= \$144 − \$100).

Note that to calculate producer surplus after the subsidy has been granted, we use the *initial supply curve*, S_1 (as we do also in the case of indirect taxes).

(The equality of producer and consumer and producer surplus, before and after the subsidy, is coincidental.)

Social surplus increased by the amount of the increase in consumer surplus plus the amount of the increase in producer surplus = \$44 + \$44 = \$88.

Welfare loss can be found by:

welfare loss =

$$\frac{(P_p-P_c)(Q_{sb}-Q^*)}{2}=\frac{(22-18)(24-20)}{2}=\frac{4\times 4}{2}=\$8$$

TEST YOUR UNDERSTANDING 4.9

1 Using the concept of allocative efficiency, explain why a subsidy creates welfare loss.

2 In the market for good alpha the *P* intercept of the demand curve is at the point where Q = 0 and P = 7 and the *P* intercept of the supply curve is at the point where Q = 0 and P = 1. The point of intersection of the demand curve and the supply curve at free market equilibrium is at the point where Q = 6 and *P* = 4.

a Draw the demand and supply curves, and identify the equilibrium price and quantity.

b Suppose that price is measured in £, and quantity in tonnes per day, and that a subsidy of £2 per tonne is granted. Draw the new supply curve, and find (through your graph) the price paid by consumers, the price received by producers and the new equilibrium quantity.

c Using your results, calculate the change in consumer expenditure, the change in firm revenue, government expenditure, the change in consumer surplus, the change in producer surplus and welfare loss.

d Identify, in your diagram, the areas that correspond to government expenditure, welfare loss, the increase in consumer surplus and the increase in producer surplus.

e Outline how the relationship between marginal benefit and marginal cost at the new (after-subsidy) equilibrium relates to allocative efficiency (or inefficiency).

THEORY OF KNOWLEDGE 4.1

Allocative efficiency: is it really value-free?

Throughout this chapter, we have used the competitive market model, explained in Chapter 2, as the basis for making assessments about government intervention in the economy. According to this model, when there is competition in the sense of many buyers and sellers who act according to their best self-interest, and when market forces are free to determine equilibrium prices, a situation is reached where there is allocative efficiency and maximum social welfare. Scarce resources are allocated in the best possible way, producing the most of what people mostly want, and it is not possible to make anyone better off without making someone worse off, a condition called *Pareto optimality*.

CONTINUED

The concept of Pareto optimality emerged in the late 19th century after a period when economists were trying to make economics more scientific in its approach. 'Scientific' meant economics should get rid of any value judgements about things that 'ought to be' and base itself entirely on positive thinking (see Chapter 1). The famous classical economists of earlier times (Adam Smith, Thomas Robert Malthus, David Ricardo, John Stuart Mill, Karl Marx and many others) openly discussed their ideas about what ought to happen in society (normative ideas) together with their positive ideas of things that 'are' or 'will be'. Yet by the late 19th century, it was believed that a true science is value-free, and economists set out to imitate the methods of the natural sciences, especially physics.

The concept of Pareto optimality, developed by Vilfredo Pareto (an Italian engineer, sociologist, economist and philosopher), was welcomed as being truly free of normative aspects. It simply stated that under certain assumptions, numerous freely acting consumers and producers behaving according to their individual preferences, give rise to an outcome of maximum efficiency and maximum social welfare (defined as $MB = MC$ or maximum social surplus).

On the surface, this sounds like a value-free, positive statement. Yet in later years (and up to the present), many economists criticised it on the grounds that it is actually heavily based on normative ideas, and is therefore not value-free. The difficulty in detecting these values is that these are implicit; they are not explicit, or *openly stated* values.

Economists who question the value-free nature of Pareto optimality point out that the concept of 'welfare' is defined in relation to individual preferences of consumers and producers (decisions made on the basis of rational self-interest) that determine consumer and producer surplus. Individual preferences become the standard, or measuring stick, by which economists evaluate real-world situations and government intervention in markets. As we saw in this chapter, a free competitive market is 'best'; government intervention in this market reduces welfare.

But surely there are other definitions of welfare, such as equality of opportunity, freedom from hunger and disease, human rights, fairness and many more. All these are normative concepts, but then, by the same token, welfare defined as maximum social surplus is also a normative concept, on the basis of which a judgement is made about how well or how poorly the economy works.

Some economists take these ideas further, and argue that not only Pareto optimality but the body of microeconomics on which it rests is also not value-free. The idea that societies should pursue allocative efficiency follows from *the definition of economics as the social science that tries to determine how to make the best use of scarce resources*. Yet, remember from Chapter 1, all societies must answer three basic economic questions: *what/how much, how* and *for whom to produce*. The definition of economics covers only the first two of these and ignores the third. The reason for neglecting the *for whom to produce* question is that economists consider the first two questions to be part of positive thinking and the third a part of normative thinking (see Theory of knowledge 2.1 in Chapter 2). However, if efficiency (*what* and *how to produce*) leads to maximum welfare, where the definition of welfare is based on a value judgment, economists have come full circle and have based their so-called positive analysis on a value judgement.

In view of the above, some economists argue that the focus of government policies on efficiency, defined as Pareto optimality, diverts attention away from the problem of income distribution, and justifies government inaction in this area. What is the point of realising (or coming close to realising) Pareto optimality in the real world, if a large portion of the population is starving because they have no income?

CONTINUED

According to Gunnar Myrdal, a Swedish economist who won the Nobel Prize in 1974, the social sciences are inevitably based on values, but these values should be made explicit:

> The only way we can strive for 'objectivity' in theoretical analysis is to expose the valuations into full light, making them conscious, specific and explicit, and permit them to determine the theoretical research [...] there is nothing wrong, *per se*, with value-loaded concepts if they are clearly defined in terms of explicitly stated value premises.[1]

Thinking points

- In your view, is Pareto optimality value-free, or is it implicitly based on a value judgement?
- Does language in the expression 'maximum social welfare' convey values? Is it possible to have value-free language?
- Does the inevitable use of language in the pursuit of economic knowledge complicate the job of economists as social *scientists* in pursuit of value-free knowledge?
- Do you think the natural sciences are value-free?
- Do you agree with Myrdal's claim that it may be possible to reach 'objectivity' in theoretical social science by making values explicit?
- Do you think economics is, or ever can be, completely value-free?

INQUIRY AND REFLECTION

The following questions will help you reflect on your learning and enhance your understanding of key topics in this chapter. They are suggestions for inquiries that you can undertake on your own or in groups in order to revise the learning objectives of this chapter.

1 Find out if the city or area you live in uses rent controls. If it doesn't, search online and find a city or area where rent controls are in place. Research the effects of the rent controls and evaluate their usefulness. Consider subsidies on housing, which is alternative policy that could have been used to make housing affordable to people on low incomes. Compare and contrast the two policies and try to determine the advantages and disadvantages of each. Using your knowledge of economics, determine which of the two policies should be preferred. Justify your conclusion.

2 Find out if subsidies are granted in your country. If not, find a real-world example of one or more subsidies, research the stakeholders involved and the reasons why the subsidies are given. Evaluate the merits of the subsidies by considering the effects of the subsidies on markets and stakeholders.

EXAM STYLE QUESTIONS

You can find questions in the style of IB exams in the digital coursebook.

[1] G. Myrdal (1970) *Objectivity in Social Research*, Gerald Duckworth.



› Chapter 5

Market failure and socially undesirable outcomes I:

Common pool resources and negative externalities

BEFORE YOU START

- Can you think of some goods or services that when consumed or produced cause harm to others even though they didn't consume or produce them?
- What do you think should be the role of government to reduce the harm caused by such harmful actions?

In this chapter we will discover why the market economy fails to achieve many of its promises. In particular we will see how some actions of individuals or groups of individuals may have negative impacts on others and on the environment. We will also study how government intervention can help markets overcome their shortcomings.

Photo: Borneo, Malaysia. Long tailed macaque monkey eating plastic bag in Bako National Park

5.1 The meaning of common pool resources

LEARNING OBJECTIVES

After studying this section you will be able to:

- define all the terms appearing in **orange bold** in the text (AO1)
- explain the meaning of common pool resources in terms of (AO2)
  - the concepts of rivalry and non-excludability
  - the tragedy of the commons
- explain the meaning of unsustainable production (AO2)

Environmental problems can be studied by examining a special category of resources known as common pool resources.[1] **Common pool resources** are resources that are not owned by anyone, do not have a price and are available for anyone to use without payment or any other restriction. Examples include clean air, lakes, rivers, fish in the open seas, wildlife, hunting grounds, forests, biodiversity, the fertility of the soil that occurs in nature, open grazing land, the ozone layer, the stable global climate, and many more.

Understanding common pool resources

Common pool resources: rivalrous but non-excludable

Common pool resources differ from any other kind of resource or good, because they possess a special combination of characteristics: they are rivalrous and non-excludable. To understand what these terms mean, it is useful to consider the definition of private goods. A *private good* has two characteristics:

- It is **rivalrous**: its consumption by one person reduces its availability for someone else; for example, your computer, textbook, pencils and clothes are rivalrous, because when you buy them, another person cannot buy the same ones; most goods are rivalrous.
- It is **excludable**: it is possible to exclude people from using the good; exclusion is usually achieved by charging a price for the good; if someone is unwilling or unable to pay the price, he or she will not have the benefit of using it; most goods are excludable.

As noted above, most goods are rivalrous. Common pool resources are also rivalrous. If we use up clean air, there is less left over for use by others; when we catch fish in the open sea, there are fewer fish left over for others to catch; if we destroy the stability of the global climate, it will not be available for use by future generations.

Also, it was noted that most goods are excludable. However, open pool resources differ because *they have no price or any other means of excluding users; anyone can use them without payment or other restriction*; therefore they are **non-excludable**. Non-excludable means it is not possible to exclude someone from using a good or resource.

> Common pool resources are *rivalrous* but *non-excludable*. This combination poses serious threats to the environment. *Rivalry* means that the use of resources reduces their availability for others. *Non-excludability* means that the resources can be used abundantly without restrictions and therefore may be overused, degraded and depleted.

There is no end to examples of overuse, depletion and degradation of common pool resources. When factories, homes or cars use fossil fuels that emit pollutants into the atmosphere or into oceans, rivers and lakes, they 'overuse' a portion of these natural resources without paying for them. Some of these activities give rise to global warming, with likely devastating effects on agriculture, health and ecosystems; this involves 'overusing' the benefits provided by a stable global climate. When fish are overfished, the fishing industry uses up an excessive amount of the global stock of fish and disrupts the marine ecosystem. Similarly, when forests are cleared to create land for use in agriculture or for the sale of timber by the lumber industry, there

[1] It may be noted that in the previous syllabus these were known as *common access resources*. The term was changed as *common pool resources* is more commonly used.

are huge consequences in terms of loss of biodiversity and threats to wildlife, the ozone layer and the global climate. Land is being overgrazed because of excessive grazing; arable land is lost because of soil erosion and salinisation; wildlife is endangered because of the destruction of natural habitats due to the encroachment of settlers and agriculture. *In all these cases, common pool resources are used and overused, leading to serious environmental degradation and depletion.*

Common pool resources: The tragedy of the commons

The **tragedy of the commons** is a story about cattle that feed on a fertile pasture that is owned in common by a group of herders (cattle owners). In the beginning each herder had a small number of cattle and all the animals had plenty of space and grass on which to feed. As this was a profitable business, each herder began to increase the number of cattle grazing on the pasture. But as the cattle increased, after some time the pasture became overfilled with grazing cattle that had to increasingly compete with each other for food that was becoming more and more scarce. In the end the grass was all gone, the soil was eroded and the pasture could no longer be used for grazing.

This story allows us to better understand the concepts of rivalry and non-excludability. The fertile pasture is rivalrous because whatever grass is eaten by one animal is not available for another. It is also non-excludable since one herder cannot exclude others from using it. This story is used by environmentalists to illustrate the overuse of a resource when there are no restrictions on its use.

The tragedy of the commons has been challenged by the ideas of Elinor Ostrom, that we will study below under the topic *Collective self-governance* as well as in Theory of knowledge 5.1.

Sustainability and common pool resources

As we know from Chapter 1, *sustainability* in connection with the environment refers to the use of resources in ways that do not result in fewer or lower-quality resources for future generations. Sustainable production is production that uses resources in a sustainable way, in other words by not degrading or depleting them. By contrast, **unsustainable production** refers to production that uses resources unsustainably, depleting or degrading them.

Sustainable production: maximum sustainable yield of common pool resources (Supplementary material)

A simple example shown in Figure 5.1(a) illustrates the meaning of sustainable and unsustainable production and resource use. Fish in the open seas are a common pool resource that anyone has access to without payment. The horizontal axis measures the number of fishing boats, and the vertical axis measures the quantity of fish caught in tonnes. The first, second and third boats each catch 4 tonnes; therefore, in this range of 'constant average yield' (yield refers to the amount of output), the three boats together catch 12 tonnes, or 4 tonnes each on average.

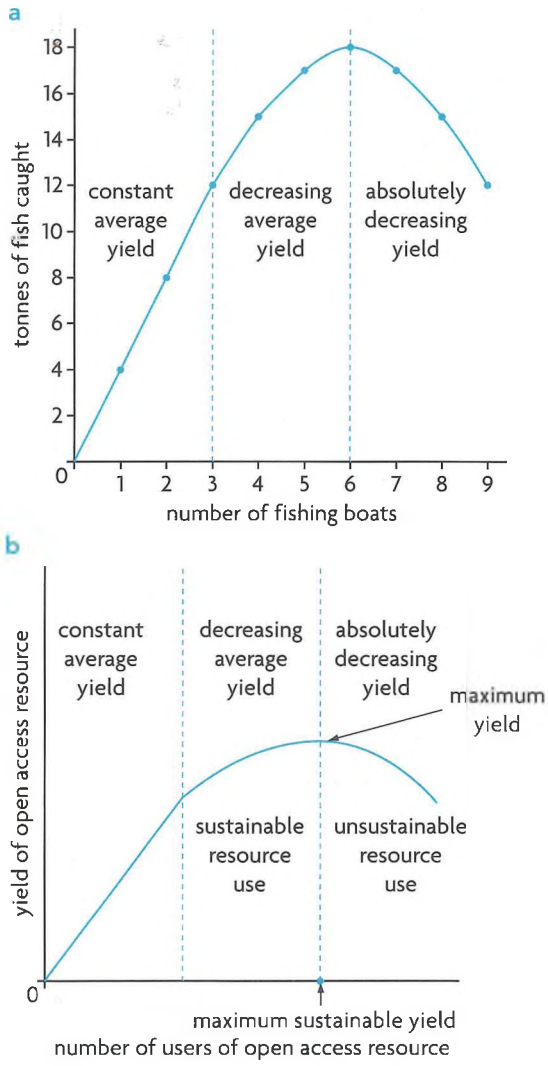


Figure 5.1: Illustrating sustainable and unsustainable resource use

When a fourth boat goes out to sea, it brings back only 3 tonnes of fish; this translates into a smaller quantity of fish caught by each boat on average. The four boats together have caught 15 tonnes, or an average of 3.75 tonnes ($= \frac{15}{4}$) instead of 4 tonnes.

When the fifth boat is added, the five boats catch 17 tonnes, and the average catch falls further to 3.4 tonnes ($= \frac{17}{5}$). With the sixth boat, the total is only 19 tonnes or 3.2 tonnes for each boat on average. This is the range of 'decreasing average yield', meaning that each boat that goes out brings back a smaller amount of fish than the previous one.

What happens if a seventh boat goes out? The total amount of fish caught by the seven boats together (17 tonnes) is *less* than what was caught by 6 boats (18 tonnes). As the graph indicates, in this range of 'absolutely decreasing yield', as more and more boats go fishing, the total amount of fish they bring back becomes less and less.

This example illustrates that the fish were plentiful for the first three boats, but with the addition of the fourth, fishing became more difficult because it began to put pressure on the supply of fish in the ocean. As the supply of fish was more and more depleted, it became increasingly difficult to catch fish, so the average quantity of fish brought back fell with the addition of each boat. Finally, with the addition of the seventh boat, the fish supply was *overused*; the fish population was no longer able to reproduce itself, and therefore the quantity of fish in the ocean began to drop

Figure 5.1(b) shows that the point of maximum yield of a common pool resource is the resource's *maximum sustainable yield. This is the maximum use that can be made of the resource that is also sustainable, in that the resource can reproduce itself.* All points to the left of the maximum sustainable yield indicate sustainable levels of use; points to the right indicate unsustainable use, meaning that the resource is being depleted or degraded. The further to the right, the greater the resource depletion or degradation. In the real world, many common pool resources are used unsustainably, i.e. to the right of their maximum sustainable yield.

Note that while it is an easy matter to discuss the maximum sustainable yield of a resource in theoretical terms as we have done here, it is very difficult in practice to determine what this actually is for any resource.

Sustainable resource use means that resources are used at a rate that allows them to reproduce themselves, so that they do not become degraded or depleted.

A note on renewable and non-renewable resources, and sustainability

Non-renewable resources are those resources that do not last indefinitely, because they have a finite supply (they need tens of thousands or millions of years to reproduce themselves). Examples include metals, minerals and fossil fuels, such as oil, natural gas and coal. Many of these resources, with the exception of fossil fuels, do not get destroyed through their use, and so through effective recycling could be made to last indefinitely. By contrast, fossil fuels are destroyed when used, and moreover have devastating effects on the earth's atmosphere, the global climate and the ozone layer.

Renewable resources are those resources that can last indefinitely if they are managed properly (not overused), because they are reproduced over relatively short periods of time by natural processes. Examples include forests, wildlife, fish, biomass, water resources, geothermal power, soil fertility and biodiversity. The idea of sustainable resource use applies mainly to *renewable resources*, because given appropriate management, these resources can be made to last forever. On the other hand, through mismanagement or overuse, these resources become depleted and degraded, indicating unsustainability.

The idea of sustainable resource use does not apply to non-renewable resources, such as fossil fuels. If resources are non-renewable, they could be used sustainably only if they were not used at all. On the other hand, as we will see in the pages that follow the idea of sustainability is relevant to fossil fuels when referring to the negative externalities that are created by their use.

TEST YOUR UNDERSTANDING 5.1

1 Provide examples of common pool resources, making reference to their overuse.

2 a Define common pool resources using the concepts of rivalry and non-excludability.

b Explain how these characteristics pose a threat to the environment.

c Outline how these characteristics can be illustrated by the tragedy of the commons.

3 Outline the meaning of sustainable production.

4 a In discussions of common pool resources, there is an emphasis on their *overuse* rather than their *use*. Explain why.

b Explain why cutting down a small amount of forest over an extended period of time may be consistent with the concept of environmental sustainability.

5.2 Market failure and externalities: diverging private and social benefits and costs

LEARNING OBJECTIVES

After studying this section you will be able to:

- define all the terms appearing in **orange bold** in the text (AO1)
- explain market failure as the failure of the market to achieve allocative efficiency (AO2)

CONTINUED

- explain that socially optimum output is produced when there is allocative efficiency; this occurs when MSC=MSB and social (community) surplus is maximum (AO2)
- draw a diagram illustrating allocative efficiency and socially optimum output (AO4)
- explain the meaning of externalities as an example of market failure (AO2)

Introducing market failure: the failure of the market to achieve allocative efficiency

Our discussion in previous chapters has shown that a free, competitive market economy gives rise to a number of desirable outcomes. In Chapter 2 we learned that in a free competitive market, when the price of a good adjusts to make quantity demanded equal to quantity supplied, the equilibrium quantity reflects the 'best' or optimal allocation of resources to the production of that good. This condition is known as allocative efficiency, achieved when marginal benefit equals marginal cost ($MB = MC$), or when social surplus is maximum.

However, the achievement of these outcomes depends on very strict and unrealistic conditions that are practically never met in the real world. Therefore, in reality, the free market most often fails to achieve these highly desirable results. The study of market failure focuses on one particular failing: the free market's inability to realise allocative efficiency in a variety of circumstances.

Market failure does not necessarily lessen the market's significance as a mechanism that can advance the well-being of societies; instead, it suggests that for markets to realise their potential, they must be supported by appropriate government policies. Allocative efficiency is a concept used by economists to identify real-world situations that differ from the ideal of a perfect allocation of resources. Once these are identified, it is possible to design government policies aimed at reducing the extent of the inefficiencies.

Market failure refers to the failure of the market to allocate resources efficiently. Market failure results in **allocative inefficiency**, where too much or too little of goods or services are produced and consumed from the point of view of what is socially most desirable. Overprovision of a good means too many resources are allocated to its production (overallocation); underprovision means that too few resources are allocated to its production (underallocation).

TEST YOUR UNDERSTANDING 5.2

1 Using a diagram, and the concepts of consumer and producer surplus, and marginal benefits and marginal costs, explain the meaning of allocative efficiency.

2 Explain, in a general way, the meaning of market failure.

The meaning of externalities: diverging private and social benefits and costs

Understanding externalities

When a consumer buys and consumes a good, she or he derives some benefits. When a firm produces and sells a good, it incurs costs. Sometimes the benefits or costs spill over onto other consumers or producers or society as a whole, who have nothing to do with consuming or producing the good. When this happens, there is an externality.

An **externality** occurs when the actions of consumers or producers give rise to negative or positive side-effects on other people who are not part of these actions, and whose interests are not taken into consideration.

The other people feeling the effects of an externality are often referred to as 'third parties'. If the side-effects on third parties involve benefits, there arises a **positive externality**, also known as external (or spillover) benefit; if they involve costs, in the form of negative side-effects, there arises a **negative externality**, also known as external (or spillover) costs.

Externalities can result either from consumption activities (consumption externalities) or from production activities (production externalities).

Marginal private benefits and costs, and marginal social benefits and costs

To fully understand externalities, let's return to the demand and supply curves we studied in Chapter 2. As we know, the demand curve is also a 'marginal benefit curve' where marginal benefit is the benefit received by consumers for consuming one more unit of the good (see Figure 2.17, Chapter 2). Since the benefits derived from consuming the good go to private individuals, who are the consumers buying the good, the demand curve represents marginal *private* benefits, shown as *MPB* in Figure 5.2.

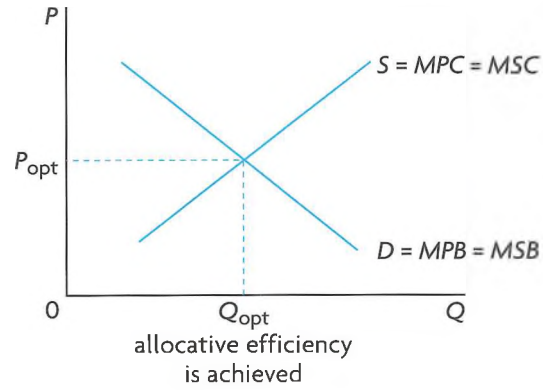


Figure 5.2: Demand, supply and allocative efficiency with no externalities

The standard supply curve reflects firms' costs of production, specifically marginal costs (see Figure 2.17). Marginal cost is the cost to producers of producing one more unit of the good. The supply curve therefore represents marginal *private* costs, appearing as *MPC* in Figure 5.2.

Now, if there are no externalities, so the actions of buyers and sellers do not produce side-effects on third parties, the marginal private benefit (D) curve and marginal private cost (S) curve determine an equilibrium price and quantity that reflect a *social optimum*, where there is allocative efficiency. In Figure 5.2, these are P_{opt} and Q_{opt}. A social optimum refers to a 'best' situation from the point of view of allocative efficiency. Q_{opt} is known as **socially optimum output**.

If, however, there is an externality, additional benefits or additional costs affecting third parties arise, and the full benefits or full costs to society differ from the private ones. These involve *marginal social benefits* (MSB) that differ from marginal private benefits; or *marginal social costs* (MSC) that differ from the marginal private costs.

When this occurs, the equilibrium price and quantity determined by the intersection of the demand (MPB) curve and supply (MPC) curve is no longer a social optimum, because *allocative inefficiency* is introduced by social benefits or costs that differ from private ones.

In a diagram, social benefits appear as a marginal social benefit curve, MSB, representing the full benefits to society from the consumption of a good, and social costs as a marginal social cost curve, MSC, representing the full costs to society of producing the good. When MSB and MSC are equal to each other, there is a social optimum in which allocative efficiency is realised.

Figure 5.2 shows the case where there are no external benefits or external costs (no externalities). Therefore $D = MPB = MSB$, and $S = MPC = MSC$.

marginal private costs (MPC) refer to costs to producers of producing one more unit of a good.

marginal social costs (MSC) refer to costs to society of producing one more unit of a good.

marginal private benefits (MPB) refer to benefits to consumers from consuming one more unit of a good.

marginal social benefits (MSB) refer to benefits to society from consuming one more unit of a good.

When $MSC = MSB$, there is allocative efficiency and *socially optimum output* is produced. When there is no externality, the competitive free market leads to an outcome where $MPC = MSC = MPB = MSB$, as in Figure 5.2, indicating allocative efficiency. An externality creates a divergence between MPC and MSC or between MPB and MSB. When there is an externality, the free market leads to an outcome where $MPB = MPC$, but where MSB is not equal to MSC, indicating allocative inefficiency. Either too much or too little is produced relative to the social optimum.

You should note that the condition $MSC = MSB$ is the same as $MC = MB$ when there are no externalities. In Chapter 4 we referred to $MC = MB$ as the condition for allocative efficiency because we were considering markets with no market failures.

We will examine four types of externalities:

- negative production externalities
- negative consumption externalities
- positive production externalities
- positive consumption externalities.

The first two of these, negative externalities of production and consumption, will be examined in the present chapter. The last two, positive externalities of production and consumption, will be examined in Chapter 6.

The present chapter will also examine in more detail the topic of *common pool resources*, introduced above, which are very closely related to negative externalities.

TEST YOUR UNDERSTANDING 5.3

1 a Outline the meaning of an externality.
  b Use the concept of allocative efficiency to explain how externalities relate to market failure.
2 Explain the difference:
  a between marginal private benefit and marginal social benefit, and
  b between marginal private cost and marginal social cost.
3 State what condition of perfect markets is violated, leading to allocative inefficiency, when there is an externality.

5.3 Negative production externalities

LEARNING OBJECTIVES

After studying this section you will be able to:

- define all the terms appearing in **orange bold** in the text (AO1)
- explain negative externalities of production and the resulting welfare loss (AO2)
- draw a diagram illustrating negative externalities of production and welfare loss (AO4)
- calculate welfare loss that arises from negative externalities of production (HL only) (AO4)
- explain that negative production externalities can be used to illustrate overuse of common pool resources (AO2)
- explain government intervention to correct negative externalities of production and prevent overuse of common pool resources including: (AO2)
  - indirect (Pigouvian) taxes
  - carbon taxes
  - tradable permits
  - legislation and regulation
  - collective self-governance
  - education-awareness creation
  - international agreements
- draw diagrams to illustrate the above government responses (AO4)
- discuss strengths and limitations of the above government policies with respect to: (AO3)
  - difficulties in measurement of externalities
  - degree of effectiveness
  - consequences for stakeholders

Explaining and illustrating negative production externalities

Negative production externalities refer to external costs created by producers. The problem of environmental pollution, created as a side-effect of production activities, is very commonly analysed as a negative production externality.

Consider a cement factory that emits smoke into the air and disposes its waste by dumping it into the ocean. There is a production externality, because over and above the firm's private costs of production, there are additional costs that spill over onto society due to the polluted air and ocean, with negative consequences for the local inhabitants, swimmers, sea life, the fishing industry and the marine ecosystem. This is shown in Figure 5.3, where the supply curve, $S = MPC$, reflects the firm's private costs of production, and the marginal social cost curve given by MSC represents the full cost to society of producing cement. For each level of output, Q, the social costs of producing cement given by MSC are greater than the firm's private costs. The vertical difference between MSC and MPC represents the external costs. Since the externality involves only production (the supply curve), the demand curve represents both marginal private benefits and marginal social benefits.

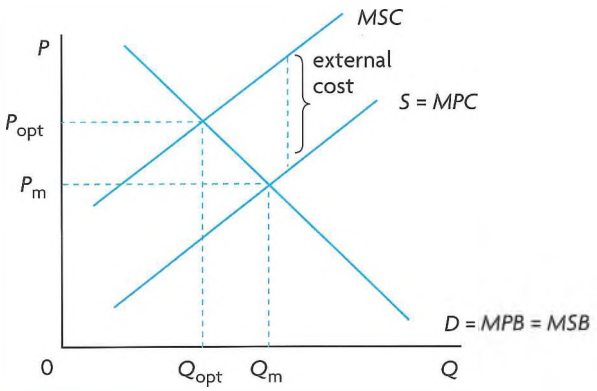


Figure 5.3: Negative production externality

Figure 5.3 illustrates a general point that you should keep in mind whenever you examine (or draw) an externality diagram: the free market outcome is determined by the intersection of MPB and MPC, resulting in quantity Q_m and price P_m. The socially optimum (or 'best') outcome is given by the intersection of MSB with MSC, which determines quantity Q_{opt} and price P_{opt}.

We can draw an important conclusion from the negative externality in Figure 5.3:

> When there is a negative production externality, the free market overallocates resources to the production of the good and too much of it is produced relative to the social optimum. This is shown by $Q_m > Q_{opt}$ and $MSC > MSB$ at the point of production, Q_m, in Figure 5.3.

The welfare loss of negative production externalities

Welfare loss

Whenever there is an externality, there is a welfare loss, involving a reduction in social benefits, due to the misallocation of resources.

In Figure 5.4(a), the shaded area represents the welfare loss arising from the negative production externality. For all units of output greater than Q_{opt}, $MSC > MSB$, meaning that society would be better off if less were produced. The welfare loss is equal to the difference between MSC and MSB for the amount of output that is overproduced ($Q_m - Q_{opt}$). It is a loss of social benefits due to overproduction of the good caused by the externality. If the externality were corrected, so that the economy reaches the social optimum, the loss of benefits would disappear. It may be useful to note that the point of the welfare loss triangle always lies at the Q_{opt} quantity of output.

Calculating welfare loss (HL only)

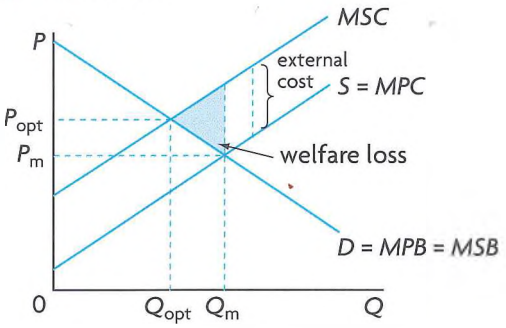
Figure 5.4(b) is similar to 5.4(a) except that it includes figures that allow us to calculate welfare loss. To find the area of the shaded triangle, we take the height times the width of the triangle and divide it by 2. Note that the height of the triangle is equal to the external cost per unit, or $MSC - MPC$, and the width is equal to the amount of overproduction by the market or $Q_m - Q_{opt}$:

$$\text{Welfare loss} = \frac{(6-4)\times(100-70)}{2} = \frac{2\times 30}{2} = \$30$$

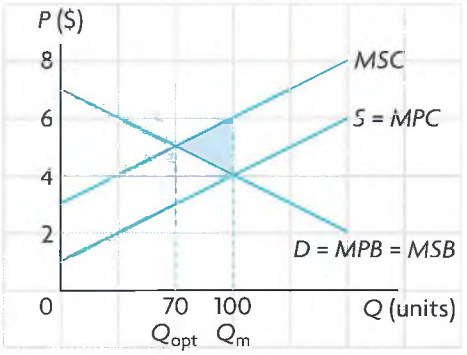
Welfare loss in relation to consumer and producer surplus (Supplementary material)

We can use the concepts of consumer and producer surplus to understand the welfare loss due to the externality. In Figure 5.4(c), in market equilibrium,

a Welfare loss



b Calculating welfare loss (HL only)



c Supplementary material

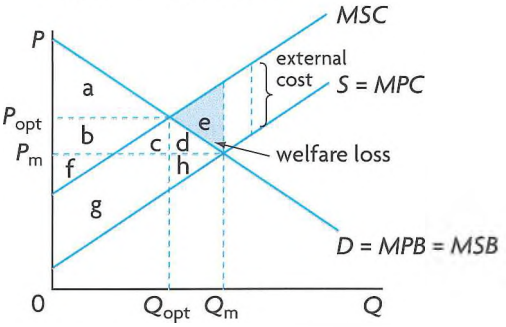


Figure 5.4: Welfare loss in a negative production externality

consumer surplus is equal to areas a + b + c + d, while producer surplus is equal to areas f + g + h. The value of the external cost is the difference between the MSC and MPC curves up to Q_m (the quantity produced by the market), and is therefore equal to c + d + e + g + h. The total social benefits in market equilibrium are equal to consumer surplus plus producer surplus minus the external cost:

$$(a + b + c + d) + (f + g + h) - (c + d + e + g + h)$$
$$= a + b + f - e$$

At the social optimum, or at Q_{opt} and P_{opt}, consumer surplus is equal to area a, and producer surplus is equal to area b + f. The external cost is now equal to zero. Therefore, the total social benefits are equal to consumer surplus plus producer surplus:

a + b + f

Comparing total social benefits at the market equilibrium and at the social optimum, we find that they are smaller at the market equilibrium by the area e. This is the welfare loss.

Negative production externalities and overuse of common pool resources

The concept of negative externalities that we studied above can be used to illustrate the problem of overuse of common pool resources and its effects on sustainability. For example, in the negative production externality diagram of Figure 5.3 the difference between the *MPC* and *MSC* curves can be interpreted as the external cost arising from the cement factory's overuse of clean air, water and sea life on account of its dependence on fossil fuels; it can also be interpreted as the cost to society of causing global warming (destroying the stability of the global climate, which is also a common pool resource). The burning of fossil fuels creates external costs in terms of overuse of common pool resources.

If it were possible to make the cement factory pay for the overuse of these resources, the producer would not necessarily stop its polluting activities entirely, and would not stop *using* common pool resources. However, it would stop *overusing* them, thus leading to a sustainable use of common pool resources.

Figure 5.3 can be used to illustrate the overuse of many common pool resources as a negative production externality. For example, if the S = *MPC* curve represents the private costs of a fishing firm that fishes in the open seas, the external costs would be depletion of the stock of fish, and environmental damage due to disruption of the marine ecosystem, or the common pool resources that the fishing firm has overused but not paid for. The costs to society of the firm's fishing activities are given by *MSC*, which are the private costs plus the external costs.

TEST YOUR UNDERSTANDING 5.4

1 a Using a diagram, show how marginal private costs and marginal social costs differ when there is a negative production externality.

b Explain the difference between the equilibrium quantity determined by the market and the quantity that is optimal from the point of view of society's preferences.

c Describe the problem with the allocation of resources achieved by the market when there is a negative production externality.

d Show the welfare loss created by the negative production externality in your diagram, and explain what this means.

2 a Provide examples of negative production externalities.

b Using diagrams and examples, explain how negative production externalities can be used to analyse the overuse of common pool resources.

Policies to correct negative production externalities and prevent overuse of common pool resources and their evaluation

We will examine and evaluate a variety of policies to deal with negative production externalities and overuse of common pool resources.

Market-based policies I: indirect (Pigouvian) taxes

An important group of policies that can be pursued by governments rely on the market to correct negative production externalities and promote sustainable use of common pool resources. Market-based policies work by changing the incentives faced by firms.

In one such approach the government could impose an indirect tax on the firm per unit of output produced.

This is known as a **Pigouvian tax** (or **Pigovian tax**), after the English economist Arthur Cecil Pigou who was the first to propose the idea of imposing a tax in order to correct a negative externality. In Figure 5.5(a), the tax results in an upward shift of the supply curve, from $S = MPC$ to MSC ($=MPC$ + tax). The optimal (or best) tax policy is to impose a tax that is exactly equal to the external cost, so the MPC curve shifts upward until it overlaps with MSC. The new, after-tax equilibrium is given by the intersection of MSC and the demand curve, $D = MPB = MSB$, *resulting in the lower, optimal quantity* of the good produced, Q_{opt}, and higher, optimal price, P_{opt}. Note that whereas the indirect taxes discussed in Chapter 4 *introduced allocative inefficiency*, indirect taxes in the present context are intended to *lead to allocative efficiency.*

Bearing in mind our discussion of indirect taxes in Chapter 4, you may note that P_{opt} is the price paid by consumers, or P_c, while the price received by producers is P_p, which is equal to P_c minus tax per unit.

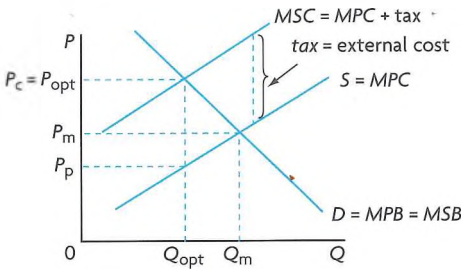
Market-based policies II: carbon taxes

Carbon taxes are a kind of tax designed to deal with what is perhaps the single most pressing and complex threat to the global ecosystem: global warming, caused by emissions of greenhouse gases, the most important of which is carbon dioxide. When we speak of the contribution of greenhouse gases to global warming, we refer to those gases emitted by man-made processes, and specifically by the burning of fossil fuels (oil, coal and natural gas).

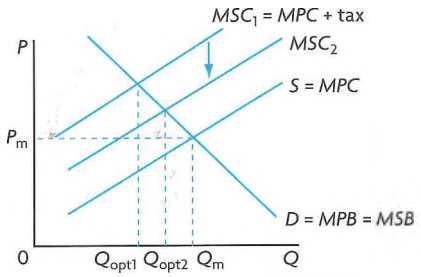
There are two key measures under discussion in the global community to deal with the problem of carbon dioxide emissions: carbon taxes and tradable permits (see below).

The **carbon tax** is a tax per unit of carbon emissions of fossil fuels. Fossil fuels do not all emit the same amounts of carbon when burned, therefore the carbon tax is calculated on the basis of how much carbon the fuel emits: *the more carbon emitted, the higher the tax*. This can be illustrated by the same diagram used to show the effects of a tax per unit of output, in Figure 5.5(a). Following the imposition of the tax, firms must pay the higher price to buy the fossil fuel. This appears in Figure 5.5(a) as the familiar upward shift in $S = MPC$ toward MSC because of the firm's higher costs of production, but this has further consequences. Since there are other substitute energy sources with lower carbon emissions (thus taxed at a lower rate), or that do not emit carbon (if they are not fossil fuels, thus not taxed at all), the increase in the price of the high-carbon fuel creates incentives for firms to switch to other, less polluting or non-polluting energy sources.

a Imposing an indirect tax on output or on pollutants



b Effects on external costs of a tax on emissions (carbon tax)



c Tradable permits

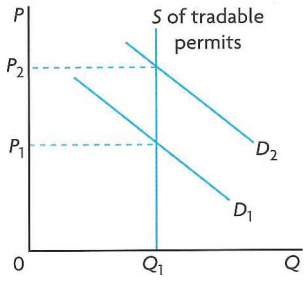


Figure 5.5: Market-based policies to correct negative production externalities and promote sustainable use of common pool resources

The result is that if the firm switches to alternative, less polluting resources, Q_{opt} *in Figure 5.3 will increase, because the external costs of producing the output will become smaller.* This can be seen in Figure 5.5(b), where the MSC curve shifts from MSC_1 to MSC_2, indicating that the external costs are lower due to the use of the less polluting resources. With the fall in external costs, the optimum quantity of output increases from Q_{opt1} to Q_{opt2}. (Note that this also involves a lower tax on pollutants, shown by the smaller distance between the demand curve and MSC_2.)

Carbon taxes are used in many countries as a method to reduce pollution, for example, Denmark, Finland,

France, Ireland, Japan, Mexico, Poland, Sweden, as well as some states in Canada and the United States and more.

A tax on carbon (or on emissions generally) has the effect of creating incentives for producers to reduce the amount of pollution they create by purchasing less polluting resources (such as fossil fuels), and to switch to less polluting technologies (alternative energy resources). This reduces the size of the negative externality and increases the optimum quantity of output. A tax on the output of the polluter does not have this effect; it corrects the overallocation of resources to the good, reducing the quantity of output produced.

Market-based policies III: tradable permits

Tradable permits, also known as *cap and trade schemes*, are a policy involving *permits to pollute* issued to firms by a government or an international body. These permits to pollute can be traded (bought and sold) in a market. Consider a number of firms whose production pollutes the environment. The government grants each firm a particular number of permits (or rights) to produce a particular level of pollutants over a given time period. The permits to pollute can be bought and sold among interested firms, with the price of permits being determined by supply and demand. If a firm can produce its product by emitting a lower level of pollutants than the level set by its permits, it can sell its extra permits in the market. If a firm needs to emit more pollutants than the level set by its permits, it can buy more permits in the market.

Figure 5.5(c) shows a market for tradable pollution permits. The supply of permits is perfectly inelastic (i.e. the supply curve is vertical), as it is fixed at a particular level by the government (or an international authority if several countries are participating). For this policy to effectively reduce the level of pollution, the total pollution that is permitted based on the pollution permits must be *less than the amount of pollution created with no permits*. The fixed supply of permits is distributed to firms. The position of the demand-for-permits curve determines the equilibrium price. As an economy grows and the firms increase their output levels, the demand for permits is likely to increase, as shown by the rightward shift of the demand curve from D_1 to D_2. With supply fixed, the price of permits increases from P_1 to P_2.

Tradable permits are like taxes on emissions in that they provide incentives to producers to switch to less polluting resources for which it is not necessary to buy permits. If a firm finds a way to reduce its emissions, it can sell its permits thus adding to profits. Permits are therefore intended to reduce the quantity of pollutants emitted, thus reducing the size of the negative externality, and increasing the optimum quantity of output produced, by shifting the *MSC* curve upward toward *MPC*, as shown in Figure 5.5(b).

Tradable permit schemes may be set up within a country, such as Kazakhstan (2013), Switzerland (2008), New Zealand (2008); or within a groups of countries such as the European Union Emissions Trading System (EU ETS) (2005); or internationally such as the Paris Agreement (2016; see below *International agreements*). Tradable permits are hotly debated together with carbon taxes; see Real World Focus 5.1.

You may note in your reading about tradable permits that they are referred to as an Emissions Trading System (ETS).

Advantages of market-based policies

Economists usually prefer the market-based solutions to government regulations to deal with negative production externalities and overuse of common pool resources, as long as the particular externality permits the use of such policies. Both taxes and tradable permits have the effect of *internalising the externality*, meaning that the costs that were previously external are made internal, because they are now paid for by producers and consumers who are parties to the transaction.

In the case of taxes, *taxes on emissions such as the carbon tax are superior to taxes on output*. Taxes on output only provide incentives to producers to reduce the quantity of output produced with a given technology and given polluting resources, but not to reduce the amount of pollution they create or to switch to less polluting resources.

Taxes on pollutants emitted provide incentives to firms to economise on the use of polluting resources (such as fossil fuels) and use production methods that pollute less. Firms do not all face the same costs of reducing pollution; for some, the costs of reducing pollution are lower than for others, and these will be the ones most likely to cut their pollution emissions to avoid paying the tax. Firms that face the highest costs of reducing pollution will be the ones least likely to cut their pollutants, and so will pay the tax. The result is that

taxation leads to lower pollution levels at a lower overall cost to society since the firms that will switch to clean forms of energy are the ones that can do it more cheaply.

Similarly, in the case of tradable permits, the system creates incentives for firms to cut back on their pollution if they can do so at relatively low cost. If it is a relatively low-cost procedure for a firm to reduce its pollutant emissions, it will be in its interests to do so and sell excess permits. Firms that can only reduce pollution at high cost will be forced to buy additional permits. Therefore, both taxes and tradable permits are methods to reduce pollution more efficiently (at a lower cost).

Disadvantages of market-based policies

Whereas taxes and tradable permits are simple in theory, in practice they are faced with numerous technical difficulties. While it is known with a reasonable degree of certainty that man-made greenhouse gases cause global warming, there is tremendous uncertainty in calculating the precise contribution of each of these to increases in global temperatures. This gives rise to difficulties in designing effective carbon taxes and tradable permit schemes.

Taxes

Taxes face serious practical difficulties that involve designing a tax equal in value to the amount of the pollution. An effective tax policy requires answers to the following questions:

- **What production methods produce pollutants?** Different production methods create different pollutants. It is necessary to identify what methods produce which pollutants, which is technically very difficult.
- **Which pollutants are harmful?** It is necessary to identify the harmful pollutants, which is also technically difficult, and there is much controversy among scientists over the extent of harm done by each type of pollutant.
- **What is the value of the harm?** It is then necessary to attach a monetary value to the harm: how much is the harm done by each pollutant worth? This raises questions that have no easy answers: who or what is harmed; how is the value of harm to be measured?
- **What is the appropriate amount of tax?** It is necessary to determine the size of the tax to make it equal to the value of the harm.
- **How will consumers be affected?** Indirect taxes are *regressive* (as you will learn in in Chapter 12) meaning that lower income people have to pay a higher proportion of their income in tax than higher income people, which is considered inequitable (unfair).

A serious problem with carbon taxes is that they are usually set too low to make a significant impact. According to the OECD Secretary General:

> *The gulf between today's carbon prices and the actual cost of emissions to our planet is unacceptable. Pricing carbon correctly is a concrete and cost-effective way to slow climate change. We are wasting an opportunity to steer our economies along a low-carbon growth path and losing precious time with every day that passes.*[2]

The reason is that it is politically difficult to impose carbon taxes that are high enough to make the necessary difference.

Aside from the technical difficulties, there is also a risk that even if taxes are imposed some polluting firms may not lower their pollution levels, continuing to pollute even though they pay a tax.

Tradable permits

Tradable permits face the technical limitations regarding production methods and pollutants noted above for taxes. In addition, tradable permits require the government (or international body) to set a maximum acceptable level for each type of pollutant, called a 'cap'. This task demands having technical information on quantities of each pollutant that are acceptable from an environmental point of view, which is often not available. If the maximum level is set too high, it will not have the desired effect on cutting pollution levels. If it is set too low, the permits become very costly, causing hardship for firms that need to buy them. To date, tradable permits have been developed for just a few pollutants (CO_2, SO_2).

In addition, a method must be found to distribute permits to polluting firms in a fair way. Issues of political favouritism may come into play, as governments give preferential treatment to their 'friends' and supporters.

In practice, the most that can be hoped for is a shift of the *MPC* curve toward the *MSC* curve, as well some reduction in the size of the externality, but it is unlikely that these policies can achieve the optimal results.

[2] OECD, Few countries are pricing carbon high enough to meet climate target http://www.oecd.org/tax/few-countries-are-pricing-carbon-high-enough-to-meet-climate-targets.htm

REAL WORLD FOCUS 5.1

Carbon taxes versus tradable permits: how to best limit carbon dioxide emissions

There is a growing momentum around the world to implement carbon pricing in some form. As of 2019 over 60 countries, states or cities had implemented some form of carbon pricing, either through carbon taxes or through tradable permits. A survey of people in five countries (Australia, India, South Africa, the United Kingdom and the United States) found that between 60% (in the United States) and 80% (in India) supported carbon taxes provided these were on a global scale and the revenues were returned to the people or spent on climate projects.



Figure 5.6: Klitten, Germany. Lignite-fired power station

However, carbon pricing remains unpopular among both consumers and businesses. Businesses are often opposed because it raises the costs of production. Others say that it is unfair in the event that they trade with countries that do not have carbon pricing. In Australia, a carbon tax was repealed in 2014 on the grounds that it was destroying jobs. It is often opposed by consumers because it has the effect of raising prices. It has a stronger effect on the poor because it raises household energy prices.[4]

As market-based methods to reduce emissions, both carbon taxes and tradable permits provide incentives to firms to switch to less polluting forms of energy. However, as we have seen they differ in how they attempt to do this. Carbon taxes fix the price of the pollutant in the form of a tax on carbon and allow the quantity of carbon emitted to vary, depending on how firms respond to the tax; cap and trade schemes fix the quantity of the permissible pollutant, and allow its price to vary, depending on supply and demand.

Carbon taxes versus tradable permits: an evaluation

- **Carbon taxes make energy prices more predictable.** Fossil fuel prices in global markets fluctuate according to demand and supply. Under tradable permits, the price of fossil fuels might fluctuate even more due to fluctuations in the price of carbon permits. Price predictability is important for businesses that need to plan their costs ahead of time.
- **Carbon taxes are easier to design and implement.** Tradable permits are difficult to design and implement as they involve complicated decisions such as setting the cap at the right level and distributing the permits among all interested users.
- **Carbon taxes can be applied to all users of fossil fuels.** Tradable permit schemes often target one particular industry, or small group of industries. Carbon taxes can be applied to all users of fossil fuels, including all producers and consumers.
- **Carbon taxes do not offer opportunities for manipulation by governments and interest groups.** Politicians often prefer tradable permit schemes to carbon taxes, and it is believed that this may be because it is easy to manipulate the distribution of permits for the benefit of preferred groups and supporters, without affecting the impacts on the environment (because of the cap).

[3] https://www.sciencedaily.com/releases/2019/01/190117092556.htm

[4] https://www.bloomberg.com/news/articles/2018-10-14/why-pricing-carbon-is-still-more-theory-than-reality-quicktake

CONTINUED

- **Carbon taxes do not require as much monitoring for enforcement.** Tradable permit schemes require monitoring of emissions, otherwise firms may try to cheat by emitting more pollutants than they are permitted. Carbon taxes are easier to monitor as they only involve payment of a tax depending on the type and quantity of fossil fuels purchased.
- **Tradable permit schemes face political pressures to set the cap too high.** If the cap on pollutants is set too high, it would have a very limited or no impact on reducing carbon emissions.

There are also some arguments against carbon taxes and in favour of tradable permit schemes:

- **Carbon taxes face political pressures to be set too low.** Governments may be unwilling to set carbon taxes high enough for these to provide the necessary incentives for users to switch to less polluting energy sources.
- **Carbon taxes cannot target a particular level of carbon reduction.** Since carbon taxes cannot fix (or cap) the permissible level of carbon emissions, they lead to uncertain carbon-reducing outcomes.
- **Carbon taxes are regressive.** A regressive tax is one where the tax as a fraction of income is higher for low-income earners than it is for higher-income earners, and go against the principle of equity (see Chapter 12). A carbon tax on a firm is an indirect tax that is paid partly by producers and partly by consumers. Therefore, consumers would also be affected, and lower-income consumers would be affected proportionately more.

Applying your skills

1 Compare and contrast some key issues surrounding the debate on carbon taxes versus cap and trade schemes.

2 The World Bank regularly updates the following site with information on carbon pricing (carbon taxes or tradable permits) that countries around the world are either implementing or planning to implement. Select one or more countries of your choice and investigate the carbon pricing that has been selected. Research the experience of your country or countries with respect to (a) political acceptability, (b) effectiveness with respect to reducing carbon emissions, (c) any future plans for tackling the carbon emissions problem.

Source: *https://carbonpricingdashboard.worldbank.org*

Government legislation and regulation

Government legislation and regulations rely on the 'command' approach, where the government uses its authority to enact legislation and regulations in the public's interest (see Chapter 1 for a discussion of command decision-making).

Legislation and regulations intended to reduce the effects of production externalities and limit environmental damage typically involve emissions standards, quotas, licences, permits or outright restrictions. Examples include:

- restrictions on emissions of pollutants from factories and industrial production by setting a maximum level of pollutants permitted
- requirements for steel mills and electricity generating plants to install smokestack scrubbers to reduce emissions
- banning the use of harmful substances (e.g. asbestos, a group of minerals that are very toxic that is banned in many countries)
- issuing licences or permits for particular activities (such as hunting)

- prohibiting construction (such as housing) or industry or agriculture in protected areas
- restrictions on the quantity of logging
- restrictions in the form of quotas for fishing (maximum permissible quantity of fish that can be caught) or in the form of the size of shipping fleets, or total bans for specific areas or specific times of the year
- establishment of protected areas for the protection of biodiversity and endangered ecosystems.

The impact is to lower the quantity of the good produced and bring it closer to Q_{opt} in Figure 5.7 by shifting the *MPC* curve upward towards the *MSC* curve. Pollutant and output restrictions achieve this by forcing the firm to produce less. Requirements to install technologies reducing emissions achieve this by imposing higher costs of production due to the purchase of the non-polluting technologies. Ideally, the higher costs of production would be equal to the value of the negative externality. The government's policy objective is to make the *MPC* curve shift upwards until it coincides with the *MSC* curve, in which case Q_{opt} is produced, price increases from P_m to P_{opt}, and the problem of overallocation of resources to the production of the good is corrected. If polluting firms do not comply with the regulations, they would have to pay fines.

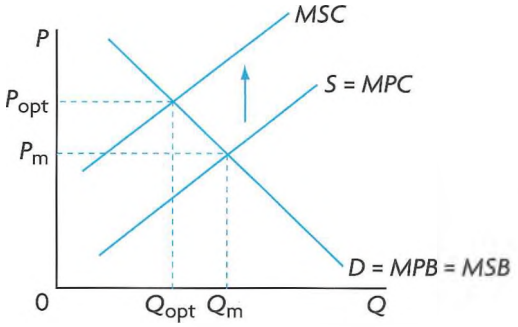


Figure 5.7: Government regulations to correct negative production externalities and promote sustainable use of common pool resources

Advantages

Legislation and regulations, including restrictions such as in the examples above, have the advantage that they are simple to put into effect and oversee. They are easier to implement compared to market-based polices and avoid the technical difficulties that arise in the use of market-based solutions. They can also be quite effective. For example, banning harmful substances, or prohibiting hunting in certain areas, or restricting the quantity of logging (chopping down trees) may be the most effective way to deal with certain problems. Moreover, regulations force firms to comply and reduce their harmful activities (which market-based policies may not always do). For these reasons, regulations are far more commonly used as a method to limit negative externalities of pollution in countries around the world.

Disadvantages

However, they also face limitations. In the case of emissions of pollutants, they do not offer incentives to reduce emissions by using less polluting resources, to increase energy efficiency and to switch to alternative fuels. They cannot distinguish between firms that have lower or higher costs of reducing pollution, which would limit the overall cost of reducing pollution (explained above). The result is that pollution is reduced at a higher overall cost.

In addition, although they can be implemented more easily, they suffer from similar limitations as the market-based policies (lack of sufficient technical information on types and amounts of pollutants emitted), and so can in most cases be only partially effective in reducing the pollution created. Finally, there are costs of monitoring and supervision to detect possible violations, leading to opportunity costs, and there may be problems with enforcement. Therefore, such measures can only attempt to partially correct the problem.

Overall, the effectiveness of legislation must be assessed in relation to the particular use for which it is intended, as it can be more effective in some situations than in others.

Correction of negative production externalities by market-based approaches or government legislation and regulation usually involve shifting the *MPC* curve upward toward the *MSC* curve through a variety of policies. For allocative efficiency to be achieved, the quantity of the good produced and consumed must fall to Q_{opt} as price increases to P_{opt}.[5]

[5] See 'Quantitative techniques' chapter in the digital coursebook for an explanation of the equivalence of upward and leftward shifts of the supply curve.

Collective self-governance: the contribution of Elinor Ostrom

The concept of **collective self-governance** refers to a solution to the use of common pool resources where the users take control of the resources and use them in a sustainable way. This concept runs counter to the idea of the *tragedy of the commons*, discussed above, where each user makes use of the resource at the expense of others with the result to ultimately degrade and deplete it.

Collective self-governance, as the term suggests, is not a policy that is imposed by the government. It is rather an approach to manage resources undertaken by communities of resource users by themselves, because they realize that it is in their own best interests to work collectively for the preservation of resources that are vital to their livelihood.

Collective self-governance as an approach to common pool resources was made famous by Elinor Ostrom, an American political scientist who became the first woman to receive the Nobel Prize in Economics (2009) for her work on the management of common pool resources. Ostrom spent decades studying how communities organise themselves to manage common resources such as fish stocks, woods, lakes, pastures and groundwater basins. She discovered that communities often develop complex mechanisms on how to make decisions and how to enforce rules, resulting in outcomes that differ widely from those predicted by standard economic theory. In an interview in 2010, she said:

> *If you are in a fishery or have a pasture and you know your family's long-term benefit is that you don't destroy it, and if you can talk with the other people who use that resource, then you may well figure out rules that fit that local setting and organise to enforce them. But if the community doesn't have a good way of communicating with each other or the costs of self-organisation are too high, they won't organise, and there will be failures.*[6]

Ostrom and her colleagues performed hundreds of experiments, which show that when people are users of a common pool resource, they are often able to find solutions on how to manage it sustainably, without overusing and destroying it, *provided they have good methods of communication between them*.[7] They make rules about how each user should behave and contribute to the management of the resource, and rules of enforcement and sanctions against those who violate the rules. Many studies by Ostrom and other scholars of real-life situations, involving the management of common pool resources, confirm the results of these experiments.

However, one requirement for sustainable resource management is that there must be boundaries of an area (such as a pasture, a wood or a lake, and so on) corresponding to the area that the resource users are managing. This raises the question whether Ostrom's approach is suitable for dealing with global problems such as the oceans and climate change. She herself notes the following:

> *I really despair over the oceans … It's so tempting to go along the coast and scoop up all the fish you can and then move on. With very big boats you can do that. I think we could move toward solving that problem, but right now there are not many instrumentalities for doing that.*

Advantages

Elinor Ostrom's work has shown that people do not always act in the self-interested, narrow manner presupposed by the tragedy of the commons. Instead, by working cooperatively, people can find solutions to the problem of overuse of common pool resources without top-down solutions imposed by governments. These cooperative solutions can be achieved in the absence of private ownership of resources (emphasised by supporters of free market economies where this is an essential characteristic) as well as in the absence of government-owned property (emphasised by supporters of command or planned economies; see Chapter 1 for these distinctions). However, it is important to have a legal system of *land rights* in place (see Chapters 19 and 20).

Disadvantages

For people to be able to manage common pool resources on their own there are two important conditions that must be satisfied, they must be able to communicate with each other in order to create rules for the use of the common pool resources, and there must be a boundary for the resource. As Ostrom herself admits this is difficult for resources such as the oceans.

Education and awareness creation

Education of the public and provision of information regarding the polluting activities of firms (or other activities with negative external effects) often makes consumers turn away from the products, with negative effects on the firms' sales. As a result the firms are forced to take consumers'

[6] Interview with Elinor Ostrom, by Fran Korten, Yesmagazine.org, 27 February 2010.
[7] The experiments are carried out at the Workshop in Political Theory and Policy Analysis at Indiana University where Ostrom was a professor.

opinions into consideration and change their production methods in order to reduce the externalities.

Advantages

The advantage here is that firms are very much influenced by the opinions of their customers and want to keep them happy, otherwise they will suffer drops in sales.

Disadvantages

The difficulty with this approach is that it can only make a small difference in terms of solving the problem of production externalities and sustainability. For example, if there is information about a firm creating significant environmental damage in a localised area (such as causing an oil spill that affects the livelihood and health of the local inhabitants), consumers may become concerned and boycott the firm for a while. However problems of a more general and broader nature, such as the use of fossil fuels that causes climate change plus a host of other environmental problems, require solutions on a far broader scale. In fact carbon taxes and tradable permits (discussed above) are far more effective in tackling problems of this kind.

International agreements

Policies are made mainly by national governments. However, negative production externalities and the overuse of common pool resources very often have international repercussions, in which case co-operation among governments and international agreements are crucially important to control and prevent negative consequences on certain resources, such as the global climate and the ozone layer. In addition, co-operation among governments is very important for the development and diffusion of new technologies intended to deal with global environmental issues. Co-operation between governments may be global or regional.

For example, the ozone layer has suffered ozone depletion, leading to reduced protection against the sun's ultraviolet radiation. This resulted from human activities involving the production of nitrogen oxides and chlorofluorocarbons (CFCs). The ozone layer is a common pool resource. No one owns it, and no one can claim damages for its destruction. The responsibility for its destruction lies with polluting activities within virtually every country, and the consequences of its destruction are felt globally. The same considerations apply to the global climate.

One of most successful examples of international collaboration for the environment is the Montreal Protocol, signed in 1987 and coming into effect in 1989, intended to phase out substances that have caused depletion of the ozone layer. By 2009, all member states of the United Nations had ratified the agreement, and significant progress has been made in the area of phasing out ozone-depleting substances.

Another successful example of a regional collaborative arrangement is the European Union's tradable permits scheme for carbon, known as the European Union Emissions Trading System (EU ETS), which was initiated in January 2005. The scheme covers the sectors of power and heat generation, oil refineries, metals, pulp and paper, and energy intensive industry. In this system, one permit, or EU Allowance (EUA) permits the holder to release one tonne of carbon dioxide. Each emitter of carbon is allocated EUAs, which are traded in a rapidly growing carbon market. The EU ETS is the cornerstone of the European Union's policy on climate change. According to a major study, while this has helped in reducing carbon emissions it has not had negative impacts on the economic performance of firms in terms of revenues, profits and employment.[8]

Another major, but less successful international agreement for the environment was the Kyoto Protocol of 2005–2012. Its objective was to make signatory countries commit themselves to reduce emissions of carbon dioxide and other greenhouse gases to slow down climate change. It also contained provisions for the development of a market of tradable emissions permits. The countries that signed were divided into developed and developing. Only the developed countries had emissions restrictions. The developing countries participated by investing in projects that were supposed to lower their emissions. The United States, which was the country with the highest greenhouse gas emissions, did not sign the Kyoto protocol on the grounds that it was facing unfair competition since developing countries did not face restrictions. China, the second highest greenhouse emitter, did not participate as it was not in the group of developed countries. Many environmental specialists argued that the agreed reductions in emissions were too small to have sufficient impact on the problem of global warming.

In 2016, the Paris Agreement came into effect, initially signed by 55 parties, which reached 197 by 2019. The purpose of the agreement is to strengthen international co-operation on climate change, based on the goal of limiting the global temperature increase to 1.5%. In addition it aims to increase the ability of countries to adapt to negative effects of climate change. It establishes binding agreements on all parties to pursue measures

[8] http://www.oecd.org/officialdocuments/publicdisplaydocumentpdf/?cote=ECO/WKP(2018)63&docLanguage=En

domestically that will reduce greenhouse emissions. Countries are free to pursue measures of their choice but targets for emission reductions must be more ambitious than earlier ones. The members have agreed to track their progress and report this to each other, to co-operate and to provide support for climate action to developing countries. In June 2017, the United States announced its intention to withdraw from the Paris agreement with November 2020 the effective date of withdrawal.

TEST YOUR UNDERSTANDING 5.5

1 For each of the examples you provided in question 2(a) in Test your understanding 5.4, state and explain some method(s) that could be used to correct the externality.

2 Outline the three types of market-based policies that can be used to correct a negative production externality.

3 Using a diagram, show how a negative production externality created by the use of fossil fuels can be corrected by use of
   - a taxes on output,
   - b taxes on emissions,
   - c tradable permits, and
   - d legislation and regulation.
   - e Discuss some advantages and disadvantages of each of these types of policy measures.

4 a Outline the meaning of and identify policies that 'internalise an externality'.
   - b Compare and contrast market-based methods and command methods (such as legislation and regulations) to deal with negative production externalities.
   - c Identify some difficulties that governments face in designing market-based methods.

5 a Define collective self-governance and discuss this as a method to deal with overuse of common pool resources.
   - b Outline some possible limitations that this approach faces.

6 a Explain why taxes on emissions such as carbon taxes and tradable permits are similar with respect to their objectives. (*Hint:* think about incentives.)
   - b Describe how they differ from taxes on output of the firms creating negative externalities.
   - c Identify the policy that is preferable from the point of view of reducing the external costs of a negative production externality: a tax on output or a tax on emissions.

7 Using examples, explain under what circumstances international co-operation among governments is essential for the preservation of the environment.

REAL WORLD FOCUS 5.2

Effects of pesticides on bees

A pesticide known as sulfoxaflor is widely used to kill insects in agriculture. In 2013 it was approved by the Environmental Protection Agency (EPA) to be used in the United States. However, as sulfoxaflor was known to be toxic to bees, the EPA recommended that it should not be used when crops are flowering, which is when they attract bees.

Soon after, a number of beekeeping groups sued the EPA arguing that the use of sulfoxaflor would worsen the problem of the declining bee population. As a result, a US federal (national) court cancelled the EPA's approval of sulfoxaflor on the grounds that there was insufficient



Figure 5.8: Dead bees that died by pesticides

CONTINUED

evidence that it was not harmful to bees. In 2015 the EPA stopped allowing the sale of sulfoxaflor in the United States.

However, in 2016, the EPA reapproved the use of sulfoxaflor, but excluded crops that are attractive to bees and that flower at various times throughout the year. In addition, the EPA allowed the more general use of sulfoxaflor on an 'emergency' basis. The EPA has the authority to grant such exemptions even for pesticides that are not officially approved, in the event of a sudden appearance of insects that damage crops.

In the period 2017–2018, the EPA approved more than 100 'emergency exemptions' for sulfoxaflor. In 2018, it was used on over 16 million acres of farmland for cotton and sorghum alone.

Yet it is well known that the use of sulfoxaflor is harmful to bees, with serious consequences for their reproductive abilities. According to a senior scientist at the Center for Biological Diversity, 'Spraying 16 million acres of bee-attractive crops with a bee-killing pesticide in a time of global insect decline is beyond the pale* . . . The EPA is routinely misusing the "emergency" process to get sulfoxaflor approved because it's too toxic to make it through normal pesticide reviews.'

According to a study published in the journal *Biological Conservation*, there is a dangerous decline around the world in more than 40% of insect species and one third are endangered. The rate of extinction of insects is eight times faster than that of mammals, birds and reptiles. The cause is intensive agriculture and especially the heavy use of pesticides.

* This phrase means unacceptable, intolerable, outrageous

Source: *www.ecowatch.com; www.acs.org; www.theguardian.com*

Applying your skills

1 The extinction of insects represents a loss of *biodiversity*, which is a common pool resource. Discuss how the concepts of rivalry and non-excludability can be used to explain depletion of this resource due to the heavy use of pesticides.

2 a Using a diagram explain what kind of externality is involved in the case of the harm being caused to bees.

b Identify and evaluate policies that could be used to deal with this externality.

c Using a diagram, show how this policy could work to correct the externality.

THEORY OF KNOWLEDGE 5.1

Economic thinking on sustainability and Elinor Ostrom, winner of the 2009 Nobel Prize in Economics

In our study of sustainability, we have seen that in economists' way of thinking, environmental destruction is analysed in terms of negative externalities. In the case of common pool resources, their overuse is simply the *external cost of* people's ordinary transactions that do not take account of consequences on the environment. In other words, the destruction of the environment is something that is thought of as being *outside the normal sphere of events*.

An important reason for thinking this way about the environment is that microeconomic theory is based on the idealised and fictional world of perfect markets with no failures, driven by economic decision-makers who make choices according to their best self-interest, responding to prices as signals and incentives, all of which leads to maximum social welfare.

CONTINUED

Economists' understanding of externalities as the divergence between private and social costs and private and social benefits is based on just this assumption of narrow self-interest: consumers and firms consume and produce taking only their own private interests into account, ignoring possible external costs that their actions give rise to. External costs arise *because consumers and producers are assumed to behave in ways that ignore the interests of society at large.*

The two main categories of measures to deal with environmental externalities we discussed, government regulations and market-based policies, are both based on this assumption about self-interested behaviour. Regulations use the command approach to force producers and consumers to reduce the external costs of their self-interested actions, and market-based policies create price incentives that try to bring the self-interested behaviour of economic decision-makers in line with society's best interests.

Yet all this raises some interesting questions. Does it make sense to view environmental destruction purely as the by-product of people's indifference toward the environment because they are self-interested beings? Are there situations when people do not display the narrow self-interested behaviour assumed by standard economic theory? If so, what are the implications, and what would be the appropriate policy responses?

These are the kinds of questions posed by Elinor Ostrom, who won the Nobel Prize in Economics (2009) for her work on the collective self-governance of common pool resources (discussed above). In one of her major works, Ostrom writes that her central question

> *'is how a group of principals who are in an interdependent situation can organise and govern themselves to obtain continuing joint benefits when all face temptations to free-ride, shirk, or otherwise act opportunistically.'*[9]

As discussed in the text, Ostrom's hundreds of experiments show that people can find solutions to managing resources sustainably by making rules regarding their behaviour and management of the common pool resources that must be followed by everyone. Her findings do not run entirely counter to the conclusions of standard economic theory, since lack of good methods of communication between people do result in the failures we studied in this chapter. However, her findings depart from standard economic theory in a very important respect: she has found that *these failures are not inevitable*, because rational economic behaviour does not always mean acting in one's own best self-interest; it often means acting in what is in the *group's best interests.* This kind of action arises as a result of institutions that permit communication between the resource users, leading to binding agreements, with monitoring and enforcement rules that ensure sustainable resource use.

Ostrom's conclusions focus on the point that people often behave *co-operatively rather than competitively,* and this has very important policy implications. Sometimes, the best method of preserving common pool resources is by allowing the resource users themselves to manage them, rather than through centralised government interference. When the conditions for co-operative solutions are present, the government's role should be to promote institutions that enable the users to manage the resource, such as a court system for resolving disputes and institutions that provide scientific knowledge for resource management. Often, this involves recognising that local co-operative institutions may be superior to institutions that are imposed on people from above or from outside.

If you decide to read further about Ostrom's work, you will find that she uses the term 'common pool resources'; other scholars use 'common property resources'. Both these terms mean common pool resources that are governed by a set of rules.

[9] Elinor Ostrom (1990) *Governing the Commons: The Evolution of Institutions for Collective Action*, Cambridge University Press.

CONTINUED

Thinking points

- How realistic do you think is the assumption that economic decision-makers are motivated by rational self-interest in making economic decisions?
- Many economists argue that even if rational self-interest is not a realistic assumption, it does not matter as long as the predictions of a theory fit with what happens in the real world (see also Theory of knowledge 7.1). What does Ostrom's work tell us about this perspective?
- If people sometimes behave co-operatively rather than competitively, what are the implications for the idea that environmental destruction is caused by externalities?
- Ostrom suggests that many people would change their behaviour if they understood that certain choices would be in their own best interests as well as in society's best interests (for example, biking rather than using a car). But many people may not know about such joint personal and social benefits. What can be done about this?

5.4 Negative consumption externalities

LEARNING OBJECTIVES

After studying this section, you will be able to:

- define all the terms appearing in **orange bold** in the text (AO1)
- explain negative externalities of consumption and the resulting welfare loss (AO2)
- draw a diagram illustrating negative externalities of consumption and welfare loss (AO4)
- explain the meaning of demerit goods (AO2)
- calculate welfare loss that arises from negative externalities of consumption (HL only) (AO4)
- explain government intervention to correct negative externalities of consumption: (AO2)
  - indirect (Pigouvian) taxes
  - legislation and regulation
  - education and awareness creation
  - nudges (HL only)
- draw diagrams to illustrate the above government responses (AO4)
- discuss strengths and limitations of the above government policies with respect to: (AO3)
  - difficulties in measurement of externalities
  - degree of effectiveness
  - consequences for stakeholders

Explaining and illustrating negative consumption externalities

Negative consumption externalities refer to external costs created by consumers. For example, when consumers smoke in public places, there are external costs that spill over onto society in the form of costs to non-smokers due to passive smoking. In addition, smoking-related diseases result in higher than necessary health care costs that are an additional burden upon society. When there is a consumption externality, the marginal private benefit (demand) curve does not reflect social benefits. In Figure 5.9, the buyers of cigarettes have a demand curve, *MPB*,

but, when smoking, create external costs for non-smokers. These costs can be thought of as 'negative benefits', which therefore cause the *MSB* curve to lie below the *MPB* curve. The vertical difference between *MPB* and *MSB* represents the external costs. Note that since the externality involves consumption (i.e. the demand curve), the supply curve represents both marginal private costs and marginal social costs. The market determines an equilibrium quantity, Q_m, and price P_m, given by the intersection of the *MPB* and *MPC* curves, but the social optimum is Q_{opt} and P_{opt}, determined by the intersection of the *MSB* and *MSC* curves.

Other examples of negative consumption externalities include heating homes and driving cars by use of fossil fuels that pollute the atmosphere.

When there is a negative consumption externality, the free market overallocates resources to the production of the good, and too much of it is produced relative to what is socially optimum. This is shown by $Q_m > Q_{opt}$ and $MSC > MSB$ at Q_m in Figure 5.9.

In general, negative externalities, whether these arise from production or consumption activities, lead to allocative inefficiency arising from an overallocation of resources to the good and to its overprovision.

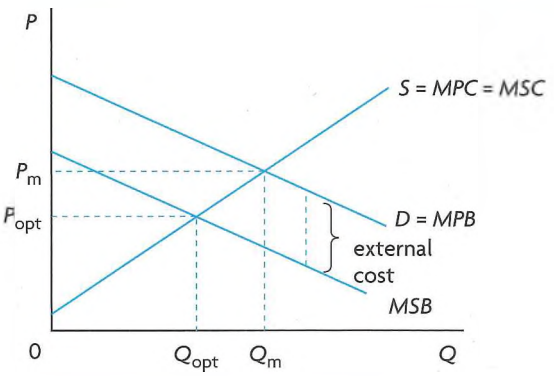


Figure 5.9: Negative consumption externality

The welfare loss of negative consumption externalities

Welfare loss

The welfare loss resulting from negative consumption externalities is the shaded area in Figure 5.10(a) and represents the reduction in benefits for society due to the overallocation of resources to the production of the good. For all units of output greater than Q_{opt}, $MSC > MSB$, indicating that too much of the good is produced. The welfare loss is equal to the difference between the *MSC* and *MSB* curves for the amount of output that is overproduced relative to the social optimum ($Q_m - Q_{opt}$). It represents the loss of social benefits from overproduction due to the externality. If this externality were corrected, society would gain the benefits represented by the shaded area. Note that, once again, the point of the welfare loss triangle lies at the Q_{opt} quantity of output (as in the case of negative production externalities).

Calculating welfare loss (HL only)

Figure 5.10(b) is the same as part (a) only with figures so we can calculate welfare loss. The area of the shaded triangle is the height times the width of the triangle divided by 2. Note that the height of the triangle is equal to the external cost per unit, or MPB – MSB, and the width is equal to the amount of overproduction by the market or $Q_m - Q_{opt}$:

$$\text{Welfare loss} = \frac{(6-4)\times(100-70)}{2} = \frac{2\times 30}{2} = \$30$$

Welfare loss in relation to consumer and producer surplus (Supplementary material)

Figure 5.10(c) shows how the welfare loss of a negative consumption externality is related to consumer and producer surplus and the external cost. In market equilibrium, consumer surplus is equal to the areas a + b, while producer surplus is equal to the areas c + d + f. The cost of the externality is represented by a + d + e (it is the difference between the *MPB* and *MSB* curves up to Q_m). The total social benefits are therefore consumer surplus plus producer surplus minus the external cost:

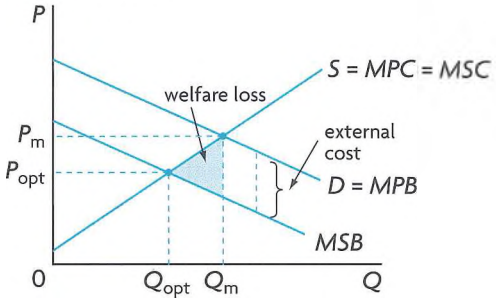
(a + b) + (c + d + f) – (a + d + e) = b + c + f – e

At the social optimum, consumer surplus is equal to b + c, and producer surplus is equal to f, while external costs are zero. Therefore, the total social surplus is equal to producer plus consumer surplus:

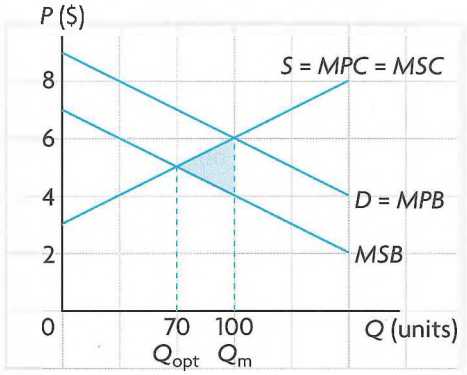
b + c + f

Comparing the total social benefits at market equilibrium and at the social optimum, we see they are smaller at market equilibrium by the area e, which is the welfare loss.

a Welfare loss



b Calculating welfare loss (HL only)



c Supplementary material

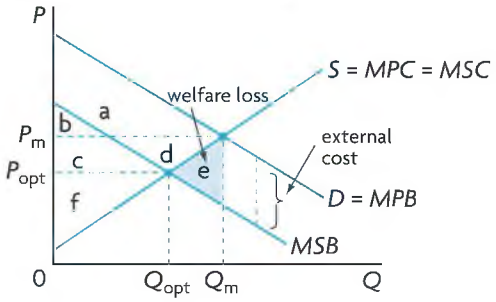


Figure 5.10: Welfare loss in a negative consumption externality

The case of demerit goods

Demerit goods are goods that are considered to be undesirable for consumers, but which are overprovided by the market. Examples of demerit goods include cigarettes, alcohol and gambling. One important reason for overprovision is that the good may have negative consumption externalities, in which case the market overallocates resources to its production. This could occur because of consumer ignorance about its negative effects or indifference: consumers may not be aware of the harmful effects upon others of their actions, or they may not care.

Negative consumption externalities and overuse of common pool resources

Overuse of common pool resources is seen as resulting more from production activities than consumption activities, therefore relating more to negative production rather than consumption externalities. However, overuse of common pool resources also results from negative consumption externalities, shown in Figure 5.9. Take the demand for heating oil, represented by the demand curve *MPB*. The overuse of clean air (the common pool resource) is the external cost that causes the marginal social benefit curve (*MSB*) to lie below the *MPB* curve. The same applies to the use of cars that run on gasoline (petrol). More importantly, air travel is a consumption activity that results in significant and rapidly increasing greenhouse gases that cause global warming. These activities on the part of consumers result in overuse of the common pool resource of clean air, seriously threatening the stability of the global climate.

TEST YOUR UNDERSTANDING 5.6

1 a Using a diagram, show how marginal private benefits and marginal social benefits differ when there is a negative consumption externality.

b Explain the difference between the equilibrium quantity determined by the market and the quantity that is optimal from the point of view of society's preferences.

c Describe the problem with the allocation of resources achieved by the market when there is a negative consumption externality.

d Show the welfare loss created by the negative consumption externality in your diagram, and explain what this means.

2 Provide some examples of negative consumption externalities.

Policies to correct negative consumption externalities and prevent overuse of common pool resources

Market-based policies

A key market-based policy to correct negative consumption externalities (including demerit goods) involves the imposition of indirect Pigouvian taxes, as in the case of negative production externalities. Indirect taxes can be imposed on the good whose consumption creates external costs (for example, cigarettes and petrol/gasoline).

The effects of an indirect tax are shown in Figure 5.11(a). When such a tax is imposed on the good whose consumption creates the external cost, the result is a decrease in supply and an upward shift of the supply curve from MPC to MPC + tax. If the tax equals the external cost, the MPC + tax curve intersects MPB at the Q_{opt} level of output, and quantity produced and consumed drops to Q_{opt}. (The demand curve does not shift but remains at $D = MPB$.) Q_{opt} is the socially optimum quantity, and price increases from P_m to P_c. The tax therefore permits allocative efficiency to be achieved.[10]

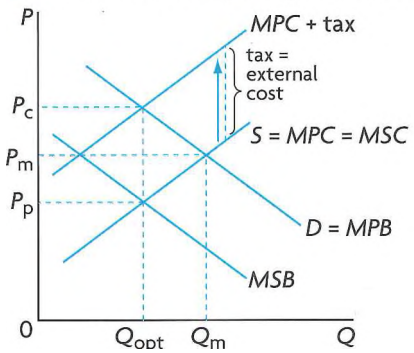
Advantages and disadvantages

As with negative production externalities, economists prefer market-based solutions to the problem of negative consumption externalities, as long as the situation permits the use of market-based policy. Therefore, indirect taxes are the preferred measure, as they internalise the externality (as in the case of negative production externalities). By changing relative prices, indirect taxes create incentives for consumers to change their consumption patterns; the good that is taxed becomes relatively more expensive and consumption is reduced.

However, there are a number of difficulties in this approach. The first involves difficulties in measuring the value of the external costs. Take, for example, the case of passive smoking, an external cost created by smokers, or the case of petrol (gasoline) consumption, which creates external costs in the form of environmental pollution. There are many technical difficulties involved in trying to assess who and what is affected, as well as to determine the value of the external costs, on the basis of which a tax can be designed.

A further difficulty is that some of the goods whose consumption leads to negative consumption externalities (for example, petrol/gasoline and cigarettes) have an inelastic demand. As you may remember from Chapter 3, when demand is inelastic, the percentage decrease in quantity demanded is smaller than the percentage increase in price (due to the tax). Therefore, it is possible that imposing taxes on such goods as petrol (gasoline) and cigarettes (both of which have an inelastic demand) works to increase government tax revenues while not significantly decreasing the quantity demanded of these goods. This could mean that in order to achieve Q_{opt}, a very high indirect tax would have to be imposed, which would very likely be politically unacceptable. On the other hand, the large tax revenues can be used to finance education programmes to discourage consumption of particular goods.

a Market-based: imposing an indirect tax



b Government regulations and advertising

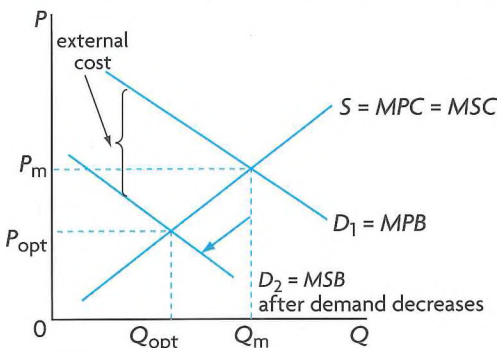


Figure 5.11: Correcting negative consumption externalities

[10] Note that the new equilibrium price, P_c, is the price paid by consumers; the price received by producers is $P_p = P_c$ minus tax per unit (see Chapter 4, section 4.2).

A tax on producers or consumers? (Supplementary material)

You may be wondering why the tax to correct negative consumption externalities should affect producers, shifting the supply curve upward in Figure 5.11(a), and not consumers by shifting the demand curve downward, who after all are the ones creating the externality through consumption. You will find the answer to this question in the digital coursebook as Supplementary material.

Government legislation and regulation

If negative consumption externalities were corrected, Q_{opt} quantity of the good would be produced, reflecting allocative efficiency. Regulations can be used to prevent or limit consumer activities that impose costs on third parties, such as legal restrictions on activities like smoking in public places or age restrictions forcing sellers to do business only with adults. This has the effect of shifting the $D_1 = MPB$ curve towards the MSB curve in Figure 5.11(b), until D_2 overlaps with MSB. This would eliminate the externality, with production and consumption occurring at Q_{opt} and price falling to P_{opt}.

Advantages and disadvantages

Some regulations can be very effective in reducing the external costs of certain consumption activities. Examples include prohibiting smoking in public places and placing restrictions on cars entering city centres. However, they cannot be used to deal with other kinds of negative consumption externalities. For example, it would be very difficult to regulate petrol (gasoline) consumption; on the other hand, imposing indirect taxes on such goods may be more effective (though subject to the limitations noted above on inelastic demand).

Education and awareness-creation

Educating the public and creating awareness by the government can be used to try to persuade consumers to buy fewer goods with negative externalities, such as anti-smoking campaigns, campaigns to avoid consumption of unhealthy foods, or campaigns to reduce the consumption of goods and services based on fossil fuel use. Examples include using public transportation to economise on petrol (gasoline) use or improving home insulation to reduce oil consumption (for heating) or providing information to consumers on the amount of carbon produced by air travel.

The objective is to try to decrease demand for goods giving rise to negative consumption externalities, and the effects are the same as with government regulations, shown in Figure 5.11(b). The MPB curve shifts to D_2 after the campaign, where it coincides with MSB, where Q_{opt} is produced and consumed, and the price falls from P_m to P_{opt}.

Advantages and disadvantages

Education and creation awareness have the advantage that they are simpler than other methods, but they too have their disadvantages. One of these involves the cost to the government of campaigns, which are funded out of tax funds, meaning there are less funds available for use elsewhere in the economy (there are opportunity costs). There is also the possibility that such methods may not be effective enough in reducing the negative externality.

Nudges (HL only)

Nudges were introduced in Chapter 2 where we saw that they form a part of behavioural economics, involving the design of methods to influence consumers' behaviour. They can be used in ways similar to education and consumer awareness to encourage consumers to rely less on goods with negative externalities, such as unhealthy foods and cigarettes. For example, unhealthy foods can be placed in less accessible places in shops, or graphic pictures of the consequences of smoking may be placed on cigarette packs. In such cases demand for the product falls so that the MPB curve shifts toward MSB as in Figure 5.11(b). In addition, nudges can be used to encourage desirable behaviour, such as creating bicycle lanes to motivate car drivers to use bicycles instead.

Advantages and disadvantages

There may be difficulties in designing effective nudges as these are still a new method intended to influence consumer behaviour, and not enough is known yet about how consumers respond to particular nudges and choice architecture. There is, too, the issue that particular nudges may not have the same effects across income groups or cultural groups.

Correction of negative consumption externalities involves either decreasing supply and shifting the *MPC* curve upward by imposing an indirect (Pigouvian) tax; or by decreasing demand and shifting the *MPB* curve toward the *MSB* curve through regulations, education and awareness creation or nudges. Both supply decreases and demand decreases are intended to lead to production and consumption at Q_{opt} and the achievement of allocative efficiency.

In general, given the limitations above for all the various policies it is only possible to move the economy in a direction towards correction of the externality, rather than achieving a precise allocation of resources where Q_{opt} is produced and consumed. Very often, the use of several policies together may be the best way to address the externality. For example, tax revenues from the imposition of a Pigouvian tax may be used to finance education and awareness creation programmes as well as nudges. Governments must therefore be selective in the methods they use to reduce consumption externalities, depending on the particular good that creates the external costs.

TEST YOUR UNDERSTANDING 5.7

1 For each of the examples you provided in question 2 of Test your understanding 5.6, explain some methods that could be used to correct the externality.

2 Using diagrams, show how a negative consumption externality can be corrected by use of
   - a legislation and regulations that limit the external costs,
   - b education and awareness creation,
   - c (HL only) nudges, and
   - d indirect taxes.
   - e Identify some advantages and disadvantages of each of these policy measures.

3 Explain the meaning of a demerit good and provide examples.

4 Explain how a negative consumption externality differs from a negative production externality.

5 a Explain what kinds of measures economists prefer to correct negative consumption externalities.
   - b Identify situations where these might not be very effective.

REAL WORLD FOCUS 5.3

Policies to reduce sugar consumption

Following a 2015 report by the World Health Organization (of the United Nations) detailing the negative health effects of sugar, more and more countries around the world are imposing a tax on the sugar content of foods and drinks in an effort to reduce obesity and other health problems like diabetes arising from sugar overconsumption. These taxes have generally met with little opposition from the public. The taxes are imposed on foods and/or drinks with sugar, which are taxed in accordance with the quantity of sugar they contain.

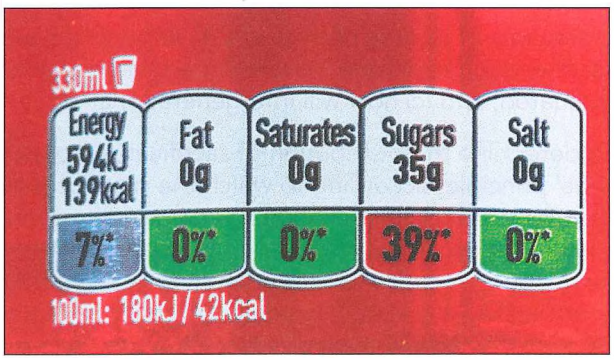


Figure 5.12: Classic coke food label decoder showing high sugar level

In response to the implementation of these policies, many major food producers around the world have responded by creating new products with no added sugar, or reduced sugar, while at the same time trying to maintain the original taste. In some reduced sugar items, the sugar content is lowered enough so that the tax can be avoided altogether. In addition, some governments are working with food producers to establish voluntary agreements with producers to lower the sugar content of their food products.

Another approach involves banning the advertisement of unhealthy foods during the times when children watch television. A still different approach involves the use of electronic ordering systems that offer consumers healthy goods as the default options.

CONTINUED

Applying your skills

1 Using a diagram, explain what kind of externality the article is referring to.

2 Using a diagram, explain how a tax on the sugar content of foods and drinks can help reduce the externality.

3 Outline a likely reason that the taxes have generally met with little public opposition.

4 a Identify and explain the other types of policies (other than indirect taxes) that the article refers to, that could help to reduce the externality.

b Using diagrams, show how these policies are expected to work.

Source: *www.kerry.com*

THEORY OF KNOWLEDGE 5.2

The ethical dimensions of sustainability and preserving the global climate

In Chapter 1 we saw that solutions to the problem of sustainability face major technical difficulties due to uncertainties and incomplete knowledge of social and natural scientists regarding the complex relationships between environmental, economic, social and institutional variables. These kinds of technical difficulties are also responsible for the uncertainties surrounding both regulatory and market-based economic policies to address environmental externalities discussed in the present chapter.

Over and above the technical difficulties, the problem of sustainability faces major ethical issues of fairness and justice, relating to intergenerational equity (running from generation to generation), as well as equity across nations and social groups within nations of the present generation.

In the area of climate change alone, important issues include (a) how will the burden of having to make sacrifices in the present be distributed among countries; (b) how will the impacts of climate change be evaluated; and (c) how will intergenerational equity be accounted for?[11]

To determine the distribution of sacrifices, a possible ethical principle that can be used is 'the polluter pays' principle, according to which the sacrifice is distributed according to how much each country contributes to climate change. In one variant of this principle, it would be necessary to take into account cumulative (historical) contributions to greenhouse gas emissions. This would place an extra burden on the developed countries of today, which over time, have contributed far more to emissions than developing countries. As a counterargument, opponents refer to 'excusable ignorance', meaning it should not be necessary to pay for past emissions if these were made without knowledge of their effects on the global climate. According to a different ethical principle, the past would be ignored and future emissions rights would be distributed to all countries on a *per capita* basis.

On the second issue, concerning evaluation of impacts of climate change, one approach involves welfare analysis. This has given rise to disagreements about how to calculate welfare and add it up across individuals in the present as well as in the future. Another approach focuses on human rights as the basis for evaluating impacts, such as the rights to food, water and shelter, which may be threatened by climate change.

[11] The World Bank (2009) *World Development Report 2010: Development and Climate Change.*

CONTINUED

Intergenerational equity, the third issue, is closely related to the evaluation of impacts of climate change, as these must account for impacts not only on the present generation but future generations as well.

These kinds of questions clearly belong to the normative realm of thought. Given the technical difficulties as well, it is no wonder that there are broad disagreements over sustainability, and no easy solutions appear on the horizon.

Thinking points

- What do you think should be the role of science and social science in providing answers to these kinds of questions?
- To what extent do you think market forces can be relied upon, if at all, to deal with problems of environmental sustainability?
- Market economies are based upon human behaviour motivated by rational self-interest (see Chapter 2). To what extent do you think this self-interest is the root cause of the environmental problems that beset the human race today? (See also Theory of knowledge 5.1.)
- Given that, historically, economically more developed countries have been mainly responsible for today's environmental problems, do you agree with the view that economically less developed countries should simply ignore calls for them to limit their growth rates to prevent further global warming?

INQUIRY AND REFLECTION

The following questions will help you reflect on your learning and enhance your understanding of key topics in this chapter. They are suggestions for inquiries that you can undertake on your own or in groups in order to revise the learning objectives of this chapter.

1 Identify a good or service in your country or a country you are interested in that gives rise to a particular negative production externality. Examine the extent of the external costs and how they affect various stakeholders. Investigate the solutions that are being used to reduce its production and consumption. Discuss the strengths and limitations of these solutions, and try to identify the effects on various stakeholders.

2 See question 1 above and apply it to a negative consumption externality.

3 Identify a scheme of tradable permits that has been implemented. Research and discuss its strengths and limitations.

4 Identify a common pool resource that is of interest to you and investigate what, if any, measures are being taken in your country of residence or a country you are interested in to deal with its possible overuse.

5 As indicated in Real world focus 5.3, a number of countries around the world have imposed or are planning to impose indirect (Pigouvian) taxes on various food items (sugar and/or fats) that are considered to be harmful to human health. Examine whether such a tax is used or is under consideration in your country or a country you are interested in, and examine the external costs of consumption of such foods, as well as the effects of the tax on various stakeholders. Investigate whether any studies have been done to evaluate the impact of the tax, and the extent to which it has been successful with respect to attaining its objective of reducing consumption of such foods. Consider also additional, complementary policies that are in use for this purpose in your country of choice.

CONTINUED

6 Air travel is the fastest-growing contributor to carbon emissions. Responsible for about 5% of global warming in 2018, by 2050 emissions may grow by more than 700%, taking up one-quarter of the global carbon budget for limiting the temperature rise to 1.5%. Yet the global airline industry rejects policies such as carbon pricing and regulation of aviation emissions by governments or international bodies. Instead it has opted for a scheme called Carbon Offsetting and Reduction Scheme for International Aviation (CORSIA) which has been criticised as being ineffective. (a) Research CORSIA and identify some advantages and disadvantages as a method to tackle the problem of airline emissions. (b) Research and outline steps that travellers can take in order to reduce their carbon footprint when they travel by air.

EXAM STYLE QUESTIONS

You can find questions in the style of IB exams in the digital coursebook.



› Chapter 6

Market failure and socially undesirable outcomes II:

Positive externalities, public goods, asymmetric information and inability to achieve equity

BEFORE YOU START

- Can you think of some goods or services that when consumed or produced bring benefits to others even though they didn't consume or produce them?
- You may have noticed that some goods and services are provided free of charge, such as roads, parks, street lighting, public schools, and many more. Who pays for these and what might be the reasons people can enjoy them without having to pay for them?

In this chapter we continue the discussion begun in Chapter 5 on the inability of the market to fulfill some of its promises. We will begin by examining positive externalities of production and consumption. We will then discover public goods, which are not produced at all by the market even though they are socially desirable. We will then examine the consequences of information asymmetries between buyers and sellers, and will conclude with a discussion of the market's inability to achieve equity in the distribution of income and wealth.

Photo: Tooth wheel mechanism

6.1 Positive production externalities

LEARNING OBJECTIVES

After studying this section you will be able to:

- define all the terms appearing in **orange bold** in the text (AO1)
- explain positive externalities of production and the resulting welfare loss (AO2)
- draw a diagram illustrating positive externalities of production and welfare loss (AO4)
- calculate welfare loss that arises from positive externalities of production (HL only) (AO4)
- explain government intervention to correct positive externalities of production: (AO2)
  - government provision
  - subsidies
- draw diagrams to illustrate the above government responses (AO4)
- discuss strengths and limitations of the above government policies with respect to: (AO3)
  - difficulties in measurement of externalities
  - degree of effectiveness
  - consequences for stakeholders

Explaining and illustrating positive production externalities

Externalities as a form of market failure were introduced in Chapter 5, where we discussed negative externalities of production and consumption. We now turn our attention to positive externalities.

Positive production externalities refer to external benefits created by producers. If, for example, a firm engages in research and development, and succeeds in developing a new technology that spreads throughout the economy, there are external benefits because not only the firm but also society benefits from widespread adoption of the new technology. Therefore, the social costs of research and development are lower than the private costs. In Figure 6.1, the *MSC* curve lies below the *MPC* curve, and the difference between the two curves is the value of the external benefits (these can be thought of as 'negative costs'). The demand curve represents both *MPB* and *MSB* since the externality involves only production. The market gives rise to equilibrium quantity Q_m and price P_m, determined by the intersection of the *MPB* and *MPC* curves, while the social optimum is given by Q_{opt} and P_{opt}, determined by the intersection of the *MSB* with *MSC* curves. Since $Q_m < Q_{opt}$, the market underallocates resources to research and development activities that lead to new technologies, and not enough of them are undertaken.

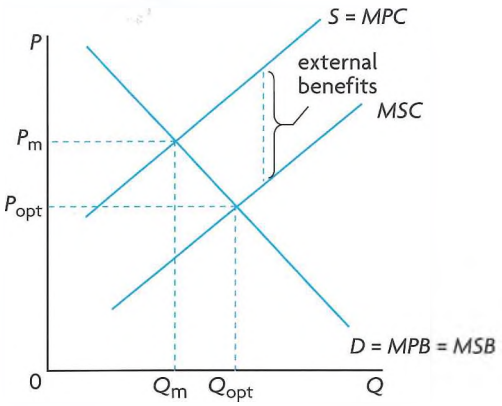


Figure 6.1: Positive production externality

When there is a positive production externality, the free market underallocates resources to the production of the good: too few resources are allocated to its production, and too little of it is produced. This is shown by $Q_m < Q_{opt}$ and *MSB* > *MSC* at Q_m in Figure 6.1.

More examples of positive production externalities include:

- firms train workers who later switch jobs and work elsewhere; external benefits are created as the new employers and society benefit from the trained workers
- a firm provides first aid classes to employees to improve work safety; external benefits are created as this knowledge is applied also to people outside the workplace.

The welfare loss of positive production externalities

Welfare loss

The underallocation of resources to the production of a good with a positive production externality leads to a welfare loss, shown in Figure 6.2(a) as the shaded area. This loss is equal to the difference between the *MSB* and *MSC* curves for the amount of output that is underproduced relative to the social optimum ($Q_{opt} - Q_m$). It involves external benefits for society that are lost because not enough of the good is produced. If the externality were corrected, society would gain the benefits represented by the shaded area. Note that the point of the welfare loss triangle lies at the Q_{opt} quantity of output.

Calculating welfare loss (HL only)

Figure 6.2(b) is similar to part (a) with figures so we can calculate welfare loss. The area of the welfare loss triangle is the height times the width of the triangle divided by 2. The height of the triangle is equal to the external benefit per unit, or $MPC - MSC$, and the width is equal to the amount of underproduction by the market or $Q_{opt} - Q_m$:

$$\text{Welfare loss} = \frac{(6-4) \times (90-60)}{30} = \frac{2 \times 30}{2} = \$30$$

Welfare loss in relation to consumer and producer surplus (Supplementary material)

Figure 6.2(c) shows the welfare loss in relation to consumer and producer surplus and the externality. At market equilibrium, consumer surplus is area a, producer surplus is area b + e, and the external benefits are c + f (the difference between the *MPC* and *MSC* curves up to the point of production by the market, Q_m). The total benefits are therefore consumer surplus plus producer surplus plus external benefits:

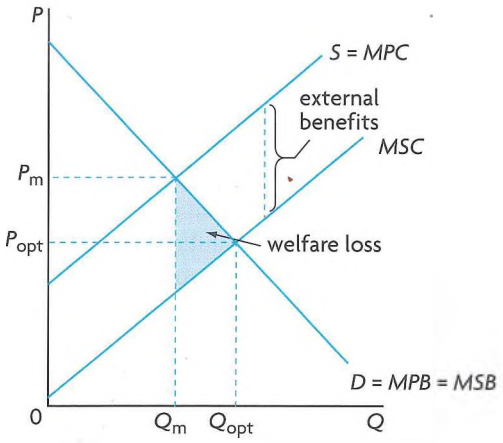
a + (b + e) + (c + f) = a + b + e + c + f

At the social optimum, consumer surplus is a + b + c + d, producer surplus is e + f + g, and external benefits are zero, making a total of:

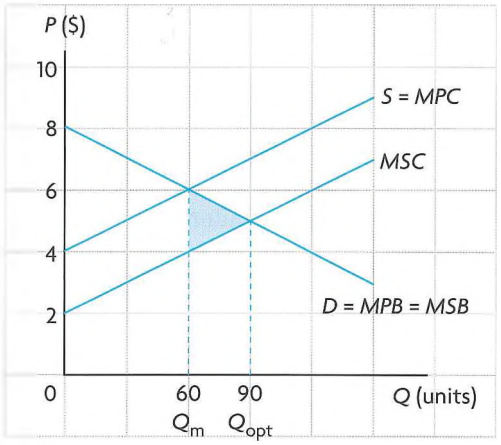
(a + b + c + d) + (e + f + g) = a + b + c + d + e + f + g

Comparing total benefits at market equilibrium and at the social optimum, we find that at the social optimum there are additional benefits of the amount d + g, corresponding to the shaded area in the figure. This is the amount of welfare that is lost at market equilibrium due to underallocation of resources arising from the positive production externality.

a Welfare loss



b Calculating welfare loss (HL only)



c Welfare loss in relation to consumer and producer (Supplementary material)

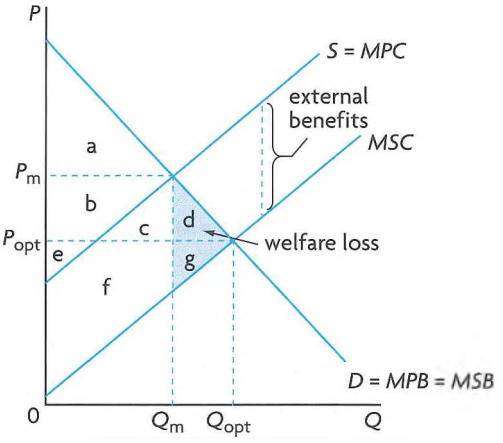


Figure 6.2: Welfare loss in a positive production externality

Correcting positive production externalities

Direct government provision

A solution often pursued by governments involves direct government provision of the good or service creating the positive production externality. For example, governments often engage in research and development (R&D) for new technology, for medicine and pharmaceuticals, and many other areas. The government can also directly provide training for workers. Governments pay for such activities with government funds, raised through taxes. Figure 6.3(a) shows that when the government intervenes by providing goods and services itself, this has the effect of shifting the supply curve (= *MPC* curve) downward (or to the right), toward the *MSC* curve so that the optimum quantity of the good, Q_{opt}, will be produced, with price falling from P_m to P_{opt}.

Subsidies

We studied subsidies and their effects in Chapter 4, where we saw how their introduction into a perfect market (with no market failures) creates allocative inefficiency. Now, we will see how subsidies can correct allocative inefficiency by correcting a market failure.

If the government provides a subsidy to a firm per unit of the good produced that is equal to the external benefit, then the marginal private cost (*MPC* = supply) curve shifts downward (or rightward[1]) until it coincides with the *MSC* curve, as shown in Figure 6.3(b). The result is to increase quantity produced to Q_{opt} and to lower the price from P_m to P_{opt}. The problem of underallocation of resources and underprovision of the good is corrected, and allocative efficiency is achieved.

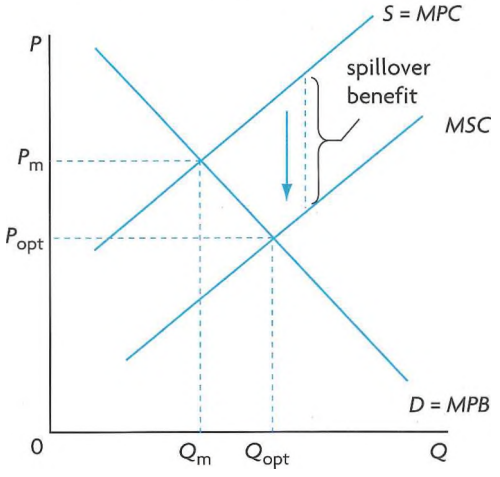
You may note that direct government provision and subsidies have the same market outcomes.

> Correction of positive production externalities involves shifting the *MPC* curve downward toward the *MSC* curve through direct government provision or by subsidies. For allocative efficiency to be achieved, the quantity produced and consumed must increase to Q_{opt} as price falls to P_{opt}.

Evaluating policies to correct positive production externalities

This topic will be discussed together with policies to correct positive consumption externalities below (Section 6.2) because of similarities of the policies involved.

a Direct government provision



b Granting a subsidy

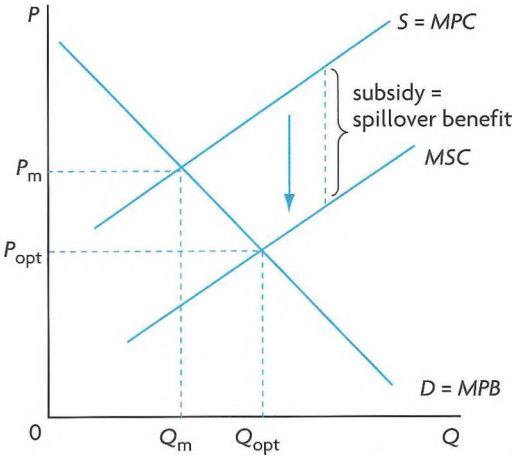


Figure 6.3: Correcting positive production externalities

[1] See 'Quantitative techniques' in the digital coursebook for an explanation of the equivalence of downward and rightward shifts of the supply curve.

TEST YOUR UNDERSTANDING 6.1

1 a Using a diagram, show how marginal private costs and marginal social costs differ when there is a positive production externality.

b Explain the difference between the equilibrium quantity determined by the market and the quantity that is optimal from the point of view of society's preferences.

c Describe what this tells you about the allocation of resources achieved by the market when there is a positive production externality.

d Show the welfare loss created by the positive production externality in your diagram, and explain what this means.

2 Provide some examples of positive production externalities.

3 For each of the examples you provided in question 2, explain some methods that can be used to correct the externality.

6.2 Positive consumption externalities

LEARNING OBJECTIVES

After studying this section you will be able to:

- define all the terms appearing in **orange bold** in the text (AO1)
- explain positive externalities of consumption and the resulting welfare loss (AO2)
- draw a diagram illustrating positive externalities of consumption and welfare loss (AO4)
- explain the meaning of merit goods (AO2)
- calculate welfare loss that arises from positive externalities of consumption (HL only) (AO4)
- explain government intervention to correct positive externalities of consumption: (AO2)
  - legislation and regulation
  - education and awareness creation
  - nudges (HL only)
  - government provision
  - subsidies
- draw diagrams to illustrate the above government responses (AO4)
- discuss strengths and limitations of the above government policies with respect to: (AO3)
  - difficulties in measurement of externalities
  - degree of effectiveness
  - consequences for stakeholders

Explaining and illustrating positive consumption externalities

When there is a **positive consumption externality**, external benefits are created by consumers. For example, the consumption of education benefits the person who receives the education, but in addition gives rise to external benefits, involving social benefits from a more productive workforce, lower unemployment, higher rate of growth, more economic development, lower crime rate, and so on. Similarly, the consumption of health care services benefits not only the person receiving the services but also society and the economy, because a healthier population is more productive, enjoys a higher standard of living, does not pass on contagious diseases as much and may have a higher rate of economic growth. In Figure 6.4, we see that the marginal social benefit (MSB) curve lies above the marginal private benefit (MPB) curve, and the difference

between the two consists of the external benefits to society. The socially optimum quantity, Q_{opt}, is given by the point where $MSB = MSC$, and the quantity produced by the market is given by the point where $MPB = MPC$. Since $Q_{opt} > Q_m$, the market underallocates resources to the good or service, and too little of it is produced.

When there is a positive consumption externality, the free market underallocates resources to the production of the good, and too little of it is produced relative to the social optimum. This is shown by $Q_m < Q_{opt}$ and $MSB > MSC$ at Q_m in Figure 6.4.

In general, positive externalities (external benefits), whether these arise from production or consumption activities, lead to an underallocation of resources to the good in question, and therefore to its underprovision.

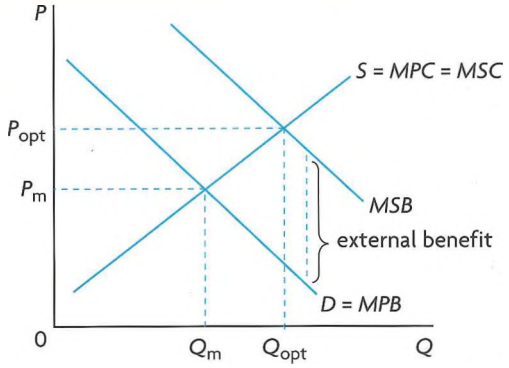


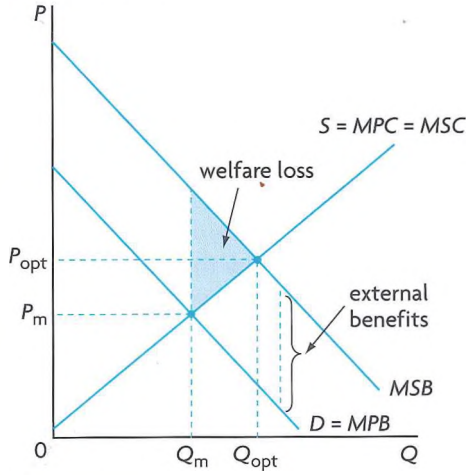
Figure 6.4: Positive consumption externality

The welfare loss of positive consumption externalities

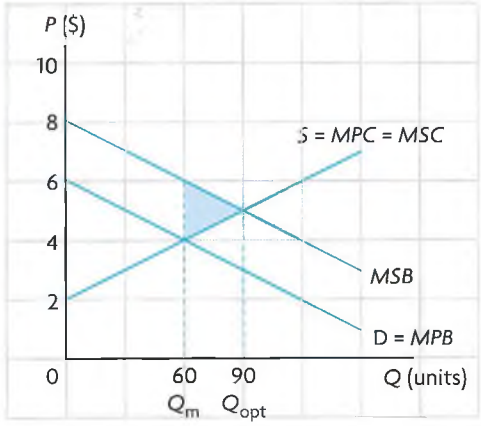
Welfare loss

The welfare loss arising from a positive consumption externality is the shaded area in Figure 6.5(a), and is the difference between the *MSB* and *MSC* curves for the amount of output that is underproduced relative to the social optimum ($Q_{opt} - Q_m$). It represents the loss of social benefits due to underproduction of the good. If this externality were corrected, society would gain the benefits represented by the shaded area. Once again, we see that the point of the welfare loss triangle lies at the Q_{opt} quantity of output.

a Welfare loss



b Calculating welfare loss (HL only)



c Welfare loss in relation to consumer and producer surplus (Supplementary material)

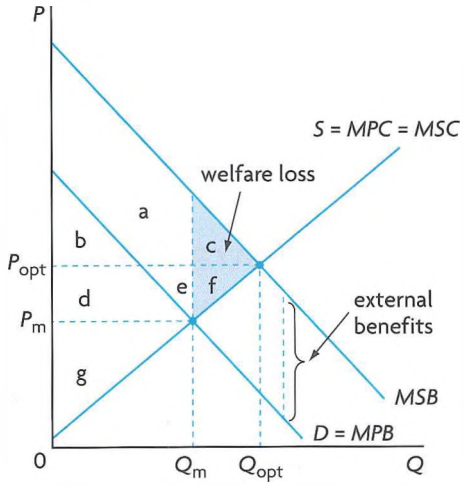


Figure 6.5: Welfare loss in a positive consumption externality

You may have noticed that in the case of negative externalities of production and consumption, where $Q_{opt} < Q_m$, the welfare loss triangle always points leftward toward Q_{opt}. By contrast, in the case of positive externalities of production and consumption, where $Q_{opt} > Q_m$, the welfare loss triangle always points rightward, again toward Q_{opt}.

Calculating welfare loss (HL only)

Figure 6.5(b) is similar to part (a) with figures so we can calculate welfare loss. The area of the welfare loss triangle the height times the width of the triangle divided by 2. The height of the triangle is equal to the external benefits per unit, or $MSB - MPB$, and the width is equal to the amount of underproduction by the market or $Q_{opt} - Q_m$:

$$\text{Welfare loss} = \frac{(6-4) \times (90-60)}{30} = \frac{2 \times 30}{2} = \$30$$

Welfare loss in relation to consumer and producer surplus (Supplementary material)

In Figure 6.5(c) we see how the welfare loss arises in relation to consumer and producer surplus and the external benefits. In market equilibrium, consumer surplus is equal to areas b + d, producer surplus is area g, and the external benefits are a + e (or the difference between MSB and MPB up to production at Q_m by the market). The total social benefits in market equilibrium are equal to consumer surplus plus producer surplus plus the external benefits:

$$(b + d) + g + (a + e) = b + d + g + a + e$$
$$= a + b + d + e + g$$

At the social optimum, consumer surplus is given by a + b + c, producer surplus is d + e + f + g, and the external benefits are zero. Therefore the total social benefits are:

$$(a + b + c) + (d + e + f + g) = a + b + c + d + e + f + g$$

Comparing the total social benefits at market equilibrium with those at the social optimum, we find that at the social optimum they are greater by the amount c + f. This is the welfare loss that arises when production occurs at market equilibrium as a result of an underallocation of resources due to the positive consumption externality.

The case of merit goods

Merit goods are goods that are held to be desirable for consumers, but which are underprovided by the market. (Note that the term 'good' in the expression 'merit good' applies to both goods and services.) Reasons for underprovision include:

- **The good may have positive externalities.** In this case too little is provided by the market. Examples of merit goods include education (for the reasons noted above in the discussion of externalities); immunisation programmes (which benefit not only those who have received them but also the broader population by wiping out a disease).
- **Low levels of income and poverty.** Some consumers may want certain goods or services but cannot afford to buy them. Remember demand shows the quantities of a good or service that consumers are willing and able to buy at different prices. If they have low incomes, they may be willing but not able to buy something, in which case their desire does not show up in the market, and market demand (the sum of all individual demands) is too low. Examples include health care services, medicines, education and recreational facilities, which people on low incomes often cannot afford to buy in the market.
- **Consumer ignorance.** Consumers may be better off if they consume certain goods and services but they may be ignorant of the benefits, and so do not demand them. For example, preventive health care (such as immunisation, annual health check-ups) can prevent serious diseases, but lack of knowledge about the benefits may lead consumers to demand too little of these services.

Note that more than one factor may be at work simultaneously; for example, the underprovision of health care services can result from all three reasons listed above.

Correcting positive consumption externalities

Government legislation and regulation

Legislation can be used to promote greater consumption of goods with positive externalities. For example, most countries have legislation that makes education compulsory up to a certain age (note that education is a merit good). In this case, demand for education increases, and the demand curve $D_1 = MPB$ shifts to the right (or upward), as in Figure 6.6(a). Ideally, it will shift until it reaches the MSB curve, where $D_2 = MSB$, and Q_{opt} is produced and consumed.

Education and awareness creation

Governments can use education of the public, awareness creation, to try to persuade consumers to buy more goods with positive externalities. For example, they can

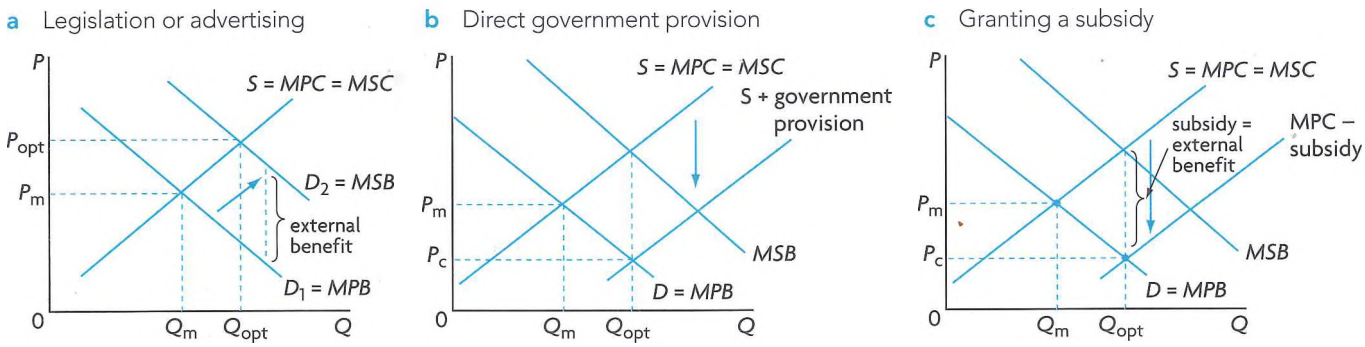


Figure 6.6: Correcting positive consumption externalities

try to encourage the use of sports facilities for improved health. The objective is to increase demand for such services, and the effect is the same as with legislation, shown in Figure 6.6(a): D_1 shifts to $D_2 = MSB$ and Q_{opt} is produced and consumed, while price increases to P_{opt}.

Nudges (HL only)

Nudges have similar effects as education and awareness creation programmes. Governments can use nudges like the creation of bicycle lanes to encourage the use of bike riding for more physical exercise. (Note that this nudge also has the effect of reducing the negative externality of gasoline (petrol) consumption from the use of cars, noted earlier.) Here, too, the objective is to increase demand shifting D_1 toward D_2.

Direct government provision

Governments are frequently involved in the direct provision of goods and services with positive consumption externalities. The most important examples include government (public) provision of education and health care in virtually all countries in the world. Education and health care are merit goods with external benefits so large and important that it is widely believed that they must not be left to private sector provision alone. In most countries where there is government provision of health care and education, there is also private sector provision of these services (though to varying degrees).

Direct government provision is shown in Figure 6.6(b), and has the effect of increasing supply and therefore shifting the supply curve S rightward (or downward) to S + government provision. To achieve the social optimum Q_{opt}, the new supply curve must intersect MPB at the level of output Q_{opt}, as seen in the figure. At the new equilibrium, price falls to P_c, Q_{opt} is produced and allocative efficiency is achieved.

Subsidies

A subsidy to the producer of the good with the positive externality has the same effects as direct government provision. It results in increasing supply and shifting the supply curve rightward (or downward), as shown in Figure 6.6(c) (which is similar to Figure 6.6(b)). If the subsidy is equal to the external benefit, the new supply curve is MPC – subsidy,[2] and it intersects MPB at the Q_{opt} level of output. Again, price falls from P_m to P_c, Q_{opt} is produced and allocative efficiency is achieved.

> Correction of positive consumption externalities involves either increasing demand and shifting the *MPB* curve towards the *MSB* curve through legislation or education and awareness creation; or increasing supply and shifting the *MPC* curve downward by direct government provision or by granting a subsidy. Both demand increases and supply increases can lead to production and consumption at Q_{opt} and the achievement of allocative efficiency. The price paid by consumers increases when demand increases, and falls when supply increases.

Note that the problem of underprovision of merit goods by the market (defined above) can be addressed by all the methods noted above: legislation, education and awareness

[2] You may note that the reason for the minus sign in *MPC* – subsidy is that the amount of the subsidy is *subtracted* from *MPC* in order to arrive at the new supply curve.

creation, direct government provision and granting of subsidies. All are intended to increase the amount of the good produced and consumed, as increased consumption of such goods is held to be desirable for society.

Evaluating policies to correct positive production and consumption externalities

Both direct government provision and subsidies are widely used as methods to deal with positive consumption externalities, and to a lesser extent also with positive production externalities. Both methods are very effective in increasing the quantity of the good produced and consumed, and both have the added advantage of lowering the price of the good to consumers.

There are, however, difficulties involved in achieving the optimum results (where $MSC = MSB$). First, both direct government provision and granting of subsidies involve the use of government funds that rely on tax revenues. Governments generally have very many possible alternative uses for these funds, each of which has an opportunity cost. As it is not possible for the government to directly provide or subsidise all goods and services with positive externalities, choices must be made on (a) which goods should be supported, and (b) by how much they should be supported.

Ideally, choices should be made on the basis of economic criteria, which would specify the amount of social benefits expected in relation to the cost of providing them, the objective being to maximise the benefits for each good and service to be provided or subsidised for a given cost. However, in practice it is very difficult to measure the size of the external benefits, and therefore to calculate precisely which goods and services should be supported and the level of support they should receive. In addition, both direct provision and subsidies are often highly political in nature, as different groups compete with each other over who will receive the most benefits. Governments are often susceptible to political pressures and sometimes make choices based on political rather than economic criteria.

(HL only) Nudges are subject to the same limitations that were noted in connection with negative consumption externalities (Chapter 5), involving difficulties in designing effective nudges in view of insufficient knowledge about how people respond to nudges and choice architecture, as well as the possible different responses these may have across income and cultural groups.

Therefore, in the real world it is very unlikely that governments are able to shift the MPC or MPB curves by the amount necessary to correct the positive externalities. The most that can be hoped for is that the policies in question will be a step in the right direction.

Legislation, education and awareness creation are subject to similar limitations concerning calculating the size of external benefits. Only sometimes can they be effective, and then can only help shift the MPB curve in the right direction, rather than achieve a demand increase that will bring the economy to the Q_{opt} level of output. For example, they can have very positive effects in certain cases (such as legislation requiring schooling up to a minimum age or education on the importance of good nutrition), but in other cases are ineffective (for example, they cannot on their own increase consumption of health care services and education to the optimum level). Moreover, they have the further effect of raising the price of the good to consumers, which may make the good unaffordable for some consumer groups. Therefore, legislation and awareness creation sometimes can be used more effectively if they are implemented together with direct provision and subsidies. A good example is education, where compulsory schooling up to a certain age (legislation) goes together with direct government provision.

TEST YOUR UNDERSTANDING 6.2

1 a Using a diagram, show how marginal private benefits and marginal social benefits differ when there is a positive consumption externality.

b Explain the difference between the equilibrium quantity determined by the market and the quantity that is optimal from the point of view of society's preferences.

c Describe the problem with the allocation of resources achieved by the market when there is a positive consumption externality.

d Show the welfare loss created by the positive consumption externality in your diagram, and explain what this means.

CONTINUED

2 Provide some examples of positive consumption externalities.

3 For each of the examples you provided in question 2, explain some methods that could be used to correct the externality.

4 Outline how a positive consumption externality differs from a positive production externality.

5 Explain the meaning of a merit good, and provide examples.

6 Discuss advantages and disadvantages of the policy measures that governments can use to correct positive externalities of production and consumption.

REAL WORLD FOCUS 6.1

Positive externalities of cash transfer programmes

A highly successful policy used in Brazil to reduce poverty and income inequalities is the Bolsa Familia programme implemented by the Brazilian government. Through this programme poor families receive cash transfers *on condition* that they send their children to school and get them vaccinated. These programmes are therefore called *conditional cash transfers* (CCTs). The programme is so successful in reducing poverty over the short term through the cash transfers and over the long term by increasing human capital that it has attracted great attention around the world.



Figure 6.7: Feijao, Acre Province, Brazil. School children in a rural school classroom

A study on crime in Brazil has found that the Bolsa Familia programme has had a significant impact on crime reduction in São Paolo. It concludes that in the areas where CCTs are implemented there is a negative impact on crime. This is not related to the increased time spent in school but rather is due to reduced poverty and inequality, resulting in a reduction primarily of robberies, hence economically motivated crimes.

Source: *Spillovers from conditional cash transfer programmes http://ftp.iza.org/dp6371.pdf IZA, DP No. 6371*

Applying your skills

You may note that this is a type of positive externality that cannot be analysed by means of the usual externality diagram as there is no market that can be easily identified here. However the idea of a positive externality resulting from government spending on CCTs is clear. Explain what kind of externality this represents, identify the external benefits and describe the implications for the government's CCT policy.

Summary of externalities

Table 6.1 summarises important information on each type of externality.

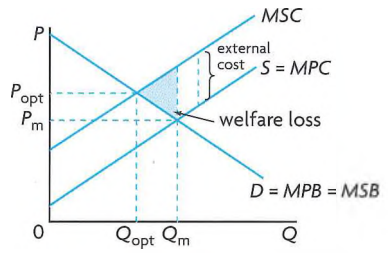
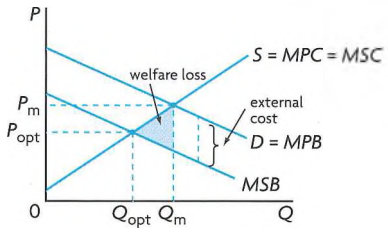
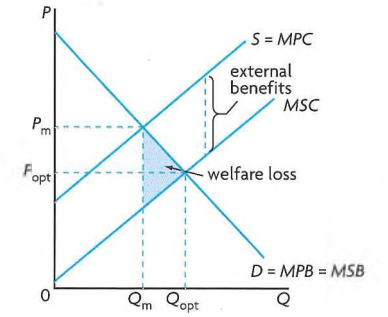
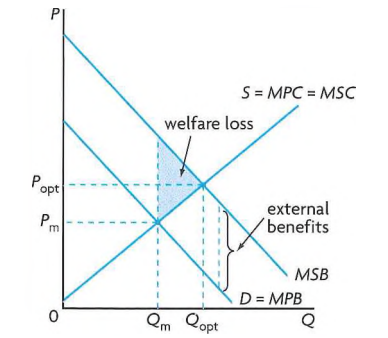
Type of externality	Examples	Policies
Negative production and many common pool resources 	Producers impose external costs on society Production by use of fossil fuels; external costs include global warming, negative effects on health, environmental pollution	• Indirect (Pigouvian) taxes • Carbon taxes • Tradable permits • Legislation, regulation • Collective self-governance • Education, awareness creation • International agreements
Negative consumption 	Consumers impose external costs on society Use of cars and heating using fossil fuels; external costs include global warming, negative effects on health, environmental pollution	• Indirect (Pigouvian) taxes • Legislation, regulation • Education, awareness creation • Nudges (HL only)
Positive production 	Producers create external benefits for society Research by private firms leads to development of new technologies that benefit the whole of society	• Government provision • Subsidies
Positive consumption 	Consumers create external benefits for society Education and health care lead to benefits for the whole of society, including lower unemployment, lower crime rates, higher economic growth	• Legislation, regulation • Education, awareness creation • Nudges (HLonly) • Government provision • Subsidies

Table 6.1: Summary of externalities

The information in this section applies to *all externalities*, including the negative ones presented in Chapter 5 and the positive ones of this chapter.

How to draw an externality diagram without memorising

Students often have difficulty learning how to draw the externality diagrams and labelling the curves correctly. The following rules will help you draw any externality diagram without memorising.

The rules

1. In a production externality, the supply curve splits into two; in a consumption externality, the demand curve splits into two.
2. Supply reflects costs; demand reflects benefits.
3. The market equilibrium quantity Q_m corresponds to private costs and benefits, *MPC* and *MPB*; the social optimum reflects social costs or benefits.
4. In a negative externality $Q_m > Q_{opt}$, meaning that *the market provides too much of a bad thing*; in a positive externality $Q_m < Q_{opt}$, meaning that *the market provides too little of a good thing*.

How to use the rules

1. Draw a demand and supply diagram and label the axes P and Q.
2. For a production externality draw two parallel upward sloping curves; for a consumption externality draw two parallel downward sloping curves. Find the two equilibrium quantities on the Q axis (but do not label them yet).
3. In a negative externality, since $Q_m > Q_{opt}$, label the larger quantity Q_m and the smaller quantity Q_{opt}; in a positive externality, since $Q_m < Q_{opt}$, label the larger quantity Q_{opt} and the smaller quantity Q_m.
4. Using rule 3 above, Q_m gives *MPC* and *MPB*, while Q_{opt} gives *MSC* and *MSB*. You can now label all the curves. (Note that the demand curve, D and supply curve, S represent *private* benefits and costs; therefore D = *MPB* and S = *MPC*.)
5. Find the triangle that points to Q_{opt}, and that lies in between the two curves that have split. This is welfare loss.

Some useful information to remember about externalities

Here are some points to bear in mind:

- All negative externalities (of production and consumption) *create external costs. When there are external costs, MSC > MSB at the point of production by the market.*
- All positive externalities (of production and consumption) *create external benefits. When there are external benefits MSB > MSC at the point of production by the market.*
- All production externalities (positive and negative) create *a divergence between private and social costs (MPC and MSC).*
- All consumption externalities (positive and negative) create *a divergence between private and social benefits (MPB and MSB).*

6.3 Market failure and public goods

LEARNING OBJECTIVES

After studying this section you will be able to:

- define all the terms appearing in **orange bold** in the text (AO1)
- explain public goods as being (AO2)
  - non-rivalrous, non-excludable
  - subject to the free-rider problem
- evaluate government intervention to provide public goods (AO3)
  - direct provision
  - contracting out to the private sector

The meaning of public goods

Public goods are non-rivalrous and non-excludable

We learned the meaning of rivalry and excludability in Chapter 5. A good is rivalrous when its consumption by one person reduces its availability for someone else. It is excludable if it is possible to exclude people from using it. Private goods as we have seen are rivalrous and excludable. Common pool resources are rivalrous and non-excludable. Public goods, by contrast, are non-rivalrous and non-excludable.

A **public good** has the following two characteristics:

- It is **non-rivalrous**; its consumption by one person does not reduce consumption by someone else.
- It is **non-excludable**; it is not possible to exclude someone from using the good.

For example, a lighthouse is non-rivalrous, because its use by one person does not make it less available for use by others. Also, it is non-excludable, because there is no way to exclude anyone from using it. Other examples of public goods include the police force, national defence, flood control, non-toll roads, fire protection, basic research, anti-poverty programmes and many others.

Public goods and the free rider problem

How do public goods relate to market failure? In the case of excludable goods, it is possible to prevent people from buying and using a good simply by charging a price for it, since those who are unable or unwilling to pay the price do not get to use it. Therefore, private firms have an incentive to provide excludable goods because they can charge a price for them, and therefore can cover their costs. Non-excludable goods differ: if a non-excludable good were produced by a private firm, people could not be prevented from using it even though they would not pay for it. Yet no profit-maximising firm would be willing to produce a good it cannot sell at some price. As a result, the market fails to produce goods that are non-excludable, giving rise to resource misallocation, as no resources are allocated to the production of public goods.

Public goods illustrate the **free rider problem**, occurring when people can enjoy the use of a good without paying for it. The free rider problem arises from non-excludability: people cannot be excluded from using the good. Public goods are a type of market failure because, due to the free rider problem, private firms do not produce these goods: the market fails to allocate resources to their production.

Table 6.2 summarises different types of goods according to their rivalry and excludability characteristics.

	Rivalrous	Non-rivalrous
Excludable	Private goods Goods with or without positive or negative externalities (both production and consumption) sold for a price. Merit goods (as long as they are produced by the market) and demerit goods Examples: computers, books, clothes, education, petrol (gasoline)	Quasi-public goods Goods that do not fall neatly into the other three categories; often (but not always) have large positive externalities so may be provided by the government Examples: uncrowded toll roads, museums, public swimming pools that charge entrance fees, cable TV See the section below for an explanation of quasi-public goods.
Non-excludable	Common pool resources Natural resources that are not owned by anyone, not sold in markets and not having a price; their lack of a price makes them subject to overuse (unsustainable use), depletion and degradation Examples: forests, rivers, lakes, soil quality, fish in the oceans	Public goods See the section below for an explanation of quasi-public goods. Socially desirable goods not produced by private firms because it is not possible to charge a price; they are subject to the *free rider problem*: people use them without having to pay; since they are socially desirable they are produced by the government and provided free of charge Examples: national defence, street lighting

Table 6.2: Summary of different kinds of goods based on rivalry and excludability

Quasi-public goods (Supplementary material)

Some goods do not fit neatly into the category of private goods or public goods. They can be considered to be 'impure' public goods, also known as 'quasi-public goods'. These goods are:

- non-rivalrous (like public goods), and
- excludable (like private goods).

Examples include public museums that charge an entrance fee and toll roads. All these are excludable because consumers must pay to use them. They are also non-rivalrous since use by one does not reduce availability for others. Since the price system can be made to work here to exclude potential users, they could be provided by private firms. However, they all have very large positive externalities, thus justifying direct government provision.

Government intervention to correct the market's failure to provide public goods

Direct government provision

We have seen that the market fails to allocate resources to the production of public goods. This means the government must step in to ensure that public goods are produced at socially desirable levels. Thus public goods are directly provided by the government, are financed out of tax revenues and are made available to the public free of charge (or nearly free of charge).

Government provision of public goods raises some issues of choice about (a) which public goods should be provided, and (b) in what quantities they should be provided. These issues are similar to what was noted earlier in connection with direct government provision and subsidies for goods with positive externalities. Limited government funds force choices on what public goods to produce, and each choice has an opportunity cost in terms of other goods and services that are foregone (sacrificed).

Here, too, the government must use economic criteria to decide which public goods will provide the greatest social benefits for a given amount of money. However, in the case of public goods, governments face a major additional difficulty in calculating expected benefits. With private goods that are provided or subsidised by the government, it is possible to make estimates of expected benefits by using the market price of the good. (Remember the market price reflects the benefits consumers receive and so reveals its value to consumers.) Therefore, the government can use the market price of private goods with positive externalities to estimate their value to consumers, but with public goods there is no such possibility as they are not produced by private firms and have no price.

This means the government must try to estimate the demand (or 'price') of public goods through such means as votes or surveys of people who are asked how much a good would be worth to them. This information is used in *cost–benefit analysis*, which compares the estimated benefits to society of a particular good with its costs. If the total benefits expected from a public good are greater than the total costs of providing it, then the good should be provided. If benefits are less than costs, then it should not be provided. Assuming that cost–benefit analysis indicates a public good should be provided, the decision on how much of it to provide is made by comparing marginal benefits with marginal costs: the public good should be provided up to the point where $MB = MC$.

Whereas the costs of providing a public good are relatively easy to estimate, there are clear difficulties in estimating benefits. A major difficulty arising with surveys is that people who really want something are likely to exaggerate its value. Therefore, cost–benefit analysis is a very rough and approximate method used to make choices about public goods.

In addition, it should be noted that just like in the case of positive externalities which invite direct government provision and subsidies for their correction, the provision of public goods is also political in nature, with different groups competing against each other, and with government sometimes susceptible to political pressures that influence their decisions.

Contracting out to the private sector

When the government provides a public good, it may either provide it itself directly, or it may contract it out to a private firm. **Contracting out** by the public sector to the private sector occurs when a government makes an agreement (or contract) with a private firm to carry out an activity that the government was previously doing itself. This is a practice that many governments around the world have been increasingly undertaking. Suppose the government would like to build a new highway system. Rather than build it itself by directly hiring engineers, workers, and supplying the materials, it may contract its construction out to a private construction firm.

Note that whether goods are provided directly by the government or contracted out, they are in both cases financed out of tax revenues.

The potential advantages of contracting out include the following:

- It is often done by competitive tendering (a competition between firms that would like to provide the government with the service) resulting in the selection of a provider that can offer the lowest cost.
- It is usually accompanied by detailed specifications regarding the activity that will be provided with criteria for measurement of the provider's performance, which may allow for better quality control.
- It provides access to a broader range of skills and technology of the private firm than the government is likely to have available itself.
- The private firm may be more flexible and innovative than the government.
- Due to all of the above it may be possible for the public goods provided to be better quality and less costly.

On the other hand, contracting out also has potential disadvantages:

- The government becomes less accountable for the public goods it provides.
- The government loses control over the services it has contracted out.
- The costs of contracting out may be greater than if the government had provided the public good itself, as the contracting private firms often charge high prices for their services.
- There is a risk that quality may be reduced because competition between firms on the basis of cost may lower the quality of the services provided.
- There is a risk of making a poor contract with a private sector firm, resulting in higher costs and lower quality, along with reduced accountability and control noted above.
- Contracting out needs to be monitored by the government, which adds to costs.

As a result of all of the above it is difficult to generalise about the effectiveness of contracting out as a method to provide public goods.

Moreover, it should be noted that contracting out only addresses issues of the quality, costs, skills, and so on discussed above; it does not address the difficulties faced by the government in making decisions about what public goods to produce and in what quantities, which as we have seen above depend on the very difficult problem of trying to determine expected benefits and costs of alternative public goods.

TEST YOUR UNDERSTANDING 6.3

1 Explain how the concepts of rivalry and excludability relate to the distinction between public goods and private goods.

2 Provide some examples of public goods, and outline how they relate to the concepts of non-rivalry and non-excludability.

3 Use the concepts of resource allocation and the free rider problem to explain how public goods are a type of market failure.

4 Discuss the difficulties of direct government provision of public goods.

5 Evaluate the government's policy option to contract out to the private sector the provision of public goods.

6.4 Asymmetric information (HL only)

LEARNING OBJECTIVES

After studying this section you will be able to:

- define all the terms appearing in **orange bold** in the text (AO1)
- distinguish between adverse selection and moral hazard (AO2)
- evaluate government responses to the problem of asymmetric information, including legislation and regulation, and provision of information (AO3)
- evaluate private responses to the problem of asymmetric information, including signalling and screening (AO3)

The competitive market mechanism presupposes that all firms and all consumers have complete information regarding products, prices, resources and methods of production. However, as we know from Chapter 2 this condition is hardly ever met in the real world where firms, consumers and resource owners find themselves in situations where information is missing. **Asymmetric information** refers to something more than just missing information; it refers to situations where buyers and sellers do not have *equal access* to information. In some cases buyers may have more information than sellers; in other cases the opposite holds where sellers have more information than buyers. As we will see, this usually results in an *underallocation of resources* to the production of goods or services, and therefore to allocative inefficiency.

We will study two different types of problems of asymmetric information: adverse selection and moral hazard.

Adverse selection

Adverse selection refers to situations where one party in a transaction has more information about the quality of the product being sold than the other party.

Adverse selection where information is available to sellers but not to buyers

Sellers often have information about the quality of a good or service that they do not make available to consumers. Sellers of used cars have information about the car's quality that they are unlikely to reveal to potential buyers if the car has a defect. In a free and unregulated market, sellers of food could sell products that are unsafe for human consumption, possibly leading to illness and even death. Sellers of medicines could sell unsafe medications that could be ineffective or dangerous. Individuals claiming to be doctors, some of whom have little or no training, could practise medicine and even surgery, resulting in huge costs in terms of human health and safety.

In a free unregulated market, the result is usually to underallocate resources to the production of the good or service. Consumers are likely to be aware of possible dangers to themselves, and will be cautious about buying the good or service, resulting in lower demand, less production and lower sales. However, if consumers are unaware of possible hidden dangers, such as with unsafe food or toys, there could result an overallocation of resources to the production of these goods and services.

Possible solutions to adverse selection where the seller knows more than the buyer

Government responses

Regulation

According to one method, governments can pass laws and regulations that ensure quality standards and safety features that must be maintained by producers and sellers of goods and services, such as food, medications, private schools, toys, buildings and all types of construction.

These methods are not without their difficulties. Legislation and regulation are time-consuming, bureaucratic procedures, which sometimes work to slow down economic activities. It takes a long time, for example, to test new medications and certify their safety, and is a costly process. Also, regulatory and quality control activities have very large opportunity costs. Just in the case of food safety control, which involves not only food and beverage products but also hygiene in restaurants, there is a huge number of products and service providers involved, who require regulation and monitoring from the level of the farm (regarding the kinds and amounts of chemical inputs) to the moment the food reaches the table.

Provision of information

Governments may also respond by directly supplying information to consumers, or by forcing producers to provide information, thus protecting consumers in their purchasing decisions. This may include information about the quality of medical care by different providers, about communicable diseases, crime rates by neighbourhoods, health hazards related to different activities, products or substances, nutritional labelling on foods, and so on. In some countries, particularly in Europe, governments provide fee schedules for services (such as legal, medical, architectural) to ensure that consumers receive a particular quality of services for a particular range of prices.

When the government is the provider of information, there are difficulties involving the collection and dissemination of all the necessary information to consumers, the accuracy of the information, as well as opportunity costs in providing the information. When a private seller/producer is the provider of the information, there are serious questions about whether information regarding all hazards in products or substances and materials used in products, or all information regarding the quality of services (whether legal, medical, financial, and so on) is accurate and complete.

Another problem is that it is sometimes not possible to eliminate an information asymmetry between sellers and buyers, because no matter what regulations and information are provided, there is still some room for the seller to hide some information from the buyer. In the areas of health care and law, doctors and lawyers have specialised, technical information about their clients that the clients themselves do not possess. Doctors and lawyers often use this information for their own private gain by selectively revealing information to their clients that causes them to demand more services than are necessary. This practice leads to what is known as 'supplier-induced demand', or demand that is induced (created) by the supplier, which would not have appeared if the client had equal access to information.

Licensure

In the case of doctors, most countries around the world have laws requiring doctors to be licensed, and a licence can only be obtained upon proof of adequate medical competence. Licensing is similarly required for many other professions in many countries, from teachers and lawyers to plumbers and electricians.

Some economists criticise licensing, because it may work to limit the supply of people in a profession, raising the price of their services and increasing their incomes at the expense of consumers who must pay higher prices (this refers to *market power* discussed in Chapter 7 at HL).

Private responses

Screening

Screening is a method used by the party with the limited information, in this case the buyer. The buyers may try to get more information about what they are buying, in other words they screen the product or the producer or seller of the product. Consumers may find information provided on the internet about reliable used car dealers, or they may informally ask friends for information about the quality of health care service providers, or legal service providers.

While screening may help in providing consumers with some missing information, it can by no means on its own solve the problem as it cannot provide systematic and complete information to match the information available on the seller's side.

Signalling

Signalling is a method used by the party that has more information, or in this case the seller. The purpose of signalling is to convince the buyers that the product being sold is of good quality. Common methods include the use of warranties, establishment of brand names that convey a feeling of reliability, and, in the case of used cars, making service records available or exhibiting cars in fancy show rooms.

The problem with signalling is that it is unlikely to provide full information to buyers, and it may even provide inaccurate or misleading information by sellers eager to promote and sell their products.

Adverse selection where information is available to buyers but not to sellers

Adverse selection where the buyer has information not available to the seller often arises in the area of insurance services, where the buyer of insurance has more information than the seller. This situation arises most often in the area of health insurance. Buyers of health insurance know more about the state of their health than sellers of insurance, and those with health problems are unlikely to tell the full truth to the insurance company. In a free unregulated market, adverse selection results in an underallocation of resources to health insurance services, as the insurance company reduces the supply of insurance to protect itself against having to provide insurance coverage to very high risks, or people who are more likely to become ill. Adverse selection also leads to high insurance costs for insurance buyers.

Possible solutions to adverse selection where the buyer knows more

Private responses

Private insurance companies usually protect themselves by offering a range of policies where the lower the cost of the insurance, the higher the deductible (out-of-pocket payments). This offers people choice, so that those who believe they have a low risk of getting sick can buy a low-cost policy with a higher deductible, while higher-cost policies with lower deductibles can be selected by people who believe they have high levels of health risk. This is actually a method of *screening* undertaken by the party with the limited information, or seller of insurance. The choice of high or low deductible given to the buyers of insurance is intended to screen them by indirectly providing information about their state of health to the seller of insurance.

However in practice, it does not work out this way. An important reason is that lower-income earners choose low-cost policies with high out-of-pocket payments because these are more affordable, *regardless of the*

state of their health. From the perspective of equity or fairness, this is undesirable because it discriminates against those on low incomes. Another reason is that in trying to protect themselves against high risks, insurance companies usually refuse to insure people above a certain age, as elderly people generally have a higher chance of becoming ill. The result is that those who mostly need health insurance coverage, who are poor people who cannot afford to buy health care in the private market and elderly people who are more likely to become ill, are left with little or no insurance coverage.

Government responses

To deal with this problem, government responses may take the form of direct provision of health care services at low or zero prices to an entire population, financed by tax revenues, thus ensuring that the entire population has health insurance coverage, such as in countries with a National Health Service (as in many European countries). Alternatively, they make take the form of social health insurance, which may cover a country's entire population (as in several European countries), or which selectively covers only certain vulnerable groups of the population (as in the United States). The benefit of these approaches, particularly in countries that offer insurance coverage to their entire population, is that no one in need of health care goes without it.

A potential problem with government-funded or social health care systems involves difficulties in controlling costs of providing health care and growing burdens on the government budget or social health insurance budget.

Moral hazard

Explaining moral hazard

Moral hazard refers to situations where one party takes risks, but does not face the full costs of these risks because the full costs of the risks are borne by the other party. It usually arises when the buyer of insurance changes his or her behaviour *after* obtaining insurance, so that the outcome works against the interests of the seller of insurance. For example, buyers of car theft insurance may be less careful about protecting their car against theft, because they know they will be reimbursed if someone steals their car. Some buyers of medical malpractice insurance (doctors) may be less careful about avoiding malpractice, because of the knowledge that malpractice costs will be covered by the insurer. Unemployment insurance may lead some people to be less hesitant about becoming unemployed, in the knowledge that their insurance will provide them with some income.

In all these cases, the buyers of insurance have information about their future intentions that is not available to the sellers of insurance. In a free, unregulated market, the result of moral hazard is to underallocate resources to the production of insurance services, as sellers of insurance try to protect themselves against higher costs due to the risky behaviour of the buyers of insurance.

Many economists have noted that the financial crisis that began in 2008 was partly a case of moral hazard. It is argued that many financial institutions made risky loans and engaged in other highly risky financial transactions because they believed that the government would support them in the event of difficulties (which, in fact, it did). (Note that this is not a matter of taking out an insurance policy in the strict sense of the term, but it is a sort of 'insurance' nonetheless, in that the government provides a kind of assurance of protection in the event that financial institutions face difficulties due to poor loan repayments.)

You may note that the term 'moral hazard' does not refer to unethical or immoral behaviour. It is simply a historical remnant of a very old insurance term that originally meant 'subjective'.

Responses to moral hazard

Problems of moral hazard in insurance are usually dealt with by the provider of insurance. This is often done by making the buyer of insurance pay for part of the cost of damages through deductibles (out-of-pocket payments). This is intended to make the insurance buyer face the consequences of risky behaviour, thus leading to less risky behaviour. It will be recalled from the discussion above that deductibles are a form of *screening*.

A problem with deductibles is that it has different effects depending on the income level of insurance buyers. As noted earlier in connection with adverse selection, private insurance companies usually offer a range of policies from which buyers can choose, where the lower the cost of the insurance, the higher the deductibles. Higher-income earners usually choose higher-cost policies with low deductibles, while lower-income earners choose low-cost policies with high deductibles because these are more affordable. This suggests that higher-income earners are more likely to engage in risky behaviour because they are offered more insurance protection, while lower-income earners are less likely to engage in risky behaviour.

In the financial area, moral hazard is dealt with through government regulation of financial institutions, intended to oversee and prevent highly risky behaviour. This

raises a whole set of issues regarding the types and degrees of government regulations that are required if these are to be effective. In general, following the onset of the global financial crisis in 2008 governments in Europe and the United States have taken steps to increase regulation of the financial sector, though there are concerns that this has not been enough, especially in the United States.

TEST YOUR UNDERSTANDING 6.4

1 a Using examples, identify two main types of information problems that give rise to market failure.

b Using the concept of allocative efficiency, explain why information asymmetries represent market failure.

2 a Provide examples of information asymmetries where information is available to sellers but not to buyers.

b Explain how governments can intervene to correct these information asymmetries.

c Explain possible private responses to information asymmetries.

d Discuss some advantages and disadvantages of both government intervention and private responses.

3 a Provide examples of information asymmetries where information is available to buyers but not to sellers.

b Explain how governments can intervene to correct these information asymmetries.

c Discuss advantages and disadvantages of these types of government intervention.

6.5 Equity in the distribution of income and wealth (HL only)

LEARNING OBJECTIVES

After studying this section you will be able to:

- define all the terms appearing in **orange bold** in the text (AO1)
- explain why the free market results in an unequal distribution of income and wealth (AO2)
- draw a circular flow of income diagram to show why free markets give rise to income inequalities (AO4)

Why the free market economy results in income inequalities

In Chapter 1, we saw that *equity* refers to the idea of being fair and just, while *equality* is the state of being equal with respect to something. Therefore *income equality* would mean that everyone in a society receives the same amount of income. We also learned that while equity and equality have different meanings, in most countries the pursuit of equity is understood to refer to efforts to reduce significant inequalities in income and wealth. The reason for this is that people around the world generally share the belief or value judgement that the free market economy results in inequalities that are considered to be unfair. We will now see why this is so.

Our study of the circular flow model in Chapter 1 shows that in a market economy the amount of goods and services that households receive depends on their income, as this determines how much they can buy. Figure 1.4, showing the simple circular flow model (with no leakages and injections), appears below as Figure 6.8. We can see here that the income of households depends on payments they receive by selling the factors of production they own. Therefore, output and income distribution in a market economy depend on how many resources consumers (households) own and are able to sell in resource markets, as well as on the prices of the factors of production they sell.

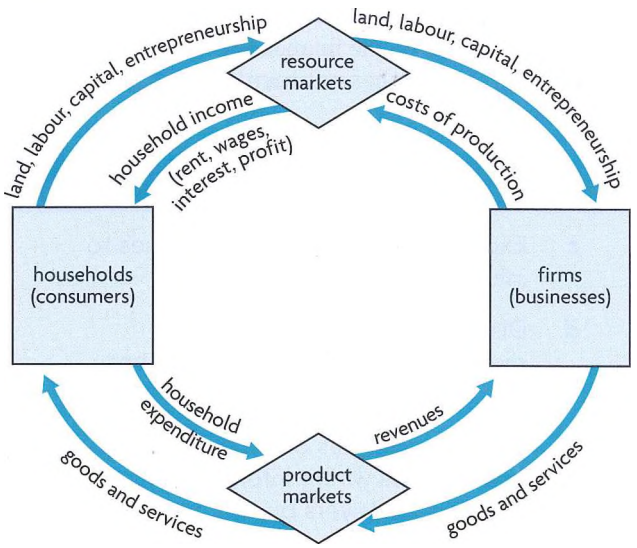


Figure 6.8: Circular flow of income model

The problem of income distribution arises because *ownership of factors of production is highly unequal, and because the prices of factors of production determined in the market vary enormously*. Most people have labour resources that they provide in labour markets, for which they receive wages. Yet some people are able to receive very high wages because of special skills and education or natural talents, while others who are less skilled, educated or talented may receive wages so low that they may be unable to cover the most basic needs for themselves and their families (food, shelter, clothing, etc.).

Further, there may be people who would like to work but cannot do so because they lack the kinds of skills firms want to hire, or because they are sick, or old, or have special needs that prevent them from working, or because there simply are not enough jobs in the economy to provide work for everyone. Then there are some individuals who possess some of the additional factors of production of land, capital and entrepreneurial abilities, for which they receive rental, interest and profit income. These additional factors of production are generally highly unequally distributed. As a result, the market-determined distribution of income in an economy is likely to be very unequal, with some people receiving far larger shares than others.

It follows that markets cannot ensure that everyone in a population will secure enough income to satisfy their basic needs. Most societies consider this to be a disadvantage of the free market economy, because of the belief held by most people that everyone in a population should be able to satisfy a minimum of basic human needs. Governments around the world therefore use a variety of methods to change the market-determined distribution of income and output, and arrive at a more socially desirable outcome. This is referred to as *redistribution*. We will come back to the topic of income distribution in Chapters 12 and 20.

Inequalities in wealth

As we have seen in connection with the circular flow model, income refers to the flow of money that is received by the owners of the factors of production. **Wealth**, on the other hand, refers to the money or things of value that people own, such as savings deposits (money saved in a bank); stocks in the stock market; bonds; land, houses and other property; valuable paintings or jewellery, and so on; minus debt to banks or other financial institutions.

Income gives rise to the possibility of saving, which can then be used to create wealth, and so wealth and income are related to each other as would be expected. Yet they are distinct and are not always closely linked together. For example, some people may have high incomes but low savings because they spend a lot, in which case they will have relatively low wealth. On the other hand, there may be people who have inherited wealth, or pensioners who have saved over a lifetime of work, who have high levels of wealth but relatively low incomes.

In general, however, we can say that the higher the income, the greater the possibilities for saving and for accumulating wealth. Therefore, just as the free market economy results in income inequalities, so too it results in wealth inequalities.

We will come back to the topic of inequalities in income and wealth in Chapter 12, where we will discover that *inequalities in wealth are in fact far greater than inequalities in income*.

Inequality in income and wealth in relation to market failure

It should be noted that the inability of the market to ensure that everyone in a population will secure enough income to satisfy their basic needs *is not strictly speaking market failure*. This follows from the way that market failure is defined. As we know, market failure is the inability of the market to achieve allocative efficiency, where $MSB = MSC$, or where social surplus is maximum. This bears no connection with income or

wealth distribution. It is in fact possible to have allocative efficiency and maximum social surplus either with complete income equality or with extreme inequality, where for example a large portion of the population is starving. This peculiarity will be discussed in the Theory of knowledge 6.1.

TEST YOUR UNDERSTANDING 6.5

1 Explain, using examples, the difference between equity and equality in income wealth distribution.

2 Using the circular flow of income diagram, explain why the market system cannot ensure that everyone in a society will be able to secure an adequate income to satisfy basic needs.

THEORY OF KNOWLEDGE 6.1

Inequality in relation to market failure: the possibility of maximum social surplus in the presence of extreme inequalities

It was noted above that inequalities in income and wealth are not a type of market failure, because it is possible to have any degree of inequality where there is at the same time allocative efficiency with maximum social surplus.

Yet you may wonder, how can there be maximum social surplus with extreme inequality? The answer to this lies partly in the positive-normative distinction we discussed in Chapter 2. If you revisit Theory of knowledge 2.1 in Chapter 2 (at the end of Section 2.5), you will be reminded that maximum social surplus (or welfare) refers to the *what/how much to produce* and *how to produce* questions of economics, and the idea of making the best possible use of resources, which are in the sphere of positive economics. Equality and inequality on the other hand refer to the *for whom to produce* question, which belongs to the sphere of normative economics.

Yet this by itself is not a satisfactory explanation, especially if we remember that allocative efficiency can be defined as *producing the combination of goods mostly wanted by society*. In Chapter 2, Section 2.5, we saw that allocative efficiency is achieved when the economy allocates its resources *so that the benefits from consumption are maximised for the whole of society*.

This clearly gives rise to a puzzle. How can benefits from consumption be maximised for the whole of society if there is extreme inequality with a large part of the population starving?

We can find the answer to this puzzle in the definition of *demand*, which we studied in Chapter 2 (Section 2.1). As you may recall, demand shows the various quantities of a good that a consumer is willing and *able to buy* at different possible prices, *ceteris paribus*. The answer to our puzzle lies in the *able to buy* part of this definition. If consumers have little or no income, they are not able to buy the good, and therefore they do not have any demand for it. In other words, no demand will show up in the market for the good from consumers with little or no income, however much they may want to buy it.

We now have the answer to the puzzle. As we know, the condition $MSB = MSC$ is equivalent to maximum social surplus (or welfare). Alternatively, we can say that $MB = MC$ is equivalent to maximum social surplus when there are no externalities. But MB is simply the demand curve! And MSB is also simply a demand curve for the whole of society that accounts for any possible external costs or benefits! This means that when we say that there is allocative efficiency and maximum social surplus when *the benefits from consumption are maximised for the whole of society*, we are simply talking about that part of society that has a demand because consumers are not only willing but also *able to buy* the good! Consumers from the rest of society with little or no income and hence no demand are bypassed, ignored and forgotten.

CONTINUED

In other words, allocative efficiency is defined on the basis of demand from those consumers who have enough income to make their preferences felt in the market.

Thinking points

The work of the classical economists, such as Adam Smith who we met in Chapter 1, was not based on the positive-normative distinction. Economists at the time, as you may remember were moral philosophers, and Adam Smith was deeply concerned with issues of ethics. It was only in the early 20th century that economists became influenced by the methods of the natural sciences and the idea that *facts* should be kept distinct from *values*. Yet in more recent years, some economists, such as Amartya Sen, an Indian Nobel Prize winning economist for his work on Economic Development, argue that it does not make sense for economists to maintain this strict separation between facts and values. Sen's work reintroduces ethics and values into economic analysis and mixes them to create a unified whole. We will encounter Amartya Sen in Chapter 18.

- In your study of economics, you are encouraged to consider the positive-normative distinction. Do you agree with the idea that facts (positive) should be kept distinct from values (normative) in the study of economics? Is it even possible to always make this distinction, or do values creep into the work of economists? We will come back to this point in Theory of knowledge features 9.1, 10.1, 15.1 and 18.1.
- Consider what redistribution of income leading to reduced inequalities in an economy implies for the concept of allocative efficiency. In particular, consider that people who previously had no demand for a good or service may now have such a demand based on their higher income. What is likely to happen to the efficient level of output of particular goods and services?

INQUIRY AND REFLECTION

The following questions will help you reflect on your learning and enhance your understanding of key topics in this chapter. They are suggestions for inquiries that you can undertake on your own or in groups in order to revise the learning objectives of this chapter.

1 Identify a good or service in your country or a country you are interested in that gives rise to a particular positive consumption externality. Examine the external benefits and their effects on stakeholders, investigate solutions in use, evaluate the solutions and identify effects on stakeholders.

2 Identify a public good that has been contracted out to the private sector in a country you are interested in and investigate the advantages and disadvantages that have emerged.

3 Investigate a situation where asymmetric information involves adverse selection or moral hazard and identify public and private solutions that have been used to address it.

EXAM STYLE QUESTIONS

You can find questions in the style of IB exams in the digital coursebook.



› Chapter 7

Market failure and socially undesirably outcomes III:

Market power (HL only)

BEFORE YOU START

- In what ways do firms compete with each other in order to increase their sales?
- When firms compete with each other to increase sales, some firms might have advantages over others. Can you think of what some of these advantages may be?
- From the consumer's point of view can you think of some advantages and disadvantages of firms that grow to a very large size?

This chapter is concerned with the relationship between the behaviour of firms and market failure. We will begin by introducing the fundamental concepts of revenues, costs and profits needed to study firm behaviour. We will then see how firm behaviour within particular markets gives rise to market power, which is linked to an important kind of market failure.

Photo: London, England. Piccadilly at night

7.1 Introduction to firms, industries and market structures

LEARNING OBJECTIVES

After studying this section you will be able to:

- define all the terms appearing in **orange bold** in the text (AO1)
- explain the main characteristics of perfect competition – many firms, free entry, homogeneous products (AO2)
- explain the main characteristics of monopoly – single or dominant firm, high barriers to entry, no close substitutes (AO2)
- explain the main characteristics of market structures in imperfect competition (AO2)
  - varying degrees of market power, firm is price maker
  - monopolistic competition – many firms, free entry, product differentiation
  - oligopoly – few large firms, high barriers to entry, interdependence
- explain the meaning of market power (AO2)

Market power and market failure

As you know from past chapters, a *firm* (or business) is an organisation that employs factors of production to produce and sell a good or service. A group of one or more firms producing identical or similar goods or services is an *industry*. The car industry consists of car manufacturers (Ford, Honda, Mercedes, etc.); the shoe industry consists of shoe manufacturers; the banking industry consists of providers of banking services (banks). Industries are analysed by use of models called *market structures*, which describe characteristics of a market organisation that determine the behaviour of firms.

A fundamental distinction between market structures is *the extent to which each individual firm in the industry is able to control the price at which it sells its product*. This is known as **market power**.

On the basis of this idea, we can distinguish between:

- the extreme of **perfect competition**, where firms have no ability to control the price of their products; such firms have *zero market power*
- the extreme of **monopoly**, where there is single firm in the market; this firm has the greatest ability to control the price of its product, and therefore *the greatest amount of market power*.

In between the two extremes are the two other market structures where *firms face some competition but also have market power*, and for this reason they are known as **imperfect competition**; they include:

- **monopolistic competition**
- **oligopoly**.

Market power is important because it gives rise to market failure, or the failure to achieve allocative efficiency. The greater the market power, the greater the allocative inefficiency.

> Firms in perfect competition, which have zero market power, do not result in market failure, and are the only market structure where firms achieve allocative efficiency. Firms in the other three market structures result in market failure to varying degrees, with monopoly resulting in the greatest degree of allocative inefficiency.

Introduction to market structures

Apart from differences in their ability to control price (market power), market structures can be defined on the basis of three main characteristics:

- the *number of firms* in the industry
- how similar or different are the goods or services produced, known as **product differentiation**, meaning that the products can be made different from each other, or not identical
- how easy or difficult it is for new firms to enter the industry and begin producing, which depends on whether there are **barriers to entry**; these include anything that can prevent a firm from entering an industry and beginning production. When there are no barriers to entry into an industry we say there is **free entry**.

Characteristics of perfect competition

- There is a very large number of firms in the industry.
- All the firms in the industry sell homogeneous products; this means the products are identical (undifferentiated); there are no brand names
- There are no barriers to entry; any firm that wants to enter the industry and begin producing the good or service can do so freely.

Perfect competition is highly unrealistic and it is difficult to find markets in the real world that meet the characteristics above. Examples that come the closest include the international markets for agricultural commodities such as wheat, rice, corn and livestock, other commodities like silver and gold, stock and bond markets, and the foreign exchange market (in which currencies of different countries are bought and sold). In these cases, each firm is small relative to the total number of firms so that each one does not have much ability to influence its price. Further, the product sold by all firms is identical or nearly identical, and any new firm is free to enter the industry.

Characteristics of monopoly

- There is a single seller or dominant firm in the industry, so the single firm is the entire industry. In the real world, a monopolistic industry may consist of one firm that dominates the market with a very large market share. For example, Monsanto controls 80% of the genetically modified corn market and is considered to be a monopoly.
- The firm produces and sells a unique good or service, with no close substitutes; if substitute goods existed, then consumers could switch to buying the substitute, and there would no longer be a monopoly for the good.
- There are high barriers to entry in the industry; the monopolist owes its dominance in the market partly to the inability of other firms to enter.

Examples of monopolies include telephone, water and electricity companies in areas where they operate as a single supplier. (These are known as natural monopolies to be discussed below.) These may be in local areas in which case the firm is a local monopoly, or it may involve a national market if it covers all residents of a country. A postal service is another example of a monopoly with a national market. On the other hand, Microsoft Corporation has a monopoly in operating system software with Windows, with roughly 75% of laptops and desktops running Windows in an international market. In all these cases, there is a single firm providing the entire, or the largest part of the market; and it is extremely difficult for other firms to enter the industry. Being the sole producer, the monopolist has significant control over the price of its good or service. Monopolies are not frequently encountered in the real world.

Characteristics of monopolistic competition

- There is a fairly large number of small (or medium-sized) firms in the industry.
- There are no barriers to entry; any new firm can enter the industry.
- There is product differentiation; each firm tries to make its product different from those of the other firms in the industry in terms of various characteristics like the quality, servicing or packaging.

Examples of monopolistic competition include the shoe, clothing, detergent, computer, publishing, furniture and restaurant industries. Monopolistic competition is like perfect competition in that there are many firms in the industry. It differs from perfect competition mainly because of product differentiation. By trying to make its good or service different from any other, each firm tries to be like a little monopoly. This is possible because each firm is the only producer of that particular version of the product. For example, Nike is a monopolist of Nike shoes, and Reebok is a monopolist of Reebok shoes. These firms use product differentiation to gain some control over the price of their products. However, the existence of other similar products (such as other brands of shoes) limits the degree of its power to control the market prices.

Characteristics of oligopoly

- There is a small number of large firms in the industry; because of their small number, the firms are *interdependent*, because the actions of one firm affect the others. This means that each firm tries to predict what the rival firms will do.
- Products may be either differentiated (cars) or undifferentiated (oil).
- There are high barriers to entry; it is difficult for a new firm to enter the industry.

Examples of oligopolies include the car industry, airlines, electrical appliances, electronic equipment (differentiated products) and the oil, steel, aluminum, copper and cement industries (identical or undifferentiated products). Oligopolies have significant control over price. Most industries in the real world are monopolistically competitive or oligopolistic.

Market power and competition

As you learned in Chapter 2, Section *The meaning of competitive markets,* competition occurs when there are many buyers and sellers acting independently, so that no one has the ability to influence the price of a product. Therefore, by definition, market power is the opposite of competition. This can be seen in Table 7.1 which summarises the characteristics of the four market structures, listed in order of increasing market power. The greater the market power, the lower the competition, and the greater the control over price.

Why we study perfect competition

We have seen that perfect competition is a highly unrealistic market structure. Why then do we bother to study it?

It is because it is the only market structure where allocative efficiency is achieved, in other words the only one where the market does not fail. Also, perfect competition sheds light on the benefits of competition in a market economy. Even though competition may not be perfect in the real world, as long as it exists in some form it offers important benefits to the workings of the market system. Perfect competition therefore provides a standard that we can use to evaluate the other three market structures.

Yet firms in perfect competition are also subject to important limitations. By contrast, the other three market structures, while not achieving allocative efficiency, often offer advantages which perfect competition cannot provide. It is therefore important to examine the advantages and disadvantages of each of the four market structures.

TEST YOUR UNDERSTANDING 7.1

1 Explain the relationship between competition and market power, and how the four market structures relate to each other with respect to these features.

2 Define each of the market structures on the basis of their key characteristics.

3 Identify examples of industries that belong to each of the four market structures.

	Number of firms	Type of product	Ease of entry into the industry	Examples	Market power/ control over price	Degree of competition
Perfect competition	very many, very small	homogeneous/ undifferentiated	very easy	agriculture	none	perfect
Monopolistic competition	relatively many, relatively small	differentiated	very easy	restaurants, computer games, books, furniture	some	a good amount
Oligopoly	few large	differentiated or undifferentiated	very difficult	steel, aluminum, cars, household appliances	significant	some (but far less than monopolistic competition)
Monopoly	one, large	one product with no close substitutes	difficult to impossible	public utilities	very significant	none

Table 7.1: Characteristics of market structures listed in order of increasing market power

7.2 Profit maximisation by the rational producer

LEARNING OBJECTIVES

After studying this section you will be able to:

- define all the terms appearing in **orange bold** in the text (AO1)
- explain cost, revenue and profit concepts (AO2)
  - total revenue, average revenue, marginal revenue
  - total cost, average cost, marginal cost
  - abnormal profit, normal profit, loss
- calculations from data of profit, marginal cost, marginal revenue, average cost, average revenue (AO4)

To understand market structures, we must first examine some important concepts relating to revenues, costs and profit.

Revenues

Total revenue, average revenue and marginal revenue

Revenues are the payments firms receive when they sell the goods and services they produce. There are three fundamental revenue concepts: total, average and marginal revenue.

The firm's **total revenue** (TR) is obtained by multiplying the price at which a good is sold (P) by the number of units of the good sold (Q):

$$TR = P \times Q$$

The firm's **average revenue** (AR) is revenue per unit of output sold, or total revenue (TR) divided by units of output (Q):

$$AR = \frac{TR}{Q}$$

Note that AR *is always equal to* P, or the price of the product. The reason is that since

$$TR = P \times Q, P = \frac{TR}{Q}, \text{therefore } P = AR.$$

The firm's **marginal revenue** (MR) is the additional revenue arising from the sale of an additional unit of output.

$$MR = \frac{\Delta TR}{\Delta Q}$$

Revenues in two different types of markets

Whereas the *definitions* of revenues apply to all firms, the *analysis* of revenues is not the same, because this depends on whether or not the firm has any ability to control its price. We must therefore make a distinction regarding revenues where:

- the firm is unable to control price; price is constant as output varies: perfect competition
- the firm has control over price; price varies with output: monopolistic competition, oligopoly, monopoly.

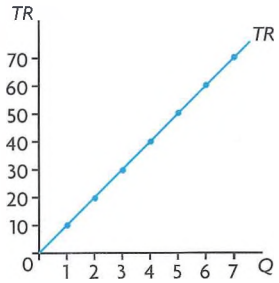
Revenue curves for the firm that cannot control price: perfect competition

Table 7.2 shows total, marginal and average revenue based on information on the price and quantity of the good in situations where a firm cannot control price. Columns 3, 4 and 5 are calculated from the data in columns 1 and 2, using the definitions given above. Note that *the price at which the good is sold does not change*; this occurs only under perfect competition. Figure 7.1 plots the data of Table 7.2.

1 Units of output (Q)	2 Product price (P) (€)	3 Total revenue $TR = P \times Q$ (€)	4 Marginal revenue $MR = \frac{\Delta TR}{\Delta Q}$ (€)	5 Average revenue $MR = \frac{TR}{Q}$ (€)
0	10	–	–	–
1	10	10	10	10
2	10	20	10	10
3	10	30	10	10
4	10	40	10	10
5	10	50	10	10
6	10	60	10	10
7	10	70	10	10

Table 7.2: Calculating total, marginal and average revenue when price is constant: the firm has no control price; the case of perfect competition

a Total revenue



b Marginal and average revenue

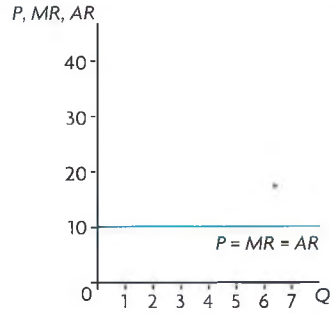


Figure 7.1: Total, marginal and average revenue curves when price is constant: the firm has no control over price; the case of perfect competition

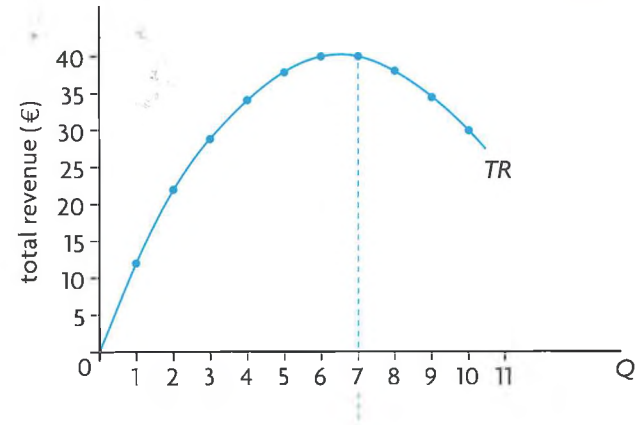
Revenue curves for the firm that can control price: monopoly, monopolistic competition, oligopoly

Table 7.3 shows total, marginal and average revenue from price and quantity information in the case where the firm has some influence over price. The method of calculation is exactly the same as in the competitive case above, where columns 3–5 are calculated using the information of the first two columns. The difference is in the price data, appearing in column 2, showing that *the price at which the good is sold changes as the quantity of output changes*. The lower the price the greater the quantity of output, illustrating the law of demand. This occurs under all market models that we will study other than perfect competition. Figure 7.2 plots the data of Table 7.3.

1 Units of output (Q)	2 Product price (P) (€)	3 Total revenue ($TR = P \times Q$) (€)	4 Marginal revenue $MR = \frac{\Delta TR}{\Delta Q}$ (€)	5 Average revenue $AR = \frac{TR}{Q}$ (€)
0	–	–	–	–
1	12	12	12	12
2	11	22	10	11
3	10	30	8	10
4	9	36	6	9
5	8	40	4	8
6	7	42	2	7
7	6	42	0	6
8	5	40	–2	5
9	4	36	–4	4
10	3	30	–6	3

Table 7.3: Calculating total, marginal and average revenue when price varies: the firm has some control over price; the cases of monopoly, monopolistic competition, oligopoly

a Total revenue



b Marginal and average revenue

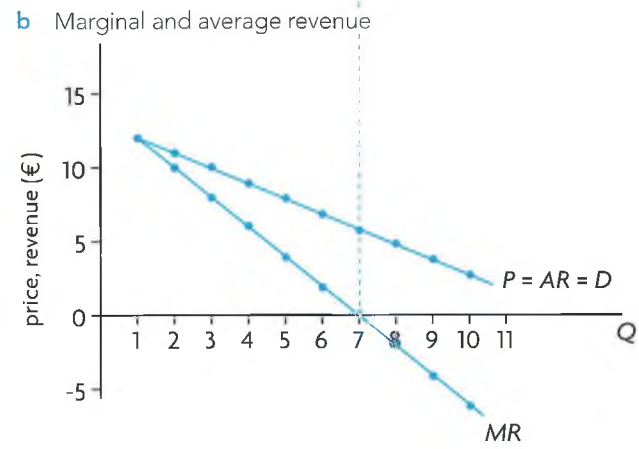


Figure 7.2: Total, marginal and average revenue curves when price varies: the firm has some control over price; the cases of monopoly, monopolistic competition, oligopoly

Calculating revenues

To calculate revenues, it is only necessary to remember and understand the relationships between the various revenue concepts, beginning with the idea that $TR = P \times Q$. If you are given data on P and Q, you can find TR, and from there you can calculate $AR = \frac{TR}{Q}$ and $MR = \frac{\Delta TR}{\Delta Q}$. You can also work backwards to find TR and MR if you know AR and Q, or to find TR and AR if you know MR and Q. Remember also that $AR = P$ in all cases, so that if you know AR, you also know the price of the product.

To calculate revenues from a diagram, we simply read off the information appearing in the graph, and apply exactly the same principles as above to calculate the revenue variable or variables we are interested in.

Costs of production

The meaning of economic costs

When firms use inputs or resources to produce, they incur **costs of production**, which include money payments to buy resources plus anything else given up by a firm for the use of resources. The resources include land, labour, capital and entrepreneurship (see Chapter 1).

When the firm uses resources it does not own, it buys them from outsiders and makes payments of money to the resource suppliers. For example, a firm hires labour and pays a wage; it purchases materials and pays the price to the seller. Such payments made by a firm to outsiders to acquire resources for use in production are known as *explicit costs*.

On the other hand, when the firm uses resources it owns, there is still a cost, which consists of the income that is sacrificed when the firm uses such a self-owned resource. For example, in the case of an office building owned and used by the firm, the cost is the rental income that could have been earned if the building were rented out. You can think of this as an opportunity cost. The hours of work a firm owner puts into his or her own business have an opportunity cost equal to what the firm owner could have earned if s/he had worked elsewhere. The entrepreneurial abilities the firm owner puts into the business (risk-taking, innovative, organisational and managerial abilities) involve a further opportunity cost equal to what these abilities could have earned elsewhere. The sacrificed income arising from the use of self-owned resources by a firm is an *implicit cost*.

TEST YOUR UNDERSTANDING 7.2

1 Define

a total revenue,

b marginal revenue, and

c average revenue.

2 Given the following price and quantity data for a product, calculate total revenue, marginal revenue and average revenue.

Price ($)	5	5	5	5	5
Quantity (thousand units)	0	1	2	3	4

3 Given the following price and quantity data for a product, calculate total revenue, marginal revenue and average revenue.

Price ($)	8	7	6	5	4	3	2
Quantity (thousand units)	2	3	4	5	6	7	8

4 a Outline what can be concluded about how price changes (or does not change) for each unit of output sold in questions 2 and 3.

b Explain the relationship between price and average revenue.

5 Given the following data, calculate total revenue and marginal revenue for each level of output. Identify the price at each level of output.

Quantity (thousand units)	1	2	3	4	5	6
Average revenue (€)	20	18	16	14	12	10

6 Given the following data, calculate total revenue and average revenue for each level of output. Identify the price at each level of output.

Quantity (thousand units)	1	2	3	4	5	6
Marginal revenue (£)	14	12	10	8	6	4

The sum of explicit and implicit costs incurred by a firm for its use of resources, whether purchased or self-owned, are known as *economic costs*. When economists refer to 'costs' they mean 'economic costs'.

Costs of production in the short run

In Chapter 2 (Section *Assumptions underlying the law of supply*), you were introduced to costs of production. (It is suggested that you reread this section before continuing further.) Very briefly, the relevant parts from that section include the following concepts:

- short run = the period of time when at least one factor of production is fixed
- long run = the period of time when all factors of production are variable
- total cost = all costs of production incurred by a firm
- marginal cost (MC) = the extra or additional cost of producing one more unit of output.

To calculate MC, we use the formula

$$MC = \frac{\Delta TC}{\Delta Q}.$$

We now need to introduce one more cost concept: **average cost** (AC), which is cost per unit of output produced, or total cost (TC) divided by units of output (Q):

$$AC = \frac{TC}{Q}$$

Table 7.4 below is the same as Table 2.5 in Chapter 2 (where you learned how to calculate marginal cost) and in addition includes a new column showing *average cost*.

The relationship between average cost and marginal cost in the short run

Figure 7.3 plots marginal cost and average cost based on the data that appear in Table 7.4. As explained in Chapter 2 the MC curve is based on the law of diminishing marginal returns (Section *Assumptions underlying the law of supply*). The same applies to the AC curve. You may recall that diminishing returns holds only in the short run when at least one factor of production is fixed. Therefore *our analysis of costs in this section clearly applies to the short run.*

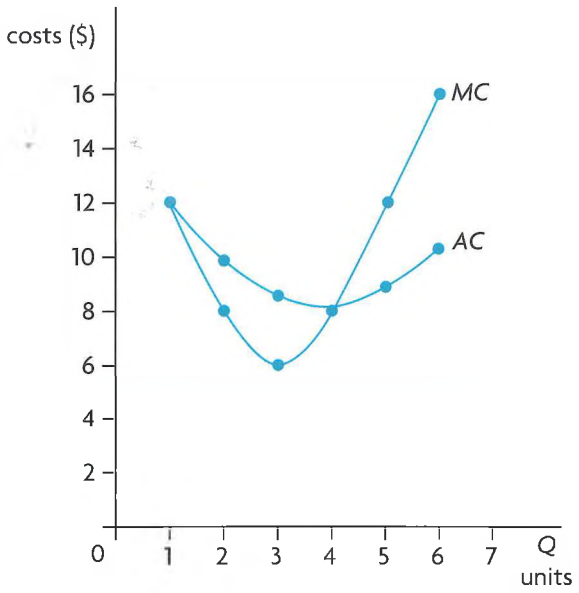


Figure 7.3: Marginal cost and average cost in the short run

Total product (number of units, Q)	Total cost (TC) ($)	Marginal cost (MC) ($)	Average cost (AC) ($)
1	12	12	12
2	20	8	10
3	26	6	8.67
4	34	8	8.5
5	46	12	9.2
6	62	16	10.33

Table 7.4: Total cost, marginal cost and average cost

It is important to note the relationship between the average cost and marginal cost curves: when the marginal cost curve lies below the average cost curve, ($MC<AC$), average cost is falling; and when the marginal cost curve lies above the average costs curve ($MC>AC$), average cost is increasing. *This means that the marginal cost curve always intersects the average cost curve when this is at its minimum*. The reason lies in the mathematical relationship between the average and marginal values of any variable.

Consider a simple example involving test scores. Say you have an average of 80 in your tests. If your next test score (the 'marginal' score) is greater than your average of 80, your average will increase. If your next test score is lower than your average of 80, your average will fall. The relationship between average and marginal test scores is exactly the same as the relationship between average and marginal costs.

Calculating short-run costs of production

You can calculate costs easily if you remember and understand the relationships between the cost concepts we have just discussed. If you have information on units of output (Q) and total cost (TC), you can find $AC = \frac{TC}{Q}$ and $MC = \frac{\Delta TC}{\Delta Q}$. You can also work backwards to find TC and MC if you know AC and Q, or to find TC and AC if you know MC and Q.

To calculate costs from a diagram, you can just read off the information appearing in the graph, and apply exactly the same principles as above to calculate the cost variable or variables you are interested in.

TEST YOUR UNDERSTANDING 7.3

1 Define

- a total cost,
- b marginal cost, and
- c average cost.

2 Given the following quantity and total cost data for a product, calculate average cost and marginal cost.

Quantity (thousand units)	2	5	9	14	18	21	23	24
Total cost (€)	300	400	500	600	700	800	900	1000

CONTINUED

3 Given the following data, calculate total cost and marginal cost for each level of output.

Quantity (thousand units)	1	2	3	4	5
Average cost (€)	180	140	133	135	140

4 Given the following data, calculate total cost and average cost for each level of output.

Quantity (thousand units)	1	2	3	4	5
Marginal cost (£)	14	12	10	12	14

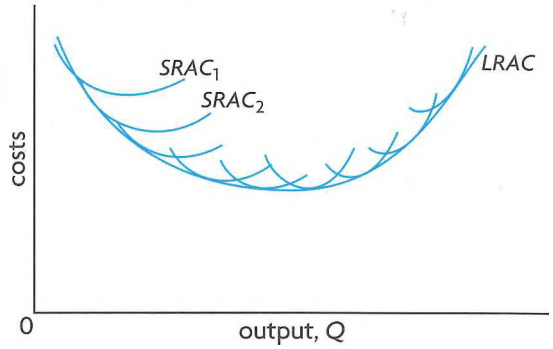
Costs of production in the long run and economies of scale

You may remember that the long run is the period of time when the firm varies (changes) all its factors of production. We are interested in examining how the firm's average costs, or its costs per unit of output, change when it grows larger by *increasing all of its factors of production*.

At any moment in time, when the firm is in the short run, it has a particular short-run AC curve; we can call this its $SRAC$. When it grows over time, we can think of it as going into the long run, increasing all its factors of production, and then going into a new short run with a new $SRAC$. Imagine this process repeating itself again and again: the firm goes from one short-run position to another and then to another, and these short run positions are connected to each other through moments when the firm enters the long run in order to increase all of its factors of production. This process is shown in Figure 7.4(a), where we see that as the firm grows bigger, it has a series of $SRAC$ curves, $SRAC_1$, $SRAC_2$, and so on. Imagine now an infinite number of $SRAC$s; they will trace out the *long run average cost curve*, or $LRAC$, which is the curve that just touches (is tangent to) each of the short-run curves.

We can see that the $SRAC$ curves keeps shifting to the right, which is what we would expect since there is more and more output (Q) being produced. But note that the $SRAC$ curves not only move to the right, but also at first move downward, and after a point they move upward. It is this U-shape of the $LRAC$ curve that we want to examine.

a Long-run average cost curve in relation to short-run average cost curves



b Economies and diseconomies of scale

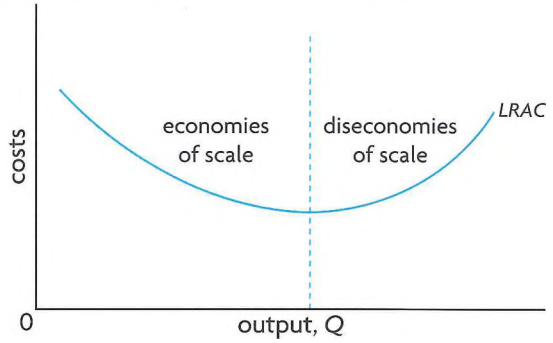


Figure 7.4: The long-run average cost curve

The reasons for the U-shape of the *LRAC* have nothing to do with diminishing marginal returns, which are a feature of short-run production and costs. The U-shape of the *LRAC* curve can be found in *economies and diseconomies of scale*.

Economies of scale are decreases in average costs of production over the long run as a firm increases all its factors of production. They explain the downward-sloping portion of the *LRAC* curve.

Some reasons why this occurs include:

- **Specialisation of labour.** As the scale of production increases, more workers must be employed, allowing for greater labour *specialisation*. Each worker specialises in performing tasks that make use of skills, interests and talents, thus increasing efficiency and allowing output to be produced at a lower average cost.
- **Specialisation of management.** Larger scales of production allow for more managers to be employed, each of whom can be specialised in a particular area (such as production, sales, finance, and so on), again resulting in greater efficiency and lower average cost.
- **Bulk buying of inputs (factors of production).** As quantities of inputs purchased increase, the price per unit drops.
- **Financing economies.** Larger firms may have lower interest rates, thus contributing to lower costs per unit of output.
- **Spreading of certain costs, such as marketing, over larger volumes of output.** Costs of certain activities such as marketing and advertising, design, research and development, result in lower average costs if they can be spread over large volumes of output.

Diseconomies of scale, on the other hand, are increases in the average costs of production in the long run as a firm increases its output by increasing all its inputs. They are responsible for the upward-sloping part of the *LRAC* curve: as a firm increases its scale of production, average costs increase.

Reasons for diseconomies of scale include:

- **Co-ordination and monitoring difficulties.** As a firm grows larger, its management may run into difficulties of co-ordination, organisation, co-operation and monitoring. The result involves growing inefficiencies causing average costs to increase as the firm expands.
- **Communication difficulties.** A larger firm size may lead to difficulties in communication between various component parts of the firm, resulting in inefficiencies and higher average costs.
- **Poor worker motivation.** If workers begin to lose their motivation, to feel bored and to care little about their work, they become less efficient, resulting in higher average costs.

Profits

The meaning of profit

Standard economic theory assumes that firms display rational behaviour by trying to maximise their profits, meaning that they try to make their profits as large as possible (Chapter 2). In general,

profit = revenues – costs of production

In economics, profit is defined as:

profit = total revenue – economic costs
= total revenue – the sum of explicit costs + implicit costs

since economic costs are the sum of explicit plus implicit costs.

Profit maximisation

Standard economic theory of the firm assumes that firm behaviour is guided by the firm's goal to maximise profit. **Profit maximisation** involves determining the level of output that the firm should produce to make profit as large as possible.

Yet firms do not always make a profit; in some cases, their total revenue is not sufficient to cover all costs, in which case they make a **loss**, which can be thought of as negative profit. If a firm is making a loss, it may eventually go out of business, but until it decides to shut down, it will be interested in producing the quantity of output that will make its loss as small as possible. Therefore, the theory of the firm is also concerned with how much output a loss-making firm should produce in order to minimise its loss.

There are two approaches to analysing profit maximisation (or loss minimisation): one involves use of *total revenues and total costs* and the other involves use of *marginal revenues and marginal costs*. Both these approaches yield the same results for the profit-maximising (or loss-minimising) level of output.

Profit maximisation based on the total revenue and total cost approach

This approach is based on the simple principle that

profit = total revenue (TR) − total cost (TC)

where TC is the firm's *economic costs* (explicit plus implicit).

The firm's profit-maximisation rule is to produce the level of output where $TR - TC$ is as large as possible.

The amount of profit made by the firm is equal to the numerical difference between TR and TC.

Positive profit: $TR > TC$; the firm earns **abnormal profit**.

Zero profit: $TR = TC$; the firm earns **normal profit**.

Negative profit: $TR < TC$; the firm makes a **loss**

We will return to the terms *abnormal profit* and *normal profit*.

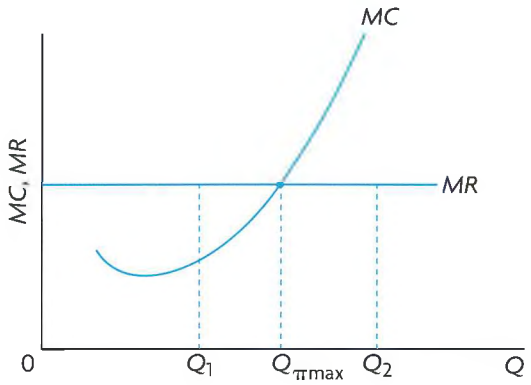
Profit maximisation based on the marginal revenue and marginal cost approach

Profit maximisation using this approach is based on a comparison of marginal revenue (MR) with marginal cost (MC) to determine the profit-maximising level of output. The firm's profit-maximisation rule (and loss-minimisation) is to choose to produce the level of output where $MC = MR$.

In Figure 7.5, both parts (a) and (b) show the standard MC curve that we studied above (as well as in Chapter 2 in connection with the firm's supply curve). As we know, there are two kinds of marginal revenue curves, depending on whether or not the firm has control over the price of its output. Part (a) shows the MR curve of the firm with no control over price. Part (b) shows the MR curve of the firm with some control over price. Both parts (a) and (b) illustrate the identical principle about profit maximisation.

According to the profit-maximising rule, $MC = MR$, the point of intersection between the MC and MR curves determines the profit-maximising level of output; this is $Q_{\pi max}$ in Figure 7.5(a) and (b). Why is this so? Consider a firm producing output Q_1 in both parts (a) and (b), where $MR > MC$. If this firm increases its output by one unit, the additional revenue it would receive (MR) will be greater

a Price constant



b Price varies with output

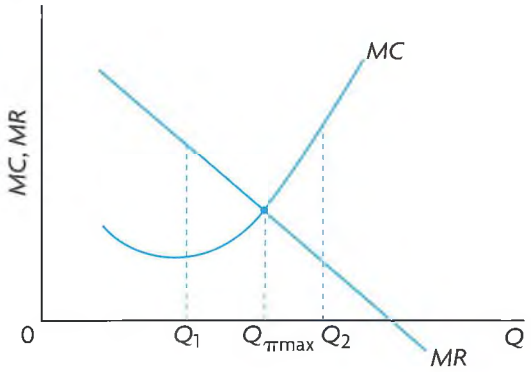


Figure 7.5: Profit maximisation using the marginal revenue and marginal cost approach

than its additional cost (MC). It is therefore in the firm's interests to increase its level of output until it reaches $Q_{\pi max}$ where $MR = MC$. If it continues to increase output beyond $Q_{\pi max}$, say to Q_2, where $MR < MC$, the additional revenue it would receive for an extra unit of output is less than the additional cost, and so it should cut back on its Q. There is only one point where the firm can do nothing to improve its position, and that is $Q_{\pi max}$, where $MR = MC$, and profit is the greatest it can be.

Firms maximise profits or minimise losses when they produce a quantity of output where $MC = MR$. This holds for all market structures.

When we are given data for MC and MR (and no information on total costs and total revenues), all we can do is find the profit-maximising level of output where $MC = MR$, but we cannot find the amount of profit (or loss) unless we have more information. We will see how this is done below when we study the market structures.

TEST YOUR UNDERSTANDING 7.4

1 a Identify the two approaches to profit maximisation by firms.

b Outline the profit-maximising rule of firms in each of the two approaches.

2 a Say a firm is producing a level of output Q where $MC > MR$. State what it should do to increase its profit (or reduce its loss).

b State what it should do if it is producing Q where $MC < MR$.

3 Assume that a firm that has no control over its price sells its output at $5 per unit.

a Given the following data, use the total revenue and total cost approach to determine the level of output at which the firm will maximise profit.

b Calculate how much profit it will make.*

c Calculate the amount of profit (or loss) when $Q = 3$, $Q = 6$, $Q = 10$.

Units of output (Q)	1	2	3	4	5	6	7	8	9	10
Total cost ($)	15	18	20	21	23	26	30	35	41	48

CONTINUED

4 Given the data in question 3

a determine the level of output at which the firm will maximise profit using the marginal revenue (MR) and marginal cost (MC) approach. (*Hint:* you must use the information in the question to find MR and MC.)

b Did you find the same profit-maximising level of output as in question 3(a)?

5 Suppose that a firm with some control over price faces the costs and prices per unit of output shown in the table below.

a Use the total revenue and total cost approach to determine the level of output at which the firm will maximise profit.

b Calculate how much profit will it make.*

c Calculate the amount of profit (or loss) when $Q = 2$, $Q = 3$, $Q = 8$.

Units of output (Q)	1	2	3	4	5	6	7	8
Total cost ($)	15	18	20	21	23	26	30	35
Price ($)	10	9	8	7	6	5	4	3

6 Given the data in question 5,

a determine the level of output at which the firm will maximise profit using the marginal revenue (MR) and marginal cost (MC) approach. (*Hint:* you must use the information in the question to find MR and MC.)

b Did you find the same profit-maximising level of output as in question 5?

*When using the TR and TC approach your results give two profit-maximising levels of output, whereas the MR and MC approach gives only one. This is because the MR and MC approach is actually more precise than the TR and TC approach. It is a good idea to use the larger of the two values of output that you get by using the TR and TC approach.

The meaning of normal profit

When profit is equal to zero, and total revenue is equal to total economic costs, the firm is said to be making normal profit.

Normal profit can be defined as the minimum amount of revenue that the firm must receive so that it will keep the business running (as opposed to shutting down). It can also be defined as the amount of revenue that covers all explicit and implicit costs. Therefore, a firm earns normal profit when total revenue = economic costs and profit = zero.

These apparently different definitions are in fact consistent: the minimum amount of revenue the firm must receive to make it worthwhile to stay in business is equal to the revenue that covers all of the firm's costs, implicit plus explicit.

Note that normal profit also *includes the payment for entrepreneurship*. Entrepreneurship, you will remember, includes the talents to organise and manage a business and take risks. Entrepreneurship receives a payment just as all other factors of production do, and this payment is included in normal profit. This means that if you are the owner of a pizza restaurant which is earning normal profit (meaning zero profit) you are still getting paid for all your work as entrepreneur in your business. Therefore you have no reason to close down your restaurant.

This means that a firm will continue to produce even if it is earning zero profit, meaning it is earning normal profit.

TEST YOUR UNDERSTANDING 7.5

1 Define profit and normal profit, and explain the difference between them.

2 Explain why profit can be positive, zero or negative.

3 A firm earns zero profit, and yet it does not shut down. Explain why.

7.3 Perfect competition

LEARNING OBJECTIVES

After studying this section you will be able to:

- define all the terms appearing in **orange bold** in the text (AO1)
- explain firms in perfect competition as price takers having no market power (AO2)
- explain profit maximisation in the short run and in the long run (AO2)
- explain the meaning of allocative efficiency in terms of its necessary conditions, $P = MC$ or $MB = MC$ or maximum social surplus (AO2)
- draw diagrams showing (AO4)
  - the perfectly competitive firm as a price taker where $P = D = AR = MR$
  - the perfectly competitive firm making abnormal profit, normal profit, loss
  - perfectly competitive market equilibrium showing allocative efficiency
- discuss the advantages and disadvantages of perfect competition (AO3)

Let's remind ourselves of the characteristics of perfect competition:

- There is a very large number of firms in the industry.
- All the firms in the industry sell a homogeneous or identical (undifferentiated) product.
- There are no barriers to entry into the industry.

Demand and revenue curves

The demand curve (average revenue curve) facing the firm

Suppose you have a farm that produces strawberries. You are one of many small strawberry producers, your strawberries are very similar to those produced by other strawberry farmers, and anyone who would like to produce strawberries can do so (there are no barriers to entry).

Table 7.5 summarises the cost, product, revenue and profit concepts we have studied.

Revenue concepts	Definition	Formula
Total revenue	The total earnings of a firm from the sale of its output.	$TR = P \times Q$
Marginal revenue	The additional revenue of a firm arising from the sale of an additional unit of output.	$MR = \frac{\Delta TR}{\Delta Q}$
Average revenue	Revenue per unit of output.	$AR = \frac{TR}{Q} = P$
Cost concepts		
Explicit cost	The monetary payment made by a firm to an outsider to acquire an input.	
Implicit cost	The income sacrificed by a firm that uses a resource it owns.	
Total cost (*TC*)	The sum of explicit and implicit costs	
Average total cost (*AC*)	Total cost per unit of output.	$AC = \frac{TC}{Q}$
Marginal cost (*MC*)	The change in cost arising from one additional unit of output.	$MC = \frac{\Delta TC}{\Delta Q}$
Long-run average cost (*LRAC*) curve	A U-shaped curve showing average costs in the long run when all of the firm's inputs are variable.	
Profit concepts		
Profit	Total revenue minus total cost (the sum of explicit plus implicit costs).	TR–TC
Normal profit	Occurs when total revenue equals total cost. It is the minimum amount of revenue required by a firm to keep running.	TR=TC
Abnormal profit	Profit that results when total revenue is greater than total cost. It is revenue that is over and above normal profit.	TR>TC
Loss	Negative profit; occurs when total revenue is less than total cost	TR<TC

Table 7.5: Summary of cost, product, revenue and profit concepts

Figure 7.6(a) shows standard market demand and supply curves for strawberries, which determine the equilibrium price, P_e. Figure 7.6(b) shows the demand curve for strawberries *as it appears to you, the strawberry producer*. It is perfectly elastic, appearing as a horizontal line at P_e determined in the market. A perfectly elastic demand curve has a price elasticity of demand (*PED*) equal to infinity throughout its range (see Chapter 3, Section 3.1). What does this mean for you?

As a strawberry producer, being small, you can do nothing to influence this price; you must accept P_e and sell the amount of strawberry output that will

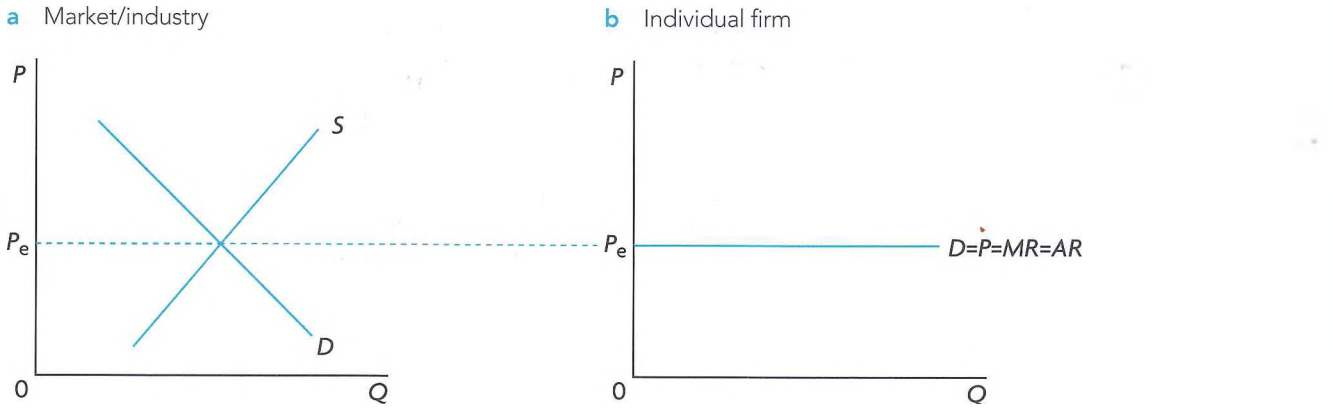


Figure 7.6: Market (industry) demand and supply determine demand faced by the perfectly competitive firm

maximise your profit. Your farm (or your firm) is therefore a **price-taker**. If you raise your price above P_e, you will not sell any strawberries because buyers will buy strawberries elsewhere at the lower price P_e. On the other hand, since you can sell all you want at price P_e, you would have nothing to gain and something to lose (some revenue) if you dropped your price below P_e. Therefore, you sell all your strawberry output at P_e.

The demand curve for a good facing the perfectly competitive firm is perfectly elastic (horizontal) at the price determined in the market for that good. This means the firm is a *price-taker*, as it accepts the price determined in the market. The firm has no ability to influence price therefore it has no market power.

The firm's revenue curves

The firm we are considering is the one studied in Section 7.2 above, when we studied revenue data and curves for the firm that is *unable to influence price*. Consider once again the example used in Table 7.2 and Figure 7.1. Assume that a perfectly competitive firm sells a good at €10 per unit. In Table 7.2, column 3 shows total revenue, calculated by multiplying units of output in column 1 by price shown in column 2. Column 4 calculates marginal revenue, by taking the change in total revenue and dividing it by the change in output. Column 5 shows average revenue, obtained by dividing total revenue by quantity of output. The data in the table reveal an interesting pattern:

No matter how much output the perfectly competitive firm sells, $P = MR = AR$ and these are constant at the level of the horizontal demand curve. This follows from the fact that price is constant regardless of the level of output sold.

This result holds only for firms operating under perfect competition, because these are the only firms that are unable to influence price and are forced to sell all their output at the single price determined in the market.

The data of Table 7.2 are plotted in Figure 7.7 below which is the same as Figure 7.1(b), where we see that since price is constant at €10, $P = MR = AR$, and they all coincide with the horizontal demand curve.

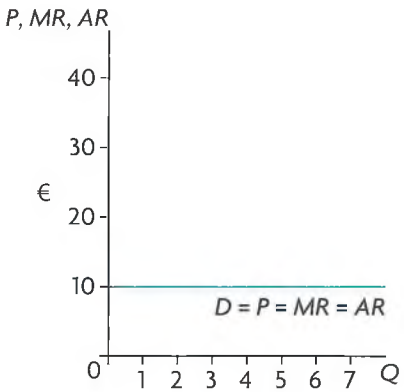


Figure 7.7: Demand, marginal revenue and average revenue in perfect competition

TEST YOUR UNDERSTANDING 7.6

1 Outline the characteristics of perfect competition.

2 a Explain why the perfectly competitive firm is a price-taker.

b Outline what would happen if this firm tried to raise its price above the market price or if it lowered its price below the market price.

3 Outline how the demand curve facing the perfectly competitive firm relates to the industry/market equilibrium.

4 a Using a diagram, explain the relationship between the firm's average revenue (*AR*) and marginal revenue (*MR*) in perfect competition. Outline how are they related to

b product price,

c the demand curve facing the firm, and

d the principle that each firm is a price-taker.

Profit maximisation in the short run

Remember, the short run is the period when the firm has at least one fixed input. This means the number of firms in the industry is also fixed. To enter or leave an industry, a firm must be able to vary *all its inputs*. Since this cannot be done in the short run, firms cannot enter or leave the industry (until they move into the long run).

In terms of your strawberry farm, this means that you have some fixed inputs such as the land you cultivate, and perhaps your farm machinery. As long as all other strawberry producers also have a fixed amount of agricultural land and farm machinery, no one can enter or leave strawberry production.

When a firm wants to maximise profit in the short run, what must it do? Since it is a price-taker, it cannot influence its selling price. It can only make a choice on how much quantity of output it should produce. We will see how the firm does this using the marginal revenue and marginal cost rule, introduced above.

Short-run profit maximisation based on the marginal revenue and marginal cost rule

The analysis consists of three steps.

i **Find the point where marginal revenue equals marginal cost to determine profit-maximising (or loss-minimising) level of output.** As we know, a firm interested in maximising profit (or minimising loss) produces output where $MR = MC$. Figure 7.5(a) shows $Q_{\pi max}$ to be the profit-maximising level of output.

ii **Compare average revenue (or price) and average cost to determine the amount of profit (or loss) per unit of output.** A comparison of average revenue (which is equal to price) with average cost shows the amount of profit (or loss) per unit of output. We know profit = $TR - TC$. If we divide this throughout by output, Q, we get an expression for profit per unit of output, in other words, in terms of averages:

$$\frac{profit}{Q} = \frac{TR}{Q} - \frac{TC}{Q}$$

Alternatively,

$$\frac{profit}{Q} = \text{AR} - \text{AC}.$$

Moreover, since $P = AR$, it follows that

$$\frac{profit}{Q} = \text{P} - \text{AC}$$

This is the key to calculating how much profit or loss per unit of output the firm is making.

iii **Find total profit (or total loss).** To do this, we multiply:

$\frac{profit}{Q}$ by Q (or $\frac{loss}{Q}$ by Q)

At the profit-maximising level of output Q:

- If $AR > AC$ (or $P > AC$), the firm makes abnormal profit (positive profit).
- If $AR = AC$ (or $P = AC$), the firm makes normal profit (zero profit)
- If $AR < AC$ (or $P < AC$), the firm makes a loss (negative profit).

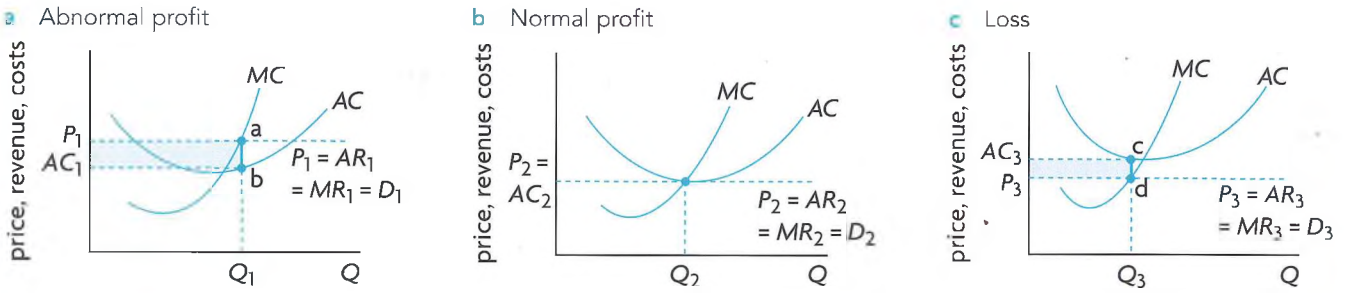


Figure 7.8: Short-run profit maximisation in perfect competition

Using the above three step approach, we will examine the behaviour of the perfectly competitive firm in the short run, making use of the diagrams in Figure 7.8. Each of these diagrams contains identical AC and MC cost curves; note that MC always intersects AC at its minimum point. What differs between the diagrams is the position of the perfectly elastic demand curve, showing different possible prices that the firm, being a price-taker, must accept.

Profit maximisation and abnormal profit

In Figure 7.8(a), price $P_1 = AR_1 = MR_1$ represent the demand curve facing the firm. Using the rule $MR = MC$, we arrive at the profit maximising level of output Q_1 (simply draw a vertical line from the point of intersection to the horizontal axis). We then compare P_1 with AC, along this same vertical line, and since $P_1 > AC$, we conclude the firm is making abnormal profit per unit equal to $P_1 - AC_1$, given by the vertical distance between points a and b. To find total profit we multiply profit per unit times the total number of units produced, given by

$\text{profit} = \frac{profit}{Q} \times Q$ and is represented by the shaded area in the diagram.

> When $P > AC$ (or $AR > AC$) at the level of output where $MC = MR$, the firms earns abnormal profit (positive profit).

Profit maximisation and normal profit

Suppose the market-determined price falls to P_2, corresponding to demand curve D_2 as in Figure 7.8(b). Applying again the $MR = MC$ rule, we find the profit-maximising level of output Q_2. Comparing P_2 with AC at output Q_2, we see they are equal to each other; therefore, profit per unit is $P_2 - AC_2 = 0$. Therefore, profit is zero and the firm is earning normal profit. When profit is zero, price equals minimum AC. The firm's total revenues are equal to its total costs.

> When P = minimum AC (or AR = minimum AC) at the level of output where $MC = MR$, the firm earns normal profit (zero profit).

Loss minimisation

If the market price falls below minimum AC, such as P_3 in Figure 7.8(c), corresponding to demand curve D_3 as in Figure 7.8(b), the firm does not earn enough revenue to cover all its costs. Using the $MC = MR$ rule, we see that the profit-maximising or loss-minimising level of output is Q_3, at which $P_3 < AC$, indicating the firm is making a negative profit, or loss. Therefore, Q_3 is the firm's loss-minimising output. $AC_3 - P_3$, or the difference between points c and d, represents the firm's loss per unit of output, or $\frac{loss}{Q}$.

If we multiply this vertical distance by Q_3, we get the firm's total loss, given by the shaded area.

> When $P <$ minimum AC (or $AR <$ minimum AC) at the level of output where $MC = MR$, the firm makes a loss (negative profit).

Profit maximisation in the long run

In the long run, all the firm's factors of production are variable; therefore, the number of firms in the industry is no longer unchanging. New firms can enter the industry, existing firms can change their size (increase or decrease all their inputs), or firms can leave the industry altogether. There is therefore free entry of firms in an industry.

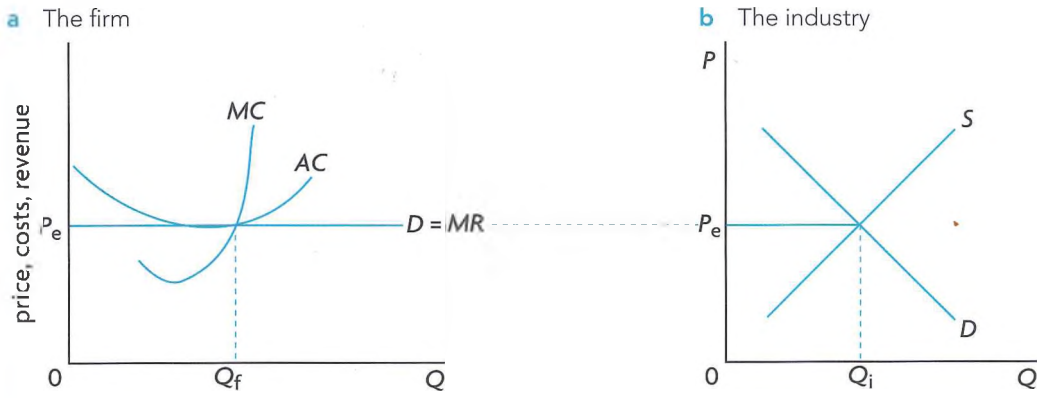


Figure 7.9: The firm and industry long-run equilibrium position in perfect competition

Normal profit in the long run

In the long-run equilibrium of perfect competition, all firms earn zero profit, in other words, they earn normal profit. The long-run equilibrium position of each firm and the industry under perfect competition is shown in Figure 7.9. The market settles at the price P_e, which is just equal to the firm's *minimum AC*, where each firm is earning normal profit, since $P = AC$ (or $AR = AC$). You may note that at this point AC must be minimum since profit maximisation occurs where $MR = MC$, and MC intersects AC at its minimum point. Each firm in the industry produces output Q_f, and the industry as a whole produces output Q_i (equal to the sum of all the firms' outputs).

Why does this happen? How is it that the firms that were in the short run earning abnormal profits, or were making losses, end up making normal profits in the long run?

Suppose the strawberry industry is profitable, meaning that strawberry farms (or firms) are making abnormal profits, so that $P>AC$ (or $AR>AC$). In the long run, when firms can vary all their inputs, new farmers are attracted into the strawberry industry since they would also like to earn abnormal profits. As more and more farms begin to produce strawberries, the supply of strawberries increases, and this has the effect of reducing the price of strawberries. But as the price of strawberries falls, the abnormal profits of strawberry producers fall. The price of strawberries continues to fall until it is just equal to minimum AC. At that point the strawberry producers, who as *price-takers* must accept the price that is determined in the market, end up earning normal profit where $P = AC$ (or $AR = AC$).

Suppose now that the strawberry industry were loss-making, in which case $P<AC$ (or $AR<AC$). In the long run some strawberry farms would close down, or else they might stop producing strawberries and begin producing another more profitable crop. As strawberry farmers leave the industry or switch out of strawberries, the supply of strawberries decreases, causing the price of strawberries to rise. This process continues until price is just equal to AC. Here too, the remaining strawberry producers in the industry will end up earning normal profit where $P = AC$ (or $AR = AC$).

The interested student may see how this process occurs using diagrams (presented as Supplementary material in the digital coursebook).

> In perfectly competitive long-run equilibrium, firms' profits and losses are eliminated, and revenues are just enough to cover all costs so that every firm earns normal profit.

This process illustrates an important principle, which is that competition leads to a process of firms opening (or closing) such that price is driven down (or up) to the lowest level that is acceptable to the firm in order for it to continue operating. This is the price that is equal to minimum average cost, and that allows the firm to earn normal profit so that it is just covering all its costs. As we will see in the pages below, *anything that restricts competition results in higher prices, abnormal profits for firms and welfare loss for society*.

TEST YOUR UNDERSTANDING 7.7

1 Using diagrams, show when a firm
   - a earns profit (show profit per unit and total profit),
   - b earns normal profit, and
   - c earns negative profit (a loss) (show loss per unit and total loss).

CONTINUED

2 For each of the following, calculate abnormal profit or loss per unit of output, and then calculate total abnormal profit or loss:

a Q = 200 units; AC = \$8, P = \$9

b Q = 250 units; AC = \$15, P = \$13

c Q = 150 units; AC = \$17, P = \$17

3 Given the information in the table,

a if price = €6, calculate how much the profit-maximising (loss-minimising) firm will produce, and how much profit or loss it will make;

b if price = €4, calculate how much the profit-maximising (loss-minimising) firm will produce, and how much profit or loss it will make.

Total product (units of output)	Average cost (€)	Marginal cost (€)
1	14.00	4
2	8.50	3
3	6.33	2
4	5.00	1
5	4.40	2
6	4.17	3
7	4.14	4
8	4.25	5
9	4.44	6
10	4.70	7

4 a Using a diagram show and explain a perfectly competitive firms' long run equilibrium position.

b Outline the process by which the firm reaches this position.

c Explain what kind of profit the firm makes in long run equilibrium.

Allocative efficiency

In Chapter 2, we saw that competitive markets achieve allocative efficiency, because equilibrium is determined by $MB = MC$, where consumer plus producer (social) surplus is maximum. This discussion focused on efficiency at the level of the *market*, or industry. We now want to see how efficiency can be analysed also at the level of the individual firm.

Review of allocative efficiency

Allocative efficiency is achieved when $MB = MC$; but since $MB = P$, it follows there is allocative efficiency when $P = MC$. Note that this condition holds only when there are no externalities, in which case it is also true that $MSB = MSC$ (this was our condition for allocative efficiency when we studied externalities in Chapters 5 and 6).

Allocative efficiency occurs when firms produce the particular combination of goods and services that consumers mostly prefer. The condition is the following:

Allocative efficiency is achieved when $P = MC$ *(or alternatively* $MB = MC$*)*

The price, P, paid by consumers to acquire a good reflects the marginal benefit they derive from consumption of one more unit of the good and shows the amount of money they are willing to pay to buy one more unit. Marginal cost, MC, measures the value of the resources used to produce one extra unit of the good. When price is equal to marginal cost, there is equality between what consumers are prepared to pay for one more unit and what it costs to produce it.

What would happen if P and MC were not equal to each other? If $P > MC$, an additional unit of the good is worth more to consumers than the cost to produce it. There is an underallocation of resources to its production, and consumers would be better off if more of it were produced. If $P < MC$, an additional unit of the good costs more to produce than it is worth to consumers; there is an overallocation of resources to the good, and consumers would be better off if output were reduced. In both these cases, allocative inefficiency results. Therefore, resources are allocated efficiently only when the price of a good is equal to the marginal cost of producing it.

Allocative efficiency and perfect competition

Figure 7.10 shows the long-run equilibrium position of a firm and industry in perfect competition. Part (a) shows the firm to be earning normal profit, and indicates that in long-run equilibrium, the perfectly competitive firm achieves allocative efficiency since at the profit-maximising level of output, Q_e, $P = MC$. Part (b) shows the industry to be achieving allocative efficiency ($MB = MC$ and social surplus is maximum). It illustrates how the efficiency at the level of the firm corresponds to efficiency at the level of the industry.

In long-run equilibrium under perfect competition, the firm achieves allocative efficiency where $P = MC$ (or where $MB = MC$). At the level of the industry, social surplus (consumer plus producer surplus) is maximum, and $MB = MC$.

The achievement of allocative efficiency in long-run equilibrium is an important result, because perfect competition is the only market structure where this occurs.

Evaluating perfect competition

Perfect competition, though not a realistic market structure, offers a number of insights into the workings of competitive markets.

Insights provided by the model

- **Allocative efficiency.** Perfect competition leads to the best or 'optimal' allocation of resources, achieved through $P = MC$ (or $MB = MC$) in long-run equilibrium.
- **Low prices for consumers.** Consumers benefit from low prices, due to the absence of abnormal profits, which would have led to a higher price. You can check this by comparing Figure 7.8(a) and (b), showing that the price when the firm earns abnormal profit is higher than the price when the firm earns only normal profit.
- **Competition leads to the closing down of inefficient producers.** Inefficient firms are those that produce at higher than necessary costs. Inefficiency could be due to factors like less productive labour, or the use of outdated technologies, or poor entrepreneurship. The revenues of inefficient firms are insufficient to cover all costs, leading to losses that force these firms to leave the industry in the long run.
- **The market responds to consumer tastes.** Changes in consumer tastes are reflected in changes in market demand and therefore market price. By creating short-run abnormal profits or losses, price changes result in long-run adjustments that make the quantity of output produced by the industry respond to consumer tastes.

Limitations of the model

- **Unrealistic assumptions.** The model rests on strict and unrealistic assumptions that are rarely met in the real world.
- **Cannot take advantage of economies of scale.** Economies of scale lead to lower average costs as a firm grows larger and larger. In perfect competition firms are too small to grow to a size large enough to have economies of scale.
- **Lack of product variety.** All firms within an industry produce identical or undifferentiated (homogeneous) products, however consumers prefer product variety.

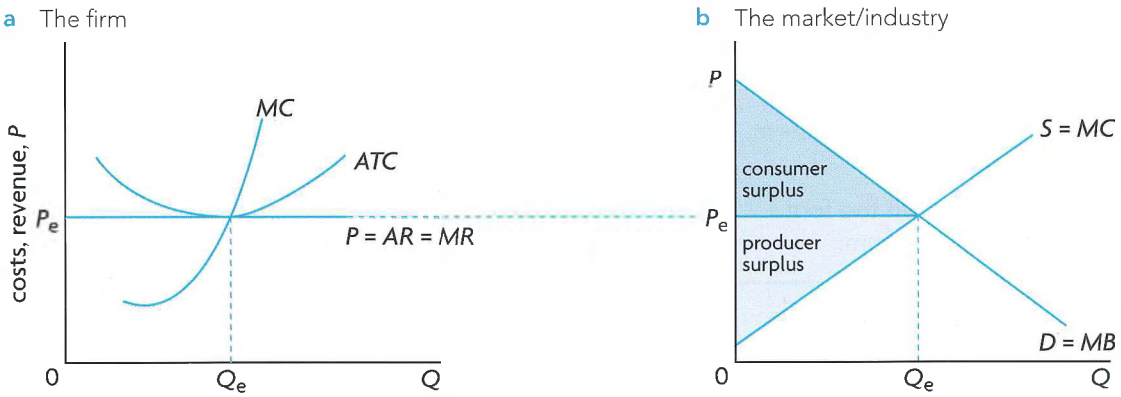


Figure 7.10: Allocative efficiency in perfect competition in the long run

- **Limited ability to engage in new product development.** The lack of abnormal profits in the long run does not offer firms the necessary funds to pursue research and development. In any case, since products are undifferentiated, firms do not have the incentive to pursue product development or improvement.

TEST YOUR UNDERSTANDING 7.8

1 Outline why we study the perfectly competitive market model extensively when this model is based on so many unrealistic assumptions.

2 a Explain the meaning of and state the conditions for allocative efficiency.

b Using diagrams, show how the perfectly competitive *firm* and *industry* achieve allocative efficiency in long-run equilibrium.

3 Evaluate the perfectly competitive market model by referring to the insights it offers and its limitations.

7.4 Monopoly

LEARNING OBJECTIVES

After studying this section you will be able to:

- define all the terms appearing in **orange bold** in the text (AO1)
- explain the monopolist's market power using a diagram where *AR>MC* (AO2, AO4)
- explain profit maximisation by the monopolist with diagrams showing abnormal profit, normal profit, loss (AO2, AO4)
- explain the monopolist's allocative inefficiency illustrating market failure (AO2)
- explain welfare loss compared to perfect competition with a diagram showing lower output and higher price and welfare loss (AO2, AO4)
- explain natural monopoly and draw a diagram for natural monopoly (AO2, AO4)
- discuss advantages and disadvantages of monopoly (AO3)

The term 'monopoly' is derived from the Greek word μονοπώλιο (monopolio) meaning 'single seller'. We begin by reminding ourselves of the characteristics of monopoly:

- there is a single firm in the industry
- the firm produces and sells a unique good or service, with no close substitutes
- there are high barriers to entry in the industry.

Monopoly lies at the opposite extreme of market structures to perfect competition. As a single seller, the monopolist faces no competition from other firms and it has substantial market power (the ability to control price).

Barriers to entry

There are several kinds of barriers to entry, described below.

Economies of scale

Economies of scale result in the downward-sloping portion of a firm's long-run average cost curve (*LRAC*), resulting in lower average costs as the firm increases its size (Figure 7.11). Large economies of scale create a barrier to entry. In Figure 7.11 the average costs of a large firm on $SRAC_1$ are substantially lower than the average costs faced by a smaller firm on $SRAC_2$. The large firm can charge a lower price than the smaller firm, and can force the smaller firm into a situation where it will not be able to cover its costs. Therefore, if new firms try to enter the industry on a small scale they will be unable to compete with the larger one.

On the other hand, a new firm attempting to enter the market on a very large scale would encounter huge start-up costs, and would be unlikely to take the risk. Economies of scale form a significant barrier to entry also in oligopolies.

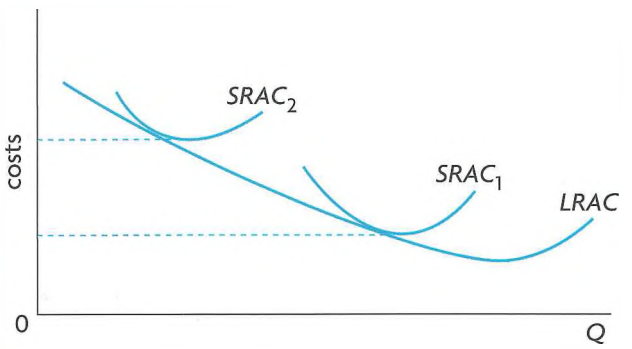


Figure 7.11: Economies of scale as a barrier to entry

Natural monopolies

Natural monopolies are firms that have economies of scale so large that they can produce for an entire market and still not exhaust their economies of scale. They will be examined in more detail below.

Branding

Branding involves the creation by a firm of a unique image and name of a product. It works through advertising campaigns that try to influence consumer tastes in favour of the product, attempting to establish consumer loyalty. If branding is successful, many consumers will be convinced of the product's superiority, and will be unwilling to switch to substitute products, even though these may be qualitatively very similar. Branding may work as a barrier to entry by making it difficult for new firms to enter a market that is dominated by a successful brand. Note that branding need not lead to a monopoly (it is a method used also by firms in monopolistic competition and oligopoly), but it does have the effect of limiting the number of new competitor firms that enter a market. Examples of branding include brand-name items (such as NIKE®, Adidas®, Coca-Cola®, etc.).

Legal barriers

Examples of legal barriers include the following:

- **Patents** are rights given by the government to a firm that has developed a new product or invention to be its sole producer for a specified period of time. For that period, the firm producing the patented product has a monopoly on production and sale of the product. Examples include patents on new pharmaceutical products, and Intel and microprocessor chips used by IBM computers.
- **Licences** are granted by governments for particular professions or particular industries. Licences may be required, for example, to operate radio or television stations, or to enter a particular profession (such as medicine, dentistry, architecture, law and others). Such licences do not usually result in a monopoly, but they do have the impact of limiting competition.
- **Copyrights** guarantee that an author (or an author's appointed person) has the sole rights to print, publish and sell copyrighted works.
- **Tariffs, quotas and other trade restrictions** limit the quantities of a good that can be imported into a country, thus reducing competition.

Not all of these legal barriers lead to monopoly, but they all have the effect of limiting competition, thus contributing to the creation of some degree of market power.

Control of essential resources

Monopolies can arise from ownership or control of an essential resource. A classic example of an international monopoly is DeBeers, the South African diamond firm, whose control of the diamond industry peaked at 90% in the 1980s, allowing it to have a significant control over the price of diamonds. (More recently DeBeer's no longer has control and diamond prices are driven by the market.) On a national level, an example is Alcoa (the Aluminum Company of America), which, following the expiration of patents in 1909, was able to maintain its monopoly position on the production of aluminium within the United States until the Second World War, because of its control of almost all the bauxite resources within the country. On a local level, professional sports leagues create a local monopoly by signing long-term contracts with the best players and securing exclusive use of sports stadiums. A local monopoly is a single producer/supplier within a particular geographical area. Local monopolies appear more commonly than national or international ones. For example, a local grocery store in a residential area located some distance from any other stores may be a local monopoly.

Aggressive tactics

If a monopolist is confronted with the possibility of a new entrant into the industry, it can create entry barriers by cutting its price, advertising aggressively, threatening a takeover of the potential entrant, or any other behaviour that can dissuade a new firm from entering the market.

Demand and revenue curves under monopoly

The demand curve (*AR* curve) facing the monopolist

Since the pure monopolist is the entire industry, the demand or *AR* curve it faces is the industry or market demand curve, which is downward-sloping. The two demand curves shown in Figure 7.12 indicate that the perfectly competitive firm is a price-taker with zero market power, while the monopolist is a **price-maker** with a significant degree of market power.

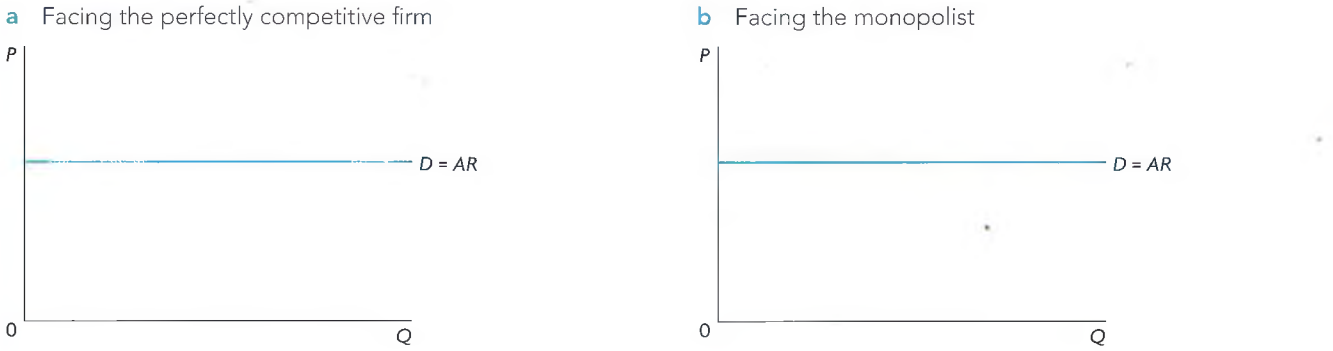


Figure 7.12: Demand curves

All firms under market structures other than perfect competition are to varying degrees price-makers, as they all face downward-sloping demand curves. Of these, the monopolist has the greatest degree of market power, or the ability to control price, because it is the sole or dominant firm in the industry.

> Market power arises whenever a firm faces a downward-sloping demand curve. Firms in all market structures except perfect competition face a downward-sloping demand curve, and therefore have varying degrees of market power, or the ability to control the price at which they sell their output; they are therefore *price-makers*.

However, whereas the monopolist has a large control over price, this control is limited by the position of the market demand curve. In Figure 7.12(b), when the monopolist chooses how much output to produce, say Q_1, it simultaneously determines the price at which the good can be sold, or P_1. It could not sell output Q_1 at a price such as P_2, since the price–quantity combination P_2 and Q_1 is at point a, lying off the demand curve.

The monopolist's revenue curves

When a firm faces a downward-sloping demand curve, price is no longer constant for all output: more output can only be sold at a lower price. Consider the example used in Table 7.3 and Figure 7.2 showing revenue data and curves for the firm that has some ability to control price. Table 7.6 provides the same data for a monopolist's total, marginal and average revenues, and Figure 7.13 plots these data.

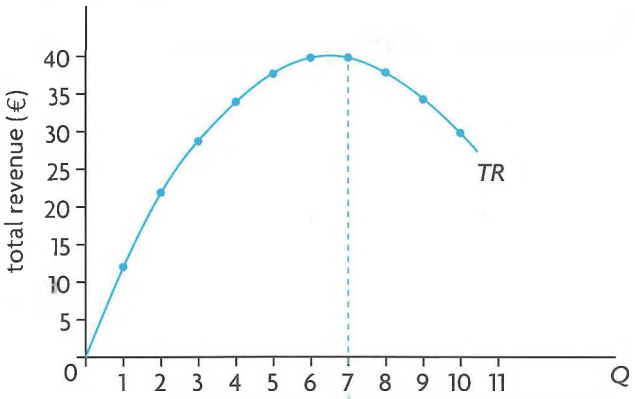
1 Units of output (Q)	2 Product price (P) (€)	3 Total revenue $TR = P \times Q$ (€)	4 Marginal revenue $MR = \frac{\Delta TR}{\Delta Q}$ (€)	5 Average revenue $AR = \frac{TR}{Q}$ (€)
0	–	–	–	–
1	12	12	12	12
2	11	22	10	11
3	10	30	8	10
4	9	36	6	9
5	8	40	4	8
6	7	42	2	7
7	6	42	0	6
8	5	40	–2	5
9	4	36	–4	4
10	3	30	–6	3

Table 7.6: Total, marginal and average revenue when price varies with output

Looking at Table 7.6 and Figure 7.13, we may note the following:

- As price (P) falls, output (Q) increases giving rise to the downward-sloping demand curve.
- Marginal revenue, showing the change in total revenue resulting from a change in output, falls continuously; MR is equal to zero when total revenue is at its maximum (at seven units of output), and becomes negative when total revenue falls. It may be noted that the MR curve shows the slope (gradient) of the TR curve. Therefore when TR is maximum, $MR = 0$. When TR is falling, MR is negative.
- Average revenue (column 5 of Table 7.6) is equal to price (see column 2).

a Total revenue



b Marginal and average revenue

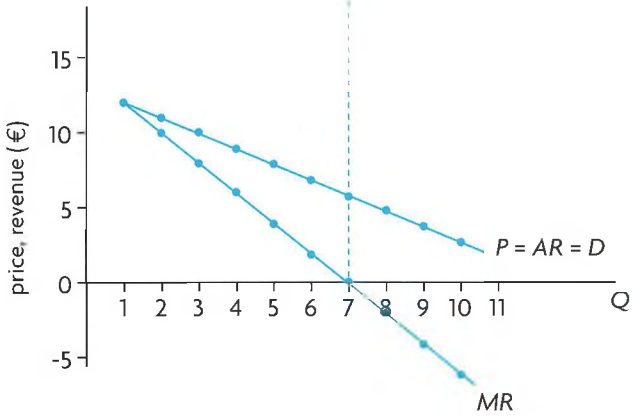


Figure 7.13: Revenue curves in monopoly

Since $TR = P \times Q$, and $AR = \frac{TR}{Q}$, it follows that *P is equal to AR.*

- *The AR and P curves represent the demand curve facing the firm.*
- *The MR curve lies below the demand curve.* The reason is that, unlike in perfect competition, where $MR = P$, here the firm must lower its price in order to sell more output. The lower price is charged not only for the last unit of output but all the previous units of output sold. Marginal revenue, or the extra revenue from selling an additional unit of output, is therefore equal to the amount of the price of the last unit sold minus what is lost by selling all the other units of output at the now lower price.[1]

TEST YOUR UNDERSTANDING 7.9

1 Outline the assumptions defining the market model of monopoly.

2 Using a diagram, explain how economies of scale can result in a monopoly market structure by posing barriers to entry.

3 Using examples, explain how branding and legal factors provide barriers to entry into an industry.

4 a Compare and contrast the demand curve facing the perfectly competitive firm and that facing the monopolist.

b Explain why one firm is a price-taker and the other a price-maker.

5 a Explain the relationship between the monopolist's average revenue (*AR*) and marginal revenue (*MR*).

Outline how they are related to

b product price, and

c the demand curve facing the firm.

Profit maximisation by the monopolist

Profit maximisation based on the marginal revenue and cost approach

The monopolist interested in maximising profit (or minimising loss) follows the same three-step approach used by the perfectly competitive firm:

i The monopolist determines the profit-maximising (or loss-minimising) level output using the $MC = MR$ rule.

ii For that level of output, it determines profit per unit or loss per unit by using

$$\frac{profit}{Q} = P - AC$$

If $AR > AC$ *(or* $P > AC$*)*, the monopolist is making abnormal profit;

[1] To understand this, consider the following numerical example. Say output increases from 3 to 4 units. Marginal revenue will be the result of a gain and a loss. The gain is €9, obtained from selling the fourth unit of output at the price of €9. The loss is equal to €1 for each of the initial 3 units of output that previously were selling for €10 and must now sell for €9, equal to €3. Marginal revenue is equal to the gain minus the loss, or 9 – 3 = 6.

if AR = AC (or P = AC) it is earning normal profit (zero profit);
if $AR < AC$ (or P < AC) it is making a loss.

iii The firm multiplies $\frac{profit}{Q}$ by Q to determine total profit, or $\frac{loss}{Q}$ by Q to determine total loss.

Figure 7.14 (a), (b) and (c) show the standard AC and MC curves; these are the same as those used for profit maximisation in perfect competition. On these cost curves, the monopolist's demand and marginal revenue curves are added. Consider first part (a).

- We first find where $MR = MC$, which determines the profit-maximising level of output, Q_e.
- At Q_e, we draw a vertical line upward to the AR (or demand) curve and from there extend a horizontal line leftward to the vertical axis; this will determine the price, P_e, at which the monopolist sells output Q_e.
- For output Q_e, we find profit per unit ($\frac{profit}{Q}$), given by $P - AC$; this is the vertical distance between the average revenue (demand) and AC curves.
- To find total profit, we multiply profit per unit times the total number of units produced, which is

$$\text{profit} = \frac{profit}{Q} \times Q$$

and is represented by the shaded area.

The monopolist need not always make profits; it may make losses if price cannot cover AC. This is shown in Figure 7.14(c), where the monopolist is minimising loss. At the level of output Q_e, determined by $MR = MC$, the monopolist's loss is minimised. The price that will be charged is given by P_e, found by extending a line upward to the demand curve at output level Q_e. Loss per unit of output ($\frac{loss}{Q}$) is given by $AC - P$, and total loss is given by the shaded area, found by multiplying loss per unit of output by the total number of units produced.

It is also possible that the monopolist earns normal profits. This is shown in Figure 7.14(b), where Q_e occurs where $P = AC$.

Note that the distinction between the short run and the long run is not important in monopoly as it is in perfect competition. In perfect competition, this distinction is crucial because as firms enter and exit an industry in the long run, abnormal profits and losses disappear, and firms are left with normal profits in long-run equilibrium. This is not possible in monopoly, due to the presence of barriers to entry.

> Under monopoly, high barriers to entry prevent potential competitor firms from entering a profit-making industry, and the monopolist can therefore continue making abnormal profits indefinitely in the long run.

Natural monopoly

A **natural monopoly** is a firm that has economies of scale so large that it is possible for the single firm alone to supply the entire market at a lower average cost than two or more firms.

A natural monopoly is illustrated in Figure 7.15. The natural monopolist's demand curve intersects its $LRAC$ curve at a point where average costs are still falling. At the level of output Q^*, there are still economies of scale. The figure shows that this firm cannot produce any output greater than Q^* and not make a loss. If it did, then P which is given by the demand curve would be lower than $LRAC$. But as we know, $P<AC$ means the firm is making a loss. So if the firm is to earn normal profit or abnormal profit, it must produce a quantity less than or equal to Q^* and charge a price greater than or equal to AC^*.

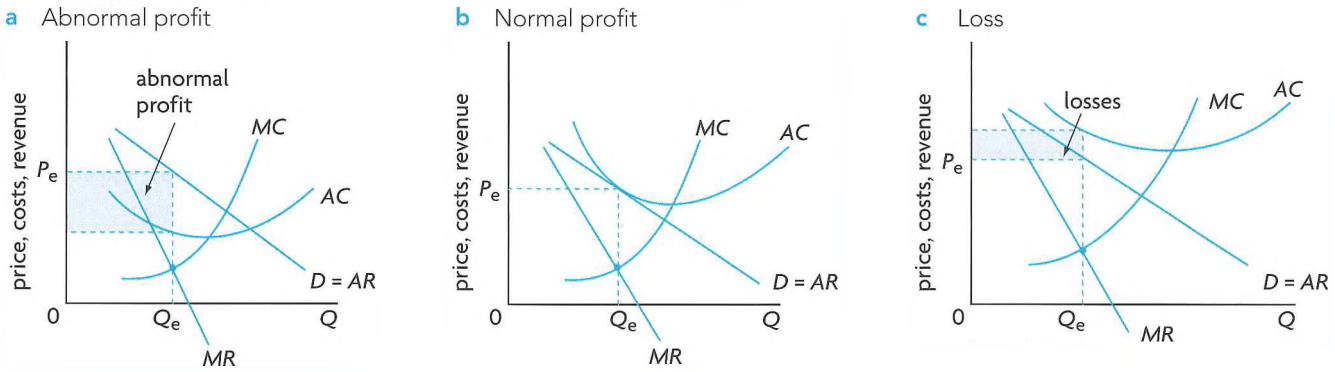


Figure 7.14: Profit maximisation and loss minimisation in monopoly: marginal revenue and cost approach

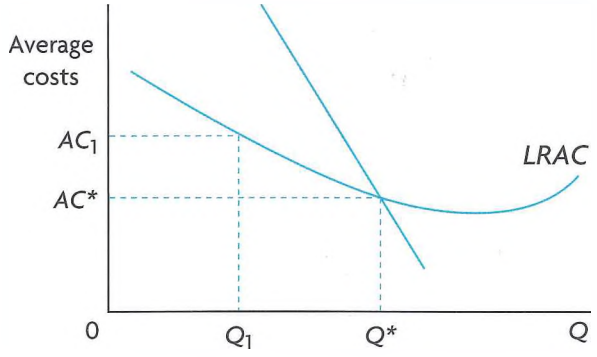


Figure 7.15: Natural monopoly

Suppose now that this firm is split into two firms of equal size. Each would produce output Q_1 and with average costs at AC_1. Clearly, average costs would be higher, and the price that would allow the firm to earn normal profit, where $P = AC$, would also be higher.

If the market demand for a product cuts *LRAC* when this is falling, this means that a single large firm can produce for the entire market at a lower average total cost than two or more smaller firms. When this occurs, the firm is called a **natural monopoly.**

Natural monopoly acts as a strong barrier to entry because potential entrants realise that it would be extremely difficult to attain the low costs of the already existing firm. High average costs would mean having to charge a high price for the product, so that the new firm would be unable to compete with the existing firm.

Examples of natural monopolies include water, gas and electricity distribution, cable television, fire protection and postal services. The falling average costs over a very large range of output often occur because of very large capital costs (such as laying pipes for water distribution, or laying cables for electricity distribution, or putting a satellite into orbit).

A natural monopoly may stop being 'natural' if changing technologies create conditions that allow new competitor firms to enter the industry and begin production at a relatively low cost. This has happened in recent years with technological change in telecommunications, forcing telephone companies that previously were natural companies to compete with new entrants into the market.

TEST YOUR UNDERSTANDING 7.10

1 Using diagrams, show the case where a monopolist
   - a earns abnormal profit (show profit per unit and total profit),
   - b earns normal profit, and
   - c incurs losses (show loss per unit and total loss).

2 a Outline the difference, if any, between the short-run and long-run equilibrium of a monopolist.
   - b Outline why a monopolist can continue to earn profits in the long run.

3 The data in the table below show the demand curve and costs (*AC* and *MC*) facing a monopolist.
   - a Calculate the monopolist's total revenue and marginal revenue for each level of output.
   - b Identify this monopolist's profit-maximising level of output.
   - c Identify the price at which this level of output be sold.
   - d Calculate the monopolist's profit per unit and total profit.

Units of output	Price ($)	Average cost ($)	Marginal cost ($)
1	10	14.0	4.0
2	9	8.5	3.0
3	8	6.3	2.0
4	7	5.0	1.0
5	6	4.4	2.0
6	5	4.2	3.0
7	4	4.1	4.0
8	3	4.3	5.0

4 a Explain the relevance of economies of scale to natural monopoly.
   - b Using a diagram and examples, explain what a natural monopoly is.
   - c Outline why a natural monopoly can be a strong barrier to entry into an industry.
   - d Suggest why governments often do not break up natural monopolies in order to increase competition.

Monopoly market outcomes and efficiency

Higher price and lower output by the monopolist compared to the industry in perfect competition

A comparison of monopoly with perfect competition at the level of the industry reveals that price is higher and quantity of output produced lower in monopoly. Figure 7.16 shows the long-run equilibrium positions of a perfectly competitive industry, composed of many small firms, and of a monopoly, which is the entire industry. Part (a) for the perfectly competitive industry shows equilibrium price and quantity to be P_{pc} and Q_{pc}, Point a, where the industry demand and supply curves intersect, appears also in part (b), showing what would happen to price and quantity if the perfectly competitive industry were organised as a monopoly. The MC curve of part (a), or the competitive industry's supply curve becomes the monopolist's marginal cost curve.[2] The demand curve remains unchanged, but the monopolist's marginal revenue (MR_m) curve lies below D. When the profit-maximising monopolist applies the $MR = MC$, the result is output Q_m and price P_m.

Since $Q_m < Q_{pc}$, the industry under monopoly produces a smaller quantity of output than the industry under perfect competition. And since $P_m > P_{pc}$, the monopolist sells output at a higher price than the perfectly competitive industry. Higher prices and lower output go against consumers' interests.

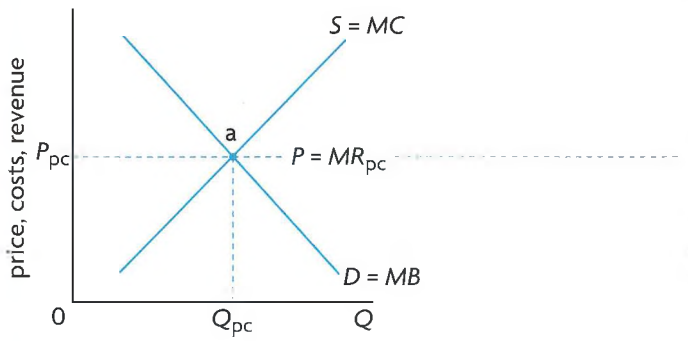
Allocative inefficiency and market failure

Loss of consumer and producer surplus

The higher price and lower output of the monopolist have important implications for consumer and producer surplus. Whereas the perfectly competitive industry achieves allocative efficiency shown by $MB = MC$ and maximum social surplus, monopoly does not. This can be seen in Figure 7.17, which is the same as Figure 7.16, only consumer and producer surplus have been added. In part (a), area A represents consumer surplus, while area B is producer surplus, with A + B showing maximum social surplus. Part (b) shows the inefficiencies that result in monopoly.

- Area C, consumer surplus in monopoly, is smaller than area A in perfect competition. Part of A was converted into producer surplus because of the higher monopoly price (P_m rather than P_{pc}), and another part of A was lost as triangle E because of the lower monopoly quantity (Q_m rather than Q_{pc}). Area E represents a welfare loss.
- Area D, producer surplus in monopoly, shows that producer surplus has increased by taking away a portion of consumer surplus (due to the monopolist's higher price), and it has also decreased by losing area F (due to the monopolist's lower quantity). Area F is also a welfare loss.
- E + F = welfare loss represents loss of social benefits (consumer and producer surplus) due to monopoly's higher price and lower quantity.

a Industry in perfect competition



b Monopoly

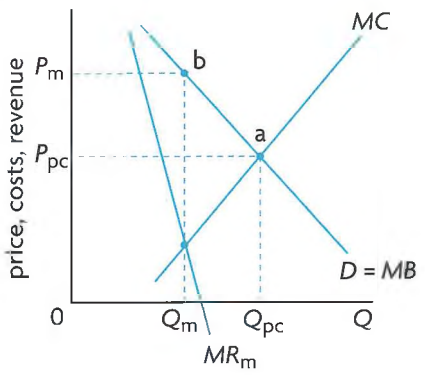


Figure 7.16: Higher price, lower output by the firm in monopoly

[2] However, note that the monopolist's *MC* curve is not its supply curve. In fact, the monopolist does not have a supply curve, because there is no single relationship between price and quantity supplied in monopoly. The reason is that the *MC = MR* rule for the monopolist may result in different prices for the same quantity, depending on demand conditions for the monopolist's product.

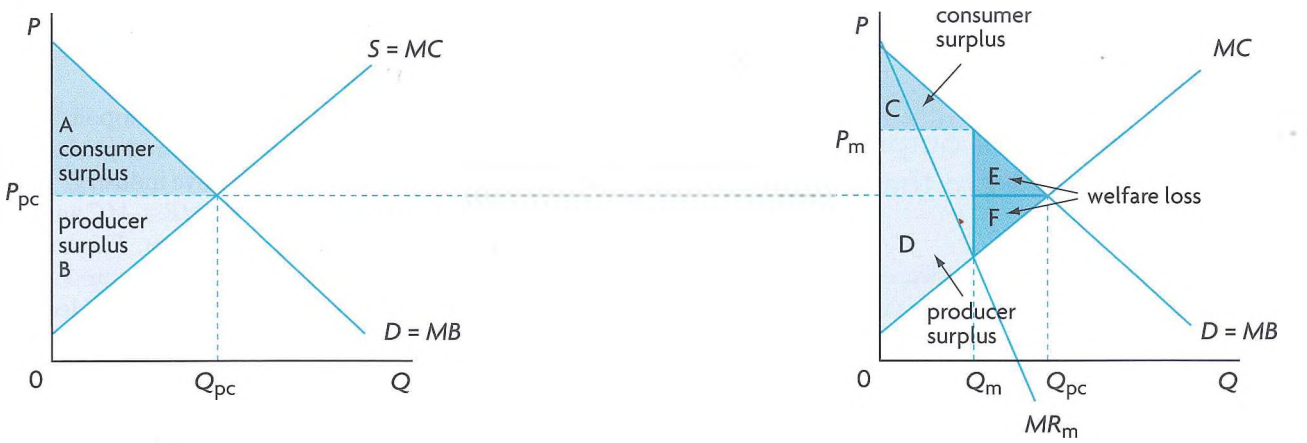


Figure 7.17: Consumer and producer surplus and welfare loss in monopoly compared with perfect competition

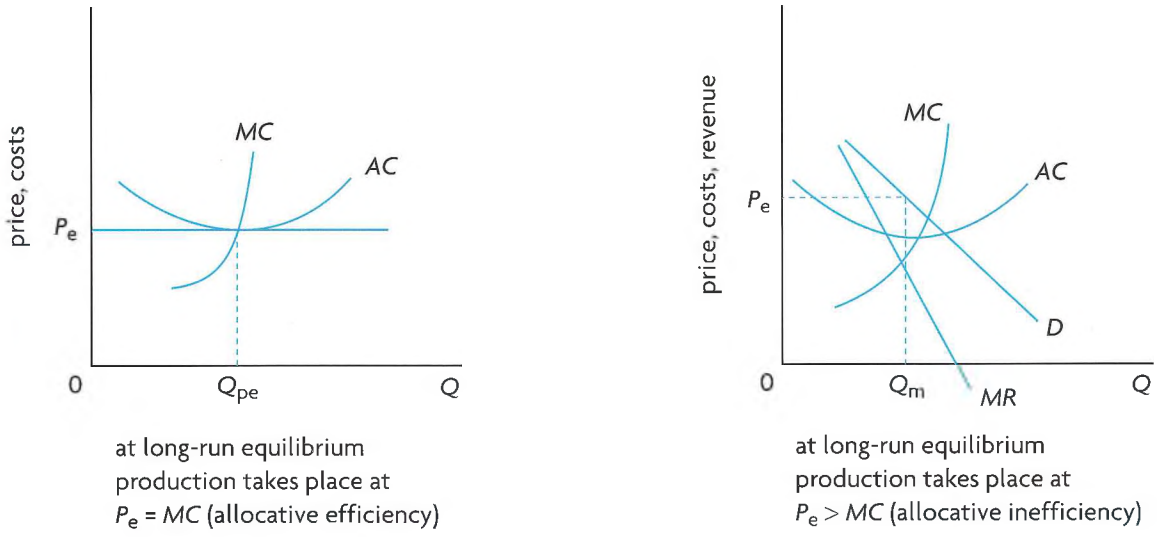
Note that the presence of welfare loss means that MC and MB are no longer equal. At the point of monopoly production, Q_m, $MB > MC$, meaning that there is an underallocation of resources to the good, and consumers are not getting as much of it as they would have liked.

The presence of welfare loss in monopoly indicates market failure: there is allocative inefficiency, shown also by $MB > MC$ at Q_m, meaning there is underallocation of resources: too little of the good is produced. Also, the monopolist gains at the expense of consumers as a portion of consumer surplus is converted into producer surplus.

Allocative inefficiency: P > MC

Figure 7.18 shows the long-run equilibrium position of the firm in perfect competition and monopoly. The condition for allocative efficiency is given by $P = MC$ at the profit-maximising level of output. As we know, this condition holds for the firm in perfect competition. In Figure 7.18(b) we can see that at the profit-maximising level of output Q_m, the monopolist's price, P_e, is greater than marginal cost. This is hardly surprising, since $P > MC$ is the same as $MB > MC$ (since $P = MB$), which we saw in Figure 7.17(b).

Therefore, we conclude once again that the monopolist does not achieve allocative efficiency.



at long-run equilibrium production takes place at $P_e = MC$ (allocative efficiency)

at long-run equilibrium production takes place at $P_e > MC$ (allocative inefficiency)

Figure 7.18: Allocative efficiency in perfect competition and allocative inefficiency in monopoly

In monopoly the underallocation of resources to the good is indicated also by $P > MC$ at the profit-maximising level of output.

Market failure and market power

The above discussion shows clearly that monopoly is a type of market failure, since it results in an underallocation of resources to the production of the good with welfare loss.

We can now also see how *market power* is connected to market failure. Earlier in this chapter market power was defined as the ability of a firm to control the price at which it sells its output. Why does this lead to market failure?

The answer lies in Figure 7.16, which compares perfect competition with monopoly. In part (a) we see the perfectly competitive firm with its horizontal demand curve accepting a P that is equal to MC, therefore there is allocative efficiency. In part (b) we see the monopolist with its downward sloping demand curve charging a P that is greater than MC, therefore there is allocative inefficiency. In fact this allocative inefficiency is the result of the downward sloping demand curve with the resulting *market power* that it provides to the firm. We can therefore redefine market power.

Market power refers to the ability of a firm to charge a price greater than marginal cost, or $P>MC$. This can only occur when a firm faces a downward sloping demand curve.

In view of their extensive market power, monopolies are held to be undesirable from a social point of view. For this reason, private, unregulated monopolies are illegal in most countries in the world.

TEST YOUR UNDERSTANDING 7.11

1 Using a diagram, compare and contrast the price and output outcomes of a perfectly competitive industry and an industry organised as a monopoly.

2 Using diagrams, compare and contrast welfare outcomes of a perfectly competitive industry and a monopolistic industry. Explain the meaning of $MB > MC$ at the monopolist's equilibrium level of output.

CONTINUED

3 Using diagrams, show whether or not the monopolist achieves allocative efficiency, and compare with the firm in perfect competition.

4 Using diagrams, explain the relationship between market power and market failure.

Evaluating monopoly and comparing with perfect competition

Criticisms of monopoly

Welfare loss, allocative inefficiency and market failure

In contrast to perfect competition, the monopolist fails to achieve allocative efficiency. Figure 7.17 shows the loss of social surplus and the appearance of welfare loss while Figure 7.18 shows that $P>MC$. Both diagrams also show that $MB>MC$, which are indications that the monopolist underallocates resources to the production of a good. Monopoly therefore represents a form of market failure.

Higher price and lower output in monopoly

The welfare loss noted above in fact arises because the monopolist produces a smaller quantity of output and sells it at a higher price than a perfectly competitive industry as shown in Figure 7.16, once again showing that monopoly is not in the interests of consumers.

Loss of consumer surplus to the monopolist

By charging a higher price than the perfectly competitive firm, such that $P>MC$ means that a portion of consumer surplus is taken away from consumers and gained by the monopolist. Figure 7.17, shows that the monopolist gains at the expense of consumers.

Negative impacts on the distribution of income

Since monopolies charge higher prices than perfectly competitive firms, there is a redistribution of income away from consumers who must pay the higher prices and toward the owners of the monopoly in the form of higher profits.

Lack of competition may give rise to higher costs

Whereas firms in perfect competition are under constant pressure to produce with the lowest possible costs to survive, under monopoly the absence of competitor firms may result in higher average costs for many possible reasons such as a poorly motivated workforce, lack of innovation, poor management or avoidance of risk. The possibility of maintaining abnormal profits over the long run due to high barriers to entry can make the monopolist less concerned about keeping costs low. This is known as *X-inefficiency*.

Possibly less innovative

While monopolies have good reasons to pursue R&D for product development and innovation, the opposite may also occur. High barriers to entry, shielding monopolies from competition, could make them less likely to innovate than firms in monopolistic competition or oligopoly (see below) which are constantly under pressure to innovate to maintain or increase their share of sales. (See Real world focus 7.2.)

Potential benefits of monopoly

Economies of scale

Economies of scale are a major argument in favour of large firms that can achieve lower costs as they grow larger. Monopolies, because of their size, are very well placed to take advantage of economies of scale. Lower average costs provide the monopolist with the possibility of lowering prices, which could possibly approach those achieved in perfect competition. Consumers can therefore gain because lower average costs may potentially translate into lower prices, as well as increased quantity of output. Society also gains from economies of scale because the lower average costs of production mean there is less waste in the use of resources. A perfectly competitive firm, due to its very small size, cannot capture economies of scale.

Natural monopoly

In the event of a natural monopoly, there are added benefits due to the achievement of very low average costs by the single firm. See the discussion above on natural monopolies.

Research and development (R&D) for product development and technological innovation

Firms in perfect competition are unlikely to engage in R&D. They have no abnormal profits in the long-run with which they can finance R&D. They sell identical products and therefore are not interested in product development that would differentiate their products. They are unable to create barriers to entry as they are too small and so have no incentive to engage in R&D.

By contrast, a number of factors suggest that monopolies have good reasons to pursue innovation:

- Their abnormal profits provide them with the ability to finance large R&D projects.
- Protection from competition due to high barriers to entry may favour innovation and product development, by offering firms the opportunity to enjoy the profits arising from their innovative activities (new inventions, new products, new technologies, etc.). This, after all, is the rationale of awarding firms patent protection for a period of time.
- Firms may use product development and technological innovation as a means of maintaining their abnormal profits over the long term, by creating barriers to entry for new potential rivals. If a firm can develop a new product that potential rivals are unable to produce, the rivals may be less likely to try to enter the industry and compete with the innovating monopolist.

TEST YOUR UNDERSTANDING 7.12

1 Using diagrams, compare and contrast the market structures of perfect competition and monopoly, emphasising the advantages and disadvantages of each.

2 Discuss

a the factors that may make monopoly a desirable market structure; and

b why monopoly as a market structure, and market power generally, come under heavy criticism and are held to be against society's best interests.

7.5 Monopolistic competition

LEARNING OBJECTIVES

After studying this section you will be able to:

- define all the terms appearing in **orange bold** in the text (AO1)
- explain that monopolistic competition has less market power due to many substitutes compared to monopoly and illustrate with a diagram showing a more elastic demand curve compared with monopoly (AO2, AO4)
- explain profit maximisation in the short run and long run, with diagrams showing abnormal profit, normal profit, loss (AO2, AO4)
- explain allocative inefficiency and market failure in monopolistic competition (AO2)
- explain that monopolistic competition has more product variety in exchange for less efficiency (AO2)
- discuss the advantages and disadvantages of monopolistic competition (AO3)

The model of *monopolistic competition* is based on the following assumptions:

- There is a large number of firms; this is similar to perfect competition.
- There are no barriers to entry, as in perfect competition.
- There is product differentiation; unlike in perfect competition where products are identical.

Product differentiation can be achieved by:

- physical differences – products may differ in size, shape, materials, texture, taste, packaging, etc. (think, for example, of the variety of clothes, shoes, books, processed foods, furniture)
- quality differences – products can differ in quality
- location – some firms attempt to differentiate their product by locating themselves in areas that allow easy access for customers, such as hotels near airports and convenience stores in residential areas
- services – some firms offer specific services to make their products more attractive, such as home delivery, product demonstrations, free support, warranties and purchase terms
- product image – some firms attempt to create a favourable image by use of celebrity advertising or endorsements, by brand names, or attractive packaging.

Examples of monopolistically competitive industries include book publishing, clothing, shoes, processed foods of all kinds, jewellery, furniture, textiles, dry cleaners, petrol (gas) stations, restaurants.

Product differentiation and the demand and revenue curves

More elastic demand curve compared with monopoly

As the term 'monopolistic competition' suggests, this market structure combines elements of both competition and monopoly. It resembles perfect competition because there are many firms in the industry and there is freedom of entry. It is like monopoly because of product differentiation. Each firm in an industry is a 'mini-monopoly' in the specific version of the good that it produces. For example, Adidas is a monopoly in Adidas® shoes, NIKE is a monopoly in NIKE® shoes, and Puma is a monopoly in Puma® shoes. This means that each of these producers faces a downward-sloping demand curve for its product. However, because each of these products is at the same time a substitute for the other, this demand curve is relatively elastic, i.e. it is more elastic than in monopoly, but less elastic than in perfect competition, as shown in Figure 7.19.

In perfect competition, if a firm raises its price, it loses all its sales to its competitors (Figure 7.19(a)). In monopoly, if a firm raises its price, it loses some but not all sales, as it is the sole producer of the good and consumers have no alternative product they can buy (Figure 7.19(b)). In monopolistic competition (Figure 7.19 (c)), if a firm raises its price, it will lose more sales than the monopolist, because consumers now do have substitutes they can switch to; but it will lose fewer sales than the perfectly competitive firm because of product differentiation – the available substitutes are not perfect substitutes, as they are in perfect competition.

This has important implications: it means that if consumers can be convinced that the product they are purchasing (for example, Puma® shoes) is superior to the available

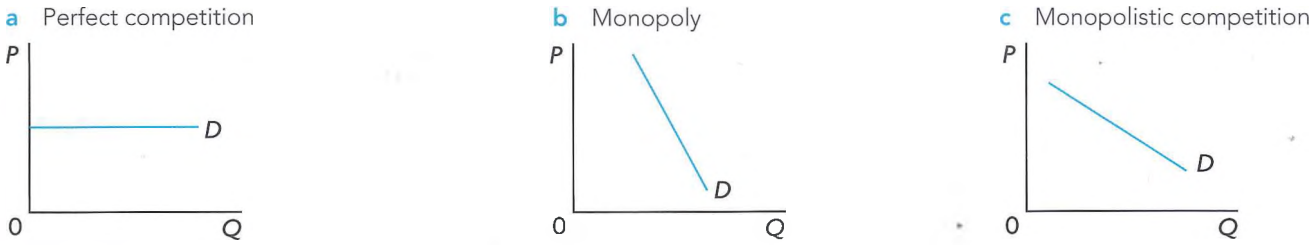


Figure 7.19: Demand curves facing the firm under three market structures

substitutes (Adidas® and NIKE® shoes), then Puma has succeeded in establishing a mini-monopoly for its product. Therefore if the price of Puma shoes increases, only some, and not all, buyers of Puma shoes will switch to other brands. Those who believe that Puma shoes are superior will continue to buy them, in spite of the higher price.

Firms in monopolistic competition face a demand curve that is less elastic than in perfect competition but more elastic than in monopoly.

The roles of price and non-price competition

Price competition occurs when a firm lowers its price to attract customers away from rival firms, thus increasing sales at the expense of other firms. **Non-price competition** occurs when firms use methods other than price reductions to attract customers from rivals. The most common forms of non-price competition are product differentiation (including all the features noted above, such as physical and quality differences, packaging, services provision, location, etc.), advertising and branding (creating brand names for products).

Monopolistically competitive firms compete with each other on the basis of both price and non-price competition. They engage heavily in product differentiation through R&D in product development, as well as in advertising and branding. Firms that can attract customers by use of these methods increase their market power and their ability to charge a higher price without risking loss of buyers to rival firms.

In general, the more differentiated the product is from its substitutes and the more successful the advertising and branding as methods of convincing consumers about the superiority of a product, the less elastic will be the demand curve facing the firm,[3] the greater the market power (the ability to control price), and the larger the firm's potential to increase short-run profits. By contrast, firms that are less able to achieve consumer loyalty for their product, and whose product is less differentiated from substitutes, may have to rely more on price competition to increase their sales.

TEST YOUR UNDERSTANDING 7.13

1 Outline the assumptions defining the market model of monopolistic competition.

2 List some examples of monopolistically competitive firms in your neighbourhood. Analyse what makes them so.

3 Describe how a monopolistically competitive firm is like a firm in perfect competition; how it is like a monopoly.

4 Explain

   a what firms in monopolistic competition try to achieve through product differentiation, advertising and branding, and

   b how these activities affect the demand and revenue curves facing the firm.

5 Explain why we never see price competition and non-price competition in

   a perfectly competitive firms, and

   b monopolies.

   c Explain their importance in monopolistic competition.

[3] Advertising and branding work by making the demand curve shift to the right and making it rotate so it becomes steeper. These two changes mean that demand increases and it becomes less elastic.

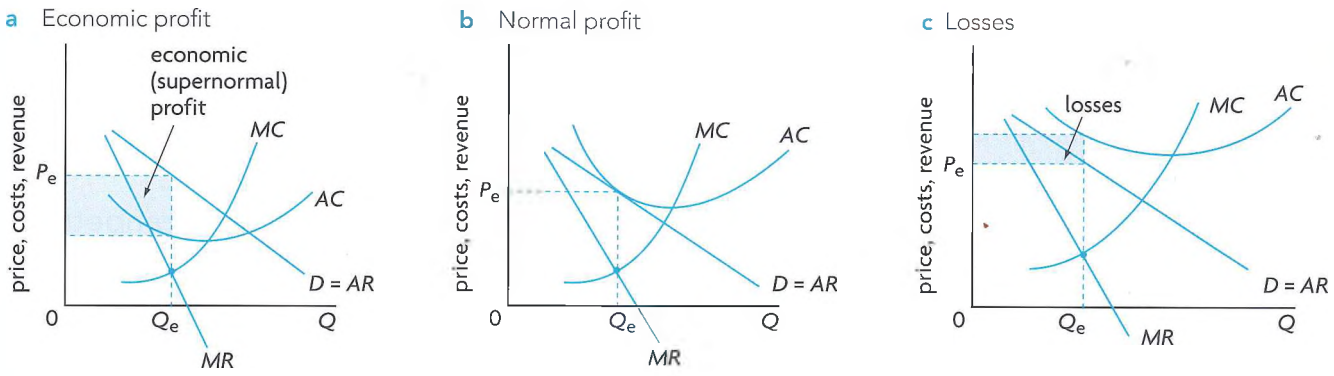


Figure 7.20: Short-run equilibrium positions of the firm in monopolistic competition

Profit maximisation

Abnormal profit and normal profit or loss in the short run

The short-run equilibrium position of the individual firm in monopolistic competition is identical to that of the monopolist, the only difference being that the demand curve is more elastic and flatter in monopolistic competition than in monopoly. In the short run, the firm can make either abnormal profit (i.e. positive profit), normal profit or losses (negative economic profit). Each of these possibilities is shown in Figure 7.20. The firm applies the $MR = MC$ rule to find the profit-maximising or loss-minimising level of output (Q_e), and then for that level of output compares price (given by the demand curve) with AC to determine profit per unit or loss per unit.

In part (a) of Figure 7.20 the firm earns abnormal profits, since $P>AC$ at Q_e; in part (b) the firm's profit is exactly zero since $P = AC$ at Q_e, and therefore the firm is earning normal profit; and in part (c), the firm is making losses because $P<AC$ at Q_e. Total profit or total loss is found by multiplying profit:

$$\frac{profit}{Q} \text{ by } Q \text{ or } \frac{loss}{Q} \text{ by } Q.$$

Normal profit in the long run

The assumption of free entry of firms in the industry is very important in determining the long-run equilibrium of the firm (like in perfect competition).

In monopolistic competition, in the long run, profit-making industries attract new entrants; in loss-making industries, some firms shut down and exit the industry. The process of entry and exit of firms in the long run ensures that economic profit or loss is zero and all firms earn normal profit.

Figure 7.21 shows the long-run equilibrium of the monopolistically competitive firm. At the level of output where $MR = MC$, $P = AC$; therefore, profit is zero and each firm is earning normal profit. This figure is the same as Figure 7.20 where it happens that the firm is earning normal profit in the short run.

The process of adjustment to normal profit in the long run is different to that of perfect competition. In monopolistic competition suppose firms are making abnormal profits in the short run. New firms will enter and they will attract customers away from the existing firms. The result will be to decrease the demand facing existing firms, so that it

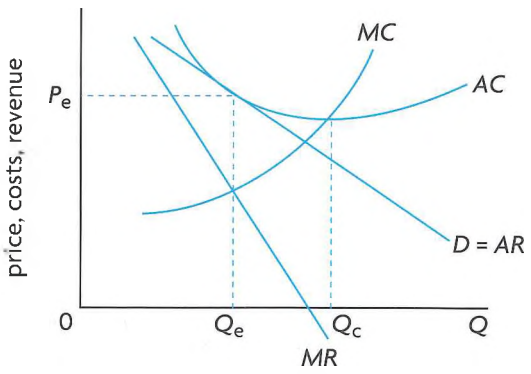


Figure 7.21: Long-run equilibrium of the firm in monopolistic competition

shifts to the left until it is just tangent to (it just touches) the AC curve. When this happens, the firms earn normal profit since $P = AC$. This is shown in Figure 7.21

On the other hand if firms are making losses in the short run, some of them will leave the industry in the long run. As they do so, their customers switch their purchases to the remaining firms, which experience an increase in demand for their product. This appears as a rightward shift of the demand curve facing them, and this process continues until losses disappear and firms are earning normal profit where $P = MC$.

Monopolistic competition, allocative efficiency and market failure

Allocative efficiency is given by the condition $P = MC$. Figure 7.21, showing the long-run equilibrium of the firm in monopolistic competition, indicates that allocative efficiency is not achieved.

Comparing price with marginal cost along the vertical line at the equilibrium level of output, Q_e, we can see that price is higher than MC, indicating that there is an underallocation of resources to the production of the good: society would have liked to have more units of the good produced.

In monopolistic competition $P>MC$ or alternatively $MB>MC$, indicating there is market failure. The market underallocates resources to the production of the good and too little of it is produced.

Inefficiency and product differentiation

Economists argue that some inefficiency in monopolistic competition is justified by the presence of product differentiation, which leads to greater product variety. In other words consumers gain product variety while giving up some efficiency. Perfect competition, by contrast achieves efficiency but at the cost of zero product variety. Because consumers enjoy product variety, it is often argued that with monopolistic competition some inefficiency may be the 'price' consumers pay for having greater product variety.

On the other hand there is much debate among economists about how much product variety is socially desirable. Some economists argue that there is too much product variety, such as for example too many brands of breakfast cereals, detergents, cars or virtually any other product. This leads to confusion among consumers, as well as high costs of advertising and new product development which result in higher prices. It is sometimes argued that consumers would be better off with less product variety and lower prices.

Comparison of monopolistic competition with other market structures

Below is a summary of the important differences between the market models.

Monopolistic competition and perfect competition

Similarities

- Large number of firms.
- Free entry of firms into an industry (no barriers to entry).
- Normal profit in the long run, abnormal profit or loss in the short run (due to no barriers).

Differences

- **Market power and the demand curve.** Firms in perfect competition have no market power; they are price-takers facing a perfectly elastic (horizontal) demand curve they face. Firms in monopolistic competition do have some market power (ability to influence price), reflected in their downward-sloping demand curve.
- **Allocative efficiency.** Whereas the perfectly competitive firm achieves allocative efficiency in long-run equilibrium, the monopolistically competitive firm does not. Fewer than optimal resources are allocated to the production of the good. Since $P>MC$, consumers pay a higher price for the good than in perfect competition. Monopolistic competition is therefore a type of market failure.
- **Product variety.** Whereas all firms in perfect competition produce the identical product, under monopolistic competition firms go to great lengths to differentiate their products. From the consumer's perspective, product variety is usually an advantage; perfect competition cannot offer this advantage. Some inefficiency in monopolistic competition may be the 'price' consumers pay for greater product variety.
- **Economies of scale.** Firms in perfect competition cannot achieve economies of scale because they are very small. Firms in monopolistic competition may have some small room for achieving economies of scale but only to a relatively small degree as these firms also tend to be relatively small.

Monopolistic competition and monopoly

Similarities

- **No allocative efficiency therefore a form of market failure.** Both these market structures face downward-sloping demand curves, and therefore both have *MR* curves that lie below the demand curve. This means that at the profit-maximising level of output (found by $MR = MC$), $P > MC$ for both (i.e. no allocative efficiency).

Differences

- **Number of producers.** While in monopolistic competition there is a large number of firms, in monopoly there is a single firm, or else the industry is dominated by one large firm.
- **Size of firms.** In monopolistic competition firms are usually small, whereas in monopoly the fact that there is a single or dominant firm suggests a very large size.
- **Barriers to entry.** Monopolistic competition is characterised by free entry whereas in monopoly there are high barriers to entry.
- **Normal and abnormal profits.** Whereas the firm under monopolistic competition earns normal profit in the long run, the monopoly can earn abnormal profits due to high barriers to entry.
- **Competition and prices.** Free entry and exit under monopolistic competition drive abnormal profits down to zero in the long run, and allow prices to be lower for the consumer than is possible under monopoly, where barriers to entry allow the firm to maintain abnormal profits over the long run. Therefore prices under monopolistic competition are likely to be lower and quantity larger than in monopoly and more in the interests of consumers.
- **Market power.** While both monopolies and firms in monopolistic competition have market power, a monopoly is likely to have more market power because there are no substitutes for the good it produces. The availability of substitutes means that consumers can switch to substitute goods, thus reducing the firm's market power. This is another reason why prices tend to be lower in monopolistic competition than in monopoly thus favouring consumers.
- **Competition and costs.** Competition between firms in monopolistic competition puts a downward pressure on costs as firms compete with each other. These competitive pressures may force less efficient firms to leave the industry. The absence of competition in monopoly does not exert such a downward pressure on costs.
- **Research and development.** The abnormal profits that monopolies can earn over the long run puts them in a better position than monopolistically competitive firms with respect to financing R&D. However, the pressures of competition faced by monopolistically competitive firms may induce them to pursue R&D for product development in order to maintain or increase their sales.
- **Economies of scale and the possibility of natural monopoly.** Some small economies of scale may be achieved by firms under monopolistic competition, but the potential for this is much greater under monopoly, which can benefit consumers through lower prices.
- **Product variety.** Whereas many monopolies sell more than one product, there is likely to be far greater product variety in monopolistic competition which is characterised by many firms producing products that are substitutes for each other.

TEST YOUR UNDERSTANDING 7.14

1 Use diagrams to explain how a firm in monopolistic competition can

   a earn abnormal profit (show profit per unit and total profit),

   b earn normal profit, and

   c incur losses (show loss per unit and total loss).

2 Outline the role of free entry of firms in monopolistic competition in the adjustment from short-run to long-run equilibrium.

3 a Using a diagram, show the firm's long-run equilibrium position in monopolistic competition.

   b Comment on whether the firm achieves abnormal profit or normal profit in long-run equilibrium.

CONTINUED

4 Explain why the firm in monopolistic competition represents a type of market failure.

5 Explain what is meant by the idea that some inefficiency is the 'price' of product variety.

6 Evaluate monopolistic competition in comparison with

a perfect competition, and

b monopoly.

7.6 Oligopoly

LEARNING OBJECTIVES

After studying this section you will be able to:

- define all the terms appearing in **orange bold** in the text (AO1)
- distinguish between collusive and non-collusive oligopoly and draw a diagram showing collusive oligopoly (AO2, AO4)
- explain features of oligopoly including interdependence, risk of price war, and incentive to collude versus incentive to cheat (AO2)
- explain the relevance of price and non-price competition for firms in oligopoly (AO2)
- explain the presence of allocative inefficiency and market failure (AO2)
- explain the simple game theory payoff matrix (AO2)
- explain the meaning of market concentration and concentration ratios (AO2)
- discuss advantages and disadvantages of oligopoly (AO3)

Oligopolies have the following characteristics:

- **There is a small number of large firms.** The term 'oligopoly' derives from the Greek word ολιγοπώλιο (oligopolio) meaning 'few sellers'. Oligopolistic industries are dominated by a small number of large firms, though in any one industry the firms are likely to vary in size.
- **There are high barriers to entry.** All the barriers to entry discussed under monopoly are relevant to oligopoly. They include economies of scale (for example, the aircraft and car industries); legal barriers such as patents (the pharmaceutical industry); control of natural resources (such as oil, copper, silver); aggressive tactics such as advertising or threats of takeovers of potential new firms. An additional barrier to entry in oligopoly involves high start-up costs (the costs of starting a new firm) associated with developing a new or differentiated product. Many established oligopolies spend enormous sums on product differentiation and advertising, making it difficult for new firms to match such expenditures.
- **There is interdependence.** Firms in perfect and monopolistic competition, due to their large numbers in an industry, behave independently of each other, so when they make decisions such as how much to produce they do not take the possible actions of other firms into consideration. By contrast, the small number of firms in oligopolistic industries makes the firms mutually interdependent; decisions taken by one firm affect other firms in the industry. If any one firm changes its behaviour, this can have a major impact on the demand curve facing the other firms. Therefore, firms are keenly aware of the actions of their rivals.

Most products of oligopolistic firms are differentiated such as pharmaceuticals, cars, aircraft, breakfast cereals, cigarettes, refrigerators and freezers, cameras, tyres, bicycles, motorcycles, soaps, detergents. Homogeneous or undifferentiated products include oil, steel, aluminium, copper, cement.

Interdependence, game theory and price and non-price competition

Interdependence

The interdependence of oligopolistic firms has important implications for their behaviour:

- **Strategic behaviour.** Strategic behaviour is based on plans of action that take into account rivals' possible courses of action. It is similar to playing

a card game, or chess, where individual players' actions are based on the expected actions and reactions of their rival(s). Strategic behaviour of oligopolistic firms is the result of their interdependence. For example, a firm plans a course of action X assuming its rivals will follow one policy, and it plans course of action Y assuming its rivals follow a different policy. Under oligopoly, firms planning their strategies make great efforts to guess the actions and reactions of their rivals in order to formulate their own strategy.

- ***Conflicting incentives.*** Firms in oligopoly face incentives that conflict, or clash with each other:
  - *Incentive to collude* – the term *collusion* refers to an agreement between firms to limit competition between them, usually by fixing price and therefore lowering quantity produced. By colluding to limit competition, they reduce uncertainties resulting from not knowing how rivals will behave, and maximise profits for the industry as a whole.
  - *Incentive to compete, or to cheat in a collusive agreement* – at the same time, each firm faces an incentive to compete with its rivals in the hope that it will capture a portion of its rivals' market shares and profits, thereby increasing profits at the expense of other firms. If firms have formed a collusive agreement, they face an incentive to cheat on their 'partners' in the agreement in order to increase their profits at their expense.

Explaining oligopolistic behaviour by use of game theory: collude or compete/cheat

The characteristics of interdependence, strategic behaviour, and conflicting incentives are illustrated very effectively by **game theory**, a mathematical technique analysing the behaviour of decision-makers who are dependent on each other, and who display strategic behaviour. Game theory has become an important tool in microeconomics, and is based heavily on the work of American mathematician and economist John F. Nash (the subject of the 2001 film, *A Beautiful Mind*), who together with John Harsanyi and Reinhard Selten, received the 1994 Nobel Prize in Economics.

The game we will use here illustrates the *prisoner's dilemma*, showing how two rational decision-makers, who use strategic behaviour to maximise profits by trying to guess the rival's behaviour, may end up being collectively worse off. The final position that results from the game is called a *Nash equilibrium*.

Suppose there are two oligopolistic firms in the space travel industry: Intergalactic Space Travel (IST) and Universal Space Line (USL). Each firm must decide on a pricing strategy, i.e. what price to charge consumers for its space travel services, and can choose either a high-price or a low-price strategy. Each firm is interested in making its own profit as large as possible, but its profit will depend on the particular combination of pricing strategies that the two firms choose.

Figure 7.22 shows four possible combinations of pricing strategies and their corresponding profit outcomes (called 'payoffs') for the two firms. This figure represents a **payoff matrix**. For example, if both IST and USL choose the high-price strategy, in box 4, each will have profit of 40 million Zelninks (abbreviated as Zs). Box 3 shows the profit outcomes of differing price strategies; USL with a low-price strategy makes 70 million Zs, and IST with a high-price strategy makes 10 million Zs. The reason why the low-price firm makes much higher profits is that by charging a low price it captures a large portion of sales from its rival.

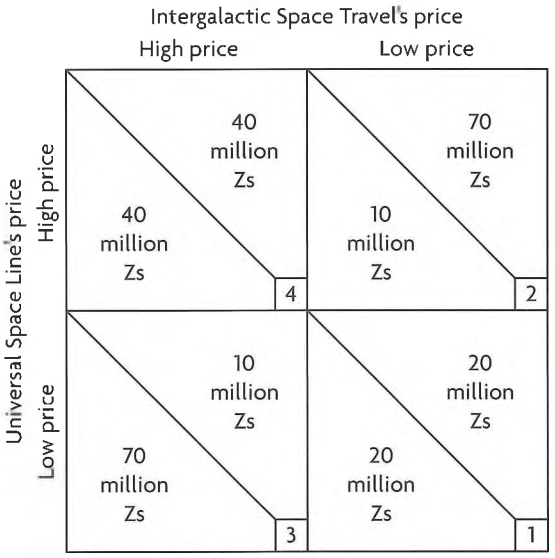


Figure 7.22: Game theory: the prisoner's dilemma

Suppose the two firms begin in box 1, where they are competing with each other on the basis of price (price competition) and therefore have a low price, leading to a low profit of 20 million Zs each. Realising that they will both be better off if they enter into a collusive agreement and charge a high price, they collaborate and agree to adopt a high-price strategy, thus entering box 4 where each one earns profits of 40 million Zs.

Now each firm faces a dilemma. Let's look at the dilemma from IST's point of view (though USL is thinking along the same lines). IST realises that by sticking to the agreement, it will continue to earn 40 million Zs, along with USL. On the other hand, IST also realises that by cheating, in other words secretly breaking the agreement, and charging a lower price, it can earn the much higher profit of 70 million Zs, while USL earns only 10 million Zs. In addition, IST realises that USL might break the agreement, in which case IST will find itself making only 10 million Zs (worse than even when it was in competition with USL, making 20 million Zs).

What should IST do? As it tries to 'outguess' USL, it is likely to cut its price to beat USL to the higher profits, but since USL is thinking along exactly the same lines, they are both likely to adopt the low-price strategy, in which case they will end up in box 1 where they both have low prices and low profits. This is the Nash equilibrium, in which both firms become worse off.

> The Nash equilibrium shows that there is sometimes a conflict between the pursuit of individual self-interest and the collective firm interest. This conflict is the prisoner's dilemma. Although the firms could be better off by cooperating, each firm, trying to make itself better off, ends up making both itself and its rival worse off.

This game illustrates many real-world aspects of oligopolistic firms, which:

- are interdependent – what happens to the profits of one firm depends on the strategies adopted by other firms; they therefore try to predict the actions of their rivals in order to plan out their own strategy
- display strategic behaviour – they plan their actions based on guesses about what their competitors are likely to do
- face conflicting incentives – they face the incentive to collude (agree to fix prices and move to box 4 where they both earn high profits); and they face incentives to compete, or in this case to 'cheat' on the agreement, by lowering their price
- become worse off as a result of price competition (trying to capture sales from their rivals by cutting prices) – since the rivals are likely to match the price cuts, all firms end up with lower prices and lower profits (box 1); this is called a **price war**
- have a strong interest in avoiding price wars, because they realise that every one will become worse off through price cutting – this creates a strong incentive for them to compete on the basis of factors other than price (non-price competition).

The roles of price and non-price competition in oligopoly

Unlike firms in monopolistic competition that compete on the basis of both price and non-price competition, oligopolistic firms go to great lengths to *avoid price competition*. They are very careful not to trigger a price war, where one firm's price cut is matched by a retaliatory price cut by another firm. As our discussion of game theory showed, a price war makes all the firms of an industry collectively worse off due to lower prices and lower profits. A price war may even lead to prices lower than average costs, and therefore losses.

However, oligopolistic firms usually do engage in intense *non-price competition*, involving efforts by firms to increase market share by methods other than price, which typically include the following:

- product development
- advertising
- branding
- numerous services such as quality customer service, warranties, provision of credit, discounts on upgrades and others.

Non-price competition is very important in oligopoly for the following reasons:

- Oligopolistic firms often have considerable financial resources (due to large profits) that they can devote to both R&D and advertising and branding. Whereas monopolistically competitive firms also engage in non-price competition, their resources for these purposes are generally not as large.
- The development of new products provides firms with a competitive edge; they increase their market power, demand for the firm's product becomes less elastic, and successful products give rise to opportunities for substantially increased sales and profits.

- Product differentiation can increase a firm's profit position without creating risks for immediate retaliation by rivals. It takes time and resources for rival firms to develop new competitive products.

TEST YOUR UNDERSTANDING 7.15

1 Identify the main characteristics of an oligopoly.

2 a Identify the conflicting incentives faced by oligopolistic firms.

b Explain how they relate to their interdependence.

3 a Referring to the conflicting incentives faced by oligopolistic firms, explain why the payoff matrix shown below illustrates the 'prisoner's dilemma' confronting the players of this game.

b Explain the possible profit outcomes of the two firms.

c Describe how these outcomes are related to the firms' interdependence.



4 Provide examples of non-price competition, and explain why it is important to firms in oligopoly.

5 a Outline why oligopolistic firms avoid price competition.

b Define a price war, and outline why oligopolistic firms avoid it.

Collusive and non-collusive oligopoly

There are two predominant types of oligopolies: collusive and non-collusive.

Collusive oligopoly

Collusive oligopoly refers to situations where firms agree to *collude*, which means they form an agreement between themselves to limit competition, increase market power and increase profits. The most common form of **collusion** involves agreements to fix prices such as by holding prices constant at some level, raising prices by some fixed amount, fixing price differences between different products, and others.

Collusion is illegal in most countries, because it works to limit competition.

Cartels

A common type of collusion involves the formation of a *cartel*, which is a formal agreement between firms in an industry to take actions to limit competition in order to increase profits. The agreement may involve limiting and fixing the quantity to be produced by each firm, which results in an increase in price; fixing the price at which output can be sold; dividing the market according to geographical or other factors; or agreeing to set up barriers to entry. Whatever the case, the objective is to limit competition, increase the market power of the firms, and increase profits.

Suppose the firms of an industry decide to form a cartel by fixing price. Figure 7.23 illustrates how the cartel maximises profit. Note that this figure is identical to Figure 7.14(a), which illustrates profit maximisation for a monopolist.

> The key objective of a cartel is to limit competition between the member firms and attempt to maximise joint profits. Cartel members collectively behave like a monopoly.

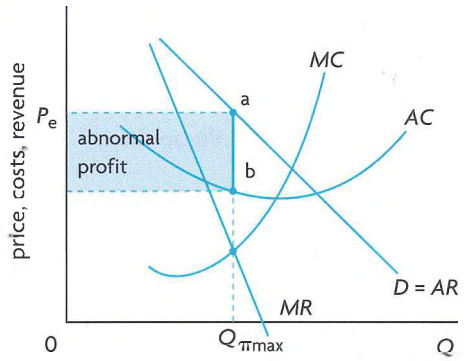


Figure 7.23: Profit maximisation by a price-fixing cartel

In Figure 7.23 the demand curve and marginal revenue curve shown are for the industry as a whole. The *MC* curve is the sum of all the *MC* curves of all the firms in the cartel. The cartel equates *MR* with *MC* to find the cartel's profit-maximising level of output, $Q_{\pi max}$, and then determines price P_e (given by the demand curve). It is then a question of dividing up industry output $Q_{\pi max}$ between all the firms, or deciding how much of the total quantity will be produced by each firm. One way this can be done is to agree on what share of the market each firm will have based on historical market shares. Another way is that firms may agree to compete with each other for market shares using non-price competition (product differentiation and advertising).

The best-known example of a cartel is OPEC (Organization of the Petroleum Exporting Countries), composed of a group of 13 oil-producing countries. OPEC periodically tries to raise the world price of oil by cutting back on its total output. Each member country is assigned an output level (quota) that it is permitted to produce. The restricted quantity of oil results in a higher price.

Firms participating in a cartel have much to gain:

- increased market power and hence the ability to control price of the product
- increased profits due to higher prices
- elimination of competition between the firms, and therefore no more uncertainty or need to outguess their rivals.

However, collusive oligopoly is illegal in most countries, for the same reasons that private unregulated monopolies are also illegal: they restrict competition, raise prices, reduce quantities of output and are held to be against consumers' and society's best interests. However, in practice it is difficult for authorities to discover and prove the existence of a cartel. It is suspected that there may be far more cases of collusive oligopoly than what are discovered.

While cartels bring important benefits to their members, they are not easy to create and maintain for several reasons.

- **The incentive to cheat.** Every firm in a cartel faces an incentive to cheat on the agreement, by offering to secretly lower the price for some buyers. But if many firms cheat, or if cheating is discovered by other firms in the cartel, then the cartel may collapse.
- **Cost differences between firms.** Since the price agreed upon by the cartel is common to all the firms, firms with higher average costs have lower profits, while lower-cost firms enjoy higher profits. Cost differences between firms lead to difficulties in agreeing on a common price.
- **Number of firms.** The larger the number of firms, the more difficult it is to arrive at an agreement regarding price and the allocation of output, as the greater number of differing views make agreement and compromise more difficult to achieve.
- **The possibility of a price war.** A possible outcome of one or more firms cheating on the cartel agreement is a price war, where one firm's price cut is matched by retaliatory price cuts by other firms. The result of a price war is to make all the firms of an industry collectively worse off due to lower prices and lower profits (in the example of game theory).

Other types of collusion

The difficulties involved in establishing and maintaining cartels as well as their illegality sometimes make firms turn towards informal types of collusion. *Informal collusion* refers to co-operation that is implicit or understood between the co-operating firms, without a formal agreement. The objectives of informal collusion are also to co-ordinate prices, avoid competitive price-cutting, limit competition, reduce uncertainties and increase profits.

One type of informal collusion is *price leadership*, where a dominant firm in the industry (which may be the largest, or the one with lowest costs) sets a price and also initiates any price changes. The remaining firms in the industry become price-takers, accepting the price that has been established by the leader. The implicit agreement (as there is no formal agreement) binds the firms as far as price goes, but they are free to engage in non-price competition. A characteristic of price leadership arrangements is that price changes tend to be infrequent, and are undertaken by the leader only when major demand or cost changes occur. Examples include the airline industry, supermarkets and fast food restaurants.

Non-collusive oligopoly

Non-collusive oligopoly refers to oligopolistic firms that do not collude in any way in order to fix or coordinate prices and limit competition. Each firm behaves independently; however, they are still aware of each other in their pricing decisions and display strategic behaviour in that they take the possible actions of their rivals into consideration.

It can be observed that in the real world, prices of oligopolistic industries tend to be rigid or inflexible; once a particular price is reached, it tends to be relatively stable over long periods of time. Moreover, in situations when prices do change, they tend to change together for all the firms in an industry. Such price rigidities can be easily explained by collusive oligopoly, since firms behave like a monopoly in such cases and so tend to have similar prices, but how can they be explained in situations where firms do not collude?

We can understand the observed price stability of oligopolistic firms that do not collude using the idea of strategic behaviour. Suppose there are three oligopolistic firms, A, B and C producing a similar product. A, B and C do not collude or communicate with each other in any way, instead their pricing behaviour is strategic, and is strongly influenced by their expectations of how rival firms will react if they undertake a price change.

Suppose firm A considers a price change, but before changing (increasing or decreasing) its price, it tries to predict how firms B and C will react, and what will be the consequences of their reaction. Firm A's reasoning is as follows:

- If I raise my price, what will B and C do? They are unlikely to increase their price, because if they continue to sell at their lower price they will take away a portion of my sales, they will be better off and I will be worse off. Therefore I should *not increase my price*.
- If I drop my price, what will B and C do? They are likely to drop their price as well, and so as a result, I am unlikely to be much better off than I am now, and I may be worse off with a lower price, so I should *not drop my price*.

This line of reasoning is the same for all three firms, A, B and C, and tends to result in price stability.

This simple idea illustrates three important points:

- **Firms that do not collude are forced to take into account the actions of their rivals in making pricing decisions.** Otherwise they risk lowering their revenues and profits, which in turn could lead to price instability.
- **Even though the firms do not collude, there is still price stability.** Firms are reluctant to change their price because of the likely actions of their rivals, which could result in lower profits for the firm initiating price changes.
- **Firms do not compete with each other on the basis of price.** They do not try to increase their sales by attracting customers through lower prices. A lower price not only invites price cuts by rivals, with resulting lower profits for all the firms, but also risks setting off a *price war*.

The concentration ratio

As oligopolies involve a small number of large firms that dominate an industry, it is important to know how 'concentrated' the industry's output is among the industry's largest firms, as this information may provide clues on whether oligopolies have too much market power. A **concentration ratio** provides an indication of the percentage of output produced by the largest firms in an industry; it measures **market concentration**. There is no fixed number of firms for which a concentration is calculated. For example, we could say that the 3-firm concentration ratio of industry X is 78%, which means that the three largest firms of industry X produce 78% of the industry's total output; or the 4-firm concentration ratio of industry Y is 45%, which means that the four largest firms in industry Y produce 45% of the industry's total output. Table 7.7 provides some examples of concentration ratios in the United States. We can see there are wide variations from industry to industry, with the most concentrated industry of those appearing in the table being transportation equipment and the least concentrated being furniture.

Industry	4-firm	8-firm
Food	35.0%	46.7%
Chemicals	44.8%	58.7%
Transportation equipment	64.3%	77.7%
Furniture	18.8%	26.5%
Textile mill products	32.0%	44.9%

Source: *http://www.swarthmore.edu/SocSci/Economics/fpryor1/Concentration_ratios.pdf*

Table 7.7: Selected concentration ratios in domestic US manufacturing

Concentration ratios provide an indication of the degree of competition in an industry. They suggest that the higher the concentration ratio, the lower the degree of competition, while a low concentration ratio would indicate a greater degree of competition.

In general, an industry is considered to be oligopolistic if the four largest firms control 40% of output. (This is an arbitrary cut-off point, as there is nothing special about a concentration ratio of 40%.)

Concentration ratios have several weaknesses that limit their usefulness as a measure of the degree of competition:

- Whereas concentration ratios reflect concentration in a national market, they do not reflect competition from abroad, arising from imports.
- Concentration ratios provide no indication of the importance of firms in the global market; there may be some competition in a domestic market, but the firms may have a very strong, or dominant position in the global market.
- Concentration ratios do not account for competition from other industries, which may be important in the case of substitute goods, such as in the case of different metals. Whereas there may be a high concentration ratio in the aluminium industry, for example, this would be lower if considered together with copper, with which aluminium competes.
- Concentration ratios do not distinguish between different possible sizes of the largest firms. For example, a three-firm concentration ratio of 90% could consist of three firms with 30% of the market each, or of three firms, one of which has 60% of the market and the other two have 15% each.

REAL WORLD FOCUS 7.1

Pharmaceutical oligopoly faces lawsuit

Forty-four states in the United States filed a lawsuit in 2019 against 20 pharmaceutical firms for fixing prices of over 100 common generic (non-patented) medicines, including treatments for diabetes and cancer. Following a five-year investigation, the firms have been accused of illegal conspiracies to either stop prices from falling or to raise prices, in some cases by more than 1000%. The Attorney General of the State of Connecticut states that:

'We have hard evidence that the generic drug industry perpetrated a multi-billion-dollar fraud on the American people. We have emails, text messages, telephone records and former company insiders that we believe will prove a multi-year conspiracy to fix prices and divide market share for huge numbers of generic drugs.'



Figure 7.24: Capsules containing business and economics formulas

The legal action seeks damages, civil penalties and court actions to restore competition to the generic drug market. Generic drugs are lower-priced alternatives to brand-name drugs and as a result save drug buyers and taxpayers billions of dollars a year. Yet according to the lawsuit:

'Prices for hundreds of generic drugs has risen – while some have skyrocketed, without explanation, sparking outrage from politicians, payers and consumers across the country.'

Sources: *www.cnbc.comw; www.bmj.com*

Applying your skills

1 Identify this pharmaceutical oligopoly as collusive or non-collusive, offering reasons why.

2 Draw a diagram and use it to explain how this oligopoly fails to achieve allocative efficiency.

3 Normally, it is very difficult to determine the presence of collusion. Outline why this may not be the case here.

Evaluating oligopoly

Oligopoly, allocative efficiency and market failure

As firms in collusive oligopoly behave like a monopoly, it is clear that there is welfare loss, allocative inefficiency and market failure, as Figure 7.22 demonstrates.

Yet the same is also true of oligopolistic firms that do not collude. They also face a downward sloping demand curve, and the market produces a level of output that is below the level where social surplus is maximum. Therefore, there is welfare loss here too as in the case of collusive oligopoly.

Criticisms of oligopoly

To the extent that oligopolistic firms succeed in avoiding price competition, they achieve a considerable degree of market power, and therefore face similar criticisms as monopoly:

- Welfare loss, allocative inefficiency and market failure.
- Higher prices and lower quantities of output than under competitive conditions.
- Loss of consumer surplus to the oligopolists due to higher prices resulting in $P>MC$.
- Negative impacts on the distribution of income.
- There may be higher production costs due to lack of price competition.
- Possibly less innovative.

In addition, there is a further argument against oligopoly:

- Whereas many countries have anti-monopoly legislation that protects against the abuse of market power, the difficulties of detecting and proving collusion among oligopolistic firms means that such firms may actually behave like monopolies by colluding and yet may get away with it.

Benefits of oligopoly

The benefits of oligopoly are also similar to the benefits of monopoly:

- Economies of scale can be achieved due to the large size of oligopolistic firms, leading to lower production costs to the benefit of society and the consumer (through lower prices).
- Product development and technological innovations can be pursued due to the high abnormal profits from which research funds can be drawn. This benefit of oligopoly is more important than in the case of monopoly, since non-price competition forces firms to be innovative in order to increase their market share and profits.
- Technological innovations that improve efficiency and lower costs of production may be passed to consumers in the form of lower prices.

Over and above the benefits of oligopoly that are similar to monopoly, oligopoly also offers the following advantage:

- Product development leads to increased product variety, thus providing consumers with greater choice (monopoly does not offer much product differentiation and variety).

TEST YOUR UNDERSTANDING 7.16

1 a Outline the meaning of collusion.
  b Identify what firms in an oligopoly try to achieve through collusion.

2 a Define a cartel.
  b Using a diagram, show how a cartel resembles a monopoly.
  c Explain why cartels are illegal in most countries.

3 a Outline why the cartel members face the incentive to cheat.
  b Identify some obstacles to forming and maintaining cartels.

4 Explain how non-collusive oligopoly differs from collusive oligopoly.

5 a Explain the meaning of 'concentration ratio' and provide examples (they may be hypothetical).
  b State the purpose of calculating concentration ratios.
  c Identify some shortcomings of concentration ratios.

6 'Oligopoly is a type of market failure.' Justify this statement.

7 Discuss potential advantages and disadvantages of oligopolistic firms.

THEORY OF KNOWLEDGE 7.1

Perfect competition and the real world

The model of perfect competition is the basis of an idealised free-market economy, where price is the rationing system that supplies answers to the *what* and *how to produce* questions.[4] The market and the price mechanism are the means by which allocative efficiency is achieved, thus avoiding waste of resources.

Many economists have strongly criticised preoccupation with the perfectly competitive model. They note that the real world is dominated by large oligopolistic firms and monopolistically competitive firms, and the model of perfect competition bears no relationship to either of these. Trying to test the perfectly competitive model would be meaningless given the real-world context of firm behaviour. The very concept of 'competition' has vastly different meanings, with competition in perfect competition being anonymous and equivalent to complete absence of market power (no ability to influence price), and competition in the other two market models involving efforts to capture market shares from rival firms, either through price cuts or through product differentiation and advertising (non-price competition). These real-world types of competition are completely irrelevant for the perfectly competitive firm that can neither gain anything by lowering its price nor can it differentiate and advertise its product.

In addition, many economists question the notion of equilibrium (an idea borrowed from physics), and point out that equilibrium for the firm makes sense only under conditions of complete certainty and perfect knowledge. As there is no such thing in the real world, there can be no equilibrium.

On the other hand, defenders of the perfectly competitive market model argue that this can be used as a 'parable' that can approximate (or describe very roughly) the outcomes of real-world firm behaviour, regardless whether these firms are oligopolistic or monopolistically competitive. They also argue that in spite of its lack of realism, this model serves as a tool for assessing real-world situations that depart from the ideal of allocative efficiency, thus helping governments prescribe policy measures to deal with issues in the industrial sector.

Thinking points

- How useful do you think is the model of perfect competition?
- Do you think it matters that it is based on highly unrealistic assumptions (see also Theory of knowledge 2.3 in Chapter 2)?
- Do you think economists should focus more on developing and using more realistic market models, based on monopolistic competition and oligopoly?

[4] It also answers the *for whom to produce* question on income distribution. However, the answer provided is generally highly unsatisfactory, and for this reason becomes a normative issue about how governments should intervene in markets to change market-determined income distribution (see Chapter 12).

7.7 Government intervention in response to abuse of market power

LEARNING OBJECTIVES

After studying this section you will be able to:

- define all the terms appearing in **orange bold** in the text (AO1)
- discuss advantages of large firms with significant market power, including economies of scale and investments in R&D financed by abnormal profits, innovation (AO3)
- discuss risks with respect to output, price, consumer choice in markets dominated by large firms (AO3)
- discuss advantages and disadvantages of government intervention to deal with abuse of market power including legislation and regulation, government ownership, fines (AO3)

Advantages and risks of large firms with significant market power

In the discussion above on monopoly and oligopoly we saw that large firms with significant market power offer a number of potential advantages, including:

- advantages arising from large economies of scale and natural monopoly
- the ability of such firms to carry out R&D on new product and technology development.

We have also seen that there are a number of disadvantages and risks in terms of:

- allocative inefficiency and welfare loss
- higher prices and lower output
- loss of consumer surplus to the large firm or firms
- negative impact on the distribution of income
- higher than necessary average costs due to lack of price competition
- possibly less innovative.

Government intervention to address abuse of market power: an evaluation

The meaning of *abuse* of market power

We have seen that all market structures other than perfect competition represent market failure, resulting in a price above marginal cost ($P>MC$), and welfare loss to a lesser or greater degree. However **abuse of market power**, also known as *anti-competitive practices*, refers to situations where firms engage in activities that result in reduced competition. According to the European Commission (of the EU):

> *'A company can restrict competition if it is in a position of strength on a given market. A dominant position is not in itself anti-competitive, but if the company exploits this position to eliminate competition, it is considered to have* ***abused*** *it.'*

Examples include:

- charging unreasonably high prices
- depriving smaller competitors of customers by selling at artificially low prices they can't compete with
- obstructing competitors in the market (or in another related market) by forcing consumers to buy a product which is artificially related to a more popular, in-demand product
- refusing to deal with certain customers or offering special discounts to customers who buy all or most of their supplies from the dominant company
- making the sale of one product conditional on the sale of another product.[5]

In general we can say the following:

- *Monopolies* on the whole have the highest degree of market power which is abusive due to the lack of competition; this is why private, unregulated monopolies are illegal most everywhere.
- *Firms in oligopolies* may or may not abuse their market power. The market power of a collusive

[5] http://ec.europa.eu/competition/consumers/abuse_en.html

REAL WORLD FOCUS 7.2

Risks of increasing market concentration

In 2019 the International Monetary Fund (IMF, see Chapter 20) published a study providing evidence in support of the very popular notion that competition is weakening as markets are becoming more and more concentrated. The study examined markups over marginal costs (the amount added to cost to obtain the price) of 900 000 firms in 27 countries. Markups over marginal cost are an indicator of market power. It found that markups increased by 8% on average in the period 2000–2015, increasing most in the United States and to a lesser extent in Europe.

But the study also found that the rising markups are due to only a small share of the firms. The 10% of firms with the highest markups increased these by more than 30% compared to their competitor firms. Moreover, mergers and acquisitions, which clearly result in increased concentration, are followed by significantly higher markups.

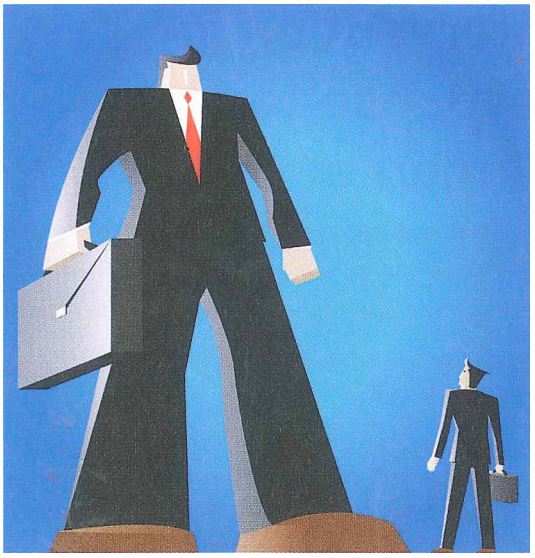


Figure 7.25: David vs Goliath business competition, illustrating great difficulties for small businesses against large corporations

In addition it found that higher markups are linked with less investment in physical capital, which is estimated to have resulted in lower GDP growth by one percentage point in advanced economies. Further, the firms with the highest markups pay workers a smaller share of the value they create, contributing to income inequality. Finally, the study warns that market power may result in less innovation as firms become too comfortable in their position due to decreasing competitive pressures.

These negative trends may worsen if market power is not checked.

Source: *www.economist.com*

Applying your skills

1 Outline the meaning of increasing market concentration

2 Describe the evidence used by the IMF to conclude that there is increasing market concentration.

3 Explain the risks to consumers and the economy of increasing concentration of markets.

oligopoly is similar to that of a monopoly, and represents a form of abuse; therefore collusion is illegal most everywhere. Non-collusive oligopolies may or may not abuse their market power, depending on the situation.

- *Firms in monopolistic competition* have less market power which for the most part they do not abuse because there is significant competition among firms that produce substitute goods.

Therefore government intervention in response to abuse of market power occurs mainly in the cases of monopoly and oligopoly.

Legislation to protect competition

Most countries have laws that try to promote competition by preventing *collusion* between oligopolistic firms, as well as preventing anti-competitive behaviour by a single firm that dominates a market (see Real world focus 7.1). This is known as *competition policy*. Firms that are found guilty of anticompetitive behaviour are usually asked to pay fines (see Real world focus 7.5), or may be broken up into smaller firms.

Difficulties that may arise in connection with competition policies include:

- Possible difficulties in interpreting the legislation in connection with the behaviour of the offending firms. Different people may have different views on what actions involve anti-competitive behaviour. The laws themselves may be vague, allowing much room for different interpretations. According to the OECD, 'Determining when a firm's behaviour is an abuse of market power, as opposed to a competitive action, is one of the most complex and controversial areas in competition policy.'[6]
- Laws in a particular country may be enforced to varying degrees, with some governments enforcing them more strictly than others, depending on their priorities or their political and ideological views. Some governments may accept the principle that government intervention in the market in the form of strict enforcement of competition policies is necessary to protect consumers against monopolistic practices and to achieve allocative efficiency. Other governments may accept the principle that government intervention in the market is not necessary to achieve consumer protection and allocative efficiency, because over long periods of time, the market and competitive forces on their own accomplish these functions. There is no 'right' or 'wrong' answer to this issue, as it depends on normative ideas about the economy.
- If firms collude, it is difficult to discover evidence of the collusion and to prove it, as collusion occurs secretly, since it is illegal.

Legislation in the case of mergers

A merger is an agreement between two or more firms to join together and become a single firm. Mergers may occur for a number of reasons, such as an interest in capturing economies of scale (a single larger firm may be able to produce at lower average costs), or an interest in firm growth (the firms would like to become larger), or interest in acquiring market power, which is made possible by the larger size of the new, larger firm.

Mergers are an issue in competition policy because of the possibility that the single firm created from the merger have too much market power. Legislation usually involves limits on the size of the combined firms. (See Real world focus 7.4.)

Difficulties with merger policies include questions and uncertainties about what firms should be allowed to merge and what firms should not, related to issues of interpreting the legislation as well as ideological differences among different governments on the desirability or not of a high degree of market power.

The imposition of fines

Fines are often imposed if a government agency responsible for investigating anti-competitive behaviour discovers some wrongdoing (see Real world focus 7.5 below for examples).

A problem with fines is that firms will often get their lawyers to calculate whether breaking the law or complying with the law is more costly (or more profitable). Often, the profits of firms that abuse market power are great enough that they are better off illegally abusing their power and paying fines, than not abusing their power and not paying fines. Moreover, often they are not caught for anti-competitive behaviour, possibly because of the political climate, or because they are careful to cover their actions.

Big firms often neglect the ethics of wrongful behaviour if they believe that getting caught is not as costly as compliance.

[6] http://www.oecd.org/competition/abuse

REAL WORLD FOCUS 7.3

Conflicting views regarding regulation

In 2018 the European Commission (of the EU) imposed a fine of $5 billion on Google for violating the EU's competition rules by requiring phone manufacturers to preinstall Google Search and its browser app, Chrome, in order to access Google Play, Google's app store. According to the EU, 'These practices have denied rivals the chance to innovate and compete on the merits. They have denied European consumers the benefits of effective competition in the important mobile sphere.'[7]

Soon after President Donald Trump tweeted 'The European Union just slapped a Five Billion Dollar fine on one of our great companies, Google. They truly have taken advantage of the U.S., but not for long!' Some years earlier President Barack Obama had stated:

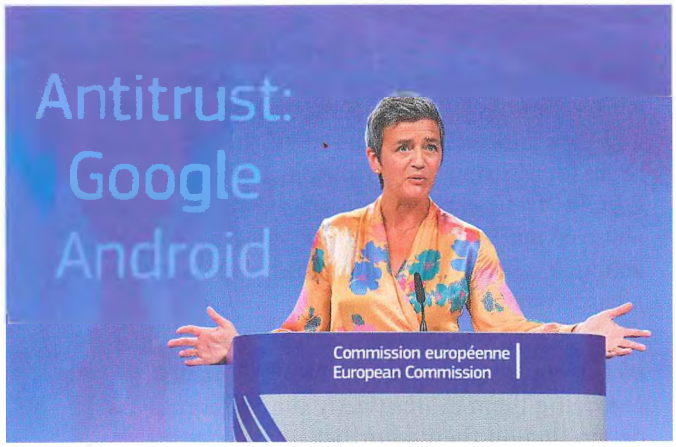


Figure 7.26: European Commissioner Margrethe Vestager, 18 July 2018, when the EU gave Google 90 days to end illegal practices or face further fines (after being fined $5billion)

Sometimes the European response here is more commercially driven than anything else. We have owned the internet, our companies have created it, expanded it, perfected it in ways that they [European firms] can't compete. And oftentimes what is portrayed as high-minded positions on issues sometimes is just designed to carve out some of their commercial interests.[8]

Later US Senator Elizabeth Warren argued:

Today's big tech companies have too much power—too much power over our economy, our society, and our democracy. They've bulldozed competition, used our private information for profit, and tilted the playing field against everyone else. And in the process, they have hurt small businesses and stifled innovation.[9]

To deal with this, she proposes breaking up the big tech companies.

Tyler Cowen, a US economics professor disagrees, arguing that large tech companies are major innovators, and breaking them up could weaken them and therefore hurt innovation.

Sources: *https://europa.eu; www.newyorker.com; https://medium.com/*

Applying your skills

1 Describe why the European Commission thought Google was abusing its market power.

2 To what extent do you think the European Commission's concerns are legitimate?

3 Compare and contrast the policies of an imposition of a fine versus legislation to break up Google.

[7] https://europa.eu/rapid/press-release_IP-18-4581_en.htm
[8] https://www.newyorker.com/news/our-columnists/why-did-the-european-commission-fine-google-five-billion-dollars
[9] https://medium.com/@teamwarren/heres-how-we-can-break-up-big-tech-9ad9e0da324c

REAL WORLD FOCUS 7.4

Competition authorities reject UK supermarket merger

In 2019, Britain's Competition and Markets Authority (CMA) blocked a merger between Asda, the United Kingdom's second largest supermarket, with Sainsbury's, the third largest. The two supermarkets combined would have had a market share of 30.7%, and would have become the largest supermarket in the country. In their proposed merger, Asda and Sainsbury's argued that the merger would have allowed them to cut costs with resulting price decreases of as much as 10%. However, the CMA claimed 'we have found this deal would lead to increased prices, reduced quality and choice of products, or a poorer shopping experience for all of their UK wshoppers.'



Figure 7.27: Flint, UK. Asda and Sainsbury's

There is growing concern in the United Kingdom that there is increasing concentration of firms in the domestic economy, where there have been $2 trillion worth of mergers and acquisitions in the past decade. There are similar concerns in the United States that the economy has become too concentrated, limiting competition and reducing welfare (see also Real world focus 7.2). Therefore the CMA's decision on the supermarket merger 'could yet prove to be an important moment in the reform of Britain's over-concentrated economy'.

Source: *www.economist.com*

Applying your skills

1 Use a diagram to explain why the merger of the two supermarkets might have resulted in lower average costs.

2 Explain the theoretical arguments behind the CMA's reasoning that the merger would likely result in higher prices and reduced quality and choice of products.

The case of natural monopoly

While most countries around the world do not encourage monopoly, an exception is made if there is a natural monopoly, because it is not in society's interests to break it up into smaller firms, as this would result in higher average costs and a waste of resources.

Government ownership of natural monopolies

One possible solution to natural monopoly is to nationalise it, which involves transfer of ownership from the private sector to the public sector (the government). Government ownership allows the government to regulate natural monopolies, forcing them to lower prices and increase quantities produced in the interests of consumers, thereby reducing allocative inefficiency and welfare loss.

However government ownership sometimes leads to inefficiencies and higher than necessary costs of production, as governments are not driven by the goal to maximise profits. In general since the 1990s there has been a trend around to world to privatise (the opposite of nationalise) government enterprises.

Government regulation of natural monopolies

Governments usually regulate natural monopolies, to ensure more socially desirable price and quantity outcomes, and this can be done even when the monopoly remains under private ownership. There are two ways this can be done.

REAL WORLD FOCUS 7.5

Fines imposed by the European Union

In 2019 a group of banks (Barclays, J.P. Morgan, MUFG and Royal Bank of Scotland) were fined $1.2 billion for rigging the multi-trillion-dollar foreign exchange market for 11 currencies. The Swiss bank UBS was exempted from the fine because it alerted the existence of two cartels to the European Union.

At the same time another group of banks (Barclays, BNB Paribas, Citigroup, J.P. Morgan, Royal Bank of Scotland and UBS) were fined $2.8 billion by U.S. regulators also for rigging exchange rates.

Source: https://www.cnbc.com/2019/05/16/eu-regulators-fine-five-banks-for-forex-rigging.html



Figure 7.28: Mountain View, California, USA. One of the Google buildings

- In 2019, Google was fined €1.5 for restricting competition by imposing anticompetitive contractual restrictions on third-party websites for over ten years.
- In 2018, Google was fined €5 billion for requiring phone manufacturers to preinstall Google Search and its browser app, Chrome, in order to access Google Play, Google's app store.
- In 2018, Qualcomm was fined $1.2 billion for making 'significant payments' to Apple for using Qualcomm's chips rather than competing chips.
- In 2017, Google was fined €2.7 billion for giving its Google Shopping Service an illegal advantage in the search results.
- In 2009, Intel was fined $1.45 billion for offering customers price reductions if they used Intel's microprocessors instead of its rival AMD.

Sources: *www.theguardian.com; www.businessinsider.com; www.reuters.com;*

Applying your skills

Outline how each of these violations matches the European Commission's definition and examples of abuse of market power (see Section *The meaning of abuse of market power* above).

- **Marginal cost pricing**

The government can force the monopoly to charge a price equal to marginal cost, since with $P = MC$ the monopolist will achieve allocative efficiency, with P falling and Q increasing to the socially desirable level. This is called *marginal cost pricing*.

However, marginal cost pricing leads to losses for the natural monopolist. The reason is that $P = MC$ results in a price that is too low for the firm to be able to cover its average costs. As a result, the firm will either go out of business, otherwise the government would have to subsidise it in order to cover its losses. (This need not occur in a monopoly that is not a natural monopoly.) For these reasons marginal cost pricing is not a popular way to regulate natural monopolies.

(The interested reader may refer to the digital coursebook, illustrating a natural monopoly with marginal cost pricing, as Supplementary material.)

- **Average cost pricing**

To avoid creating losses for the natural monopolist, governments can force the firm to charge a price equal to its average costs, where $P = AC$, meaning that the firm earns normal profit. This is called *average cost pricing*. This results in a higher price and lower quantity than marginal cost pricing. However, it leads to a price and quantity combination that is superior to that of the unregulated monopolist, meaning that price is lower and quantity greater.

Although allocative efficiency is not achieved through average cost pricing, this policy offers two very important advantages: (a) the monopolist makes normal profit and is not in danger of having to shut down; and (b) it is more efficient than the market solution.

Yet, average cost pricing also has disadvantages. A monopolist in a free, unregulated market faces incentives to keep its average costs low, in order to maximise profits. If, through regulation, it is guaranteed a price equal to its average costs, it loses this incentive. If average costs increase due to inefficiency, it will still receive a price covering its costs.

Another possible disadvantage is that the regulated monopoly may continue to survive as a monopoly, even though it may stop being a natural monopoly (if technological improvements change cost conditions, such as in telecommunications). Continued regulation provides protection to the firm from new competitors that would have been able to produce more efficiently.

TEST YOUR UNDERSTANDING 7.17

1 Discuss
   - a potential advantages of large firms with significant market power, and
   - b potential risks in markets dominated by large firms.

2 Outline the meaning of abuse of market power.

3 Discuss advantages and disadvantages of possible government interventions to deal with abuse of market power.

4 a Using a diagram, explain the meaning of natural monopoly.
   - b Discuss policy options for governments to deal with natural monopoly.

INQUIRY AND REFLECTION

The following questions will help you reflect on your learning and enhance your understanding of key topics in this chapter. They are suggestions for inquiries that you can undertake on your own or in groups in order to revise the learning objectives of this chapter.

1 Research and find an example of collusive oligopoly in your country of residence or in any other country of your choice. Try to discover how the firms colluded, how their collusion was discovered, what if any were the fines imposed, and what were the consequences.

2 Research a case of competition policy in a country and industry of your choice where firms have been accused of anti-competitive behaviour. Explain why the behaviour runs against the interests of consumers and society, and how the actions taken by the competition authority would correct the problem.

3 There is a major ongoing debate about market power concentration in the large tech firms like Amazon, Google and Facebook. According to many observers this is excessively high and growing. Research and explain the arguments that favour regulation and legislation to limit the market power of such firms. Explain the possible risks to consumers and society of the rising market power of these firms. Describe the various policies that are being proposed to regulate this industry.

EXAM STYLE QUESTIONS

You can find questions in the style of IB exams in the digital coursebook.